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(54) **BREAST CANCER DETECTION KIT OR
DEVICE, AND DETECTION METHOD**

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(58) **Field of Classification Search**

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(57) **ABSTRACT**

It is intended to provide a kit or a device for the detection of
breast cancer and a method for detecting breast cancer. The
present invention provides a kit or a device for the detection
of breast cancer, comprising nucleic acid(s) capable of
specifically binding to a miRNA in a sample of a subject, and
a method for detecting breast cancer, comprising measuring
the miRNA in vitro.

6 Claims, 4 Drawing Sheets

Specification includes a Sequence Listing.

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Fig. 1

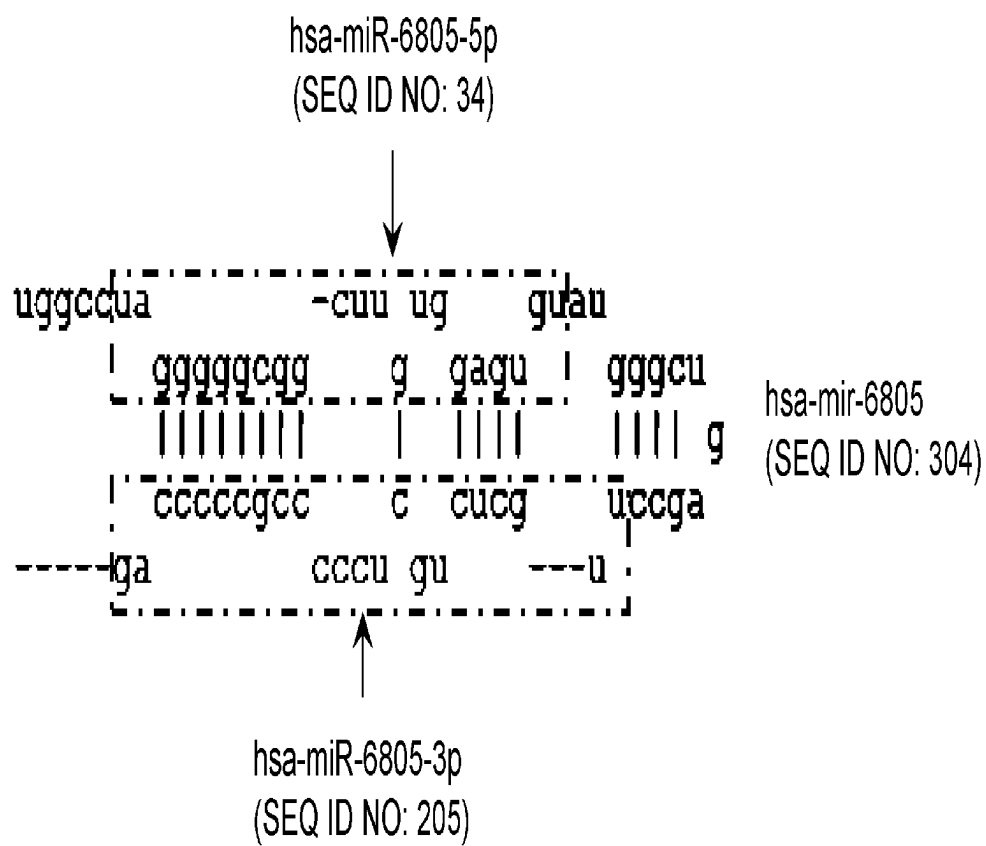


Fig. 2

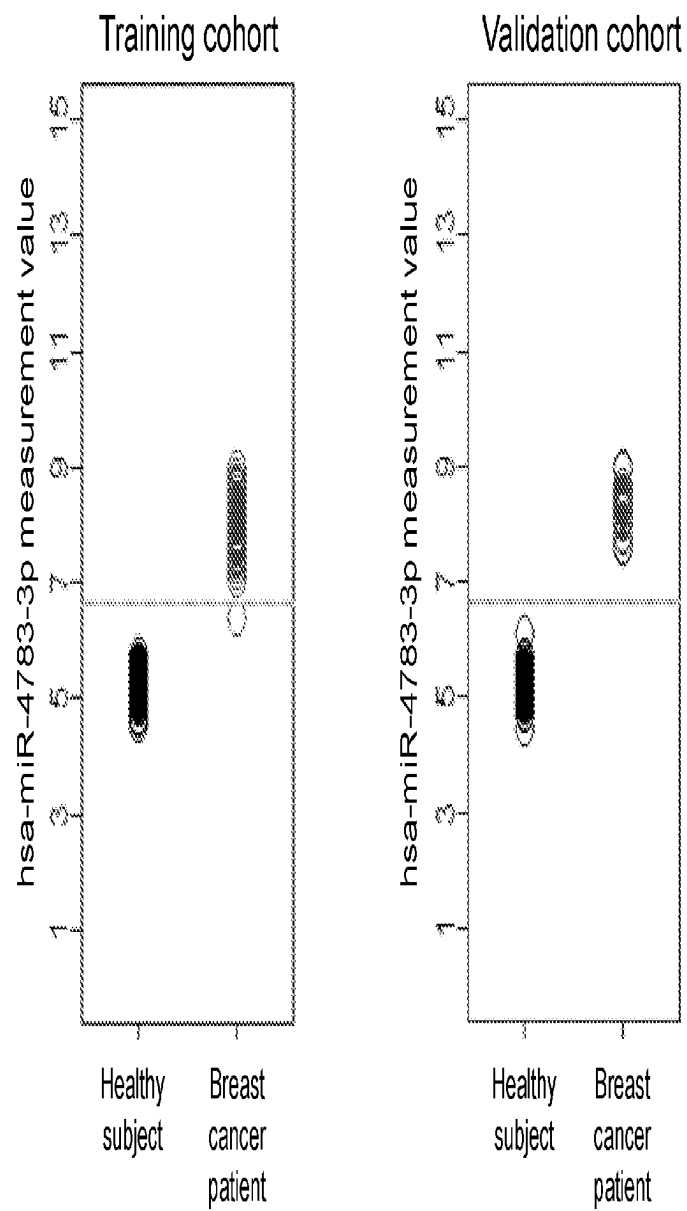


Fig. 3

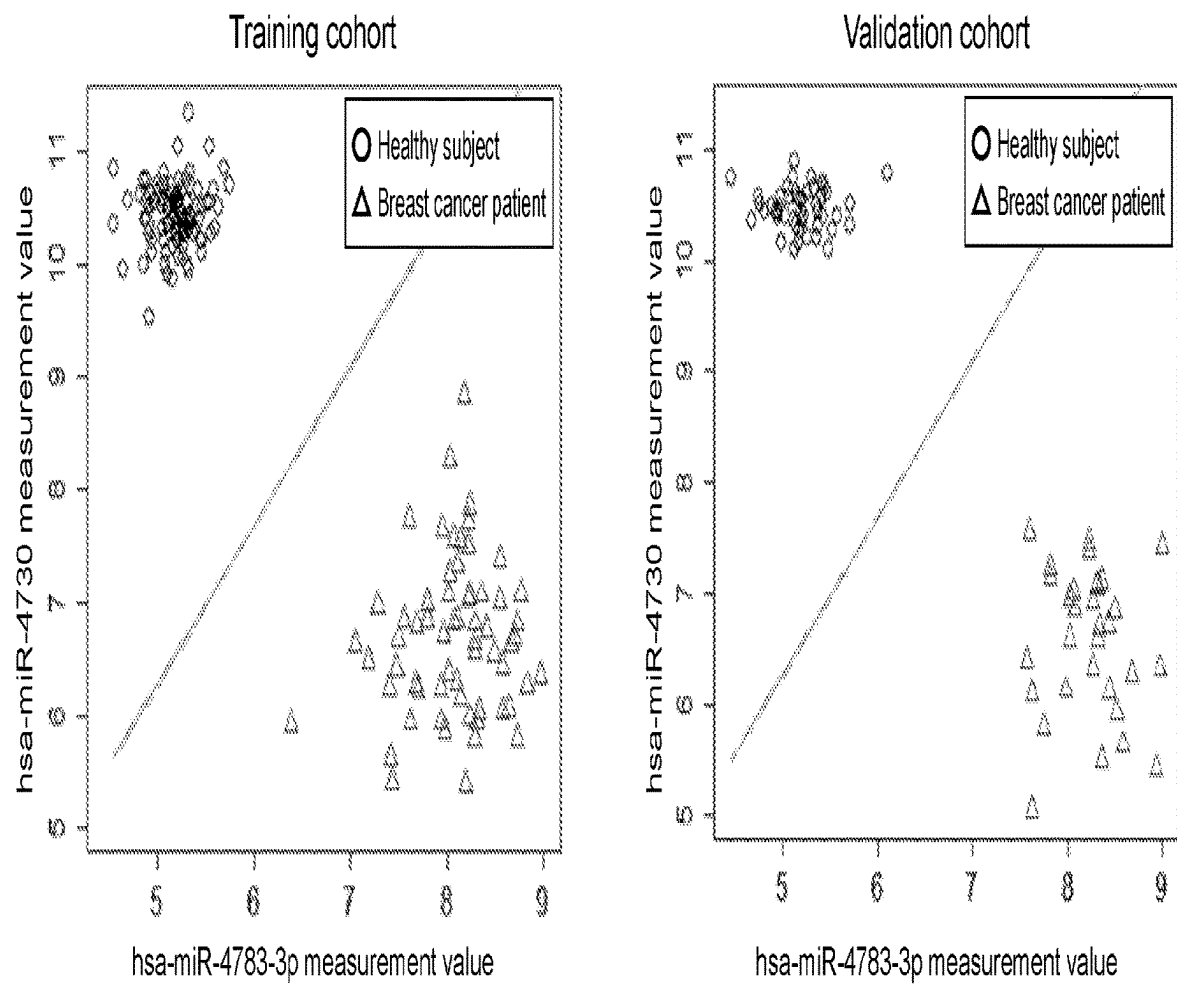
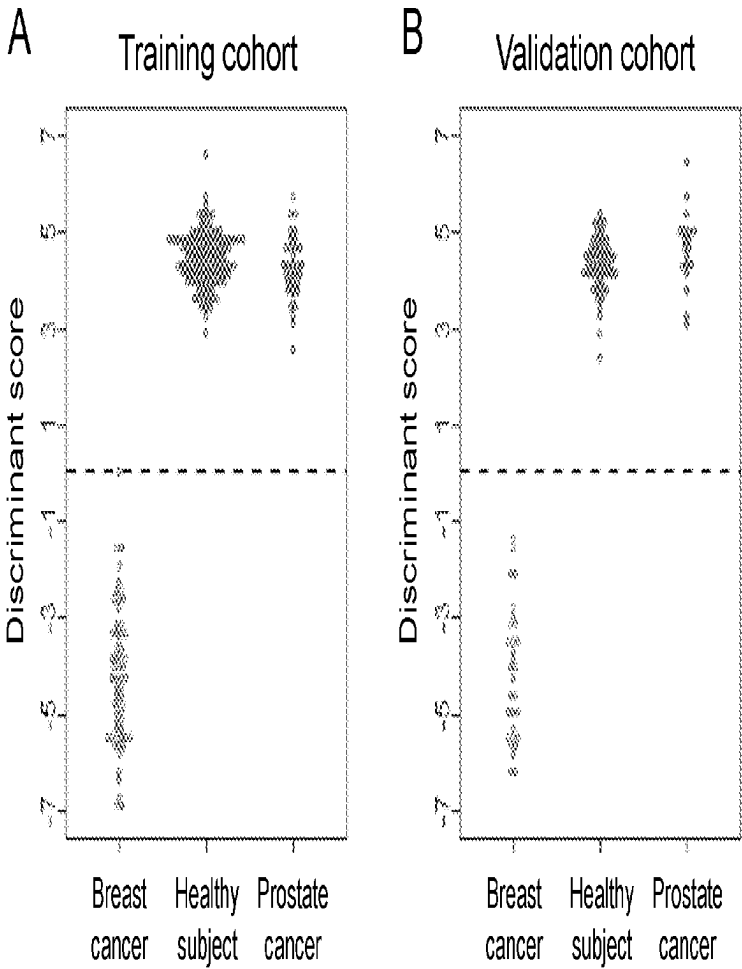


Fig. 4



**BREAST CANCER DETECTION KIT OR
DEVICE, AND DETECTION METHOD****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a Divisional of U.S. application Ser. No. 17/944,779 filed Sep. 14, 2022, which is a Divisional of U.S. application Ser. No. 16/797,625 filed Feb. 21, 2020 (now U.S. Pat. No. 11,479,822), which is a Divisional of U.S. application Ser. No. 15/318,328, filed on Dec. 12, 2016 (now U.S. Pat. No. 10,597,726), which is the National Phase under 35 U.S.C. § 371 of International Application No. PCT/JP2015/066986 filed on Jun. 12, 2015, which claims the benefit under 35 U.S.C. § 119(a) to Japanese Patent Application Nos. 2014-122672 filed Jun. 13, 2014, and 2015-069321 filed Mar. 30, 2015, all of which are hereby expressly incorporated by reference into the present application.

**REFERENCE TO ELECTRONIC SEQUENCE
LISTING**

The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on Sep. 12, 2022, is named "PH-6234-PCT-US-DIV1-DIV1 Sequence Listing ST26" and is 784,115 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present invention relates to a kit or a device for the detection of breast cancer, comprising nucleic acid(s) capable of specifically binding to a particular miRNA, which is used for examining the presence or absence of breast cancer in a subject, and a method for detecting breast cancer, comprising measuring an expression level of the miRNA using the nucleic acid.

BACKGROUND ART

The breast is constituted by a mammary gland which produces mother milk, lobules which arise from the mammary gland, mammary ducts which arise from the lobules and deliver milk, and fat which supports these constituents, etc. Approximately 90% of breast cancer cases originate in the mammary ducts, while approximately 5 to 10% of the breast cancer cases originate in the lobules (Non-Patent Literature 1). According to the 2011 statistics of cancer type-specific mortality in Japan disclosed by the Center for Cancer Control and Information Services, National Cancer Center, the number of breast cancer deaths climbed to 12,731 people. It is estimated that one out of 14 Japanese females will experience breast cancer. The number of incidences of this cancer in females takes the 1st place by cancer type. It is estimated that one out of 8 American females will experience breast cancer. The estimated number of American individuals affected by breast cancer climbed to 232,670 people in 2014, among which approximately 40,000 people reportedly died (Non-Patent Literature 1).

The stages of breast cancer progression are defined in Non-Patent Literature 2 and classified into stages 0, IA, IB, IIA, IIB, IIIA, IIIB, IIIC, and IV according to tumor size, infiltration, lymph node metastasis, distant metastasis, etc.

The 5-year relative survival rate of breast cancer largely depends on the stages of cancer progression and is reportedly 100% for stage 0 and stage I, 93% for stage II, 72% for stage III, and 22% for stage IV (Non-Patent Literature 1). Thus, the early detection of breast cancer leads to improvement in the survival rate. Therefore, an approach that permits the early detection is strongly desired.

The treatment of breast cancer is basically surgical treatment, which is used in combination with drug therapy or radiotherapy depending on the progressed stage, metastasis, general health conditions, and breast cancer classification. Particularly, for early breast cancer of stage 1 or 2, breast conservation therapy may be selected with a combined use with radiotherapy (Non-Patent Literature 1).

According to Non-Patent Literature 1, initial diagnostic tests of breast cancer include inspection and palpation as well as imaging tests such as mammography, which is breast-dedicated X-ray examination, and ultrasonography (echo examination). When there are findings on suspected breast cancer by the initial test, pathological examination which involves inserting a needle into a lesion and collecting cells or tissues to be examined under a microscope, is carried out as a secondary test. If necessary, imaging tests such as CT, MRI, abdominal ultrasonography, bone scintigraphy, and PET are also carried out in order to examine the state or spread of the lesion.

For example, CEA, CA-15-3, and CA27-29 are known as tumor markers for the detection of breast cancer. These tumor markers in blood have been reported to elevate when breast cancer has metastasized to other organs such as the bone or the liver. However, these tumor markers do not elevate in some patients and may thus be limited by their usefulness (Non-Patent Literature 1).

As shown in Patent Literatures 1 to 4, there are reports, albeit at a research stage, on the detection of breast cancer using the expression levels of microRNAs (miRNAs) or combinations of the expression levels of miRNAs and the expression levels of additional protein markers in biological samples including blood.

Specifically, Patent Literature 1 discloses a method for detecting prostate cancer or other cancers including breast cancer by combining hsa-miR-602 or hsa-miR-135a-3p with known protein markers in blood.

Patent Literature 2 discloses a method for detecting various cancers including breast cancer by combining hsa-miR-23b-3p or hsa-miR-135a-3p with 5 or more other miRNAs in blood or tissues.

Patent Literature 3 discloses a method for detecting breast cancer using hsa-miR-92a-3p, hsa-miR-92a-2-5p, hsa-miR-92b-5p, and the like in blood cells.

Patent Literature 4 discloses a method for detecting breast cancer using hsa-miR-451a, hsa-miR-296-5p, hsa-miR-16-5p, and the like in tissues.

Non-Patent Literature 3 discloses that hsa-miR-760 and the like in blood are significantly expressed in breast cancer patients.

Non-Patent Literature 4 discloses that hsa-miR-423-5p, hsa-miR-486-5p, and the like in blood are decreased after surgery of breast cancer.

Non-Patent Literature 5 discloses that hsa-miR-4257, hsa-miR-1915-3p, hsa-miR-718, and the like in blood are significantly expressed in breast cancer patients.

Non-Patent Literature 6 discloses that hsa-miR-940 and the like in blood are significantly expressed in breast cancer patients.

CITATION LIST

Patent Literature

- Patent Literature 1: JP Patent Publication (Kohyo) No. 2012-507300 A (2012)
 Patent Literature 2: JP Patent Publication (Kohyo) No. 2008-500837 A (2008)
 Patent Literature 3: International Publication No. WO 10/123043
 Patent Literature 4: Published U.S. Patent Application No. 2008/0076674

Non-Patent Literature

- Non-Patent Literature 1: American Cancer Society, "Breast Cancer", 2013, p. 6-9, 13, 27-28, 41-46, 52-54, 63-64, and 106
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SUMMARY OF INVENTION

Problem to be Solved by Invention

An object of the present invention is to find novel tumor markers for breast cancer and to provide a method that can effectively detect breast cancer using nucleic acid(s) capable of specifically binding to the markers. As described in Non-Patent Literature 1, initial diagnostic tests of breast cancer include inspection and palpation as well as imaging tests such as mammography, which is breast-dedicated X-ray examination, and ultrasonography. The mammography is reportedly effective as breast cancer examination targeting women aged 40 or older, and the American Cancer Society recommends that women in this age range take mammography every year (Non-Patent Literature 1). The mammography, however, has been reported to have limitations in the visualization of breast cancer present in the dense breast before menopause or a very small tumor of early breast cancer (Non-Patent Literature 1). In Japan, the mammography rate was only 24.3% in 2010, and a challenge to improvement in breast cancer survival rate will be to increase this mammography rate (Non-Patent Literature 7).

For example, CEA, CA-15-3, and CA27-29 mentioned above are known as tumor markers for the detection of breast cancer. These tumor markers, however, are helpful in confirming therapeutic effects on recurrent breast cancer, but rarely elevate in early breast cancer. Therefore, the tumor markers may not be useful for the purpose of breast cancer examination (Non-Patent Literature 1). According to Non-Patent Literature 8, the specific sensitivity of CEA and CA15-3 is uselessly 6.4% and 12.2%, respectively, for stage

1 and is only 25.0% and 62.5%, respectively, even for stage 4. Thus, the tumor marker measurement is less significant as a preoperative test. Since these blood tumor markers may elevate for reasons other than breast cancer, the presence or absence of breast cancer is difficult to determine. The false diagnosis of other cancers as breast cancer wastes appropriate therapeutic opportunity or places unnecessary economical and physical burdens on patients due to the application of wrong medicine.

As described below, there are reports, albeit at a research stage, on the determination of breast cancer using the expression levels of microRNAs (miRNAs) in biological samples including blood, none of which, however, have yet been brought into practical use.

Patent Literature 1 discloses a method for detecting prostate cancer or other cancers including breast cancer by combining hsa-miR-602 or hsa-miR-135a-3p with known protein markers in blood. The measurement of both miRNA and protein markers, however, brings about increase in examination costs and a complicated process and is therefore not favorable. This detection method does not describe specific detection performance such as accuracy, sensitivity, or specificity for determining breast cancer and is thus industrially less practical.

Patent Literature 2 discloses a method for detecting various cancers including breast cancer by combining hsa-miR-23b-3p or hsa-miR-135a-3p with 5 or more other miRNAs in blood or tissues. This detection method does not describe specific detection performance such as accuracy, sensitivity, or specificity for determining breast cancer and is thus industrially less practical.

Patent Literature 3 describes a method for detecting breast cancer using hsa-miR-92a-3p, hsa-miR-92a-2-5p, hsa-miR-92b-5p, and the like. This detection method does not describe specific detection performance such as accuracy, sensitivity, or specificity for determining breast cancer and is thus industrially less practical. In addition, these miRNA markers were not validated in an independent sample group and are therefore less reliable.

Patent Literature 4 discloses a method for detecting breast cancer using hsa-miR-451a, hsa-miR-296-5p, hsa-miR-16-5p, and the like in tissues. For this detection method, however, tissue resection by surgical operation is essential for obtaining samples, and this step places a heavy physical burden on patients. Therefore, this method is not favorable as an examination method. In addition, this detection method does not describe specific detection performance such as accuracy, sensitivity, or specificity for determining breast cancer and is thus industrially less practical.

Non-Patent Literature 3 discloses that hsa-miR-760 and the like in blood are significantly expressed in breast cancer patients. This literature, however, neither describes detection performance such as accuracy, sensitivity, or specificity for determining breast cancer nor describes a specific method for detecting breast cancer. Therefore, this approach is industrially less practical.

Non-Patent Literature 4 discloses that hsa-miR-423-5p, hsa-miR-486-5p, and the like in blood are decreased after surgery of breast cancer. This literature, however, neither describes detection performance such as accuracy, sensitivity, or specificity for determining breast cancer nor describes a specific method for detecting breast cancer. Therefore, this approach is industrially less practical.

Non-Patent Literature 5 discloses that hsa-miR-4257, hsa-miR-1915-3p, hsa-miR-718, and the like in blood are significantly expressed in breast cancer patients. This approach, however, employed as many as 240 miRNAs for

detecting breast cancer and might cause increase in examination cost and complicated discriminant algorithms. Thus, this approach is not industrially practical.

Non-Patent Literature 6 discloses that hsa-miR-940 and the like in blood are significantly expressed in breast cancer patients. The authors, however, concluded that this marker is less reproducible, and finally abandoned the marker in the study. In addition, this literature neither describes detection performance such as accuracy, sensitivity, or specificity for determining breast cancer nor describes a specific method for detecting breast cancer. Therefore, this approach is industrially less practical.

As mentioned above, the existing tumor markers exhibit low performance in the detection of breast cancer, or neither performance nor detection methods are specifically shown as to the markers at a research stage. Therefore, use of these markers might impose implementation of needless extra examination due to the false detection of healthy subjects as being breast cancer patients, or might waste therapeutic opportunity because of overlooking breast cancer patients. In addition, the measurement of dozens of miRNAs increases examination cost and is therefore difficult to use in large-scale screening such as medical checkup. Furthermore, the collection of breast tissues for measuring the tumor markers is highly invasive to patients and is not favorable. Hence, there is a demand for a highly accurate breast cancer marker that is detectable from blood, which can be collected with limitedly invasiveness, and is capable of correctly determining a breast cancer patient as a breast cancer patient and a healthy subject as a healthy subject. Particularly, the early detection and treatment of breast cancer can drastically reduce the risk of recurrence and also permit breast conservation therapy. Therefore, a highly sensitive breast cancer marker capable of detecting breast cancer even at a low progressed stage is desired. Moreover, the mammography rate is presumably increased by providing a more convenient initial screening of breast cancer.

Solution to Problem

The present inventors have conducted diligent studies to attain the object and consequently completed the present invention by finding several genes usable as markers for the detection of breast cancer from blood, which can be collected with limitedly invasiveness, and finding that breast cancer can be significantly detected by using nucleic acids capable of specifically binding to any of these markers.

SUMMARY OF INVENTION

The present invention has the following features:

(1) A kit for the detection of breast cancer, comprising nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of the following breast cancer markers: miR-4783-3p, miR-4730, miR-1307-3p, miR-4634, miR-663a, miR-4532, miR-7704, miR-3178, miR-6729-5p, miR-6090, miR-4732-5p, miR-3184-5p, miR-6727-5p, miR-6088, miR-4674, miR-8073, miR-4787-5p, miR-1469, miR-125a-3p, miR-1233-5p, miR-885-3p, miR-6802-5p, miR-328-5p, miR-6787-5p, miR-8069, miR-6875-5p, miR-1246, miR-4734, miR-6757-5p, miR-6756-5p, miR-3665, miR-6836-3p, miR-6821-5p, miR-6805-5p, miR-4728-5p, miR-6726-5p, miR-197-5p, miR-149-3p, miR-6850-5p, miR-4476, miR-6858-5p, miR-564, miR-4763-3p, miR-575, miR-6771-5p, miR-1231, miR-1908-3p, miR-150-3p, miR-3937, miR-887-3p, miR-3940-5p, miR-4741, miR-6808-5p, miR-6869-5p, miR-

5090, miR-615-5p, miR-8072, miR-128-1-5p, miR-1238-5p, miR-365a-5p, miR-204-3p, miR-4492, miR-6785-5p, miR-6511a-5p, miR-4525, miR-1915-5p, miR-3180, miR-6879-5p, miR-1199-5p, miR-6746-5p, miR-711, miR-663b, miR-4707-3p, miR-6893-5p, miR-4675, miR-4638-5p, miR-4651, miR-6087, miR-4665-5p, miR-4758-5p, miR-6887-5p, miR-3620-5p, miR-1909-3p, miR-7641, miR-6724-5p, miR-1343-3p, miR-6780b-5p, miR-4484, miR-4690-5p, miR-4429, miR-1227-5p, miR-4725-3p, miR-6861-5p, miR-6812-5p, miR-3197, miR-8059, miR-3185, miR-4706, miR-4497, miR-3131, miR-6806-5p, miR-187-5p, miR-3180-3p, miR-6848-5p, miR-6820-5p, miR-6800-5p, miR-6717-5p, miR-6795-5p, miR-4632-5p, miR-665, miR-6778-5p, miR-3663-3p, miR-4689, miR-211-3p, miR-6511b-5p, miR-4750-5p, miR-6126, miR-614, miR-7110-5p, miR-744-5p, miR-6769a-5p, miR-4792, miR-5787, miR-6798-5p, miR-6781-5p, miR-4419b, miR-4446-3p, miR-4259, miR-5572, miR-6075, miR-296-3p, miR-6891-5p, miR-4745-5p, miR-6775-5p, miR-6870-5p, miR-920, miR-4530, miR-6819-5p, miR-6825-5p, miR-7847-3p, miR-6131, miR-4433-3p, miR-1228-5p, miR-6743-5p, miR-1268a, miR-3917, miR-6786-5p, miR-3154, miR-638, miR-6741-5p, miR-6889-5p, miR-6840-3p, miR-6510-5p, miR-3188, miR-551b-5p, miR-5001-5p, miR-1268b, miR-7107-5p, miR-6824-5p, miR-6732-5p, miR-371a-5p, miR-6794-5p, miR-6779-5p, miR-4271, miR-5195-3p, miR-6762-5p, miR-939-5p, miR-1247-3p, miR-6777-5p, miR-6722-3p, miR-3656, miR-4688, miR-3195, miR-6766-5p, miR-4447, miR-4656, miR-7108-5p, miR-3191-3p, miR-1273g-3p, miR-4463, miR-2861, miR-3196, miR-6877-5p, miR-3679-5p, miR-4442, miR-6789-5p, miR-6782-5p, miR-486-3p, miR-6085, miR-4746-3p, miR-619-5p, miR-937-5p, miR-6803-5p, miR-4298, miR-4454, miR-4459, miR-7150, miR-6880-5p, miR-4449, miR-8063, miR-6495-5p, miR-6132, miR-6829-5p, miR-4486, miR-6805-3p, miR-6826-5p, miR-4508, miR-1343-5p, miR-7114-5p, miR-3622a-5p, miR-6765-5p, miR-7845-5p, miR-3960, miR-6749-5p, miR-1260b, miR-6799-5p, miR-4723-5p, miR-6784-5p, miR-5100, miR-6769b-5p, miR-1207-5p, miR-642a-3p, miR-4505, miR-4270, miR-6721-5p, miR-7111-5p, miR-6791-5p, miR-7109-5p, miR-4258, miR-6515-3p, miR-6851-5p, miR-6125, miR-4749-5p, miR-4726-5p, miR-4513, miR-6089, miR-6816-5p, miR-4466, miR-4488, miR-6752-5p and miR-4739.

(2) The kit according to (1), wherein miR-4783-3p is hsa-miR-4783-3p, miR-4730 is hsa-miR-4730, miR-1307-3p is hsa-miR-1307-3p, miR-4634 is hsa-miR-4634, miR-663a is hsa-miR-663a, miR-4532 is hsa-miR-4532, miR-7704 is hsa-miR-7704, miR-3178 is hsa-miR-3178, miR-6729-5p is hsa-miR-6729-5p, miR-6090 is hsa-miR-6090, miR-4732-5p is hsa-miR-4732-5p, miR-3184-5p is hsa-miR-3184-5p, miR-6727-5p is hsa-miR-6727-5p, miR-6088 is hsa-miR-6088, miR-4674 is hsa-miR-4674, miR-8073 is hsa-miR-8073, miR-4787-5p is hsa-miR-4787-5p, miR-1469 is hsa-miR-1469, miR-125a-3p is hsa-miR-125a-3p, miR-1233-5p is hsa-miR-1233-5p, miR-885-3p is hsa-miR-885-3p, miR-6802-5p is hsa-miR-6802-5p, miR-328-5p is hsa-miR-328-5p, miR-6787-5p is hsa-miR-6787-5p, miR-8069 is hsa-miR-8069, miR-6875-5p is hsa-miR-6875-5p, miR-1246 is hsa-miR-1246, miR-4734 is hsa-miR-4734, miR-6757-5p is hsa-miR-6757-5p, miR-6756-5p is hsa-miR-6756-5p, miR-3665 is hsa-miR-3665, miR-6836-3p is hsa-miR-6836-3p, miR-6821-5p is hsa-miR-6821-5p, miR-6805-5p is hsa-miR-6805-5p, miR-4728-5p is hsa-miR-4728-5p, miR-6726-5p is hsa-miR-6726-5p, miR-197-5p is hsa-miR-197-5p, miR-149-3p is hsa-miR-149-3p, miR-6850-5p is hsa-miR-6850-5p, miR-4476 is hsa-miR-4476,

miR-6858-5p is hsa-miR-6858-5p, miR-564 is hsa-miR-564, miR-4763-3p is hsa-miR-4763-3p, miR-575 is hsa-miR-575, miR-6771-5p is hsa-miR-6771-5p, miR-1231 is hsa-miR-1231, miR-1908-3p is hsa-miR-1908-3p, miR-150-3p is hsa-miR-150-3p, miR-3937 is hsa-miR-3937, miR-887-3p is hsa-miR-887-3p, miR-3940-5p is hsa-miR-3940-5p, miR-4741 is hsa-miR-4741, miR-6808-5p is hsa-miR-6808-5p, miR-6869-5p is hsa-miR-6869-5p, miR-5090 is hsa-miR-5090, miR-615-5p is hsa-miR-615-5p, miR-8072 is hsa-miR-8072, miR-128-1-5p is hsa-miR-128-1-5p, miR-1238-5p is hsa-miR-1238-5p, miR-365a-5p is hsa-miR-365a-5p, miR-204-3p is hsa-miR-204-3p, miR-4492 is hsa-miR-4492, miR-6785-5p is hsa-miR-6785-5p, miR-6511a-5p is hsa-miR-6511a-5p, miR-4525 is hsa-miR-4525, miR-1915-5p is hsa-miR-1915-5p, miR-3180 is hsa-miR-3180, miR-6879-5p is hsa-miR-6879-5p, miR-1199-5p is hsa-miR-1199-5p, miR-6746-5p is hsa-miR-6746-5p, miR-711 is hsa-miR-711, miR-663b is hsa-miR-663b, miR-4707-3p is hsa-miR-4707-3p, miR-6893-5p is hsa-miR-6893-5p, miR-4675 is hsa-miR-4675, miR-4638-5p is hsa-miR-4638-5p, miR-4651 is hsa-miR-4651, miR-6087 is hsa-miR-6087, miR-4665-5p is hsa-miR-4665-5p, miR-4758-5p is hsa-miR-4758-5p, miR-6887-5p is hsa-miR-6887-5p, miR-3620-5p is hsa-miR-3620-5p, miR-1909-3p is hsa-miR-1909-3p, miR-7641 is hsa-miR-7641, miR-6724-5p is hsa-miR-6724-5p, miR-1343-3p is hsa-miR-1343-3p, miR-6780b-5p is hsa-miR-6780b-5p, miR-4484 is hsa-miR-4484, miR-4690-5p is hsa-miR-4690-5p, miR-4429 is hsa-miR-4429, miR-1227-5p is hsa-miR-1227-5p, miR-4725-3p is hsa-miR-4725-3p, miR-6861-5p is hsa-miR-6861-5p, miR-6812-5p is hsa-miR-6812-5p, miR-3197 is hsa-miR-3197, miR-8059 is hsa-miR-8059, miR-3185 is hsa-miR-3185, miR-4706 is hsa-miR-4706, miR-4497 is hsa-miR-4497, miR-3131 is hsa-miR-3131, miR-6806-5p is hsa-miR-6806-5p, miR-187-5p is hsa-miR-187-5p, miR-3180-3p is hsa-miR-3180-3p, miR-6848-5p is hsa-miR-6848-5p, miR-6820-5p is hsa-miR-6820-5p, miR-6800-5p is hsa-miR-6800-5p, miR-6717-5p is hsa-miR-6717-5p, miR-6795-5p is hsa-miR-6795-5p, miR-4632-5p is hsa-miR-4632-5p, miR-665 is hsa-miR-665, miR-6778-5p is hsa-miR-6778-5p, miR-3663-3p is hsa-miR-3663-3p, miR-4689 is hsa-miR-4689, miR-211-3p is hsa-miR-211-3p, miR-6511b-5p is hsa-miR-6511b-5p, miR-4750-5p is hsa-miR-4750-5p, miR-6126 is hsa-miR-6126, miR-614 is hsa-miR-614, miR-7110-5p is hsa-miR-7110-5p, miR-744-5p is hsa-miR-744-5p, miR-6769a-5p is hsa-miR-6769a-5p, miR-4792 is hsa-miR-4792, miR-5787 is hsa-miR-5787, miR-6798-5p is hsa-miR-6798-5p, miR-6781-5p is hsa-miR-6781-5p, miR-4419b is hsa-miR-4419b, miR-4446-3p is hsa-miR-4446-3p, miR-4259 is hsa-miR-4259, miR-5572 is hsa-miR-5572, miR-6075 is hsa-miR-6075, miR-296-3p is hsa-miR-296-3p, miR-6891-5p is hsa-miR-6891-5p, miR-4745-5p is hsa-miR-4745-5p, miR-6775-5p is hsa-miR-6775-5p, miR-6870-5p is hsa-miR-6870-5p, miR-920 is hsa-miR-920, miR-4530 is hsa-miR-4530, miR-6819-5p is hsa-miR-6819-5p, miR-6825-5p is hsa-miR-6825-5p, miR-7847-3p is hsa-miR-7847-3p, miR-6131 is hsa-miR-6131, miR-4433-3p is hsa-miR-4433-3p, miR-1228-5p is hsa-miR-1228-5p, miR-6743-5p is hsa-miR-6743-5p, miR-1268a is hsa-miR-1268a, miR-3917 is hsa-miR-3917, miR-6786-5p is hsa-miR-6786-5p, miR-3154 is hsa-miR-3154, miR-638 is hsa-miR-638, miR-6741-5p is hsa-miR-6741-5p, miR-6889-5p is hsa-miR-6889-5p, miR-6840-3p is hsa-miR-6840-3p, miR-6510-5p is hsa-miR-6510-5p, miR-3188 is hsa-miR-3188, miR-551b-5p is hsa-miR-551b-5p, miR-5001-5p is hsa-miR-5001-5p, miR-1268b is hsa-miR-1268b, miR-7107-5p is hsa-miR-7107-5p, miR-6824-5p is hsa-miR-6824-5p,

miR-6732-5p is hsa-miR-6732-5p, miR-371a-5p is hsa-miR-371a-5p, miR-6794-5p is hsa-miR-6794-5p, miR-6779-5p is hsa-miR-6779-5p, miR-4271 is hsa-miR-4271, miR-5195-3p is hsa-miR-5195-3p, miR-6762-5p is hsa-miR-6762-5p, miR-939-5p is hsa-miR-939-5p, miR-1247-3p is hsa-miR-1247-3p, miR-6777-5p is hsa-miR-6777-5p, miR-6722-3p is hsa-miR-6722-3p, miR-3656 is hsa-miR-3656, miR-4688 is hsa-miR-4688, miR-3195 is hsa-miR-3195, miR-6766-5p is hsa-miR-6766-5p, miR-4447 is hsa-miR-4447, miR-4656 is hsa-miR-4656, miR-7108-5p is hsa-miR-7108-5p, miR-3191-3p is hsa-miR-3191-3p, miR-1273g-3p is hsa-miR-1273g-3p, miR-4463 is hsa-miR-4463, miR-2861 is hsa-miR-2861, miR-3196 is hsa-miR-3196, miR-6877-5p is hsa-miR-6877-5p, miR-3679-5p is hsa-miR-3679-5p, miR-4442 is hsa-miR-4442, miR-6789-5p is hsa-miR-6789-5p, miR-6782-5p is hsa-miR-6782-5p, miR-486-3p is hsa-miR-486-3p, miR-6085 is hsa-miR-6085, miR-4746-3p is hsa-miR-4746-3p, miR-619-5p is hsa-miR-619-5p, miR-937-5p is hsa-miR-937-5p, miR-6803-5p is hsa-miR-6803-5p, miR-4298 is hsa-miR-4298, miR-4454 is hsa-miR-4454, miR-4459 is hsa-miR-4459, miR-7150 is hsa-miR-7150, miR-6880-5p is hsa-miR-6880-5p, miR-4449 is hsa-miR-4449, miR-8063 is hsa-miR-8063, miR-4695-5p is hsa-miR-4695-5p, miR-6132 is hsa-miR-6132, miR-6829-5p is hsa-miR-6829-5p, miR-4486 is hsa-miR-4486, miR-6805-3p is hsa-miR-6805-3p, miR-6826-5p is hsa-miR-6826-5p, miR-4508 is hsa-miR-4508, miR-1343-5p is hsa-miR-1343-5p, miR-7114-5p is hsa-miR-7114-5p, miR-3622a-5p is hsa-miR-3622a-5p, miR-6765-5p is hsa-miR-6765-5p, miR-7845-5p is hsa-miR-7845-5p, miR-3960 is hsa-miR-3960, miR-6749-5p is hsa-miR-6749-5p, miR-1260b is hsa-miR-1260b, miR-6799-5p is hsa-miR-6799-5p, miR-4723-5p is hsa-miR-4723-5p, miR-6784-5p is hsa-miR-6784-5p, miR-5100 is hsa-miR-5100, miR-6769b-5p is hsa-miR-6769b-5p, miR-1207-5p is hsa-miR-1207-5p, miR-642a-3p is hsa-miR-642a-3p, miR-4505 is hsa-miR-4505, miR-4270 is hsa-miR-4270, miR-6721-5p is hsa-miR-6721-5p, miR-7111-5p is hsa-miR-7111-5p, miR-6791-5p is hsa-miR-6791-5p, miR-7109-5p is hsa-miR-7109-5p, miR-4258 is hsa-miR-4258, miR-6515-3p is hsa-miR-6515-3p, miR-6851-5p is hsa-miR-6851-5p, miR-6125 is hsa-miR-6125, miR-4749-5p is hsa-miR-4749-5p, miR-4726-5p is hsa-miR-4726-5p, miR-4513 is hsa-miR-4513, miR-6089 is hsa-miR-6089, miR-6816-5p is hsa-miR-6816-5p, miR-4466 is hsa-miR-4466, miR-4488 is hsa-miR-4488, miR-6752-5p is hsa-miR-6752-5p, and miR-4739 is hsa-miR-4739.

(3) The kit according to (1) or (2), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (a) to (e):

- (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (b) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t,
- (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,

(d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and

(e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

(4) The kit according to any of (1) to (3), wherein the kit further comprises nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of other breast cancer markers miR-760, miR-602, miR-423-5p, miR-92a-2-5p, miR-16-5p, miR-451a, miR-135a-3p, miR-486-5p, miR-4257, miR-92b-5p, miR-1915-3p, miR-718, miR-940, miR-296-5p, miR-23b-3p and miR-92a-3p.

(5) The kit according to (4), wherein miR-760 is hsa-miR-760, miR-602 is hsa-miR-602, miR-423-5p is hsa-miR-423-5p, miR-92a-2-5p is hsa-miR-92a-2-5p, miR-16-5p is hsa-miR-16-5p, miR-451a is hsa-miR-451a, miR-135a-3p is hsa-miR-135a-3p, miR-486-5p is hsa-miR-486-5p, miR-4257 is hsa-miR-4257, miR-92b-5p is hsa-miR-92b-5p, miR-1915-3p is hsa-miR-1915-3p, miR-718 is hsa-miR-718, miR-940 is hsa-miR-940, miR-296-5p is hsa-miR-296-5p, miR-23b-3p is hsa-miR-23b-3p, and miR-92a-3p is hsa-miR-92a-3p.

(6) The kit according to (4) or (5), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (f) to (j):

(f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,

(g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251,

(h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,

(i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and

(j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

(7) The kit according to any of (1) to (6), wherein the kit further comprises nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of other breast cancer markers miR-658, miR-6842-5p, miR-6124, miR-6765-3p, miR-7106-5p, miR-4534, miR-92b-3p, miR-3135b, miR-4687-3p, miR-762, miR-3619-3p, miR-4467, miR-557, miR-1237-5p, miR-1908-5p, miR-4286, miR-6885-5p and miR-6763-5p.

(8) The kit according to (7), wherein miR-658 is hsa-miR-658, miR-6842-5p is hsa-miR-6842-5p, miR-6124 is hsa-miR-6124, miR-6765-3p is hsa-miR-6765-3p, miR-7106-5p is hsa-miR-7106-5p, miR-4534 is hsa-miR-4534, miR-92b-3p is hsa-miR-92b-3p, miR-3135b is hsa-miR-3135b, miR-4687-3p is hsa-miR-4687-3p, miR-762 is hsa-miR-762, miR-3619-3p is hsa-miR-3619-3p, miR-4467 is hsa-miR-4467, miR-557 is hsa-miR-557, miR-1237-5p is hsa-miR-1237-5p, miR-1908-5p is hsa-miR-1908-5p, miR-

4286 is hsa-miR-4286, miR-6885-5p is hsa-miR-6885-5p, and miR-6763-5p is hsa-miR-6763-5p.

(9) The kit according to (7) or (8), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (k) to (o):

(k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,

(l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269,

(m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,

(n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and

(o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).

(10) The kit according to any of (1) to (9), wherein the kit comprises at least two nucleic acids capable of specifically binding to at least two polynucleotides, respectively, selected from the group consisting of all of the breast cancer markers according to (1) or (2).

(11) A device for the detection of breast cancer, comprising nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of the following breast cancer markers: miR-4783-3p, miR-4730, miR-1307-3p, miR-4634, miR-663a, miR-4532, miR-7704, miR-3178, miR-6729-5p, miR-6090, miR-4732-5p, miR-3184-5p, miR-6727-5p, miR-6088, miR-4674, miR-8073, miR-4787-5p, miR-1469, miR-125a-3p, miR-1233-5p, miR-885-3p, miR-6802-5p, miR-328-5p, miR-6787-5p, miR-8069, miR-6875-5p, miR-1246, miR-4734, miR-6757-5p, miR-6756-5p, miR-3665, miR-6836-3p, miR-6821-5p, miR-6805-5p, miR-4728-5p, miR-6726-5p, miR-197-5p, miR-149-3p, miR-6850-5p, miR-4476, miR-6858-5p, miR-564, miR-4763-3p, miR-575, miR-6771-5p, miR-1231, miR-1908-3p, miR-150-3p, miR-3937, miR-887-3p, miR-3940-5p, miR-4741, miR-6808-5p, miR-6869-5p, miR-5090, miR-615-5p, miR-8072, miR-128-1-5p, miR-1238-5p, miR-365a-5p, miR-204-3p, miR-4492, miR-6785-5p, miR-6511a-5p, miR-4525, miR-1915-5p, miR-3180, miR-6879-5p, miR-1199-5p, miR-6746-5p, miR-711, miR-663b, miR-4707-3p, miR-6893-5p, miR-4675, miR-4638-5p, miR-4651, miR-6087, miR-4665-5p, miR-4758-5p, miR-6887-5p, miR-3620-5p, miR-1909-3p, miR-7641, miR-6724-5p, miR-1343-3p, miR-6780b-5p, miR-4484, miR-4690-5p, miR-4429, miR-1227-5p, miR-4725-3p, miR-6861-5p, miR-6812-5p, miR-3197, miR-8059, miR-3185, miR-4706, miR-4497, miR-3131, miR-6806-5p, miR-187-5p, miR-3180-3p, miR-6848-5p, miR-6820-5p, miR-6800-5p, miR-6717-5p, miR-6795-5p, miR-4632-5p, miR-665, miR-6778-5p, miR-3663-3p, miR-4689, miR-211-3p, miR-6511b-5p, miR-4750-5p, miR-6126, miR-614, miR-7110-5p, miR-744-5p, miR-6769a-5p, miR-4792, miR-5787, miR-6798-5p, miR-6781-5p, miR-4419b, miR-4446-3p, miR-4259, miR-5572, miR-6075, miR-296-3p, miR-6891-5p, miR-4745-5p, miR-6775-5p, miR-6870-5p, miR-920,

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miR-4530, miR-6819-5p, miR-6825-5p, miR-7847-3p, miR-6131, miR-4433-3p, miR-1228-5p, miR-6743-5p, miR-1268a, miR-3917, miR-6786-5p, miR-3154, miR-638, miR-6741-5p, miR-6889-5p, miR-6840-3p, miR-6510-5p, miR-3188, miR-551b-5p, miR-5001-5p, miR-1268b, miR-7107-5p, miR-6824-5p, miR-6732-5p, miR-371a-5p, miR-6794-5p, miR-6779-5p, miR-4271, miR-5195-3p, miR-6762-5p, miR-939-5p, miR-1247-3p, miR-6777-5p, miR-6722-3p, miR-3656, miR-4688, miR-3195, miR-6766-5p, miR-4447, miR-4656, miR-7108-5p, miR-3191-3p, miR-1273g-3p, miR-4463, miR-2861, miR-3196, miR-6877-5p, miR-3679-5p, miR-4442, miR-6789-5p, miR-6782-5p, miR-486-3p, miR-6085, miR-4746-3p, miR-619-5p, miR-937-5p, miR-6803-5p, miR-4298, miR-4454, miR-4459, miR-7150, miR-6880-5p, miR-4449, miR-8063, miR-4695-5p, miR-6132, miR-6829-5p, miR-4486, miR-6805-3p, miR-6826-5p, miR-4508, miR-1343-5p, miR-7114-5p, miR-3622a-5p, miR-6765-5p, miR-7845-5p, miR-3960, miR-6749-5p, miR-1260b, miR-6799-5p, miR-4723-5p, miR-6784-5p, miR-5100, miR-6769b-5p, miR-1207-5p, miR-642a-3p, miR-4505, miR-4270, miR-6721-5p, miR-7111-5p, miR-6791-5p, miR-7109-5p, miR-4258, miR-6515-3p, miR-6851-5p, miR-6125, miR-4749-5p, miR-4726-5p, miR-4513, miR-6089, miR-6816-5p, miR-4466, miR-4488, miR-6752-5p and miR-4739.

(12) The device according to (11), wherein miR-4783-3p is hsa-miR-4783-3p, miR-4730 is hsa-miR-4730, miR-1307-3p is hsa-miR-1307-3p, miR-4634 is hsa-miR-4634, miR-663a is hsa-miR-663a, miR-4532 is hsa-miR-4532, miR-7704 is hsa-miR-7704, miR-3178 is hsa-miR-3178, miR-6729-5p is hsa-miR-6729-5p, miR-6090 is hsa-miR-6090, miR-4732-5p is hsa-miR-4732-5p, miR-3184-5p is hsa-miR-3184-5p, miR-6727-5p is hsa-miR-6727-5p, miR-6088 is hsa-miR-6088, miR-4674 is hsa-miR-4674, miR-8073 is hsa-miR-8073, miR-4787-5p is hsa-miR-4787-5p, miR-1469 is hsa-miR-1469, miR-125a-3p is hsa-miR-125a-3p, miR-1233-5p is hsa-miR-1233-5p, miR-885-3p is hsa-miR-885-3p, miR-6802-5p is hsa-miR-6802-5p, miR-328-5p is hsa-miR-328-5p, miR-6787-5p is hsa-miR-6787-5p, miR-8069 is hsa-miR-8069, miR-6875-5p is hsa-miR-6875-5p, miR-1246 is hsa-miR-1246, miR-4734 is hsa-miR-4734, miR-6757-5p is hsa-miR-6757-5p, miR-6756-5p is hsa-miR-6756-5p, miR-3665 is hsa-miR-3665, miR-6836-3p is hsa-miR-6836-3p, miR-6821-5p is hsa-miR-6821-5p, miR-6805-5p is hsa-miR-6805-5p, miR-4728-5p is hsa-miR-4728-5p, miR-6726-5p is hsa-miR-6726-5p, miR-197-5p is hsa-miR-197-5p, miR-149-3p is hsa-miR-149-3p, miR-6850-5p is hsa-miR-6850-5p, miR-4476 is hsa-miR-4476, miR-6858-5p is hsa-miR-6858-5p, miR-564 is hsa-miR-564, miR-4763-3p is hsa-miR-4763-3p, miR-575 is hsa-miR-575, miR-6771-5p is hsa-miR-6771-5p, miR-1231 is hsa-miR-1231, miR-1908-3p is hsa-miR-1908-3p, miR-150-3p is hsa-miR-150-3p, miR-3937 is hsa-miR-3937, miR-887-3p is hsa-miR-887-3p, miR-3940-5p is hsa-miR-3940-5p, miR-4741 is hsa-miR-4741, miR-6808-5p is hsa-miR-6808-5p, miR-6869-5p is hsa-miR-6869-5p, miR-5090 is hsa-miR-5090, miR-615-5p is hsa-miR-615-5p, miR-8072 is hsa-miR-8072, miR-128-1-5p is hsa-miR-128-1-5p, miR-1238-5p is hsa-miR-1238-5p, miR-365a-5p is hsa-miR-365a-5p, miR-204-3p is hsa-miR-204-3p, miR-4492 is hsa-miR-4492, miR-6785-5p is hsa-miR-6785-5p, miR-6511a-5p is hsa-miR-6511a-5p, miR-4525 is hsa-miR-4525, miR-1915-5p is hsa-miR-1915-5p, miR-3180 is hsa-miR-3180, miR-6879-5p is hsa-miR-6879-5p, miR-1199-5p is hsa-miR-1199-5p, miR-6746-5p is hsa-miR-6746-5p, miR-711 is hsa-miR-711, miR-663b is hsa-miR-663b, miR-4707-3p is hsa-miR-4707-3p, miR-6893-5p is hsa-miR-6893-5p,

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miR-4675 is hsa-miR-4675, miR-4638-5p is hsa-miR-4638-5p, miR-4651 is hsa-miR-4651, miR-6087 is hsa-miR-6087, miR-4665-5p is hsa-miR-4665-5p, miR-4758-5p is hsa-miR-4758-5p, miR-6887-5p is hsa-miR-6887-5p, miR-3620-5p is hsa-miR-3620-5p, miR-1909-3p is hsa-miR-1909-3p, miR-7641 is hsa-miR-7641, miR-6724-5p is hsa-miR-6724-5p, miR-1343-3p is hsa-miR-1343-3p, miR-6780b-5p is hsa-miR-6780b-5p, miR-4484 is hsa-miR-4484, miR-4690-5p is hsa-miR-4690-5p, miR-4429 is hsa-miR-4429, miR-1227-5p is hsa-miR-1227-5p, miR-4725-3p is hsa-miR-4725-3p, miR-6861-5p is hsa-miR-6861-5p, miR-6812-5p is hsa-miR-6812-5p, miR-3197 is hsa-miR-3197, miR-8059 is hsa-miR-8059, miR-3185 is hsa-miR-3185, miR-4706 is hsa-miR-4706, miR-4497 is hsa-miR-4497, miR-3131 is hsa-miR-3131, miR-6806-5p is hsa-miR-6806-5p, miR-187-5p is hsa-miR-187-5p, miR-3180-3p is hsa-miR-3180-3p, miR-6848-5p is hsa-miR-6848-5p, miR-6820-5p is hsa-miR-6820-5p, miR-6800-5p is hsa-miR-6800-5p, miR-6717-5p is hsa-miR-6717-5p, miR-6795-5p is hsa-miR-6795-5p, miR-4632-5p is hsa-miR-4632-5p, miR-665 is hsa-miR-665, miR-6778-5p is hsa-miR-6778-5p, miR-3663-3p is hsa-miR-3663-3p, miR-4689 is hsa-miR-4689, miR-211-3p is hsa-miR-211-3p, miR-6511b-5p is hsa-miR-6511b-5p, miR-4750-5p is hsa-miR-4750-5p, miR-6126 is hsa-miR-6126, miR-614 is hsa-miR-614, miR-7110-5p is hsa-miR-7110-5p, miR-744-5p is hsa-miR-744-5p, miR-6769a-5p is hsa-miR-6769a-5p, miR-4792 is hsa-miR-4792, miR-5787 is hsa-miR-5787, miR-6798-5p is hsa-miR-6798-5p, miR-6781-5p is hsa-miR-6781-5p, miR-4419b is hsa-miR-4419b, miR-4446-3p is hsa-miR-4446-3p, miR-4259 is hsa-miR-4259, miR-5572 is hsa-miR-5572, miR-6075 is hsa-miR-6075, miR-296-3p is hsa-miR-296-3p, miR-6891-5p is hsa-miR-6891-5p, miR-4745-5p is hsa-miR-4745-5p, miR-6775-5p is hsa-miR-6775-5p, miR-6870-5p is hsa-miR-6870-5p, miR-920 is hsa-miR-920, miR-4530 is hsa-miR-4530, miR-6819-5p is hsa-miR-6819-5p, miR-6825-5p is hsa-miR-6825-5p, miR-7847-3p is hsa-miR-7847-3p, miR-6131 is hsa-miR-6131, miR-4433-3p is hsa-miR-4433-3p, miR-1228-5p is hsa-miR-1228-5p, miR-6743-5p is hsa-miR-6743-5p, miR-1268a is hsa-miR-1268a, miR-3917 is hsa-miR-3917, miR-6786-5p is hsa-miR-6786-5p, miR-3154 is hsa-miR-3154, miR-638 is hsa-miR-638, miR-6741-5p is hsa-miR-6741-5p, miR-6889-5p is hsa-miR-6889-5p, miR-6840-3p is hsa-miR-6840-3p, miR-6510-5p is hsa-miR-6510-5p, miR-3188 is hsa-miR-3188, miR-551b-5p is hsa-miR-551b-5p, miR-5001-5p is hsa-miR-5001-5p, miR-1268b is hsa-miR-1268b, miR-7107-5p is hsa-miR-7107-5p, miR-6824-5p is hsa-miR-6824-5p, miR-6732-5p is hsa-miR-6732-5p, miR-371a-5p is hsa-miR-371a-5p, miR-6794-5p is hsa-miR-6794-5p, miR-6779-5p is hsa-miR-6779-5p, miR-4271 is hsa-miR-4271, miR-5195-3p is hsa-miR-5195-3p, miR-6762-5p is hsa-miR-6762-5p, miR-939-5p is hsa-miR-939-5p, miR-1247-3p is hsa-miR-1247-3p, miR-6777-5p is hsa-miR-6777-5p, miR-6722-3p is hsa-miR-6722-3p, miR-3656 is hsa-miR-3656, miR-4688 is hsa-miR-4688, miR-3195 is hsa-miR-3195, miR-6766-5p is hsa-miR-6766-5p, miR-4447 is hsa-miR-4447, miR-4656 is hsa-miR-4656, miR-7108-5p is hsa-miR-7108-5p, miR-3191-3p is hsa-miR-3191-3p, miR-1273g-3p is hsa-miR-1273g-3p, miR-4463 is hsa-miR-4463, miR-2861 is hsa-miR-2861, miR-3196 is hsa-miR-3196, miR-6877-5p is hsa-miR-6877-5p, miR-3679-5p is hsa-miR-3679-5p, miR-4442 is hsa-miR-4442, miR-6789-5p is hsa-miR-6789-5p, miR-6782-5p is hsa-miR-6782-5p, miR-486-3p is hsa-miR-486-3p, miR-6085 is hsa-miR-6085, miR-4746-3p is hsa-miR-4746-3p, miR-619-5p is hsa-miR-619-5p, miR-937-5p is hsa-miR-937-5p, miR-6803-5p is

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hsa-miR-6803-5p, miR-4298 is hsa-miR-4298, miR-4454 is hsa-miR-4454, miR-4459 is hsa-miR-4459, miR-7150 is hsa-miR-7150, miR-6880-5p is hsa-miR-6880-5p, miR-4449 is hsa-miR-4449, miR-8063 is hsa-miR-8063, miR-4695-5p is hsa-miR-4695-5p, miR-6132 is hsa-miR-6132, miR-6829-5p is hsa-miR-6829-5p, miR-4486 is hsa-miR-4486, miR-6805-3p is hsa-miR-6805-3p, miR-6826-5p is hsa-miR-6826-5p, miR-4508 is hsa-miR-4508, miR-1343-5p is hsa-miR-1343-5p, miR-7114-5p is hsa-miR-7114-5p, miR-3622a-5p is hsa-miR-3622a-5p, miR-6765-5p is hsa-miR-6765-5p, miR-7845-5p is hsa-miR-7845-5p, miR-3960 is hsa-miR-3960, miR-6749-5p is hsa-miR-6749-5p, miR-1260b is hsa-miR-1260b, miR-6799-5p is hsa-miR-6799-5p, miR-4723-5p is hsa-miR-4723-5p, miR-6784-5p is hsa-miR-6784-5p, miR-5100 is hsa-miR-5100, miR-6769b-5p is hsa-miR-6769b-5p, miR-1207-5p is hsa-miR-1207-5p, miR-642a-3p is hsa-miR-642a-3p, miR-4505 is hsa-miR-4505, miR-4270 is hsa-miR-4270, miR-6721-5p is hsa-miR-6721-5p, miR-7111-5p is hsa-miR-7111-5p, miR-6791-5p is hsa-miR-6791-5p, miR-7109-5p is hsa-miR-7109-5p, miR-4258 is hsa-miR-4258, miR-6515-3p is hsa-miR-6515-3p, miR-6851-5p is hsa-miR-6851-5p, miR-6125 is hsa-miR-6125, miR-4749-5p is hsa-miR-4749-5p, miR-4726-5p is hsa-miR-4726-5p, miR-4513 is hsa-miR-4513, miR-6089 is hsa-miR-6089, miR-6816-5p is hsa-miR-6816-5p, miR-4466 is hsa-miR-4466, miR-4488 is hsa-miR-4488, miR-6752-5p is hsa-miR-6752-5p, and miR-4739 is hsa-miR-4739.

(13) The device according to (11) or (12), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (a) to (e):

- (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (b) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t,
- (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

(14) The device according to any of (11) to (13), wherein the device further comprises nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of other breast cancer markers miR-760, miR-602, miR-423-5p, miR-92a-2-5p, miR-16-5p, miR-451a, miR-135a-3p, miR-486-5p, miR-4257, miR-92b-5p, miR-1915-3p, miR-718, miR-940, miR-296-5p, miR-23b-3p and miR-92a-3p.

(15) The device according to (14), wherein miR-760 is hsa-miR-760, miR-602 is hsa-miR-602, miR-423-5p is hsa-miR-423-5p, miR-92a-2-5p is hsa-miR-92a-2-5p, miR-16-5p is hsa-miR-16-5p, miR-451a is hsa-miR-451a, miR-

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135a-3p is hsa-miR-135a-3p, miR-486-5p is hsa-miR-486-5p, miR-4257 is hsa-miR-4257, miR-92b-5p is hsa-miR-92b-5p, miR-1915-3p is hsa-miR-1915-3p, miR-718 is hsa-miR-718, miR-940 is hsa-miR-940, miR-296-5p is hsa-miR-296-5p, miR-23b-3p is hsa-miR-23b-3p, and miR-92a-3p is hsa-miR-92a-3p.

(16) The device according to (14) or (15), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (f) to (j):

- (f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251,
- (h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

(17) The device according to any of (11) to (16), wherein the device further comprises nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of other breast cancer markers miR-658, miR-6842-5p, miR-6124, miR-6765-3p, miR-7106-5p, miR-4534, miR-92b-3p, miR-3135b, miR-4687-3p, miR-762, miR-3619-3p, miR-4467, miR-557, miR-1237-5p, miR-1908-5p, miR-4286, miR-6885-5p and miR-6763-5p.

(18) The device according to (17), wherein miR-658 is hsa-miR-658, miR-6842-5p is hsa-miR-6842-5p, miR-6124 is hsa-miR-6124, miR-6765-3p is hsa-miR-6765-3p, miR-7106-5p is hsa-miR-7106-5p, miR-4534 is hsa-miR-4534, miR-92b-3p is hsa-miR-92b-3p, miR-3135b is hsa-miR-3135b, miR-4687-3p is hsa-miR-4687-3p, miR-762 is hsa-miR-762, miR-3619-3p is hsa-miR-3619-3p, miR-4467 is hsa-miR-4467, miR-557 is hsa-miR-557, miR-1237-5p is hsa-miR-1237-5p, miR-1908-5p is hsa-miR-1908-5p, miR-4286 is hsa-miR-4286, miR-6885-5p is hsa-miR-6885-5p, and miR-6763-5p is hsa-miR-6763-5p.

(19) The device according to (17) or (18), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (k) to (o):

- (k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269,
- (m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the

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- replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).
- (20) The device according to any one of (11) to (19), wherein the device is for measurement based on a hybridization technique.
- (21) The device according to (20), wherein the hybridization technique is a nucleic acid array technique.
- (22) The device according to any one of (11) to (21), wherein the device comprises at least two nucleic acids capable of specifically binding to at least two polynucleotides, respectively, selected from all of the breast cancer markers according to (11) or (12).
- (23) A method for detecting breast cancer, comprising measuring an expression level(s) of a target nucleic acid(s) in a sample from a subject using a kit according to any one of (1) to (10) or a device according to any one of (11) to (22), and evaluating in vitro whether or not the subject has breast cancer using both of the measured expression level(s) and a control expression level(s) in a sample from a healthy subject measured in the same way.
- (24) The method according to (23), wherein the subject is a human.
- (25) The method according to (23) or (24), wherein the sample is blood, serum, or plasma.

DEFINITION OF TERMS

The terms used herein are defined as described below.

Abbreviations or terms such as nucleotide, polynucleotide, DNA, and RNA used herein abide by "Guidelines for the preparation of specification which contain nucleotide and/or amino acid sequences" (edited by Japan Patent Office) and common use in the art.

The term "polynucleotide" used herein refers to a nucleic acid including any of RNA, DNA, and RNA/DNA (chimeras). The DNA includes any of cDNA, genomic DNA, and synthetic DNA. The RNA includes any of total RNA, mRNA, rRNA, miRNA, siRNA, snoRNA, snRNA, non-coding RNA and synthetic RNA. The "synthetic DNA" and the "synthetic RNA" used herein refer to DNA and RNA artificially prepared using, for example, an automatic nucleic acid synthesizer, on the basis of predetermined nucleotide sequences (which may be any of natural and non-natural sequences). The "non-natural sequence" used herein is intended to be used in a broad sense and includes, for example, a sequence comprising substitution, deletion, insertion, and/or addition of one or more nucleotide(s) (i.e., a variant sequence) and a sequence comprising one or more modified nucleotide(s) (i.e., a modified sequence), which are different from the natural sequence. The term "polynucleotide" used herein is used interchangeably with the term "nucleic acid".

The term "fragment" used herein refers to a polynucleotide (including oligonucleotides) having a nucleotide sequence that consists of a consecutive portion of a polynucleotide and desirably has a length of 15 or more nucleotides, preferably 17 or more nucleotides, more preferably 19 or more nucleotides.

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The term "gene" used herein is intended to include not only RNA and double-stranded DNA but also each single-stranded DNA such as a plus strand (or a sense strand) or a complementary strand (or an antisense strand) constituting the duplex. The gene is not particularly limited by its length. Thus, The "gene" used herein includes any of double-stranded DNA including human genomic DNA, single-stranded DNA (plus strand), single-stranded DNA having a sequence complementary to the plus strand (complementary strand) including cDNA, microRNA (miRNA), and their fragments, and their transcripts, unless otherwise specified. The "gene" includes not only a "gene" represented by a particular nucleotide sequence (or SEQ ID NO) but also "nucleic acids" encoding RNAs having biological functions equivalent to an RNA encoded by the gene, for example, a congener (i.e., a homolog or an ortholog), a variant (e.g., a genetic polymorph), and a derivative. Specific examples of such a "nucleic acid" encoding a congener, a variant, or a derivative can include a "nucleic acid" having a nucleotide sequence hybridizing under stringent conditions described later to a complementary sequence of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 871 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t. Regardless whether or not there is a difference in functional region, the "gene(s)" can comprise, for example, expression regulatory region(s), coding region(s), exon(s), or intron(s). The "gene" may be contained in a cell or may exist alone after being released into the outside of a cell. Alternatively, the "gene" may be in a state enclosed in a vesicle called exosome.

The term "exosome" used herein refers to a vesicle that is encapsulated by a lipid bilayer and secreted from a cell. The exosome is derived from a multivesicular endosome and may incorporate biomaterials such as "gene(s)" (e.g., RNA or DNA) or protein(s) when released into an extracellular environment. The exosome is known to be contained in a body fluid such as blood, serum, plasma, or lymph.

The term "transcript" used herein refers to RNA synthesized with the DNA sequence of a gene as a template. RNA polymerase binds to a site called a promoter located upstream of the gene and adds ribonucleotides complementary to the nucleotide sequence of the DNA to the 3' end to synthesize an RNA. This RNA contains not only the gene itself but also the whole sequence from a transcription initiation site to the end of a polyA sequence, including expression regulatory region(s), coding region(s), exon(s), or intron(s).

Unless otherwise specified, the term "microRNA (miRNA)" used herein is intended to mean a 15- to 25-nucleotide non-coding RNA that is transcribed as an RNA precursor which has a hairpin-like structure, cleaved by a dsRNA-cleaving enzyme which has RNase III cleavage activity, and integrated into a protein complex called RISC, and is involved in the suppression of translation of mRNA. The term "miRNA" used herein includes not only a "miRNA" represented by a particular nucleotide sequence (or SEQ ID NO) but also a precursor of the "miRNA" (pre-miRNA or pri-miRNA), and miRNAs that have biological functions equivalent thereto, for example, a congener (i.e., a homolog or an ortholog), a variant (e.g., a genetic polymorph), and a derivative. Such a precursor, a congener, a variant, or a derivative can be specifically identified using miRBase Release 20 (<http://www.mirbase.org/>), and examples thereof can include a "miRNA" having a nucleotide sequence hybridizing under stringent conditions described later to a complementary sequence of any particular nucleotide sequence represented by any of SEQ ID

NOs: 1 to 871. The term “miRNA” used herein may be a gene product of a miR gene. Such a gene product includes a mature miRNA (e.g., a 15- to 25-nucleotide or 19- to 25-nucleotide non-coding RNA involved in the suppression of translation of mRNA as described above) or a miRNA precursor (e.g., pre-miRNA or pri-miRNA as described above).

The term “probe” used herein includes a polynucleotide that is used for specifically detecting an RNA resulting from the expression of a gene or a polynucleotide derived from the RNA, and/or a polynucleotide complementary thereto.

The term “primer” used herein includes a polynucleotide that specifically recognizes and amplifies RNA resulting from the expression of a gene or a polynucleotide derived from the RNA, and/or a polynucleotide complementary thereto. In this context, the “complementary polynucleotide (complementary strand or reverse strand)” means a polynucleotide in a complementary relationship of A:T (U) and G:C base pairs with the full-length sequence of a polynucleotide consisting of a nucleotide sequence defined by any of SEQ ID NOs: 1 to 871 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, or a partial sequence thereof (here, this full-length or partial sequence is referred to as a plus strand for the sake of convenience). However, such a complementary strand is not limited to a sequence completely complementary to the nucleotide sequence of the target plus strand and may have a complementary relationship to an extent that permits hybridization under stringent conditions to the target plus strand.

The term “stringent conditions” used herein refers to conditions under which a nucleic acid probe hybridizes to its target sequence to a larger extent (e.g., a measurement value equal to or larger than “(a mean of background measurement values)+(a standard deviation of the background measurement values) $\times$ 2”) than that for other sequences. The stringent conditions are dependent on a sequence and differ depending on an environment where hybridization is performed. A target sequence complementary 100% to the nucleic acid probe can be identified by controlling the stringency of hybridization and/or washing conditions. Specific examples of the “stringent conditions” will be mentioned later.

The term “Tm value” used herein means a temperature at which the double-stranded moiety of a polynucleotide is denatured into single strands so that the double strands and the single strands exist at a ratio of 1:1.

The term “variant” used herein means, in the case of a nucleic acid, a natural variant attributed to polymorphism, mutation, or the like; a variant containing the deletion, substitution, addition, or insertion of 1 or 2 or more nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 1 to 871 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, or a partial sequence thereof; a variant that exhibits percent (%) identity of approximately 90% or higher, approximately 95% or higher, approximately 97% or higher, approximately 98% or higher, approximately 99% or higher to each of these nucleotide sequences or the partial sequence thereof; or a nucleic acid hybridizing under the stringent conditions defined above to a polynucleotide or an oligonucleotide comprising each of these nucleotide sequences or the partial sequence thereof.

The term “multiple” used herein means an integer of approximately 10, 9, 8, 7, 6, 5, 4, 3, or 2.

The “variant” used herein can be prepared by use of a well-known technique such as site-directed mutagenesis or PCR-based mutagenesis.

The term “percent (%) identity” used herein can be determined with or without an introduced gap, using a protein or gene search system based on BLAST or FASTA described above (Zheng Zhang et al., 2000, *J. Comput. Biol.*, Vol. 7, p. 203-214; Altschul, S. F. et al., 1990, *Journal of Molecular Biology*, Vol. 215, p. 403-410; and Pearson, W. R. et al., 1988, *Proc. Natl. Acad. Sci. U.S.A.*, Vol. 85, p. 2444-2448).

The term “derivative” used herein is meant to include a modified nucleic acid, for example, a derivative labeled with a fluorophore or the like, a derivative containing a modified nucleotide (e.g., a nucleotide containing a group such as halogen, alkyl such as methyl, alkoxy such as methoxy, thio, or carboxymethyl, and a nucleotide that has undergone base rearrangement, double bond saturation, deamination, replacement of an oxygen molecule with a sulfur atom, etc.), PNA (peptide nucleic acid; Nielsen, P. E. et al., 1991, *Science*, Vol. 254, p. 1497-500), and LNA (locked nucleic acid; Obika, S. et al., 1998, *Tetrahedron Lett.*, Vol. 39, p. 5401-5404) without any limitation.

The “nucleic acid” used herein capable of specifically binding to a polynucleotide selected from the breast cancer marker miRNAs described above is a synthesized or prepared nucleic acid and specifically includes a “nucleic acid probe” or a “primer”. The “nucleic acid” is utilized directly or indirectly for detecting the presence or absence of breast cancer in a subject, for diagnosing the presence or absence of breast cancer, or the severity of breast cancer, the presence or absence of amelioration of breast cancer, or the degree of amelioration of breast cancer, or the therapeutic sensitivity of breast cancer, or for screening for a candidate substance useful in the prevention, amelioration, or treatment of breast cancer. The “nucleic acid” includes a nucleotide, an oligonucleotide, and a polynucleotide capable of specifically recognizing and binding to a transcript represented by any of SEQ ID NOs: 1 to 871 or a synthetic cDNA nucleic acid thereof in vivo, particularly, in a sample such as a body fluid (e.g., blood or urine), in relation to the development of breast cancer. The nucleotide, the oligonucleotide, and the polynucleotide can be effectively used as probes for detecting the aforementioned gene expressed in vivo, in tissues, in cells, or the like on the basis of the properties described above, or as primers for amplifying the aforementioned gene expressed in vivo.

The term “detection” used herein is interchangeable with the term “examination”, “measurement”, or “detection or decision support”. The term “evaluation” used herein is meant to include diagnosing or evaluation-supporting on the basis of examination results or measurement results.

The term “subject” used herein means a mammal such as a primate including a human and a chimpanzee, a pet animal including a dog and a cat, a livestock animal including cattle, a horse, sheep, and a goat, and a rodent including a mouse and a rat. The term “healthy subject” also means such a mammal without the cancer to be detected.

The term “P” or “P value” used herein refers to a probability at which a more extreme statistic than that is actually calculated from data under a null hypothesis is observed in a statistical test. Thus, smaller “P” or “P value” is regarded as being a more significant difference between subjects to be compared.

The term “sensitivity” used herein means a value of (the number of true positives)/(the number of true positives+the number of false negatives). High sensitivity allows breast

cancer to be detected early, leading to the complete resection of cancer sites and reduction in the rate of recurrence.

The term “specificity” used herein means a value of (the number of true negatives)/(the number of true negatives+the number of false positives). High specificity prevents need-
less extra examination for healthy subjects who might have been misjudged as being breast cancer patients, leading to reduction in burden on patients and reduction in medical expense.

The term “accuracy” used herein means a value of (the number of true positives+the number of true negatives)/(the total number of cases). The accuracy indicates the ratio of samples that are identified correctly to all samples, and serves as a primary index for evaluating detection performance.

The “sample” used herein, that is subjected to determination, detection, or diagnosis refers to a tissue and a biological material in which the expression of the gene of the present invention varies as breast cancer develops, as breast cancer progresses, or as therapeutic effects on breast cancer are exerted. Specifically, the “sample” refers to a breast tissue, a perimammary vascular channel, lymph node, and organ, an organ suspected of having metastasis, the skin, a body fluid such as blood, urine, saliva, sweat, or tissue exudates, serum or plasma prepared from blood, feces, hair, and the like. The “sample” further refers to a biological sample extracted therefrom, specifically, a gene such as RNA or miRNA.

The term “hsa-miR-4783-3p gene” or “hsa-miR-4783-3p” used herein includes the hsa-miR-4783-3p gene (miRBase Accession No. MIMAT0019947) described in SEQ ID NO: 1, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4783-3p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4783” (miRBase Accession No. MI0017428, SEQ ID NO: 270) having a hairpin-like structure is known as a precursor of “hsa-miR-4783-3p”.

The term “hsa-miR-4730 gene” or “hsa-miR-4730” used herein includes the hsa-miR-4730 gene (miRBase Accession No. MIMAT0019852) described in SEQ ID NO: 2, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4730 gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4730” (miRBase Accession No. MI0017367, SEQ ID NO: 271) having a hairpin-like structure is known as a precursor of “hsa-miR-4730”.

The term “hsa-miR-1307-3p gene” or “hsa-miR-1307-3p” used herein includes the hsa-miR-1307-3p gene (miRBase Accession No. MIMAT0005951) described in SEQ ID NO: 3, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1307-3p gene can be obtained by a method described in Morin R D et al., 2008, *Genome Res*, Vol. 18, p. 610-621. Also, “hsa-mir-1307” (miRBase Accession No. MI0006444, SEQ ID NO: 272) having a hairpin-like structure is known as a precursor of “hsa-miR-1307-3p”.

The term “hsa-miR-4634 gene” or “hsa-miR-4634” used herein includes the hsa-miR-4634 gene (miRBase Accession No. MIMAT0019691) described in SEQ ID NO: 4, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4634 gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4634” (miRBase Accession No. MI0017261, SEQ ID NO: 273) having a hairpin-like structure is known as a precursor of “hsa-miR-4634”.

The term “hsa-miR-663a gene” or “hsa-miR-663a” used herein includes the hsa-miR-663a gene (miRBase Accession

No. MIMAT0003326) described in SEQ ID NO: 5, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-663a gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-663a” (miRBase Accession No. MI0003672, SEQ ID NO: 274) having a hairpin-like structure is known as a precursor of “hsa-miR-663a”.

The term “hsa-miR-4532 gene” or “hsa-miR-4532” used herein includes the hsa-miR-4532 gene (miRBase Accession No. MIMAT0019071) described in SEQ ID NO: 6, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4532 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4532” (miRBase Accession No. MI0016899, SEQ ID NO: 275) having a hairpin-like structure is known as a precursor of “hsa-miR-4532”.

The term “hsa-miR-7704 gene” or “hsa-miR-7704” used herein includes the hsa-miR-7704 gene (miRBase Accession No. MIMAT0030019) described in SEQ ID NO: 7, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7704 gene can be obtained by a method described in Swaminathan S et al., 2013, *Biochem Biophys Res Commun*, Vol. 434, p. 228-234. Also, “hsa-mir-7704” (miRBase Accession No. MI0025240, SEQ ID NO: 276) having a hairpin-like structure is known as a precursor of “hsa-miR-7704”.

The term “hsa-miR-3178 gene” or “hsa-miR-3178” used herein includes the hsa-miR-3178 gene (miRBase Accession No. MIMAT0015055) described in SEQ ID NO: 8, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3178 gene can be obtained by a method described in Stark M S et al., 2010, *PLoS One*, Vol. 5, e9685. Also, “hsa-mir-3178” (miRBase Accession No. MI0014212, SEQ ID NO: 277) having a hairpin-like structure is known as a precursor of “hsa-miR-3178”.

The term “hsa-miR-6729-5p gene” or “hsa-miR-6729-5p” used herein includes the hsa-miR-6729-5p gene (miRBase Accession No. MIMAT0027359) described in SEQ ID NO: 9, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6729-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6729” (miRBase Accession No. MI0022574, SEQ ID NO: 278) having a hairpin-like structure is known as a precursor of “hsa-miR-6729-5p”.

The term “hsa-miR-6090 gene” or “hsa-miR-6090” used herein includes the hsa-miR-6090 gene (miRBase Accession No. MIMAT0023715) described in SEQ ID NO: 10, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6090 gene can be obtained by a method described in Yoo J K et al., 2012, *Stem Cells Dev*, Vol. 21, p. 2049-2057. Also, “hsa-mir-6090” (miRBase Accession No. MI0020367, SEQ ID NO: 279) having a hairpin-like structure is known as a precursor of “hsa-miR-6090”.

The term “hsa-miR-4732-5p gene” or “hsa-miR-4732-5p” used herein includes the hsa-miR-4732-5p gene (miRBase Accession No. MIMAT0019855) described in SEQ ID NO: 11, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4732-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4732” (miRBase Accession No. MI0017369, SEQ ID NO: 280) having a hairpin-like structure is known as a precursor of “hsa-miR-4732-5p”.

The term “hsa-miR-3184-5p gene” or “hsa-miR-3184-5p” used herein includes the hsa-miR-3184-5p gene (miRBase

Accession No. MIMAT0015064) described in SEQ ID NO: 12, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3184-5p gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3184” (miRBase Accession No. MI0014226, SEQ ID NO: 281) having a hairpin-like structure is known as a precursor of “hsa-miR-3184-5p”.

The term “hsa-miR-6727-5p gene” or “hsa-miR-6727-5p” used herein includes the hsa-miR-6727-5p gene (miRBase Accession No. MIMAT0027355) described in SEQ ID NO: 13, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6727-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6727” (miRBase Accession No. MI0022572, SEQ ID NO: 282) having a hairpin-like structure is known as a precursor of “hsa-miR-6727-5p”.

The term “hsa-miR-6088 gene” or “hsa-miR-6088” used herein includes the hsa-miR-6088 gene (miRBase Accession No. MIMAT0023713) described in SEQ ID NO: 14, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6088 gene can be obtained by a method described in Yoo J K et al., 2012, Stem Cells Dev, Vol. 21, p. 2049-2057. Also, “hsa-mir-6088” (miRBase Accession No. MI0020365, SEQ ID NO: 283) having a hairpin-like structure is known as a precursor of “hsa-miR-6088”.

The term “hsa-miR-4674 gene” or “hsa-miR-4674” used herein includes the hsa-miR-4674 gene (miRBase Accession No. MIMAT0019756) described in SEQ ID NO: 15, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4674 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4674” (miRBase Accession No. MI0017305, SEQ ID NO: 284) having a hairpin-like structure is known as a precursor of “hsa-miR-4674”.

The term “hsa-miR-8073 gene” or “hsa-miR-8073” used herein includes the hsa-miR-8073 gene (miRBase Accession No. MIMAT0031000) described in SEQ ID NO: 16, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8073 gene can be obtained by a method described in Wang H J et al., 2013, Shock, Vol. 39, p. 480-487. Also, “hsa-mir-8073” (miRBase Accession No. MI0025909, SEQ ID NO: 285) having a hairpin-like structure is known as a precursor of “hsa-miR-8073”.

The term “hsa-miR-4787-5p gene” or “hsa-miR-4787-5p” used herein includes the hsa-miR-4787-5p gene (miRBase Accession No. MIMAT0019956) described in SEQ ID NO: 17, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4787-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4787” (miRBase Accession No. MI0017434, SEQ ID NO: 286) having a hairpin-like structure is known as a precursor of “hsa-miR-4787-5p”.

The term “hsa-miR-1469 gene” or “hsa-miR-1469” used herein includes the hsa-miR-1469 gene (miRBase Accession No. MIMAT0007347) described in SEQ ID NO: 18, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1469 gene can be obtained by a method described in Kawaji H et al., 2008, BMC Genomics, Vol. 9, p. 157. Also, “hsa-mir-1469” (miRBase Accession No. MI0007074, SEQ ID NO: 287) having a hairpin-like structure is known as a precursor of “hsa-miR-1469”.

The term “hsa-miR-125a-3p gene” or “hsa-miR-125a-3p” used herein includes the hsa-miR-125a-3p gene (miRBase Accession No. MIMAT0004602) described in SEQ ID NO: 19, a homolog or an ortholog of a different organism species,

and the like. The hsa-miR-125a-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, Curr Biol, Vol. 12, p. 735-739. Also, “hsa-mir-125a” (miRBase Accession No. MI0000469, SEQ ID NO: 288) having a hairpin-like structure is known as a precursor of “hsa-miR-125a-3p”.

The term “hsa-miR-1233-5p gene” or “hsa-miR-1233-5p” used herein includes the hsa-miR-1233-5p gene (miRBase Accession No. MIMAT0022943) described in SEQ ID NO: 20, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1233-5p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1233-1 and hsa-mir-1233-2” (miRBase Accession Nos. MI0006323 and MI0015973, SEQ ID NOs: 289 and 290) having a hairpin-like structure are known as precursors of “hsa-miR-1233-5p”.

The term “hsa-miR-885-3p gene” or “hsa-miR-885-3p” used herein includes the hsa-miR-885-3p gene (miRBase Accession No. MIMAT0004948) described in SEQ ID NO: 21, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-885-3p gene can be obtained by a method described in Berezikov E et al., 2006, Genome Res, Vol. 16, p. 1289-1298. Also, “hsa-mir-885” (miRBase Accession No. MI0005560, SEQ ID NO: 291) having a hairpin-like structure is known as a precursor of “hsa-miR-885-3p”.

The term “hsa-miR-6802-5p gene” or “hsa-miR-6802-5p” used herein includes the hsa-miR-6802-5p gene (miRBase Accession No. MIMAT0027504) described in SEQ ID NO: 22, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6802-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6802” (miRBase Accession No. MI0022647, SEQ ID NO: 292) having a hairpin-like structure is known as a precursor of “hsa-miR-6802-5p”.

The term “hsa-miR-328-5p gene” or “hsa-miR-328-5p” used herein includes the hsa-miR-328-5p gene (miRBase Accession No. MIMAT0026486) described in SEQ ID NO: 23, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-328-5p gene can be obtained by a method described in Kim J et al., 2004, Proc Natl Acad Sci USA, Vol. 101, p. 360-365. Also, “hsa-mir-328” (miRBase Accession No. MI0000804, SEQ ID NO: 293) having a hairpin-like structure is known as a precursor of “hsa-miR-328-5p”.

The term “hsa-miR-6787-5p gene” or “hsa-miR-6787-5p” used herein includes the hsa-miR-6787-5p gene (miRBase Accession No. MIMAT0027474) described in SEQ ID NO: 24, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6787-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6787” (miRBase Accession No. MI0022632, SEQ ID NO: 294) having a hairpin-like structure is known as a precursor of “hsa-miR-6787-5p”.

The term “hsa-miR-8069 gene” or “hsa-miR-8069” used herein includes the hsa-miR-8069 gene (miRBase Accession No. MIMAT0030996) described in SEQ ID NO: 25, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8069 gene can be obtained by a method described in Wang H J et al., 2013, Shock, Vol. 39, p. 480-487. Also, “hsa-mir-8069” (miRBase Accession No. MI0025905, SEQ ID NO: 295) having a hairpin-like structure is known as a precursor of “hsa-miR-8069”.

The term “hsa-miR-6875-5p gene” or “hsa-miR-6875-5p” used herein includes the hsa-miR-6875-5p gene (miRBase Accession No. MIMAT0027650) described in SEQ ID NO: 26, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6875-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6875” (miRBase Accession No. MI0022722, SEQ ID NO: 296) having a hairpin-like structure is known as a precursor of “hsa-miR-6875-5p”.

The term “hsa-miR-1246 gene” or “hsa-miR-1246” used herein includes the hsa-miR-1246 gene (miRBase Accession No. MIMAT0005898) described in SEQ ID NO: 27, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1246 gene can be obtained by a method described in Morin R D et al., 2008, *Genome Res.*, Vol. 18, p. 610-621. Also, “hsa-mir-1246” (miRBase Accession No. MI0006381, SEQ ID NO: 297) having a hairpin-like structure is known as a precursor of “hsa-miR-1246”.

The term “hsa-miR-4734 gene” or “hsa-miR-4734” used herein includes the hsa-miR-4734 gene (miRBase Accession No. MIMAT0019859) described in SEQ ID NO: 28, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4734 gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res.*, Vol. 71, p. 78-86. Also, “hsa-mir-4734” (miRBase Accession No. MI0017371, SEQ ID NO: 298) having a hairpin-like structure is known as a precursor of “hsa-miR-4734”.

The term “hsa-miR-6757-5p gene” or “hsa-miR-6757-5p” used herein includes the hsa-miR-6757-5p gene (miRBase Accession No. MIMAT0027414) described in SEQ ID NO: 29, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6757-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6757” (miRBase Accession No. MI0022602, SEQ ID NO: 299) having a hairpin-like structure is known as a precursor of “hsa-miR-6757-5p”.

The term “hsa-miR-6756-5p gene” or “hsa-miR-6756-5p” used herein includes the hsa-miR-6756-5p gene (miRBase Accession No. MIMAT0027412) described in SEQ ID NO: 30, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6756-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6756” (miRBase Accession No. MI0022601, SEQ ID NO: 300) having a hairpin-like structure is known as a precursor of “hsa-miR-6756-5p”.

The term “hsa-miR-3665 gene” or “hsa-miR-3665” used herein includes the hsa-miR-3665 gene (miRBase Accession No. MIMAT0018087) described in SEQ ID NO: 31, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3665 gene can be obtained by a method described in Xie X et al., 2005, *Nature*, Vol. 434, p. 338-345. Also, “hsa-mir-3665” (miRBase Accession No. MI0016066, SEQ ID NO: 301) having a hairpin-like structure is known as a precursor of “hsa-miR-3665”.

The term “hsa-miR-6836-3p gene” or “hsa-miR-6836-3p” used herein includes the hsa-miR-6836-3p gene (miRBase Accession No. MIMAT0027575) described in SEQ ID NO: 32, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6836-3p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6836” (miRBase Accession No. MI0022682, SEQ ID NO: 302) having a hairpin-like structure is known as a precursor of “hsa-miR-6836-3p”.

The term “hsa-miR-6821-5p gene” or “hsa-miR-6821-5p” used herein includes the hsa-miR-6821-5p gene (miRBase Accession No. MIMAT0027542) described in SEQ ID NO: 33, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6821-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6821” (miRBase Accession No. MI0022666, SEQ ID NO: 303) having a hairpin-like structure is known as a precursor of “hsa-miR-6821-5p”.

The term “hsa-miR-6805-5p gene” or “hsa-miR-6805-5p” used herein includes the hsa-miR-6805-5p gene (miRBase Accession No. MIMAT0027510) described in SEQ ID NO: 34, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6805-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6805” (miRBase Accession No. MI0022650, SEQ ID NO: 304) having a hairpin-like structure is known as a precursor of “hsa-miR-6805-5p”.

The term “hsa-miR-4728-5p gene” or “hsa-miR-4728-5p” used herein includes the hsa-miR-4728-5p gene (miRBase Accession No. MIMAT0019849) described in SEQ ID NO: 35, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4728-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res.*, Vol. 71, p. 78-86. Also, “hsa-mir-4728” (miRBase Accession No. MI0017365, SEQ ID NO: 305) having a hairpin-like structure is known as a precursor of “hsa-miR-4728-5p”.

The term “hsa-miR-6726-5p gene” or “hsa-miR-6726-5p” used herein includes the hsa-miR-6726-5p gene (miRBase Accession No. MIMAT0027353) described in SEQ ID NO: 36, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6726-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6726” (miRBase Accession No. MI0022571, SEQ ID NO: 306) having a hairpin-like structure is known as a precursor of “hsa-miR-6726-5p”.

The term “hsa-miR-197-5p gene” or “hsa-miR-197-5p” used herein includes the hsa-miR-197-5p gene (miRBase Accession No. MIMAT0022691) described in SEQ ID NO: 37, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-197-5p gene can be obtained by a method described in Lagos-Quintana M et al., 2003, *RNA*, Vol. 9, p. 175-179. Also, “hsa-mir-197” (miRBase Accession No. MI0000239, SEQ ID NO: 307) having a hairpin-like structure is known as a precursor of “hsa-miR-197-5p”.

The term “hsa-miR-149-3p gene” or “hsa-miR-149-3p” used herein includes the hsa-miR-149-3p gene (miRBase Accession No. MIMAT0004609) described in SEQ ID NO: 38, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-149-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, *Curr Biol*, Vol. 12, p. 735-739. Also, “hsa-mir-149” (miRBase Accession No. MI0000478, SEQ ID NO: 308) having a hairpin-like structure is known as a precursor of “hsa-miR-149-3p”.

The term “hsa-miR-6850-5p gene” or “hsa-miR-6850-5p” used herein includes the hsa-miR-6850-5p gene (miRBase Accession No. MIMAT0027600) described in SEQ ID NO: 39, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6850-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res.*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6850” (miRBase

Accession No. MI0022696, SEQ ID NO: 309) having a hairpin-like structure is known as a precursor of "hsa-miR-6850-5p".

The term "hsa-miR-4476 gene" or "hsa-miR-4476" used herein includes the hsa-miR-4476 gene (miRBase Accession No. MIMAT0019003) described in SEQ ID NO: 40, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4476 gene can be obtained by a method described in Lima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, "hsa-mir-4476" (miRBase Accession No. MI0016828, SEQ ID NO: 310) having a hairpin-like structure is known as a precursor of "hsa-miR-4476".

The term "hsa-miR-6858-5p gene" or "hsa-miR-6858-5p" used herein includes the hsa-miR-6858-5p gene (miRBase Accession No. MIMAT0027616) described in SEQ ID NO: 41, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6858-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6858" (miRBase Accession No. MI0022704, SEQ ID NO: 311) having a hairpin-like structure is known as a precursor of "hsa-miR-6858-5p".

The term "hsa-miR-564 gene" or "hsa-miR-564" used herein includes the hsa-miR-564 gene (miRBase Accession No. MIMAT0003228) described in SEQ ID NO: 42, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-564 gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, "hsa-mir-564" (miRBase Accession No. MI0003570, SEQ ID NO: 312) having a hairpin-like structure is known as a precursor of "hsa-miR-564".

The term "hsa-miR-4763-3p gene" or "hsa-miR-4763-3p" used herein includes the hsa-miR-4763-3p gene (miRBase Accession No. MIMAT0019913) described in SEQ ID NO: 43, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4763-3p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, "hsa-mir-4763" (miRBase Accession No. MI0017404, SEQ ID NO: 313) having a hairpin-like structure is known as a precursor of "hsa-miR-4763-3p".

The term "hsa-miR-575 gene" or "hsa-miR-575" used herein includes the hsa-miR-575 gene (miRBase Accession No. MIMAT0003240) described in SEQ ID NO: 44, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-575 gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, "hsa-mir-575" (miRBase Accession No. MI0003582, SEQ ID NO: 314) having a hairpin-like structure is known as a precursor of "hsa-miR-575".

The term "hsa-miR-6771-5p gene" or "hsa-miR-6771-5p" used herein includes the hsa-miR-6771-5p gene (miRBase Accession No. MIMAT0027442) described in SEQ ID NO: 45, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6771-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6771" (miRBase Accession No. MI0022616, SEQ ID NO: 315) having a hairpin-like structure is known as a precursor of "hsa-miR-6771-5p".

The term "hsa-miR-1231 gene" or "hsa-miR-1231" used herein includes the hsa-miR-1231 gene (miRBase Accession No. MIMAT0005586) described in SEQ ID NO: 46, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1231 gene can be obtained by a method described in Berezikov E et al., 2007, *Mol Cell*, Vol.

28, p. 328-336. Also, "hsa-mir-1231" (miRBase Accession No. MI0006321, SEQ ID NO: 316) having a hairpin-like structure is known as a precursor of "hsa-miR-1231".

The term "hsa-miR-1908-3p gene" or "hsa-miR-1908-3p" used herein includes the hsa-miR-1908-3p gene (miRBase Accession No. MIMAT0026916) described in SEQ ID NO: 47, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1908-3p gene can be obtained by a method described in Bar M et al., 2008, *Stem Cells*, Vol. 26, p. 2496-2505. Also, "hsa-mir-1908" (miRBase Accession No. MI0008329, SEQ ID NO: 317) having a hairpin-like structure is known as a precursor of "hsa-miR-1908-3p".

The term "hsa-miR-150-3p gene" or "hsa-miR-150-3p" used herein includes the hsa-miR-150-3p gene (miRBase Accession No. MIMAT0004610) described in SEQ ID NO: 48, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-150-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, *Curr Biol*, Vol. 12, p. 735-739. Also, "hsa-mir-150" (miRBase Accession No. MI0000479, SEQ ID NO: 318) having a hairpin-like structure is known as a precursor of "hsa-miR-150-3p".

The term "hsa-miR-3937 gene" or "hsa-miR-3937" used herein includes the hsa-miR-3937 gene (miRBase Accession No. MIMAT0018352) described in SEQ ID NO: 49, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3937 gene can be obtained by a method described in Liao J Y et al., 2010, *PLoS One*, Vol. 5, e10563. Also, "hsa-mir-3937" (miRBase Accession No. MI0016593, SEQ ID NO: 319) having a hairpin-like structure is known as a precursor of "hsa-miR-3937".

The term "hsa-miR-887-3p gene" or "hsa-miR-887-3p" used herein includes the hsa-miR-887-3p gene (miRBase Accession No. MIMAT0004951) described in SEQ ID NO: 50, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-887-3p gene can be obtained by a method described in Berezikov E et al., 2006, *Genome Res*, Vol. 16, p. 1289-1298. Also, "hsa-mir-887" (miRBase Accession No. MI0005562, SEQ ID NO: 320) having a hairpin-like structure is known as a precursor of "hsa-miR-887-3p".

The term "hsa-miR-3940-5p gene" or "hsa-miR-3940-5p" used herein includes the hsa-miR-3940-5p gene (miRBase Accession No. MIMAT0019229) described in SEQ ID NO: 51, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3940-5p gene can be obtained by a method described in Liao J Y et al., 2010, *PLoS One*, Vol. 5, e10563. Also, "hsa-mir-3940" (miRBase Accession No. MI0016597, SEQ ID NO: 321) having a hairpin-like structure is known as a precursor of "hsa-miR-3940-5p".

The term "hsa-miR-4741 gene" or "hsa-miR-4741" used herein includes the hsa-miR-4741 gene (miRBase Accession No. MIMAT0019871) described in SEQ ID NO: 52, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4741 gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, "hsa-mir-4741" (miRBase Accession No. MI0017379, SEQ ID NO: 322) having a hairpin-like structure is known as a precursor of "hsa-miR-4741".

The term "hsa-miR-6808-5p gene" or "hsa-miR-6808-5p" used herein includes the hsa-miR-6808-5p gene (miRBase Accession No. MIMAT0027516) described in SEQ ID NO: 53, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6808-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6808" (miRBase

Accession No. MI0022653, SEQ ID NO: 323) having a hairpin-like structure is known as a precursor of “hsa-miR-6808-5p”.

The term “hsa-miR-6869-5p gene” or “hsa-miR-6869-5p” used herein includes the hsa-miR-6869-5p gene (miRBase Accession No. MIMAT0027638) described in SEQ ID NO: 54, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6869-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6869” (miRBase Accession No. MI0022716, SEQ ID NO: 324) having a hairpin-like structure is known as a precursor of “hsa-miR-6869-5p”.

The term “hsa-miR-5090 gene” or “hsa-miR-5090” used herein includes the hsa-miR-5090 gene (miRBase Accession No. MIMAT0021082) described in SEQ ID NO: 55, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5090 gene can be obtained by a method described in Ding N et al., 2011, *J Radiat Res*, Vol. 52, p. 425-432. Also, “hsa-mir-5090” (miRBase Accession No. MI0017979, SEQ ID NO: 325) having a hairpin-like structure is known as a precursor of “hsa-miR-5090”.

The term “hsa-miR-615-5p gene” or “hsa-miR-615-5p” used herein includes the hsa-miR-615-5p gene (miRBase Accession No. MIMAT0004804) described in SEQ ID NO: 56, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-615-5p gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-615” (miRBase Accession No. MI0003628, SEQ ID NO: 326) having a hairpin-like structure is known as a precursor of “hsa-miR-615-5p”.

The term “hsa-miR-8072 gene” or “hsa-miR-8072” used herein includes the hsa-miR-8072 gene (miRBase Accession No. MIMAT0030999) described in SEQ ID NO: 57, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8072 gene can be obtained by a method described in Wang H J et al., 2013, *Shock*, Vol. 39, p. 480-487. Also, “hsa-mir-8072” (miRBase Accession No. MI0025908, SEQ ID NO: 327) having a hairpin-like structure is known as a precursor of “hsa-miR-8072”.

The term “hsa-miR-128-1-5p gene” or “hsa-miR-128-1-5p” used herein includes the hsa-miR-128-1-5p gene (miRBase Accession No. MIMAT0026477) described in SEQ ID NO: 58, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-128-1-5p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, *Curr Biol*, Vol. 12, p. 735-739. Also, “hsa-mir-128-1” (miRBase Accession No. MI0000447, SEQ ID NO: 328) having a hairpin-like structure is known as a precursor of “hsa-miR-128-1-5p”.

The term “hsa-miR-1238-5p gene” or “hsa-miR-1238-5p” used herein includes the hsa-miR-1238-5p gene (miRBase Accession No. MIMAT0022947) described in SEQ ID NO: 59, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1238-5p gene can be obtained by a method described in Berezikov E et al., 2007, *Mol Cell*, Vol. 28, p. 328-336. Also, “hsa-mir-1238” (miRBase Accession No. MI0006328, SEQ ID NO: 329) having a hairpin-like structure is known as a precursor of “hsa-miR-1238-5p”.

The term “hsa-miR-365a-5p gene” or “hsa-miR-365a-5p” used herein includes the hsa-miR-365a-5p gene (miRBase Accession No. MIMAT0009199) described in SEQ ID NO: 60, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-365a-5p gene can be obtained by a method described in Xie X et al., 2005, *Nature*, Vol. 434,

p. 338-345. Also, “hsa-mir-365a” (miRBase Accession No. MI0000767, SEQ ID NO: 330) having a hairpin-like structure is known as a precursor of “hsa-miR-365a-5p”.

The term “hsa-miR-204-3p gene” or “hsa-miR-204-3p” used herein includes the hsa-miR-204-3p gene (miRBase Accession No. MIMAT0022693) described in SEQ ID NO: 61, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-204-3p gene can be obtained by a method described in Lim L P et al., 2003, *Science*, Vol. 299, p. 1540. Also, “hsa-mir-204” (miRBase Accession No. MI0000284, SEQ ID NO: 331) having a hairpin-like structure is known as a precursor of “hsa-miR-204-3p”.

The term “hsa-miR-4492 gene” or “hsa-miR-4492” used herein includes the hsa-miR-4492 gene (miRBase Accession No. MIMAT0019027) described in SEQ ID NO: 62, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4492 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4492” (miRBase Accession No. MI0016854, SEQ ID NO: 332) having a hairpin-like structure is known as a precursor of “hsa-miR-4492”.

The term “hsa-miR-6785-5p gene” or “hsa-miR-6785-5p” used herein includes the hsa-miR-6785-5p gene (miRBase Accession No. MIMAT0027470) described in SEQ ID NO: 63, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6785-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6785” (miRBase Accession No. MI0022630, SEQ ID NO: 333) having a hairpin-like structure is known as a precursor of “hsa-miR-6785-5p”.

The term “hsa-miR-6511a-5p gene” or “hsa-miR-6511a-5p” used herein includes the hsa-miR-6511a-5p gene (miRBase Accession No. MIMAT0025478) described in SEQ ID NO: 64, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6511a-5p gene can be obtained by a method described in Joyce C E et al., 2011, *Hum Mol Genet*, Vol. 20, p. 4025-4040. Also, “hsa-mir-6511a-1, hsa-mir-6511a-2, hsa-mir-6511a-3, and hsa-mir-6511a-4” (miRBase Accession Nos. MI0022223, MI0023564, MI0023565, and MI0023566, SEQ ID NOs: 334, 335, 336, and 337) having a hairpin-like structure are known as precursors of “hsa-miR-6511a-5p”.

The term “hsa-miR-4525 gene” or “hsa-miR-4525” used herein includes the hsa-miR-4525 gene (miRBase Accession No. MIMAT0019064) described in SEQ ID NO: 65, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4525 gene can be obtained by a method described in Lima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4525” (miRBase Accession No. MI0016892, SEQ ID NO: 338) having a hairpin-like structure is known as a precursor of “hsa-miR-4525”.

The term “hsa-miR-1915-5p gene” or “hsa-miR-1915-5p” used herein includes the hsa-miR-1915-5p gene (miRBase Accession No. MIMAT0007891) described in SEQ ID NO: 66, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1915-5p gene can be obtained by a method described in Bar M et al., 2008, *Stem Cells*, Vol. 26, p. 2496-2505. Also, “hsa-mir-1915” (miRBase Accession No. MI0008336, SEQ ID NO: 339) having a hairpin-like structure is known as a precursor of “hsa-miR-1915-5p”.

The term “hsa-miR-3180 gene” or “hsa-miR-3180” used herein includes the hsa-miR-3180 gene (miRBase Accession No. MIMAT0018178) described in SEQ ID NO: 67, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3180 gene can be obtained by a

method described in Creighton CI et al., 2010, PLoS One, Vol. 5, e9637. Also, “hsa-mir-3180-4 and hsa-mir-3180-5” (miRBase Accession Nos. MI0016408 and MI0016409, SEQ ID NOs: 340 and 341) having a hairpin-like structure are known as precursors of “hsa-miR-3180”.

The term “hsa-miR-6879-5p gene” or “hsa-miR-6879-5p” used herein includes the hsa-miR-6879-5p gene (miRBase Accession No. MIMAT0027658) described in SEQ ID NO: 68, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6879-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6879” (miRBase Accession No. MI0022726, SEQ ID NO: 342) having a hairpin-like structure is known as a precursor of “hsa-miR-6879-5p”.

The term “hsa-miR-1199-5p gene” or “hsa-miR-1199-5p” used herein includes the hsa-miR-1199-5p gene (miRBase Accession No. MIMAT0031119) described in SEQ ID NO: 69, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1199-5p gene can be obtained by a method described in Salvi A et al., 2013, Int J Oncol, Vol. 42, p. 391-402. Also, “hsa-mir-1199” (miRBase Accession No. MI0020340, SEQ ID NO: 343) having a hairpin-like structure is known as a precursor of “hsa-miR-1199-5p”.

The term “hsa-miR-6746-5p gene” or “hsa-miR-6746-5p” used herein includes the hsa-miR-6746-5p gene (miRBase Accession No. MIMAT0027392) described in SEQ ID NO: 70, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6746-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6746” (miRBase Accession No. MI0022591, SEQ ID NO: 344) having a hairpin-like structure is known as a precursor of “hsa-miR-6746-5p”.

The term “hsa-miR-711 gene” or “hsa-miR-711” used herein includes the hsa-miR-711 gene (miRBase Accession No. MIMAT0012734) described in SEQ ID NO: 71, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-711 gene can be obtained by a method described in Artzi S et al., 2008, BMC Bioinformatics, Vol. 9, p. 39. Also, “hsa-mir-711” (miRBase Accession No. MI0012488, SEQ ID NO: 345) having a hairpin-like structure is known as a precursor of “hsa-miR-711”.

The term “hsa-miR-663b gene” or “hsa-miR-663b” used herein includes the hsa-miR-663b gene (miRBase Accession No. MIMAT0005867) described in SEQ ID NO: 72, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-663b gene can be obtained by a method described in Takada S et al., 2008, Leukemia, Vol. 22, p. 1274-1278. Also, “hsa-mir-663b” (miRBase Accession No. MI0006336, SEQ ID NO: 346) having a hairpin-like structure is known as a precursor of “hsa-miR-663b”.

The term “hsa-miR-4707-3p gene” or “hsa-miR-4707-3p” used herein includes the hsa-miR-4707-3p gene (miRBase Accession No. MIMAT0019808) described in SEQ ID NO: 73, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4707-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4707” (miRBase Accession No. MI0017340, SEQ ID NO: 347) having a hairpin-like structure is known as a precursor of “hsa-miR-4707-3p”.

The term “hsa-miR-6893-5p gene” or “hsa-miR-6893-5p” used herein includes the hsa-miR-6893-5p gene (miRBase Accession No. MIMAT0027686) described in SEQ ID NO: 74, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6893-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res,

Vol. 22, p. 1634-1645. Also, “hsa-mir-6893” (miRBase Accession No. MI0022740, SEQ ID NO: 348) having a hairpin-like structure is known as a precursor of “hsa-miR-6893-5p”.

5 The term “hsa-miR-4675 gene” or “hsa-miR-4675” used herein includes the hsa-miR-4675 gene (miRBase Accession No. MIMAT0019757) described in SEQ ID NO: 75, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4675 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4675” (miRBase Accession No. MI0017306, SEQ ID NO: 349) having a hairpin-like structure is known as a precursor of “hsa-miR-4675”.

15 The term “hsa-miR-4638-5p gene” or “hsa-miR-4638-5p” used herein includes the hsa-miR-4638-5p gene (miRBase Accession No. MIMAT0019695) described in SEQ ID NO: 76, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4638-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4638” (miRBase Accession No. MI0017265, SEQ ID NO: 350) having a hairpin-like structure is known as a precursor of “hsa-miR-4638-5p”.

The term “hsa-miR-4651 gene” or “hsa-miR-4651” used herein includes the hsa-miR-4651 gene (miRBase Accession No. MIMAT0019715) described in SEQ ID NO: 77, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4651 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4651” (miRBase Accession No. MI0017279, SEQ ID NO: 351) having a hairpin-like structure is known as a precursor of “hsa-miR-4651”.

The term “hsa-miR-6087 gene” or “hsa-miR-6087” used herein includes the hsa-miR-6087 gene (miRBase Accession No. MIMAT0023712) described in SEQ ID NO: 78, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6087 gene can be obtained by a method described in Yoo J K et al., 2012, Stem Cells Dev, Vol. 21, p. 2049-2057. Also, “hsa-mir-6087” (miRBase Accession No. MI0020364, SEQ ID NO: 352) having a hairpin-like structure is known as a precursor of “hsa-miR-6087”.

The term “hsa-miR-4665-5p gene” or “hsa-miR-4665-5p” used herein includes the hsa-miR-4665-5p gene (miRBase Accession No. MIMAT0019739) described in SEQ ID NO: 79, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4665-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4665” (miRBase Accession No. MI0017295, SEQ ID NO: 353) having a hairpin-like structure is known as a precursor of “hsa-miR-4665-5p”.

The term “hsa-miR-4758-5p gene” or “hsa-miR-4758-5p” used herein includes the hsa-miR-4758-5p gene (miRBase Accession No. MIMAT0019903) described in SEQ ID NO: 80, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4758-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4758” (miRBase Accession No. MI0017399, SEQ ID NO: 354) having a hairpin-like structure is known as a precursor of “hsa-miR-4758-5p”.

The term “hsa-miR-6887-5p gene” or “hsa-miR-6887-5p” used herein includes the hsa-miR-6887-5p gene (miRBase Accession No. MIMAT0027674) described in SEQ ID NO: 81, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6887-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6887” (miRBase

Accession No. MI0022734, SEQ ID NO: 355) having a hairpin-like structure is known as a precursor of “hsa-miR-6887-5p”.

The term “hsa-miR-3620-5p gene” or “hsa-miR-3620-5p” used herein includes the hsa-miR-3620-5p gene (miRBase Accession No. MIMAT0022967) described in SEQ ID NO: 82, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3620-5p gene can be obtained by a method described in Witten D et al., 2010, BMC Biol, Vol. 8, p. 58. Also, “hsa-mir-3620” (miRBase Accession No. MI0016011, SEQ ID NO: 356) having a hairpin-like structure is known as a precursor of “hsa-miR-3620-5p”.

The term “hsa-miR-1909-3p gene” or “hsa-miR-1909-3p” used herein includes the hsa-miR-1909-3p gene (miRBase Accession No. MIMAT0007883) described in SEQ ID NO: 83, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1909-3p gene can be obtained by a method described in Bar M et al., 2008, Stem Cells, Vol. 26, p. 2496-2505. Also, “hsa-mir-1909” (miRBase Accession No. MI0008330, SEQ ID NO: 357) having a hairpin-like structure is known as a precursor of “hsa-miR-1909-3p”.

The term “hsa-miR-7641 gene” or “hsa-miR-7641” used herein includes the hsa-miR-7641 gene (miRBase Accession No. MIMAT0029782) described in SEQ ID NO: 84, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7641 gene can be obtained by a method described in Yoo J K et al., 2013, Arch Pharm Res, Vol. 36, p. 353-358. Also, “hsa-mir-7641-1 and hsa-mir-7641-2” (miRBase Accession Nos. MI0024975 and MI0024976, SEQ ID NOs: 358 and 359) having a hairpin-like structure are known as precursors of “hsa-miR-7641”.

The term “hsa-miR-6724-5p gene” or “hsa-miR-6724-5p” used herein includes the hsa-miR-6724-5p gene (miRBase Accession No. MIMAT0025856) described in SEQ ID NO: 85, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6724-5p gene can be obtained by a method described in Li Y et al., 2012, Gene, Vol. 497, p. 330-335. Also, “hsa-mir-6724” (miRBase Accession No. MI0022559, SEQ ID NO: 360) having a hairpin-like structure is known as a precursor of “hsa-miR-6724-5p”.

The term “hsa-miR-1343-3p gene” or “hsa-miR-1343-3p” used herein includes the hsa-miR-1343-3p gene (miRBase Accession No. MIMAT0019776) described in SEQ ID NO: 86, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1343-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-1343” (miRBase Accession No. MI0017320, SEQ ID NO: 361) having a hairpin-like structure is known as a precursor of “hsa-miR-1343-3p”.

The term “hsa-miR-6780b-5p gene” or “hsa-miR-6780b-5p” used herein includes the hsa-miR-6780b-5p gene (miRBase Accession No. MIMAT0027572) described in SEQ ID NO: 87, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6780b-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6780b” (miRBase Accession No. MI0022681, SEQ ID NO: 362) having a hairpin-like structure is known as a precursor of “hsa-miR-6780b-5p”.

The term “hsa-miR-4484 gene” or “hsa-miR-4484” used herein includes the hsa-miR-4484 gene (miRBase Accession No. MIMAT0019018) described in SEQ ID NO: 88, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4484 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4484” (miRBase Accession No.

MI0016845, SEQ ID NO: 363) having a hairpin-like structure is known as a precursor of “hsa-miR-4484”.

The term “hsa-miR-4690-5p gene” or “hsa-miR-4690-5p” used herein includes the hsa-miR-4690-5p gene (miRBase Accession No. MIMAT0019779) described in SEQ ID NO: 89, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4690-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4690” (miRBase Accession No. MI0017323, SEQ ID NO: 364) having a hairpin-like structure is known as a precursor of “hsa-miR-4690-5p”.

The term “hsa-miR4429 gene” or “hsa-miR-4429” used herein includes the hsa-miR-4429 gene (miRBase Accession No. MIMAT0018944) described in SEQ ID NO: 90, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4429 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4429” (miRBase Accession No. MI0016768, SEQ ID NO: 365) having a hairpin-like structure is known as a precursor of “hsa-miR4429”.

The term “hsa-miR-1227-5p gene” or “hsa-miR-1227-5p” used herein includes the hsa-miR-1227-5p gene (miRBase Accession No. MIMAT0022941) described in SEQ ID NO: 91, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1227-5p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1227” (miRBase Accession No. MI0006316, SEQ ID NO: 366) having a hairpin-like structure is known as a precursor of “hsa-miR-1227-5p”.

The term “hsa-miR-4725-3p gene” or “hsa-miR-4725-3p” used herein includes the hsa-miR-4725-3p gene (miRBase Accession No. MIMAT0019844) described in SEQ ID NO: 92, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4725-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4725” (miRBase Accession No. MI0017362, SEQ ID NO: 367) having a hairpin-like structure is known as a precursor of “hsa-miR-4725-3p”.

The term “hsa-miR-6861-5p gene” or “hsa-miR-6861-5p” used herein includes the hsa-miR-6861-5p gene (miRBase Accession No. MIMAT0027623) described in SEQ ID NO: 93, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6861-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6861” (miRBase Accession No. MI0022708, SEQ ID NO: 368) having a hairpin-like structure is known as a precursor of “hsa-miR-6861-5p”.

The term “hsa-miR-6812-5p gene” or “hsa-miR-6812-5p” used herein includes the hsa-miR-6812-5p gene (miRBase Accession No. MIMAT0027524) described in SEQ ID NO: 94, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6812-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6812” (miRBase Accession No. MI0022657, SEQ ID NO: 369) having a hairpin-like structure is known as a precursor of “hsa-miR-6812-5p”.

The term “hsa-miR-3197 gene” or “hsa-miR-3197” used herein includes the hsa-miR-3197 gene (miRBase Accession No. MIMAT0015082) described in SEQ ID NO: 95, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3197 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3197” (miRBase Accession No.

MI0014245, SEQ ID NO: 370) having a hairpin-like structure is known as a precursor of "hsa-miR-3197".

The term "hsa-miR-8059 gene" or "hsa-miR-8059" used herein includes the hsa-miR-8059 gene (miRBase Accession No. MIMAT0030986) described in SEQ ID NO: 96, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8059 gene can be obtained by a method described in Wang H J et al., 2013, Shock, Vol. 39, p. 480-487. Also, "hsa-mir-8059" (miRBase Accession No. MI0025895, SEQ ID NO: 371) having a hairpin-like structure is known as a precursor of "hsa-miR-8059".

The term "hsa-miR-3185 gene" or "hsa-miR-3185" used herein includes the hsa-miR-3185 gene (miRBase Accession No. MIMAT0015065) described in SEQ ID NO: 97, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3185 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, "hsa-mir-3185" (miRBase Accession No. MI0014227, SEQ ID NO: 372) having a hairpin-like structure is known as a precursor of "hsa-miR-3185".

The term "hsa-miR-4706 gene" or "hsa-miR-4706" used herein includes the hsa-miR-4706 gene (miRBase Accession No. MIMAT0019806) described in SEQ ID NO: 98, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4706 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, "hsa-mir-4706" (miRBase Accession No. MI0017339, SEQ ID NO: 373) having a hairpin-like structure is known as a precursor of "hsa-miR-4706".

The term "hsa-miR-4497 gene" or "hsa-miR-4497" used herein includes the hsa-miR-4497 gene (miRBase Accession No. MIMAT0019032) described in SEQ ID NO: 99, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4497 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, "hsa-mir-4497" (miRBase Accession No. MI0016859, SEQ ID NO: 374) having a hairpin-like structure is known as a precursor of "hsa-miR-4497".

The term "hsa-miR-3131 gene" or "hsa-miR-3131" used herein includes the hsa-miR-3131 gene (miRBase Accession No. MIMAT0014996) described in SEQ ID NO: 100, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3131 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, "hsa-mir-3131" (miRBase Accession No. MI0014151, SEQ ID NO: 375) having a hairpin-like structure is known as a precursor of "hsa-miR-3131".

The term "hsa-miR-6806-5p gene" or "hsa-miR-6806-5p" used herein includes the hsa-miR-6806-5p gene (miRBase Accession No. MIMAT0027512) described in SEQ ID NO: 101, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6806-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, "hsa-mir-6806" (miRBase Accession No. MI0022651, SEQ ID NO: 376) having a hairpin-like structure is known as a precursor of "hsa-miR-6806-5p".

The term "hsa-miR-187-5p gene" or "hsa-miR-187-5p" used herein includes the hsa-miR-187-5p gene (miRBase Accession No. MIMAT0004561) described in SEQ ID NO: 102, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-187-5p gene can be obtained by a method described in Lim L P et al., 2003, Science, Vol. 299, p. 1540. Also, "hsa-mir-187" (miRBase Accession No. MI0000274, SEQ ID NO: 377) having a hairpin-like structure is known as a precursor of "hsa-miR-187-5p".

The term "hsa-miR-3180-3p gene" or "hsa-miR-3180-3p" used herein includes the hsa-miR-3180-3p gene (miRBase Accession No. MIMAT0015058) described in SEQ ID NO: 103, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3180-3p gene can be obtained by a method described in Creighton C J et al., 2010, PLoS One, Vol. 5, e9637. Also, "hsa-mir-3180-1, hsa-mir-3180-2, and hsa-mir-3180-3" (miRBase Accession Nos. MI0014214, MI0014215, and MI0014217, SEQ ID NOs: 378, 379, and 380) having a hairpin-like structure are known as precursors of "hsa-miR-3180-3p".

The term "hsa-miR-6848-5p gene" or "hsa-miR-6848-5p" used herein includes the hsa-miR-6848-5p gene (miRBase Accession No. MIMAT0027596) described in SEQ ID NO: 104, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6848-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, "hsa-mir-6848" (miRBase Accession No. MI0022694, SEQ ID NO: 381) having a hairpin-like structure is known as a precursor of "hsa-miR-6848-5p".

The term "hsa-miR-6820-5p gene" or "hsa-miR-6820-5p" used herein includes the hsa-miR-6820-5p gene (miRBase Accession No. MIMAT0027540) described in SEQ ID NO: 105, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6820-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, "hsa-mir-6820" (miRBase Accession No. MI0022665, SEQ ID NO: 382) having a hairpin-like structure is known as a precursor of "hsa-miR-6820-5p".

The term "hsa-miR-6800-5p gene" or "hsa-miR-6800-5p" used herein includes the hsa-miR-6800-5p gene (miRBase Accession No. MIMAT0027500) described in SEQ ID NO: 106, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6800-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, "hsa-mir-6800" (miRBase Accession No. MI0022645, SEQ ID NO: 383) having a hairpin-like structure is known as a precursor of "hsa-miR-6800-5p".

The term "hsa-miR-6717-5p gene" or "hsa-miR-6717-5p" used herein includes the hsa-miR-6717-5p gene (miRBase Accession No. MIMAT0025846) described in SEQ ID NO: 107, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6717-5p gene can be obtained by a method described in Li Y et al., 2012, Gene, Vol. 497, p. 330-335. Also, "hsa-mir-6717" (miRBase Accession No. MI0022551, SEQ ID NO: 384) having a hairpin-like structure is known as a precursor of "hsa-miR-6717-5p".

The term "hsa-miR-6795-5p gene" or "hsa-miR-6795-5p" used herein includes the hsa-miR-6795-5p gene (miRBase Accession No. MIMAT0027490) described in SEQ ID NO: 108, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6795-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, "hsa-mir-6795" (miRBase Accession No. MI0022640, SEQ ID NO: 385) having a hairpin-like structure is known as a precursor of "hsa-miR-6795-5p".

The term "hsa-miR-4632-5p gene" or "hsa-miR-4632-5p" used herein includes the hsa-miR-4632-5p gene (miRBase Accession No. MIMAT0022977) described in SEQ ID NO: 109, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4632-5p gene can be obtained by a method described in Persson H et al., 2011,

Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4632” (miR-Base Accession No. MI0017259, SEQ ID NO: 386) having a hairpin-like structure is known as a precursor of “hsa-miR-4632-5p”.

The term “hsa-miR-665 gene” or “hsa-miR-665” used herein includes the hsa-miR-665 gene (miRBase Accession No. MIMAT0004952) described in SEQ ID NO: 110, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-665 gene can be obtained by a method described in Berezikov E et al., 2006, Genome Res, Vol. 16, p. 1289-1298. Also, “hsa-mir-665” (miRBase Accession No. MI0005563, SEQ ID NO: 387) having a hairpin-like structure is known as a precursor of “hsa-miR-665”.

The term “hsa-miR-6778-5p gene” or “hsa-miR-6778-5p” used herein includes the hsa-miR-6778-5p gene (miRBase Accession No. MIMAT0027456) described in SEQ ID NO: 111, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6778-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6778” (miRBase Accession No. MI0022623, SEQ ID NO: 388) having a hairpin-like structure is known as a precursor of “hsa-miR-6778-5p”.

The term “hsa-miR-3663-3p gene” or “hsa-miR-3663-3p” used herein includes the hsa-miR-3663-3p gene (miRBase Accession No. MIMAT0018085) described in SEQ ID NO: 112, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3663-3p gene can be obtained by a method described in Liao J Y et al., 2010, PLoS One, Vol. 5, e10563. Also, “hsa-mir-3663” (miRBase Accession No. MI0016064, SEQ ID NO: 389) having a hairpin-like structure is known as a precursor of “hsa-miR-3663-3p”.

The term “hsa-miR-4689 gene” or “hsa-miR-4689” used herein includes the hsa-miR-4689 gene (miRBase Accession No. MIMAT0019778) described in SEQ ID NO: 113, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4689 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4689” (miRBase Accession No. MI0017322, SEQ ID NO: 390) having a hairpin-like structure is known as a precursor of “hsa-miR-4689”.

The term “hsa-miR-211-3p gene” or “hsa-miR-211-3p” used herein includes the hsa-miR-211-3p gene (miRBase Accession No. MIMAT0022694) described in SEQ ID NO: 114, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-211-3p gene can be obtained by a method described in Lim L P et al., 2003, Science, Vol. 299, p. 1540. Also, “hsa-mir-211” (miRBase Accession No. MI0000287, SEQ ID NO: 391) having a hairpin-like structure is known as a precursor of “hsa-miR-211-3p”.

The term “hsa-miR-6511b-5p gene” or “hsa-miR-6511b-5p” used herein includes the hsa-miR-6511b-5p gene (miR-Base Accession No. MIMAT0025847) described in SEQ ID NO: 115, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6511b-5p gene can be obtained by a method described in Li Y et al., 2012, Gene, Vol. 497, p. 330-335. Also, “hsa-mir-6511b-1 and hsa-mir-6511b-2” (miRBase Accession Nos. MI0022552 and MI0023431, SEQ ID NOs: 392 and 393) having a hairpin-like structure are known as precursors of “hsa-miR-6511b-5p”.

The term “hsa-miR-4750-5p gene” or “hsa-miR-4750-5p” used herein includes the hsa-miR-4750-5p gene (miRBase Accession No. MIMAT0019887) described in SEQ ID NO: 116, a homolog or an ortholog of a different organism

species, and the like. The hsa-miR-4750-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4750” (miR-Base Accession No. MI0017389, SEQ ID NO: 394) having a hairpin-like structure is known as a precursor of “hsa-miR-4750-5p”.

The term “hsa-miR-6126 gene” or “hsa-miR-6126” used herein includes the hsa-miR-6126 gene (miRBase Accession No. MIMAT0024599) described in SEQ ID NO: 117, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6126 gene can be obtained by a method described in Smith J L et al., 2012, J Virol. Vol. 86, p. 5278-5287. Also, “hsa-mir-6126” (miRBase Accession No. MI0021260, SEQ ID NO: 395) having a hairpin-like structure is known as a precursor of “hsa-miR-6126”.

The term “hsa-miR-614 gene” or “hsa-miR-614” used herein includes the hsa-miR-614 gene (miRBase Accession No. MIMAT0003282) described in SEQ ID NO: 118, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-614 gene can be obtained by a method described in Cummins J M et al., 2006, Proc Natl Acad Sci USA, Vol. 103, p. 3687-3692. Also, “hsa-mir-614” (miR-Base Accession No. MI0003627, SEQ ID NO: 396) having a hairpin-like structure is known as a precursor of “hsa-miR-614”.

The term “hsa-miR-7110-5p gene” or “hsa-miR-7110-5p” used herein includes the hsa-miR-7110-5p gene (miRBase Accession No. MIMAT0028117) described in SEQ ID NO: 119, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7110-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-7110” (miRBase Accession No. MI0022961, SEQ ID NO: 397) having a hairpin-like structure is known as a precursor of “hsa-miR-7110-5p”.

The term “hsa-miR-744-5p gene” or “hsa-miR-744-5p” used herein includes the hsa-miR-744-5p gene (miRBase Accession No. MIMAT0004945) described in SEQ ID NO: 120, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-744-5p gene can be obtained by a method described in Berezikov E et al., 2006, Genome Res, Vol. 16, p. 1289-1298. Also, “hsa-mir-744” (miRBase Accession No. MI0005559, SEQ ID NO: 398) having a hairpin-like structure is known as a precursor of “hsa-miR-744-5p”.

The term “hsa-miR-6769a-5p gene” or “hsa-miR-6769a-5p” used herein includes the hsa-miR-6769a-5p gene (miR-Base Accession No. MIMAT0027438) described in SEQ ID NO: 121, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6769a-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6769a” (miRBase Accession No. MI0022614, SEQ ID NO: 399) having a hairpin-like structure is known as a precursor of “hsa-miR-6769a-5p”.

The term “hsa-miR-4792 gene” or “hsa-miR-4792” used herein includes the hsa-miR-4792 gene (miRBase Accession No. MIMAT0019964) described in SEQ ID NO: 122, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4792 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4792” (miRBase Accession No. MI0017439, SEQ ID NO: 400) having a hairpin-like structure is known as a precursor of “hsa-miR-4792”.

The term “hsa-miR-5787 gene” or “hsa-miR-5787” used herein includes the hsa-miR-5787 gene (miRBase Accession No. MIMAT0023252) described in SEQ ID NO: 123, a

homolog or an ortholog of a different organism species, and the like. The hsa-miR-5787 gene can be obtained by a method described in Yoo H et al., 2011, *Biochem Biophys Res Commun*, Vol. 415, p. 567-572. Also, "hsa-mir-5787" (miRBase Accession No. MI0019797, SEQ ID NO: 401) having a hairpin-like structure is known as a precursor of "hsa-miR-5787".

The term "hsa-miR-6798-5p gene" or "hsa-miR-6798-5p" used herein includes the hsa-miR-6798-5p gene (miRBase Accession No. MIMAT0027496) described in SEQ ID NO: 124, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6798-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6798" (miRBase Accession No. MI0022643, SEQ ID NO: 402) having a hairpin-like structure is known as a precursor of "hsa-miR-6798-5p".

The term "hsa-miR-6781-5p gene" or "hsa-miR-6781-5p" used herein includes the hsa-miR-6781-5p gene (miRBase Accession No. MIMAT0027462) described in SEQ ID NO: 125, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6781-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6781" (miRBase Accession No. MI0022626, SEQ ID NO: 403) having a hairpin-like structure is known as a precursor of "hsa-miR-6781-5p".

The term "hsa-miR-4419b gene" or "hsa-miR-4419b" used herein includes the hsa-miR-4419b gene (miRBase Accession No. MIMAT0019034) described in SEQ ID NO: 126, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4419b gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, "hsa-mir-4419b" (miRBase Accession No. MI0016861, SEQ ID NO: 404) having a hairpin-like structure is known as a precursor of "hsa-miR-4419b".

The term "hsa-miR-4446-3p gene" or "hsa-miR-4446-3p" used herein includes the hsa-miR-4446-3p gene (miRBase Accession No. MIMAT0018965) described in SEQ ID NO: 127, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4446-3p gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, "hsa-mir-4446" (miRBase Accession No. MI0016789, SEQ ID NO: 405) having a hairpin-like structure is known as a precursor of "hsa-miR-4446-3p".

The term "hsa-miR-4259 gene" or "hsa-miR-4259" used herein includes the hsa-miR-4259 gene (miRBase Accession No. MIMAT0016880) described in SEQ ID NO: 128, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4259 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, "hsa-mir-4259" (miRBase Accession No. MI0015858, SEQ ID NO: 406) having a hairpin-like structure is known as a precursor of "hsa-miR-4259".

The term "hsa-miR-5572 gene" or "hsa-miR-5572" used herein includes the hsa-miR-5572 gene (miRBase Accession No. MIMAT0022260) described in SEQ ID NO: 129, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5572 gene can be obtained by a method described in Tandon M et al., 2012, *Oral Dis*, Vol. 18, p. 127-131. Also, "hsa-mir-5572" (miRBase Accession No. MI0019117, SEQ ID NO: 407) having a hairpin-like structure is known as a precursor of "hsa-miR-5572".

The term "hsa-miR-6075 gene" or "hsa-miR-6075" used herein includes the hsa-miR-6075 gene (miRBase Accession

No. MIMAT0023700) described in SEQ ID NO: 130, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6075 gene can be obtained by a method described in Voellenkle C et al., 2012, *RNA*, Vol. 18, p. 472-484. Also, "hsa-mir-6075" (miRBase Accession No. MI0020352, SEQ ID NO: 408) having a hairpin-like structure is known as a precursor of "hsa-miR-6075".

The term "hsa-miR-296-3p gene" or "hsa-miR-296-3p" used herein includes the hsa-miR-296-3p gene (miRBase Accession No. MIMAT0004679) described in SEQ ID NO: 131, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-296-3p gene can be obtained by a method described in Houbaviy H B et al., 2003, *Dev Cell*, Vol. 5, p. 351-358. Also, "hsa-mir-296" (miRBase Accession No. MI0000747, SEQ ID NO: 409) having a hairpin-like structure is known as a precursor of "hsa-miR-296-3p".

The term "hsa-miR-6891-5p gene" or "hsa-miR-6891-5p" used herein includes the hsa-miR-6891-5p gene (miRBase Accession No. MIMAT0027682) described in SEQ ID NO: 132, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6891-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6891" (miRBase Accession No. MI0022738, SEQ ID NO: 410) having a hairpin-like structure is known as a precursor of "hsa-miR-6891-5p".

The term "hsa-miR-4745-5p gene" or "hsa-miR-4745-5p" used herein includes the hsa-miR-4745-5p gene (miRBase Accession No. MIMAT0019878) described in SEQ ID NO: 133, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4745-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, "hsa-mir-4745" (miRBase Accession No. MI0017384, SEQ ID NO: 411) having a hairpin-like structure is known as a precursor of "hsa-miR-4745-5p".

The term "hsa-miR-6775-5p gene" or "hsa-miR-6775-5p" used herein includes the hsa-miR-6775-5p gene (miRBase Accession No. MIMAT0027450) described in SEQ ID NO: 134, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6775-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6775" (miRBase Accession No. MI0022620, SEQ ID NO: 412) having a hairpin-like structure is known as a precursor of "hsa-miR-6775-5p".

The term "hsa-miR-6870-5p gene" or "hsa-miR-6870-5p" used herein includes the hsa-miR-6870-5p gene (miRBase Accession No. MIMAT0027640) described in SEQ ID NO: 135, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6870-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, "hsa-mir-6870" (miRBase Accession No. MI0022717, SEQ ID NO: 413) having a hairpin-like structure is known as a precursor of "hsa-miR-6870-5p".

The term "hsa-miR-920 gene" or "hsa-miR-920" used herein includes the hsa-miR-920 gene (miRBase Accession No. MIMAT0004970) described in SEQ ID NO: 136, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-920 gene can be obtained by a method described in Novotny G W et al., 2007, *Int J Androl*, Vol. 30, p. 316-326. Also, "hsa-mir-920" (miRBase Accession No. MI0005712, SEQ ID NO: 414) having a hairpin-like structure is known as a precursor of "hsa-miR-920".

The term “hsa-miR-4530 gene” or “hsa-miR-4530” used herein includes the hsa-miR-4530 gene (miRBase Accession No. MIMAT0019069) described in SEQ ID NO: 137, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4530 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4530” (miRBase Accession No. MI0016897, SEQ ID NO: 415) having a hairpin-like structure is known as a precursor of “hsa-miR-4530”.

The term “hsa-miR-6819-5p gene” or “hsa-miR-6819-5p” used herein includes the hsa-miR-6819-5p gene (miRBase Accession No. MIMAT0027538) described in SEQ ID NO: 138, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6819-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6819” (miRBase Accession No. MI0022664, SEQ ID NO: 416) having a hairpin-like structure is known as a precursor of “hsa-miR-6819-5p”.

The term “hsa-miR-6825-5p gene” or “hsa-miR-6825-5p” used herein includes the hsa-miR-6825-5p gene (miRBase Accession No. MIMAT0027550) described in SEQ ID NO: 139, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6825-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6825” (miRBase Accession No. MI0022670, SEQ ID NO: 417) having a hairpin-like structure is known as a precursor of “hsa-miR-6825-5p”.

The term “hsa-miR-7847-3p gene” or “hsa-miR-7847-3p” used herein includes the hsa-miR-7847-3p gene (miRBase Accession No. MIMAT0030422) described in SEQ ID NO: 140, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7847-3p gene can be obtained by a method described in Pie H et al., 2012, *PLoS One*, Vol. 7, e50746. Also, “hsa-mir-7847” (miRBase Accession No. MI0025517, SEQ ID NO: 418) having a hairpin-like structure is known as a precursor of “hsa-miR-7847-3p”.

The term “hsa-miR-6131 gene” or “hsa-miR-6131” used herein includes the hsa-miR-6131 gene (miRBase Accession No. MIMAT0024615) described in SEQ ID NO: 141, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6131 gene can be obtained by a method described in Dannemann M et al., 2012, *Genome Biol Evol*, Vol. 4, p. 552-564. Also, “hsa-mir-6131” (miRBase Accession No. MI0021276, SEQ ID NO: 419) having a hairpin-like structure is known as a precursor of “hsa-miR-6131”.

The term “hsa-miR-4433-3p gene” or “hsa-miR-4433-3p” used herein includes the hsa-miR-4433-3p gene (miRBase Accession No. MIMAT0018949) described in SEQ ID NO: 142, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4433-3p gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4433” (miRBase Accession No. MI0016773, SEQ ID NO: 420) having a hairpin-like structure is known as a precursor of “hsa-miR-4433-3p”.

The term “hsa-miR-1228-5p gene” or “hsa-miR-1228-5p” used herein includes the hsa-miR-1228-5p gene (miRBase Accession No. MIMAT0005582) described in SEQ ID NO: 143, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1228-5p gene can be obtained by a method described in Berezikov E et al., 2007, *Mol Cell*, Vol. 28, p. 328-336. Also, “hsa-mir-1228” (miR-

Base Accession No. MI0006318, SEQ ID NO: 421) having a hairpin-like structure is known as a precursor of “hsa-miR-1228-5p”.

The term “hsa-miR-6743-5p gene” or “hsa-miR-6743-5p” used herein includes the hsa-miR-6743-5p gene (miRBase Accession No. MIMAT0027387) described in SEQ ID NO: 144, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6743-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6743” (miRBase Accession No. MI0022588, SEQ ID NO: 422) having a hairpin-like structure is known as a precursor of “hsa-miR-6743-5p”.

The term “hsa-miR-1268a gene” or “hsa-miR-1268a” used herein includes the hsa-miR-1268a gene (miRBase Accession No. MIMAT0005922) described in SEQ ID NO: 145, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1268a gene can be obtained by a method described in Morin R D et al., 2008, *Genome Res*, Vol. 18, p. 610-621. Also, “hsa-mir-1268a” (miRBase Accession No. MI0006405, SEQ ID NO: 423) having a hairpin-like structure is known as a precursor of “hsa-miR-1268a”.

The term “hsa-miR-3917 gene” or “hsa-miR-3917” used herein includes the hsa-miR-3917 gene (miRBase Accession No. MIMAT0018191) described in SEQ ID NO: 146, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3917 gene can be obtained by a method described in Creighton C) et al., 2010, *PLoS One*, Vol. 5, e9637. Also, “hsa-mir-3917” (miRBase Accession No. MI0016423, SEQ ID NO: 424) having a hairpin-like structure is known as a precursor of “hsa-miR-3917”.

The term “hsa-miR-6786-5p gene” or “hsa-miR-6786-5p” used herein includes the hsa-miR-6786-5p gene (miRBase Accession No. MIMAT0027472) described in SEQ ID NO: 147, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6786-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6786” (miRBase Accession No. MI0022631, SEQ ID NO: 425) having a hairpin-like structure is known as a precursor of “hsa-miR-6786-5p”.

The term “hsa-miR-3154 gene” or “hsa-miR-3154” used herein includes the hsa-miR-3154 gene (miRBase Accession No. MIMAT0015028) described in SEQ ID NO: 148, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3154 gene can be obtained by a method described in Berezikov E et al., 2006, *Genome Res*, Vol. 16, p. 1289-1298. Also, “hsa-mir-3154” (miRBase Accession No. MI0014182, SEQ ID NO: 426) having a hairpin-like structure is known as a precursor of “hsa-miR-3154”.

The term “hsa-miR-638 gene” or “hsa-miR-638” used herein includes the hsa-miR-638 gene (miRBase Accession No. MIMAT0003308) described in SEQ ID NO: 149, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-638 gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-638” (miRBase Accession No. MI0003653, SEQ ID NO: 427) having a hairpin-like structure is known as a precursor of “hsa-miR-638”.

The term “hsa-miR-6741-5p gene” or “hsa-miR-6741-5p” used herein includes the hsa-miR-6741-5p gene (miRBase Accession No. MIMAT0027383) described in SEQ ID NO: 150, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6741-5p gene can be

obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6741” (miRBase Accession No. MI0022586, SEQ ID NO: 428) having a hairpin-like structure is known as a precursor of “hsa-miR-6741-5p”.

The term “hsa-miR-6889-5p gene” or “hsa-miR-6889-5p” used herein includes the hsa-miR-6889-5p gene (miRBase Accession No. MIMAT0027678) described in SEQ ID NO: 151, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6889-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6889” (miRBase Accession No. MI0022736, SEQ ID NO: 429) having a hairpin-like structure is known as a precursor of “hsa-miR-6889-5p”.

The term “hsa-miR-6840-3p gene” or “hsa-miR-6840-3p” used herein includes the hsa-miR-6840-3p gene (miRBase Accession No. MIMAT0027583) described in SEQ ID NO: 152, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6840-3p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6840” (miRBase Accession No. MI0022686, SEQ ID NO: 430) having a hairpin-like structure is known as a precursor of “hsa-miR-6840-3p”.

The term “hsa-miR-6510-5p gene” or “hsa-miR-6510-5p” used herein includes the hsa-miR-6510-5p gene (miRBase Accession No. MIMAT0025476) described in SEQ ID NO: 153, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6510-5p gene can be obtained by a method described in Joyce C E et al., 2011, *Hum Mol Genet*, Vol. 20, p. 4025-4040. Also, “hsa-mir-6510” (miRBase Accession No. MI0022222, SEQ ID NO: 431) having a hairpin-like structure is known as a precursor of “hsa-miR-6510-5p”.

The term “hsa-miR-3188 gene” or “hsa-miR-3188” used herein includes the hsa-miR-3188 gene (miRBase Accession No. MIMAT0015070) described in SEQ ID NO: 154, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3188 gene can be obtained by a method described in Stark M S et al., 2010, *PLoS One*, Vol. 5, e9685. Also, “hsa-mir-3188” (miRBase Accession No. MI0014232, SEQ ID NO: 432) having a hairpin-like structure is known as a precursor of “hsa-miR-3188”.

The term “hsa-miR-551b-5p gene” or “hsa-miR-551b-5p” used herein includes the hsa-miR-551b-5p gene (miRBase Accession No. MIMAT0004794) described in SEQ ID NO: 155, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-551b-5p gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-551b” (miRBase Accession No. MI0003575, SEQ ID NO: 433) having a hairpin-like structure is known as a precursor of “hsa-miR-551b-5p”.

The term “hsa-miR-5001-5p gene” or “hsa-miR-5001-5p” used herein includes the hsa-miR-5001-5p gene (miRBase Accession No. MIMAT0021021) described in SEQ ID NO: 156, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5001-5p gene can be obtained by a method described in Hansen T B et al., 2011, *RNA Biol*, Vol. 8, p. 378-383. Also, “hsa-mir-5001” (miRBase Accession No. MI0017867, SEQ ID NO: 434) having a hairpin-like structure is known as a precursor of “hsa-miR-5001-5p”.

The term “hsa-miR-1268b gene” or “hsa-miR-1268b” used herein includes the hsa-miR-1268b gene (miRBase Accession No. MIMAT0018925) described in SEQ ID NO:

157, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1268b gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-1268b” (miRBase Accession No. MI0016748, SEQ ID NO: 435) having a hairpin-like structure is known as a precursor of “hsa-miR-1268b”.

The term “hsa-miR-7107-5p gene” or “hsa-miR-7107-5p” used herein includes the hsa-miR-7107-5p gene (miRBase Accession No. MIMAT0028111) described in SEQ ID NO: 158, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7107-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-7107” (miRBase Accession No. MI0022958, SEQ ID NO: 436) having a hairpin-like structure is known as a precursor of “hsa-miR-7107-5p”.

The term “hsa-miR-6824-5p gene” or “hsa-miR-6824-5p” used herein includes the hsa-miR-6824-5p gene (miRBase Accession No. MIMAT0027548) described in SEQ ID NO: 159, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6824-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6824” (miRBase Accession No. MI0022669, SEQ ID NO: 437) having a hairpin-like structure is known as a precursor of “hsa-miR-6824-5p”.

The term “hsa-miR-6732-5p gene” or “hsa-miR-6732-5p” used herein includes the hsa-miR-6732-5p gene (miRBase Accession No. MIMAT0027365) described in SEQ ID NO: 160, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6732-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6732” (miRBase Accession No. MI0022577, SEQ ID NO: 438) having a hairpin-like structure is known as a precursor of “hsa-miR-6732-5p”.

The term “hsa-miR-371a-5p gene” or “hsa-miR-371a-5p” used herein includes the hsa-miR-371a-5p gene (miRBase Accession No. MIMAT0004687) described in SEQ ID NO: 161, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-371a-5p gene can be obtained by a method described in Suh M R et al., 2004, *Dev Biol*, Vol. 270, p. 488-498. Also, “hsa-mir-371a” (miRBase Accession No. MI0000779, SEQ ID NO: 439) having a hairpin-like structure is known as a precursor of “hsa-miR-371a-5p”.

The term “hsa-miR-6794-5p gene” or “hsa-miR-6794-5p” used herein includes the hsa-miR-6794-5p gene (miRBase Accession No. MIMAT0027488) described in SEQ ID NO: 162, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6794-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6794” (miRBase Accession No. MI0022639, SEQ ID NO: 440) having a hairpin-like structure is known as a precursor of “hsa-miR-6794-5p”.

The term “hsa-miR-6779-5p gene” or “hsa-miR-6779-5p” used herein includes the hsa-miR-6779-5p gene (miRBase Accession No. MIMAT0027458) described in SEQ ID NO: 163, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6779-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6779” (miRBase Accession No. MI0022624, SEQ ID NO: 441) having a hairpin-like structure is known as a precursor of “hsa-miR-6779-5p”.

The term “hsa-miR-4271 gene” or “hsa-miR-4271” used herein includes the hsa-miR-4271 gene (miRBase Accession No. MIMAT0016901) described in SEQ ID NO: 164, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4271 gene can be obtained by a method described in Goff L A et al., 2009, PLoS One, Vol. 4, e7192. Also, “hsa-mir-4271” (miRBase Accession No. MI0015879, SEQ ID NO: 442) having a hairpin-like structure is known as a precursor of “hsa-miR-4271”.

The term “hsa-miR-5195-3p gene” or “hsa-miR-5195-3p” used herein includes the hsa-miR-5195-3p gene (miRBase Accession No. MIMAT0021127) described in SEQ ID NO: 165, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5195-3p gene can be obtained by a method described in Schotte D et al., 2011, Leukemia, Vol. 25, p. 1389-1399. Also, “hsa-mir-5195” (miRBase Accession No. MI0018174, SEQ ID NO: 443) having a hairpin-like structure is known as a precursor of “hsa-miR-5195-3p”.

The term “hsa-miR-6762-5p gene” or “hsa-miR-6762-5p” used herein includes the hsa-miR-6762-5p gene (miRBase Accession No. MIMAT0027424) described in SEQ ID NO: 166, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6762-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6762” (miRBase Accession No. MI0022607, SEQ ID NO: 444) having a hairpin-like structure is known as a precursor of “hsa-miR-6762-5p”.

The term “hsa-miR-939-5p gene” or “hsa-miR-939-5p” used herein includes the hsa-miR-939-5p gene (miRBase Accession No. MIMAT0004982) described in SEQ ID NO: 167, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-939-5p gene can be obtained by a method described in Lui W O et al., 2007, Cancer Res, Vol. 67, p. 6031-6043. Also, “hsa-mir-939” (miRBase Accession No. MI0005761, SEQ ID NO: 445) having a hairpin-like structure is known as a precursor of “hsa-miR-939-5p”.

The term “hsa-miR-1247-3p gene” or “hsa-miR-1247-3p” used herein includes the hsa-miR-1247-3p gene (miRBase Accession No. MIMAT0022721) described in SEQ ID NO: 168, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1247-3p gene can be obtained by a method described in Morin R D et al., 2008, Genome Res, Vol. 18, p. 610-621. Also, “hsa-mir-1247” (miRBase Accession No. MI0006382, SEQ ID NO: 446) having a hairpin-like structure is known as a precursor of “hsa-miR-1247-3p”.

The term “hsa-miR-6777-5p gene” or “hsa-miR-6777-5p” used herein includes the hsa-miR-6777-5p gene (miRBase Accession No. MIMAT0027454) described in SEQ ID NO: 169, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6777-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6777” (miRBase Accession No. MI0022622, SEQ ID NO: 447) having a hairpin-like structure is known as a precursor of “hsa-miR-6777-5p”.

The term “hsa-miR-6722-3p gene” or “hsa-miR-6722-3p” used herein includes the hsa-miR-6722-3p gene (miRBase Accession No. MIMAT0025854) described in SEQ ID NO: 170, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6722-3p gene can be obtained by a method described in Li Y et al., 2012, Gene, Vol. 497, p. 330-335. Also, “hsa-mir-6722” (miRBase

Accession No. MI0022557, SEQ ID NO: 448) having a hairpin-like structure is known as a precursor of “hsa-miR-6722-3p”.

The term “hsa-miR-3656 gene” or “hsa-miR-3656” used herein includes the hsa-miR-3656 gene (miRBase Accession No. MIMAT0018076) described in SEQ ID NO: 171, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3656 gene can be obtained by a method described in Meiri E et al., 2010, Nucleic Acids Res, Vol. 38, p. 6234-6246. Also, “hsa-mir-3656” (miRBase Accession No. MI0016056, SEQ ID NO: 449) having a hairpin-like structure is known as a precursor of “hsa-miR-3656”.

The term “hsa-miR-4688 gene” or “hsa-miR-4688” used herein includes the hsa-miR-4688 gene (miRBase Accession No. MIMAT0019777) described in SEQ ID NO: 172, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4688 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4688” (miRBase Accession No. MI0017321, SEQ ID NO: 450) having a hairpin-like structure is known as a precursor of “hsa-miR-4688”.

The term “hsa-miR-3195 gene” or “hsa-miR-3195” used herein includes the hsa-miR-3195 gene (miRBase Accession No. MIMAT0015079) described in SEQ ID NO: 173, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3195 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3195” (miRBase Accession No. MI0014240, SEQ ID NO: 451) having a hairpin-like structure is known as a precursor of “hsa-miR-3195”.

The term “hsa-miR-6766-5p gene” or “hsa-miR-6766-5p” used herein includes the hsa-miR-6766-5p gene (miRBase Accession No. MIMAT0027432) described in SEQ ID NO: 174, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6766-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6766” (miRBase Accession No. MI0022611, SEQ ID NO: 452) having a hairpin-like structure is known as a precursor of “hsa-miR-6766-5p”.

The term “hsa-miR-4447 gene” or “hsa-miR-4447” used herein includes the hsa-miR-4447 gene (miRBase Accession No. MIMAT0018966) described in SEQ ID NO: 175, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4447 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4447” (miRBase Accession No. MI0016790, SEQ ID NO: 453) having a hairpin-like structure is known as a precursor of “hsa-miR-4447”.

The term “hsa-miR-4656 gene” or “hsa-miR-4656” used herein includes the hsa-miR-4656 gene (miRBase Accession No. MIMAT0019723) described in SEQ ID NO: 176, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4656 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4656” (miRBase Accession No. MI0017284, SEQ ID NO: 454) having a hairpin-like structure is known as a precursor of “hsa-miR-4656”.

The term “hsa-miR-7108-5p gene” or “hsa-miR-7108-5p” used herein includes the hsa-miR-7108-5p gene (miRBase Accession No. MIMAT0028113) described in SEQ ID NO: 177, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7108-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-7108”

(miRBase Accession No. MI0022959, SEQ ID NO: 455) having a hairpin-like structure is known as a precursor of “hsa-miR-7108-5p”.

The term “hsa-miR-3191-3p gene” or “hsa-miR-3191-3p” used herein includes the hsa-miR-3191-3p gene (miRBase Accession No. MIMAT0015075) described in SEQ ID NO: 178, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3191-3p gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3191” (miRBase Accession No. MI0014236, SEQ ID NO: 456) having a hairpin-like structure is known as a precursor of “hsa-miR-3191-3p”.

The term “hsa-miR-1273g-3p gene” or “hsa-miR-1273g-3p” used herein includes the hsa-miR-1273g-3p gene (miRBase Accession No. MIMAT0022742) described in SEQ ID NO: 179, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1273g-3p gene can be obtained by a method described in Reshmi G et al., 2011, Genomics, Vol. 97, p. 333-340. Also, “hsa-mir-1273g” (miRBase Accession No. MI0018003, SEQ ID NO: 457) having a hairpin-like structure is known as a precursor of “hsa-miR-1273g-3p”.

The term “hsa-miR-4463 gene” or “hsa-miR-4463” used herein includes the hsa-miR-4463 gene (miRBase Accession No. MIMAT0018987) described in SEQ ID NO: 180, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4463 gene can be obtained by a method described in Lima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4463” (miRBase Accession No. MI0016811, SEQ ID NO: 458) having a hairpin-like structure is known as a precursor of “hsa-miR-4463”.

The term “hsa-miR-2861 gene” or “hsa-miR-2861” used herein includes the hsa-miR-2861 gene (miRBase Accession No. MIMAT0013802) described in SEQ ID NO: 181, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-2861 gene can be obtained by a method described in Li H et al., 2009, J Clin Invest, Vol. 119, p. 3666-3677. Also, “hsa-mir-2861” (miRBase Accession No. MI0013006, SEQ ID NO: 459) having a hairpin-like structure is known as a precursor of “hsa-miR-2861”.

The term “hsa-miR-3196 gene” or “hsa-miR-3196” used herein includes the hsa-miR-3196 gene (miRBase Accession No. MIMAT0015080) described in SEQ ID NO: 182, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3196 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3196” (miRBase Accession No. MI0014241, SEQ ID NO: 460) having a hairpin-like structure is known as a precursor of “hsa-miR-3196”.

The term “hsa-miR-6877-5p gene” or “hsa-miR-6877-5p” used herein includes the hsa-miR-6877-5p gene (miRBase Accession No. MIMAT0027654) described in SEQ ID NO: 183, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6877-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6877” (miRBase Accession No. MI0022724, SEQ ID NO: 461) having a hairpin-like structure is known as a precursor of “hsa-miR-6877-5p”.

The term “hsa-miR-3679-5p gene” or “hsa-miR-3679-5p” used herein includes the hsa-miR-3679-5p gene (miRBase Accession No. MIMAT0018104) described in SEQ ID NO: 184, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3679-5p gene can be obtained by a method described in Creighton C J et al., 2010, PLoS One, Vol. 5, e9637. Also, “hsa-mir-3679” (miRBase

Accession No. MI0016080, SEQ ID NO: 462) having a hairpin-like structure is known as a precursor of “hsa-miR-3679-5p”.

The term “hsa-miR-4442 gene” or “hsa-miR-4442” used herein includes the hsa-miR-4442 gene (miRBase Accession No. MIMAT0018960) described in SEQ ID NO: 185, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4442 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4442” (miRBase Accession No. MI0016785, SEQ ID NO: 463) having a hairpin-like structure is known as a precursor of “hsa-miR-4442”.

The term “hsa-miR-6789-5p gene” or “hsa-miR-6789-5p” used herein includes the hsa-miR-6789-5p gene (miRBase Accession No. MIMAT0027478) described in SEQ ID NO: 186, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6789-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6789” (miRBase Accession No. MI0022634, SEQ ID NO: 464) having a hairpin-like structure is known as a precursor of “hsa-miR-6789-5p”.

The term “hsa-miR-6782-5p gene” or “hsa-miR-6782-5p” used herein includes the hsa-miR-6782-5p gene (miRBase Accession No. MIMAT0027464) described in SEQ ID NO: 187, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6782-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6782” (miRBase Accession No. MI0022627, SEQ ID NO: 465) having a hairpin-like structure is known as a precursor of “hsa-miR-6782-5p”.

The term “hsa-miR-486-3p gene” or “hsa-miR-486-3p” used herein includes the hsa-miR-486-3p gene (miRBase Accession No. MIMAT0004762) described in SEQ ID NO: 188, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-486-3p gene can be obtained by a method described in Fu H et al., 2005, FEBS Lett, Vol. 579, p. 3849-3854. Also, “hsa-mir-486 and hsa-mir-486-2” (miRBase Accession Nos. MI0002470 and MI0023622, SEQ ID NOs: 466 and 467) having a hairpin-like structure are known as precursors of “hsa-miR-486-3p”.

The term “hsa-miR-6085 gene” or “hsa-miR-6085” used herein includes the hsa-miR-6085 gene (miRBase Accession No. MIMAT0023710) described in SEQ ID NO: 189, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6085 gene can be obtained by a method described in Voellenkle C et al., 2012, RNA, Vol. 18, p. 472-484. Also, “hsa-mir-6085” (miRBase Accession No. MI0020362, SEQ ID NO: 468) having a hairpin-like structure is known as a precursor of “hsa-miR-6085”.

The term “hsa-miR-4746-3p gene” or “hsa-miR-4746-3p” used herein includes the hsa-miR-4746-3p gene (miRBase Accession No. MIMAT0019881) described in SEQ ID NO: 190, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4746-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4746” (miRBase Accession No. MI0017385, SEQ ID NO: 469) having a hairpin-like structure is known as a precursor of “hsa-miR-4746-3p”.

The term “hsa-miR-619-5p gene” or “hsa-miR-619-5p” used herein includes the hsa-miR-619-5p gene (miRBase Accession No. MIMAT0026622) described in SEQ ID NO: 191, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-619-5p gene can be obtained by a method described in Cummins J M et al.,

2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-619” (miRBase Accession No. MI0003633, SEQ ID NO: 470) having a hairpin-like structure is known as a precursor of “hsa-miR-619-5p”.

The term “hsa-miR-937-5p gene” or “hsa-miR-937-5p” used herein includes the hsa-miR-937-5p gene (miRBase Accession No. MIMAT0022938) described in SEQ ID NO: 192, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-937-5p gene can be obtained by a method described in Lui W O et al., 2007, *Cancer Res*, Vol. 67, p. 6031-6043. Also, “hsa-mir-937” (miRBase Accession No. MI0005759, SEQ ID NO: 471) having a hairpin-like structure is known as a precursor of “hsa-miR-937-5p”.

The term “hsa-miR-6803-5p gene” or “hsa-miR-6803-5p” used herein includes the hsa-miR-6803-5p gene (miRBase Accession No. MIMAT0027506) described in SEQ ID NO: 193, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6803-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6803” (miRBase Accession No. MI0022648, SEQ ID NO: 472) having a hairpin-like structure is known as a precursor of “hsa-miR-6803-5p”.

The term “hsa-miR-4298 gene” or “hsa-miR-4298” used herein includes the hsa-miR-4298 gene (miRBase Accession No. MIMAT0016852) described in SEQ ID NO: 194, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4298 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, “hsa-mir-4298” (miRBase Accession No. MI0015830, SEQ ID NO: 473) having a hairpin-like structure is known as a precursor of “hsa-miR-4298”.

The term “hsa-miR-4454 gene” or “hsa-miR-4454” used herein includes the hsa-miR-4454 gene (miRBase Accession No. MIMAT0018976) described in SEQ ID NO: 195, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4454 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4454” (miRBase Accession No. MI0016800, SEQ ID NO: 474) having a hairpin-like structure is known as a precursor of “hsa-miR-4454”.

The term “hsa-miR-4459 gene” or “hsa-miR-4459” used herein includes the hsa-miR-4459 gene (miRBase Accession No. MIMAT0018981) described in SEQ ID NO: 196, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4459 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4459” (miRBase Accession No. MI0016805, SEQ ID NO: 475) having a hairpin-like structure is known as a precursor of “hsa-miR-4459”.

The term “hsa-miR-7150 gene” or “hsa-miR-7150” used herein includes the hsa-miR-7150 gene (miRBase Accession No. MIMAT0028211) described in SEQ ID NO: 197, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7150 gene can be obtained by a method described in Oulas A et al., 2009, *Nucleic Acids Res*, Vol. 37, p. 3276-3287. Also, “hsa-mir-7150” (miRBase Accession No. MI0023610, SEQ ID NO: 476) having a hairpin-like structure is known as a precursor of “hsa-miR-7150”.

The term “hsa-miR-6880-5p gene” or “hsa-miR-6880-5p” used herein includes the hsa-miR-6880-5p gene (miRBase Accession No. MIMAT0027660) described in SEQ ID NO: 198, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6880-5p gene can be obtained by a method described in Ladewig E et al., 2012,

Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6880” (miRBase Accession No. MI0022727, SEQ ID NO: 477) having a hairpin-like structure is known as a precursor of “hsa-miR-6880-5p”.

The term “hsa-miR-4449 gene” or “hsa-miR-4449” used herein includes the hsa-miR-4449 gene (miRBase Accession No. MIMAT0018968) described in SEQ ID NO: 199, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4449 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4449” (miRBase Accession No. MI0016792, SEQ ID NO: 478) having a hairpin-like structure is known as a precursor of “hsa-miR-4449”.

The term “hsa-miR-8063 gene” or “hsa-miR-8063” used herein includes the hsa-miR-8063 gene (miRBase Accession No. MIMAT0030990) described in SEQ ID NO: 200, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8063 gene can be obtained by a method described in Wang H J et al., 2013, *Shock*, Vol. 39, p. 480-487. Also, “hsa-mir-8063” (miRBase Accession No. MI0025899, SEQ ID NO: 479) having a hairpin-like structure is known as a precursor of “hsa-miR-8063”.

The term “hsa-miR-4695-5p gene” or “hsa-miR-4695-5p” used herein includes the hsa-miR-4695-5p gene (miRBase Accession No. MIMAT0019788) described in SEQ ID NO: 201, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4695-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4695” (miRBase Accession No. MI0017328, SEQ ID NO: 480) having a hairpin-like structure is known as a precursor of “hsa-miR-4695-5p”.

The term “hsa-miR-6132 gene” or “hsa-miR-6132” used herein includes the hsa-miR-6132 gene (miRBase Accession No. MIMAT0024616) described in SEQ ID NO: 202, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6132 gene can be obtained by a method described in Dannemann M et al., 2012, *Genome Biol Evol*, Vol. 4, p. 552-564. Also, “hsa-mir-6132” (miRBase Accession No. MI0021277, SEQ ID NO: 481) having a hairpin-like structure is known as a precursor of “hsa-miR-6132”.

The term “hsa-miR-6829-5p gene” or “hsa-miR-6829-5p” used herein includes the hsa-miR-6829-5p gene (miRBase Accession No. MIMAT0027558) described in SEQ ID NO: 203, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6829-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6829” (miRBase Accession No. MI0022674, SEQ ID NO: 482) having a hairpin-like structure is known as a precursor of “hsa-miR-6829-5p”.

The term “hsa-miR-4486 gene” or “hsa-miR-4486” used herein includes the hsa-miR-4486 gene (miRBase Accession No. MIMAT0019020) described in SEQ ID NO: 204, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4486 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4486” (miRBase Accession No. MI0016847, SEQ ID NO: 483) having a hairpin-like structure is known as a precursor of “hsa-miR-4486”.

The term “hsa-miR-6805-3p gene” or “hsa-miR-6805-3p” used herein includes the hsa-miR-6805-3p gene (miRBase Accession No. MIMAT0027511) described in SEQ ID NO: 205, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6805-3p gene can be obtained by a method described in Ladewig E et al., 2012,

Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6805” (miRBase Accession No. MI0022650, SEQ ID NO: 304) having a hairpin-like structure is known as a precursor of “hsa-miR-6805-3p”.

The term “hsa-miR-6826-5p gene” or “hsa-miR-6826-5p” used herein includes the hsa-miR-6826-5p gene (miRBase Accession No. MIMAT0027552) described in SEQ ID NO: 206, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6826-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6826” (miRBase Accession No. MI0022671, SEQ ID NO: 484) having a hairpin-like structure is known as a precursor of “hsa-miR-6826-5p”.

The term “hsa-miR-4508 gene” or “hsa-miR-4508” used herein includes the hsa-miR-4508 gene (miRBase Accession No. MIMAT0019045) described in SEQ ID NO: 207, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4508 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4508” (miRBase Accession No. MI0016872, SEQ ID NO: 485) having a hairpin-like structure is known as a precursor of “hsa-miR-4508”.

The term “hsa-miR-1343-5p gene” or “hsa-miR-1343-5p” used herein includes the hsa-miR-1343-5p gene (miRBase Accession No. MIMAT0027038) described in SEQ ID NO: 208, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1343-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-1343” (miRBase Accession No. MI0017320, SEQ ID NO: 361) having a hairpin-like structure is known as a precursor of “hsa-miR-1343-5p”.

The term “hsa-miR-7114-5p gene” or “hsa-miR-7114-5p” used herein includes the hsa-miR-7114-5p gene (miRBase Accession No. MIMAT0028125) described in SEQ ID NO: 209, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7114-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-7114” (miRBase Accession No. MI0022965, SEQ ID NO: 486) having a hairpin-like structure is known as a precursor of “hsa-miR-7114-5p”.

The term “hsa-miR-3622a-5p gene” or “hsa-miR-3622a-5p” used herein includes the hsa-miR-3622a-5p gene (miRBase Accession No. MIMAT0018003) described in SEQ ID NO: 210, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3622a-5p gene can be obtained by a method described in Witten D et al., 2010, BMC Biol, Vol. 8, p. 58. Also, “hsa-mir-3622a” (miRBase Accession No. MI0016013, SEQ ID NO: 487) having a hairpin-like structure is known as a precursor of “hsa-miR-3622a-5p”.

The term “hsa-miR-6765-5p gene” or “hsa-miR-6765-5p” used herein includes the hsa-miR-6765-5p gene (miRBase Accession No. MIMAT0027430) described in SEQ ID NO: 211, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6765-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6765” (miRBase Accession No. MI0022610, SEQ ID NO: 488) having a hairpin-like structure is known as a precursor of “hsa-miR-6765-5p”.

The term “hsa-miR-7845-5p gene” or “hsa-miR-7845-5p” used herein includes the hsa-miR-7845-5p gene (miRBase Accession No. MIMAT0030420) described in SEQ ID NO: 212, a homolog or an ortholog of a different organism

species, and the like. The hsa-miR-7845-5p gene can be obtained by a method described in Pie H et al., 2012, PLoS One, Vol. 7, e50746. Also, “hsa-mir-7845” (miRBase Accession No. MI0025515, SEQ ID NO: 489) having a hairpin-like structure is known as a precursor of “hsa-miR-7845-5p”.

The term “hsa-miR-3960 gene” or “hsa-miR-3960” used herein includes the hsa-miR-3960 gene (miRBase Accession No. MIMAT0019337) described in SEQ ID NO: 213, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3960 gene can be obtained by a method described in Hu R et al., 2011, J Biol Chem, Vol. 286, p. 12328-12339. Also, “hsa-mir-3960” (miRBase Accession No. MI0016964, SEQ ID NO: 490) having a hairpin-like structure is known as a precursor of “hsa-miR-3960”.

The term “hsa-miR-6749-5p gene” or “hsa-miR-6749-5p” used herein includes the hsa-miR-6749-5p gene (miRBase Accession No. MIMAT0027398) described in SEQ ID NO: 214, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6749-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6749” (miRBase Accession No. MI0022594, SEQ ID NO: 491) having a hairpin-like structure is known as a precursor of “hsa-miR-6749-5p”.

The term “hsa-miR-1260b gene” or “hsa-miR-1260b” used herein includes the hsa-miR-1260b gene (miRBase Accession No. MIMAT0015041) described in SEQ ID NO: 215, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1260b gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-1260b” (miRBase Accession No. MI0014197, SEQ ID NO: 492) having a hairpin-like structure is known as a precursor of “hsa-miR-1260b”.

The term “hsa-miR-6799-5p gene” or “hsa-miR-6799-5p” used herein includes the hsa-miR-6799-5p gene (miRBase Accession No. MIMAT0027498) described in SEQ ID NO: 216, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6799-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6799” (miRBase Accession No. MI0022644, SEQ ID NO: 493) having a hairpin-like structure is known as a precursor of “hsa-miR-6799-5p”.

The term “hsa-miR-4723-5p gene” or “hsa-miR-4723-5p” used herein includes the hsa-miR-4723-5p gene (miRBase Accession No. MIMAT0019838) described in SEQ ID NO: 217, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4723-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4723” (miRBase Accession No. MI0017359, SEQ ID NO: 494) having a hairpin-like structure is known as a precursor of “hsa-miR-4723-5p”.

The term “hsa-miR-6784-5p gene” or “hsa-miR-6784-5p” used herein includes the hsa-miR-6784-5p gene (miRBase Accession No. MIMAT0027468) described in SEQ ID NO: 218, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6784-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6784” (miRBase Accession No. MI0022629, SEQ ID NO: 495) having a hairpin-like structure is known as a precursor of “hsa-miR-6784-5p”.

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The term “hsa-miR-5100 gene” or “hsa-miR-5100” used herein includes the hsa-miR-5100 gene (miRBase Accession No. MIMAT0022259) described in SEQ ID NO: 219, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5100 gene can be obtained by a method described in Tandon M et al., 2012, *Oral Dis*, Vol. 18, p. 127-131. Also, “hsa-mir-5100” (miRBase Accession No. MI0019116, SEQ ID NO: 496) having a hairpin-like structure is known as a precursor of “hsa-miR-5100”.

The term “hsa-miR-6769b-5p gene” or “hsa-miR-6769b-5p” used herein includes the hsa-miR-6769b-5p gene (miRBase Accession No. MIMAT0027620) described in SEQ ID NO: 220, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6769b-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6769b” (miRBase Accession No. MI0022706, SEQ ID NO: 497) having a hairpin-like structure is known as a precursor of “hsa-miR-6769b-5p”.

The term “hsa-miR-1207-5p gene” or “hsa-miR-1207-5p” used herein includes the hsa-miR-1207-5p gene (miRBase Accession No. MIMAT0005871) described in SEQ ID NO: 221, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1207-5p gene can be obtained by a method described in Huppi K et al., 2008, *Mol Cancer Res*, Vol. 6, p. 212-221. Also, “hsa-mir-1207” (miRBase Accession No. MI0006340, SEQ ID NO: 498) having a hairpin-like structure is known as a precursor of “hsa-miR-1207-5p”.

The term “hsa-miR-642a-3p gene” or “hsa-miR-642a-3p” used herein includes the hsa-miR-642a-3p gene (miRBase Accession No. MIMAT0020924) described in SEQ ID NO: 222, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-642a-3p gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-642a” (miRBase Accession No. MI0003657, SEQ ID NO: 499) having a hairpin-like structure is known as a precursor of “hsa-miR-642a-3p”.

The term “hsa-miR-4505 gene” or “hsa-miR-4505” used herein includes the hsa-miR-4505 gene (miRBase Accession No. MIMAT0019041) described in SEQ ID NO: 223, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4505 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4505” (miRBase Accession No. MI0016868, SEQ ID NO: 500) having a hairpin-like structure is known as a precursor of “hsa-miR-4505”.

The term “hsa-miR-4270 gene” or “hsa-miR-4270” used herein includes the hsa-miR-4270 gene (miRBase Accession No. MIMAT0016900) described in SEQ ID NO: 224, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4270 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, “hsa-mir-4270” (miRBase Accession No. MI0015878, SEQ ID NO: 501) having a hairpin-like structure is known as a precursor of “hsa-miR-4270”.

The term “hsa-miR-6721-5p gene” or “hsa-miR-6721-5p” used herein includes the hsa-miR-6721-5p gene (miRBase Accession No. MIMAT0025852) described in SEQ ID NO: 225, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6721-5p gene can be obtained by a method described in Li Y et al., 2012, *Gene*, Vol. 497, p. 330-335. Also, “hsa-mir-6721” (miRBase Accession No. MI0022556, SEQ ID NO: 502) having a hairpin-like structure is known as a precursor of “hsa-miR-6721-5p”.

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The term “hsa-miR-7111-5p gene” or “hsa-miR-7111-5p” used herein includes the hsa-miR-7111-5p gene (miRBase Accession No. MIMAT0028119) described in SEQ ID NO: 226, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7111-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-7111” (miRBase Accession No. MI0022962, SEQ ID NO: 503) having a hairpin-like structure is known as a precursor of “hsa-miR-7111-5p”.

The term “hsa-miR-6791-5p gene” or “hsa-miR-6791-5p” used herein includes the hsa-miR-6791-5p gene (miRBase Accession No. MIMAT0027482) described in SEQ ID NO: 227, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6791-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6791” (miRBase Accession No. MI0022636, SEQ ID NO: 504) having a hairpin-like structure is known as a precursor of “hsa-miR-6791-5p”.

The term “hsa-miR-7109-5p gene” or “hsa-miR-7109-5p” used herein includes the hsa-miR-7109-5p gene (miRBase Accession No. MIMAT0028115) described in SEQ ID NO: 228, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7109-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-7109” (miRBase Accession No. MI0022960, SEQ ID NO: 505) having a hairpin-like structure is known as a precursor of “hsa-miR-7109-5p”.

The term “hsa-miR-4258 gene” or “hsa-miR-4258” used herein includes the hsa-miR-4258 gene (miRBase Accession No. MIMAT0016879) described in SEQ ID NO: 229, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4258 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, “hsa-mir-4258” (miRBase Accession No. MI0015857, SEQ ID NO: 506) having a hairpin-like structure is known as a precursor of “hsa-miR-4258”.

The term “hsa-miR-6515-3p gene” or “hsa-miR-6515-3p” used herein includes the hsa-miR-6515-3p gene (miRBase Accession No. MIMAT0025487) described in SEQ ID NO: 230, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6515-3p gene can be obtained by a method described in Joyce C E et al., 2011, *Hum Mol Genet*, Vol. 20, p. 4025-4040. Also, “hsa-mir-6515” (miRBase Accession No. MI0022227, SEQ ID NO: 507) having a hairpin-like structure is known as a precursor of “hsa-miR-6515-3p”.

The term “hsa-miR-6851-5p gene” or “hsa-miR-6851-5p” used herein includes the hsa-miR-6851-5p gene (miRBase Accession No. MIMAT0027602) described in SEQ ID NO: 231, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6851-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6851” (miRBase Accession No. MI0022697, SEQ ID NO: 508) having a hairpin-like structure is known as a precursor of “hsa-miR-6851-5p”.

The term “hsa-miR-6125 gene” or “hsa-miR-6125” used herein includes the hsa-miR-6125 gene (miRBase Accession No. MIMAT0024598) described in SEQ ID NO: 232, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6125 gene can be obtained by a method described in Smith J L et al., 2012, *J Virol*, Vol. 86, p. 5278-5287. Also, “hsa-mir-6125” (miRBase Accession

No. MI0021259, SEQ ID NO: 509) having a hairpin-like structure is known as a precursor of "hsa-miR-6125".

The term "hsa-miR-4749-5p gene" or "hsa-miR-4749-5p" used herein includes the hsa-miR-4749-5p gene (miRBase Accession No. MIMAT0019885) described in SEQ ID NO: 233, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4749-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, "hsa-mir-4749" (miRBase Accession No. MI0017388, SEQ ID NO: 510) having a hairpin-like structure is known as a precursor of "hsa-miR-4749-5p".

The term "hsa-miR-4726-5p gene" or "hsa-miR-4726-5p" used herein includes the hsa-miR-4726-5p gene (miRBase Accession No. MIMAT0019845) described in SEQ ID NO: 234, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4726-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, "hsa-mir-4726" (miRBase Accession No. MI0017363, SEQ ID NO: 511) having a hairpin-like structure is known as a precursor of "hsa-miR-4726-5p".

The term "hsa-miR-4513 gene" or "hsa-miR-4513" used herein includes the hsa-miR-4513 gene (miRBase Accession No. MIMAT0019050) described in SEQ ID NO: 235, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4513 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, "hsa-mir-4513" (miRBase Accession No. MI0016879, SEQ ID NO: 512) having a hairpin-like structure is known as a precursor of "hsa-miR-4513".

The term "hsa-miR-760 gene" or "hsa-miR-760" used herein includes the hsa-miR-760 gene (miRBase Accession No. MIMAT0004957) described in SEQ ID NO: 236, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-760 gene can be obtained by a method described in Berezikov E et al., 2006, *Genome Res*, Vol. 16, p. 1289-1298. Also, "hsa-mir-760" (miRBase Accession No. MI0005567, SEQ ID NO: 513) having a hairpin-like structure is known as a precursor of "hsa-miR-760".

The term "hsa-miR-602 gene" or "hsa-miR-602" used herein includes the hsa-miR-602 gene (miRBase Accession No. MIMAT0003270) described in SEQ ID NO: 237, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-602 gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, "hsa-mir-602" (miRBase Accession No. MI0003615, SEQ ID NO: 514) having a hairpin-like structure is known as a precursor of "hsa-miR-602".

The term "hsa-miR-423-5p gene" or "hsa-miR-423-5p" used herein includes the hsa-miR-423-5p gene (miRBase Accession No. MIMAT0004748) described in SEQ ID NO: 238, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-423-5p gene can be obtained by a method described in Kasashima K et al., 2004, *Biochem Biophys Res Commun*, Vol. 322, p. 403-410. Also, "hsa-mir-423" (miRBase Accession No. MI0001445, SEQ ID NO: 515) having a hairpin-like structure is known as a precursor of "hsa-miR-423-5p".

The term "hsa-miR-92a-2-5p gene" or "hsa-miR-92a-2-5p" used herein includes the hsa-miR-92a-2-5p gene (miRBase Accession No. MIMAT0004508) described in SEQ ID NO: 239, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-92a-2-5p gene can be obtained by a method described in Mourelatos Z et al., 2002, *Genes Dev*, Vol. 16, p. 720-728. Also, "hsa-mir-92a-2"

(miRBase Accession No. MI0000094, SEQ ID NO: 516) having a hairpin-like structure is known as a precursor of "hsa-miR-92a-2-5p".

The term "hsa-miR-16-5p gene" or "hsa-miR-16-5p" used herein includes the hsa-miR-16-5p gene (miRBase Accession No. MIMAT0000069) described in SEQ ID NO: 240, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-16-5p gene can be obtained by a method described in Lagos-Quintana M et al., 2001, *Science*, Vol. 294, p. 853-858. Also, "hsa-mir-16-1 and hsa-mir-16-2" (miRBase Accession Nos. MI0000070 and MI0000115, SEQ ID NOs: 517 and 518) having a hairpin-like structure are known as precursors of "hsa-miR-16-5p".

The term "hsa-miR-451a gene" or "hsa-miR-451a" used herein includes the hsa-miR-451a gene (miRBase Accession No. MIMAT0001631) described in SEQ ID NO: 241, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-451a gene can be obtained by a method described in Altuvia Y et al., 2005, *Nucleic Acids Res*, Vol. 33, p. 2697-2706. Also, "hsa-mir-451a" (miRBase Accession No. MI0001729, SEQ ID NO: 519) having a hairpin-like structure is known as a precursor of "hsa-miR-451a".

The term "hsa-miR-135a-3p gene" or "hsa-miR-135a-3p" used herein includes the hsa-miR-135a-3p gene (miRBase Accession No. MIMAT0004595) described in SEQ ID NO: 242, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-135a-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, *Curr Biol*, Vol. 12, p. 735-739. Also, "hsa-mir-135a" (miRBase Accession No. MI0000452, SEQ ID NO: 520) having a hairpin-like structure is known as a precursor of "hsa-miR-135a-3p".

The term "hsa-miR-486-5p gene" or "hsa-miR-486-5p" used herein includes the hsa-miR-486-5p gene (miRBase Accession No. MIMAT0002177) described in SEQ ID NO: 243, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-486-5p gene can be obtained by a method described in Fu H et al., 2005, *FEBS Lett*, Vol. 579, p. 3849-3854. Also, "hsa-mir-486 and hsa-mir-486-2" (miRBase Accession Nos. MI0002470 and MI0023622, SEQ ID NOs: 466 and 467) having a hairpin-like structure are known as precursors of "hsa-miR-486-5p".

The term "hsa-miR-4257 gene" or "hsa-miR-4257" used herein includes the hsa-miR-4257 gene (miRBase Accession No. MIMAT0016878) described in SEQ ID NO: 244, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4257 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, "hsa-mir-4257" (miRBase Accession No. MI0015856, SEQ ID NO: 521) having a hairpin-like structure is known as a precursor of "hsa-miR-4257".

The term "hsa-miR-92b-5p gene" or "hsa-miR-92b-5p" used herein includes the hsa-miR-92b-5p gene (miRBase Accession No. MIMAT0004792) described in SEQ ID NO: 245, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-92b-5p gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, "hsa-mir-92b" (miRBase Accession No. MI0003560, SEQ ID NO: 522) having a hairpin-like structure is known as a precursor of "hsa-miR-92b-5p".

The term "hsa-miR-1915-3p gene" or "hsa-miR-1915-3p" used herein includes the hsa-miR-1915-3p gene (miRBase Accession No. MIMAT0007892) described in SEQ ID NO: 246, a homolog or an ortholog of a different organism

species, and the like. The hsa-miR-1915-3p gene can be obtained by a method described in Bar M et al., 2008, Stem Cells, Vol. 26, p. 2496-2505. Also, “hsa-mir-1915” (miRBase Accession No. MI0008336, SEQ ID NO: 339) having a hairpin-like structure is known as a precursor of “hsa-miR-1915-3p”.

The term “hsa-miR-718 gene” or “hsa-miR-718” used herein includes the hsa-miR-718 gene (miRBase Accession No. MIMAT0012735) described in SEQ ID NO: 247, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-718 gene can be obtained by a method described in Artzi S et al., 2008, BMC Bioinformatics, Vol. 9, p. 39. Also, “hsa-mir-718” (miRBase Accession No. MI0012489, SEQ ID NO: 523) having a hairpin-like structure is known as a precursor of “hsa-miR-718”.

The term “hsa-miR-940 gene” or “hsa-miR-940” used herein includes the hsa-miR-940 gene (miRBase Accession No. MIMAT0004983) described in SEQ ID NO: 248, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-940 gene can be obtained by a method described in Lui W O et al., 2007, Cancer Res, Vol. 67, p. 6031-6043. Also, “hsa-mir-940” (miRBase Accession No. MI0005762, SEQ ID NO: 524) having a hairpin-like structure is known as a precursor of “hsa-miR-940”.

The term “hsa-miR-296-5p gene” or “hsa-miR-296-5p” used herein includes the hsa-miR-296-5p gene (miRBase Accession No. MIMAT0000690) described in SEQ ID NO: 249, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-296-5p gene can be obtained by a method described in Houbaviy H B et al., 2003, Dev Cell, Vol. 5, p. 351-358. Also, “hsa-mir-296” (miRBase Accession No. MI0000747, SEQ ID NO: 409) having a hairpin-like structure is known as a precursor of “hsa-miR-296-5p”.

The term “hsa-miR-23b-3p gene” or “hsa-miR-23b-3p” used herein includes the hsa-miR-23b-3p gene (miRBase Accession No. MIMAT0000418) described in SEQ ID NO: 250, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-23b-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, Curr Biol, Vol. 12, p. 735-739. Also, “hsa-mir-23b” (miRBase Accession No. MI0000439, SEQ ID NO: 525) having a hairpin-like structure is known as a precursor of “hsa-miR-23b-3p”.

The term “hsa-miR-92a-3p gene” or “hsa-miR-92a-3p” used herein includes the hsa-miR-92a-3p gene (miRBase Accession No. MIMAT0000092) described in SEQ ID NO: 251, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-92a-3p gene can be obtained by a method described in Mourelatos Z et al., 2002, Genes Dev, Vol. 16, p. 720-728. Also, “hsa-mir-92a-1 and hsa-mir-92a-2” (miRBase Accession Nos. MI0000093 and MI0000094, SEQ ID NOs: 526 and 527) having a hairpin-like structure are known as precursors of “hsa-miR-92a-3p”.

The term “hsa-miR-658 gene” or “hsa-miR-658” used herein includes the hsa-miR-658 gene (miRBase Accession No. MIMAT0003336) described in SEQ ID NO: 252, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-658 gene can be obtained by a method described in Cummins J M et al., 2006, Proc Natl Acad Sci USA, Vol. 103, p. 3687-3692. Also, “hsa-mir-658” (miRBase Accession No. MI0003682, SEQ ID NO: 528) having a hairpin-like structure is known as a precursor of “hsa-miR-658”.

The term “hsa-miR-6842-5p gene” or “hsa-miR-6842-5p” used herein includes the hsa-miR-6842-5p gene (miRBase Accession No. MIMAT0027586) described in SEQ ID NO:

253, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6842-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6842” (miRBase Accession No. MI0022688, SEQ ID NO: 529) having a hairpin-like structure is known as a precursor of “hsa-miR-6842-5p”.

The term “hsa-miR-6124 gene” or “hsa-miR-6124” used herein includes the hsa-miR-6124 gene (miRBase Accession No. MIMAT0024597) described in SEQ ID NO: 254, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6124 gene can be obtained by a method described in Smith J L et al., 2012, J Virol, Vol. 86, p. 5278-5287. Also, “hsa-mir-6124” (miRBase Accession No. MI0021258, SEQ ID NO: 530) having a hairpin-like structure is known as a precursor of “hsa-miR-6124”.

The term “hsa-miR-6765-3p gene” or “hsa-miR-6765-3p” used herein includes the hsa-miR-6765-3p gene (miRBase Accession No. MIMAT0027431) described in SEQ ID NO: 255, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6765-3p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6765” (miRBase Accession No. MI0022610, SEQ ID NO: 531) having a hairpin-like structure is known as a precursor of “hsa-miR-6765-3p”.

The term “hsa-miR-7106-5p gene” or “hsa-miR-7106-5p” used herein includes the hsa-miR-7106-5p gene (miRBase Accession No. MIMAT0028109) described in SEQ ID NO: 256, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7106-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-7106” (miRBase Accession No. MI0022957, SEQ ID NO: 532) having a hairpin-like structure is known as a precursor of “hsa-miR-7106-5p”.

The term “hsa-miR-4534 gene” or “hsa-miR-4534” used herein includes the hsa-miR-4534 gene (miRBase Accession No. MIMAT0019073) described in SEQ ID NO: 257, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4534 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4534” (miRBase Accession No. MI0016901, SEQ ID NO: 533) having a hairpin-like structure is known as a precursor of “hsa-miR-4534”.

The term “hsa-miR-92b-3p gene” or “hsa-miR-92b-3p” used herein includes the hsa-miR-92b-3p gene (miRBase Accession No. MIMAT0003218) described in SEQ ID NO: 258, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-92b-3p gene can be obtained by a method described in Cummins J M et al., 2006, Proc Natl Acad Sci USA, Vol. 103, p. 3687-3692. Also, “hsa-mir-92b” (miRBase Accession No. MI0003560, SEQ ID NO: 522) having a hairpin-like structure is known as a precursor of “hsa-miR-92b-3p”.

The term “hsa-miR-3135b gene” or “hsa-miR-3135b” used herein includes the hsa-miR-3135b gene (miRBase Accession No. MIMAT0018985) described in SEQ ID NO: 259, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3135b gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-3135b” (miRBase Accession No. MI0016809, SEQ ID NO: 534) having a hairpin-like structure is known as a precursor of “hsa-miR-3135b”.

The term “hsa-miR-4687-3p gene” or “hsa-miR-4687-3p” used herein includes the hsa-miR-4687-3p gene (miRBase

Accession No. MIMAT0019775) described in SEQ ID NO: 260, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4687-3p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4687” (miR-Base Accession No. MI0017319, SEQ ID NO: 535) having a hairpin-like structure is known as a precursor of “hsa-miR-4687-3p”.

The term “hsa-miR-762 gene” or “hsa-miR-762” used herein includes the hsa-miR-762 gene (miRBase Accession No. MIMAT0010313) described in SEQ ID NO: 261, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-762 gene can be obtained by a method described in Berezikov E et al., 2006, *Genome Res*, Vol. 16, p. 1289-1298. Also, “hsa-mir-762” (miRBase Accession No. MI0003892, SEQ ID NO: 536) having a hairpin-like structure is known as a precursor of “hsa-miR-762”.

The term “hsa-miR-3619-3p gene” or “hsa-miR-3619-3p” used herein includes the hsa-miR-3619-3p gene (miRBase Accession No. MIMAT0019219) described in SEQ ID NO: 262, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3619-3p gene can be obtained by a method described in Witten D et al., 2010, *BMC Biol*, Vol. 8, p. 58. Also, “hsa-mir-3619” (miRBase Accession No. MI0016009, SEQ ID NO: 537) having a hairpin-like structure is known as a precursor of “hsa-miR-3619-3p”.

The term “hsa-miR-4467 gene” or “hsa-miR-4467” used herein includes the hsa-miR-4467 gene (miRBase Accession No. MIMAT0018994) described in SEQ ID NO: 263, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4467 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4467” (miRBase Accession No. MI0016818, SEQ ID NO: 538) having a hairpin-like structure is known as a precursor of “hsa-miR-4467”.

The term “hsa-miR-557 gene” or “hsa-miR-557” used herein includes the hsa-miR-557 gene (miRBase Accession No. MIMAT0003221) described in SEQ ID NO: 264, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-557 gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-557” (miR-Base Accession No. MI0003563, SEQ ID NO: 539) having a hairpin-like structure is known as a precursor of “hsa-miR-557”.

The term “hsa-miR-1237-5p gene” or “hsa-miR-1237-5p” used herein includes the hsa-miR-1237-5p gene (miRBase Accession No. MIMAT0022946) described in SEQ ID NO: 265, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1237-5p gene can be obtained by a method described in Berezikov E et al., 2007, *Mol Cell*, Vol. 28, p. 328-336. Also, “hsa-mir-1237” (miR-Base Accession No. MI0006327, SEQ ID NO: 540) having a hairpin-like structure is known as a precursor of “hsa-miR-1237-5p”.

The term “hsa-miR-1908-5p gene” or “hsa-miR-1908-5p” used herein includes the hsa-miR-1908-5p gene (miRBase Accession No. MIMAT0007881) described in SEQ ID NO: 266, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1908-5p gene can be obtained by a method described in Bar M et al., 2008, *Stem Cells*, Vol. 26, p. 2496-2505. Also, “hsa-mir-1908” (miR-Base Accession No. MI0008329, SEQ ID NO: 541) having a hairpin-like structure is known as a precursor of “hsa-miR-1908-5p”.

The term “hsa-miR-4286 gene” or “hsa-miR-4286” used herein includes the hsa-miR-4286 gene (miRBase Accession No. MIMAT0016916) described in SEQ ID NO: 267, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4286 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, “hsa-mir-4286” (miRBase Accession No. MI0015894, SEQ ID NO: 542) having a hairpin-like structure is known as a precursor of “hsa-miR-4286”.

The term “hsa-miR-6885-5p gene” or “hsa-miR-6885-5p” used herein includes the hsa-miR-6885-5p gene (miRBase Accession No. MIMAT0027670) described in SEQ ID NO: 268, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6885-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6885” (miRBase Accession No. MI0022732, SEQ ID NO: 543) having a hairpin-like structure is known as a precursor of “hsa-miR-6885-5p”.

The term “hsa-miR-6763-5p gene” or “hsa-miR-6763-5p” used herein includes the hsa-miR-6763-5p gene (miRBase Accession No. MIMAT0027426) described in SEQ ID NO: 269, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6763-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6763” (miRBase Accession No. MI0022608, SEQ ID NO: 544) having a hairpin-like structure is known as a precursor of “hsa-miR-6763-5p”.

The term “hsa-miR-6089 gene” or “hsa-miR-6089” used herein includes the hsa-miR-6089 gene (miRBase Accession No. MIMAT0023714) described in SEQ ID NO: 851, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6089 gene can be obtained by a method described in Yoo J K et al., 2012, *Stem Cells Dev*, Vol. 21, p. 2049-2057. Also, “hsa-mir-6089-1 and hsa-mir-6089-2” (miRBase Accession Nos. MI0020366 and MI0023563, SEQ ID NOs: 857 and 858) having a hairpin-like structure are known as precursors of “hsa-miR-6089”.

The term “hsa-miR-6816-5p gene” or “hsa-miR-6816-5p” used herein includes the hsa-miR-6816-5p gene (miRBase Accession No. MIMAT0027532) described in SEQ ID NO: 852, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6816-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6816” (miRBase Accession No. MI0022661, SEQ ID NO: 859) having a hairpin-like structure is known as a precursor of “hsa-miR-6816-5p”.

The term “hsa-miR-4466 gene” or “hsa-miR-4466” used herein includes the hsa-miR-4466 gene (miRBase Accession No. MIMAT0018993) described in SEQ ID NO: 853, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4466 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4466” (miRBase Accession No. MI0016817, SEQ ID NO: 860) having a hairpin-like structure is known as a precursor of “hsa-miR-4466”.

The term “hsa-miR-4488 gene” or “hsa-miR-4488” used herein includes the hsa-miR-4488 gene (miRBase Accession No. MIMAT0019022) described in SEQ ID NO: 854, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4488 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4488” (miRBase Accession No. MI0016849, SEQ ID NO: 861) having a hairpin-like structure is known as a precursor of “hsa-miR-4488”.

The term “hsa-miR-6752-5p gene” or “hsa-miR-6752-5p” used herein includes the hsa-miR-6752-5p gene (miRBase Accession No. MIMAT0027404) described in SEQ ID NO: 855, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6752-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6752” (miRBase Accession No. MI0022597, SEQ ID NO: 862) having a hairpin-like structure is known as a precursor of “hsa-miR-6752-5p”.

The term “hsa-miR-4739 gene” or “hsa-miR-4739” used herein includes the hsa-miR-4739 gene (miRBase Accession No. MIMAT0019868) described in SEQ ID NO: 856, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4739 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4739” (miRBase Accession No. MI0017377, SEQ ID NO: 863) having a hairpin-like structure is known as a precursor of “hsa-miR-4739”.

A mature miRNA may become a variant due to the sequence cleaved shorter or longer by one to several upstream or downstream nucleotide(s) substitution when cleaved as the mature miRNA from its RNA precursor having a hairpin-like structure. This variant is called isomiR (Morin R D. et al., 2008, Genome Res., Vol. 18, p. 610-621). The miRBase Release 20 shows the nucleotide sequences represented by SEQ ID NOs: 1 to 269 and 851 to 856 as well as a large number of the nucleotide sequence variants and fragments represented by SEQ ID NOs: 545 to 850 and 864 to 871, called isomiRs. These variants can also be obtained as miRNAs that have a nucleotide sequence represented by any of SEQ ID NOs: 1 to 269 and 851 to 856. Specifically, among the variants of polynucleotides consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1, 2, 3, 4, 5, 6, 8, 11, 14, 15, 19, 20, 21, 23, 27, 28, 31, 35, 37, 38, 40, 42, 43, 47, 48, 50, 51, 52, 55, 56, 58, 60, 61, 62, 64, 65, 66, 67, 71, 72, 73, 76, 77, 78, 79, 80, 82, 83, 85, 86, 88, 89, 90, 92, 95, 97, 98, 99, 100, 102, 103, 107, 109, 110, 113, 114, 115, 116, 117, 118, 120, 122, 123, 126, 127, 129, 131, 133, 137, 141, 142, 143, 145, 146, 148, 149, 153, 154, 155, 156, 157, 161, 164, 165, 167, 168, 171, 172, 173, 178, 179, 180, 181, 182, 184, 185, 188, 191, 192, 194, 195, 196, 199, 201, 202, 204, 207, 210, 213, 215, 217, 219, 222, 223, 225, 230, 232, 233, 234, 235, 236, 238, 239, 240, 241, 242, 243, 245, 246, 247, 248, 249, 250, 251, 252, 254, 258, 259, 260, 263, 265, 266, 267, 851, 853, 854 and 856, or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t according to the present invention, examples of the longest variants registered in miRBase Release 20 include polynucleotides represented by SEQ ID NOs: 545, 547, 549, 551, 553, 555, 557, 559, 561, 563, 565, 567, 569, 571, 573, 575, 577, 579, 581, 583, 585, 587, 589, 591, 593, 595, 597, 599, 601, 603, 605, 607, 609, 611, 613, 615, 617, 619, 621, 623, 625, 627, 629, 631, 633, 635, 637, 639, 641, 643, 645, 647, 649, 651, 653, 655, 657, 659, 661, 663, 665, 667, 669, 671, 673, 675, 677, 679, 681, 683, 685, 687, 689, 691, 693, 695, 697, 699, 701, 703, 705, 707, 709, 711, 713, 715, 717, 719, 721, 723, 725, 727, 729, 731, 733, 735, 737, 739, 741, 743, 745, 747, 749, 751, 753, 755, 757, 759, 761, 763, 765, 767, 769, 771, 773, 775, 777, 779, 781, 783, 785, 787, 789, 791, 793, 795, 797, 799, 801, 803, 805, 807, 809, 811, 813, 815, 817, 819, 821, 823, 825, 827, 829, 831, 833, 835, 837, 839, 841, 843, 845, 847, 849, 864, 866, 868 and 870, respectively. Also, among the variants of polynucleotides consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1, 2, 3, 4, 5, 6, 8, 11, 14, 15, 19, 20, 21, 23, 27, 28, 31, 35, 37, 38, 40, 42, 43, 47, 48, 50,

51, 52, 55, 56, 58, 60, 61, 62, 64, 65, 66, 67, 71, 72, 73, 76, 77, 78, 79, 80, 82, 83, 85, 86, 88, 89, 90, 92, 95, 97, 98, 99, 100, 102, 103, 107, 109, 110, 113, 114, 115, 116, 117, 118, 120, 122, 123, 126, 127, 129, 131, 133, 137, 141, 142, 143, 145, 146, 148, 149, 153, 154, 155, 156, 157, 161, 164, 165, 167, 168, 171, 172, 173, 178, 179, 180, 181, 182, 184, 185, 188, 191, 192, 194, 195, 196, 199, 201, 202, 204, 207, 210, 213, 215, 217, 219, 222, 223, 225, 230, 232, 233, 234, 235, 236, 238, 239, 240, 241, 242, 243, 245, 246, 247, 248, 249, 250, 251, 252, 254, 258, 259, 260, 263, 265, 266, 267, 851, 853, 854 and 856, or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t according to the present invention, examples of the shortest variants registered in miRBase Release 20 include polynucleotides having sequences represented by SEQ ID NOs: 546, 548, 550, 552, 554, 556, 558, 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, 580, 582, 584, 586, 588, 590, 592, 594, 596, 598, 600, 602, 604, 606, 608, 610, 612, 614, 616, 618, 620, 622, 624, 626, 628, 630, 632, 634, 636, 638, 640, 642, 644, 646, 648, 650, 652, 654, 656, 658, 660, 662, 664, 666, 668, 670, 672, 674, 676, 678, 680, 682, 684, 686, 688, 690, 692, 694, 696, 698, 700, 702, 704, 706, 708, 710, 712, 714, 716, 718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748, 750, 752, 754, 756, 758, 760, 762, 764, 766, 768, 770, 772, 774, 776, 778, 780, 782, 784, 786, 788, 790, 792, 794, 796, 798, 800, 802, 804, 806, 808, 810, 812, 814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844, 846, 848, 850, 865, 867, 869 and 871, respectively. In addition to these variants and fragments, examples thereof include a large number of isomiR polynucleotides of SEQ ID NOs: 1 to 269 and 851 to 856 registered in miRBase. Examples of the polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 269 and 851 to 856 include a polynucleotide represented by any of SEQ ID NOs: 270 to 544, and 857 to 863, which are their respective precursors.

The names and miRBase Accession Nos. (registration numbers) of the genes represented by SEQ ID NOs: 1 to 871 are shown in Table 1.

The term “capable of specifically binding” used herein means that the nucleic acid probe or the primer used in the present invention binds to a particular target nucleic acid and cannot substantially bind to other nucleic acids.

TABLE 1

SEQ ID NO:	Gene name	miRBase registration No.
1	hsa-miR-4783-3p	MIMAT0019947
2	hsa-miR-4730	MIMAT0019852
3	hsa-miR-1307-3p	MIMAT0005951
4	hsa-miR-4634	MIMAT0019691
5	hsa-miR-663a	MIMAT0003326
6	hsa-miR-4532	MIMAT0019071
7	hsa-miR-7704	MIMAT0030019
8	hsa-miR-3178	MIMAT0015055
9	hsa-miR-6729-5p	MIMAT0027359
10	hsa-miR-6090	MIMAT0023715
11	hsa-miR-4732-5p	MIMAT0019855
12	hsa-miR-3184-5p	MIMAT0015064
13	hsa-miR-6727-5p	MIMAT0027355
14	hsa-miR-6088	MIMAT0023713
15	hsa-miR-4674	MIMAT0019756
16	hsa-miR-8073	MIMAT0031000
17	hsa-miR-4787-5p	MIMAT0019956
18	hsa-miR-1469	MIMAT0007347
19	hsa-miR-125a-3p	MIMAT0004602
20	hsa-miR-1233-5p	MIMAT0022943
21	hsa-miR-885-3p	MIMAT0004948
22	hsa-miR-6802-5p	MIMAT0027504

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
23	hsa-miR-328-5p	MIMAT0026486
24	hsa-miR-6787-5p	MIMAT0027474
25	hsa-miR-8069	MIMAT0030996
26	hsa-miR-6875-5p	MIMAT0027650
27	hsa-miR-1246	MIMAT0005898
28	hsa-miR-4734	MIMAT0019859
29	hsa-miR-6757-5p	MIMAT0027414
30	hsa-miR-6756-5p	MIMAT0027412
31	hsa-miR-3665	MIMAT0018087
32	hsa-miR-6836-3p	MIMAT0027575
33	hsa-miR-6821-5p	MIMAT0027542
34	hsa-miR-6805-5p	MIMAT0027510
35	hsa-miR-4728-5p	MIMAT0019849
36	hsa-miR-6726-5p	MIMAT0027353
37	hsa-miR-197-5p	MIMAT0022691
38	hsa-miR-149-3p	MIMAT0004609
39	hsa-miR-6850-5p	MIMAT0027600
40	hsa-miR-4476	MIMAT0019003
41	hsa-miR-6836-5p	MIMAT0027616
42	hsa-miR-564	MIMAT0003228
43	hsa-miR-4763-3p	MIMAT0019913
44	hsa-miR-575	MIMAT0003240
45	hsa-miR-6771-5p	MIMAT0027442
46	hsa-miR-1231	MIMAT0005586
47	hsa-miR-1908-3p	MIMAT0026916
48	hsa-miR-150-3p	MIMAT0004610
49	hsa-miR-3937	MIMAT0018352
50	hsa-miR-887-3p	MIMAT0004951
51	hsa-miR-3940-5p	MIMAT0019229
52	hsa-miR-4741	MIMAT0019871
53	hsa-miR-6808-5p	MIMAT0027516
54	hsa-miR-6869-5p	MIMAT0027638
55	hsa-miR-5090	MIMAT0021082
56	hsa-miR-615-5p	MIMAT0004804
57	hsa-miR-8072	MIMAT0030999
58	hsa-miR-128-1-5p	MIMAT0026477
59	hsa-miR-1238-5p	MIMAT0022947
60	hsa-miR-365a-5p	MIMAT0009199
61	hsa-miR-204-3p	MIMAT0022693
62	hsa-miR-4492	MIMAT0019027
63	hsa-miR-6785-5p	MIMAT0027470
64	hsa-miR-6511a-5p	MIMAT0025478
65	hsa-miR-4525	MIMAT0019064
66	hsa-miR-1915-5p	MIMAT0007891
67	hsa-miR-3180	MIMAT0018178
68	hsa-miR-6879-5p	MIMAT0027658
69	hsa-miR-1199-5p	MIMAT0031119
70	hsa-miR-6746-5p	MIMAT0027392
71	hsa-miR-711	MIMAT0012734
72	hsa-miR-663b	MIMAT0005867
73	hsa-miR-4707-3p	MIMAT0019808
74	hsa-miR-6893-5p	MIMAT0027686
75	hsa-miR-4675	MIMAT0019757
76	hsa-miR-4638-5p	MIMAT0019695
77	hsa-miR-4651	MIMAT0019715
78	hsa-miR-6087	MIMAT0023712
79	hsa-miR-4665-5p	MIMAT0019739
80	hsa-miR-4758-5p	MIMAT0019903
81	hsa-miR-6887-5p	MIMAT0027674
82	hsa-miR-3620-5p	MIMAT0022967
83	hsa-miR-1909-3p	MIMAT0007883
84	hsa-miR-7641	MIMAT0029782
85	hsa-miR-6724-5p	MIMAT0025856
86	hsa-miR-1343-3p	MIMAT0019776
87	hsa-miR-6780b-5p	MIMAT0027572
88	hsa-miR-4484	MIMAT0019018
89	hsa-miR-4690-5p	MIMAT0019779
90	hsa-miR-4429	MIMAT0018944
91	hsa-miR-1227-5p	MIMAT0022941
92	hsa-miR-4725-3p	MIMAT0019844
93	hsa-miR-6861-5p	MIMAT0027623
94	hsa-miR-6812-5p	MIMAT0027524
95	hsa-miR-3197	MIMAT0015082
96	hsa-miR-8059	MIMAT0030986
97	hsa-miR-3185	MIMAT0015065
98	hsa-miR-4706	MIMAT0019806
99	hsa-miR-4497	MIMAT0019032

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
5	100 hsa-miR-3131	MIMAT0014996
	101 hsa-miR-6806-5p	MIMAT0027512
	102 hsa-miR-187-5p	MIMAT0004561
	103 hsa-miR-3180-3p	MIMAT0015058
	104 hsa-miR-6848-5p	MIMAT0027596
	105 hsa-miR-6820-5p	MIMAT0027540
10	106 hsa-miR-6800-5p	MIMAT0027500
	107 hsa-miR-6717-5p	MIMAT0025846
	108 hsa-miR-6795-5p	MIMAT0027490
	109 hsa-miR-4632-5p	MIMAT0022977
	110 hsa-miR-665	MIMAT0004952
	111 hsa-miR-6778-5p	MIMAT0027456
	112 hsa-miR-3663-3p	MIMAT0018085
15	113 hsa-miR-4689	MIMAT0019778
	114 hsa-miR-211-3p	MIMAT0022694
	115 hsa-miR-6511b-5p	MIMAT0027547
	116 hsa-miR-4750-5p	MIMAT0019887
	117 hsa-miR-6126	MIMAT0024599
	118 hsa-miR-614	MIMAT0003282
20	119 hsa-miR-7110-5p	MIMAT0028117
	120 hsa-miR-744-5p	MIMAT0004945
	121 hsa-miR-6769a-5p	MIMAT0027438
	122 hsa-miR-4792	MIMAT0019964
	123 hsa-miR-5787	MIMAT0023252
	124 hsa-miR-6798-5p	MIMAT0027496
25	125 hsa-miR-6781-5p	MIMAT0027462
	126 hsa-miR-4419b	MIMAT0019034
	127 hsa-miR-4446-3p	MIMAT0018965
	128 hsa-miR-4259	MIMAT0016880
	129 hsa-miR-5572	MIMAT0022260
	130 hsa-miR-6075	MIMAT0023700
	131 hsa-miR-296-3p	MIMAT0004679
	132 hsa-miR-6891-5p	MIMAT0027682
	133 hsa-miR-4745-5p	MIMAT0019878
	134 hsa-miR-6775-5p	MIMAT0027450
	135 hsa-miR-6870-5p	MIMAT0027640
	136 hsa-miR-920	MIMAT0004970
35	137 hsa-miR-4530	MIMAT0019069
	138 hsa-miR-6819-5p	MIMAT0027538
	139 hsa-miR-6825-5p	MIMAT0027550
	140 hsa-miR-7847-3p	MIMAT0030422
	141 hsa-miR-6131	MIMAT0024615
	142 hsa-miR-4433-3p	MIMAT0018949
	143 hsa-miR-1228-5p	MIMAT0005582
40	144 hsa-miR-6743-5p	MIMAT0027387
	145 hsa-miR-1268a	MIMAT0005922
	146 hsa-miR-3917	MIMAT0018191
	147 hsa-miR-6786-5p	MIMAT0027472
	148 hsa-miR-3154	MIMAT0015028
	149 hsa-miR-638	MIMAT0003308
45	150 hsa-miR-6741-5p	MIMAT0027383
	151 hsa-miR-6889-5p	MIMAT0027678
	152 hsa-miR-6840-3p	MIMAT0027583
	153 hsa-miR-6510-5p	MIMAT0025476
	154 hsa-miR-3188	MIMAT0015070
	155 hsa-miR-551b-5p	MIMAT0004794
50	156 hsa-miR-5001-5p	MIMAT0021021
	157 hsa-miR-1268b	MIMAT0018925
	158 hsa-miR-7107-5p	MIMAT0028111
	159 hsa-miR-6824-5p	MIMAT0027548
	160 hsa-miR-6732-5p	MIMAT0027365
	161 hsa-miR-371a-5p	MIMAT0004687
55	162 hsa-miR-6794-5p	MIMAT0027488
	163 hsa-miR-6779-5p	MIMAT0027458
	164 hsa-miR-4271	MIMAT0016901
	165 hsa-miR-5195-3p	MIMAT0021127
	166 hsa-miR-6762-5p	MIMAT0027424
	167 hsa-miR-939-5p	MIMAT0004982
	168 hsa-miR-1247-3p	MIMAT0022721
60	169 hsa-miR-6777-5p	MIMAT0027454
	170 hsa-miR-6722-3p	MIMAT0025854
	171 hsa-miR-3656	MIMAT0018076
	172 hsa-miR-4688	MIMAT0019777
	173 hsa-miR-3195	MIMAT0015079
	174 hsa-miR-6766-5p	MIMAT0027432
65	175 hsa-miR-4447	MIMAT0018966
	176 hsa-miR-4656	MIMAT0019723

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
177	hsa-miR-7108-5p	MIMAT0028113
178	hsa-miR-3191-3p	MIMAT0015075
179	hsa-miR-1273g-3p	MIMAT0022742
180	hsa-miR-4463	MIMAT0018987
181	hsa-miR-2861	MIMAT0013802
182	hsa-miR-3196	MIMAT0015080
183	hsa-miR-6877-5p	MIMAT0027654
184	hsa-miR-3679-5p	MIMAT0018104
185	hsa-miR-4442	MIMAT0018960
186	hsa-miR-6789-5p	MIMAT0027478
187	hsa-miR-6782-5p	MIMAT0027464
188	hsa-miR-486-3p	MIMAT0004762
189	hsa-miR-6085	MIMAT0023710
190	hsa-miR-4746-3p	MIMAT0019881
191	hsa-miR-619-5p	MIMAT0026622
192	hsa-miR-937-5p	MIMAT0022938
193	hsa-miR-6803-5p	MIMAT0027506
194	hsa-miR-4298	MIMAT0016852
195	hsa-miR-4454	MIMAT0018976
196	hsa-miR-4459	MIMAT0018981
197	hsa-miR-7150	MIMAT0028211
198	hsa-miR-6880-5p	MIMAT0027660
199	hsa-miR-4449	MIMAT0018968
200	hsa-miR-8063	MIMAT0030990
201	hsa-miR-4695-5p	MIMAT0019788
202	hsa-miR-6132	MIMAT0024616
203	hsa-miR-6829-5p	MIMAT0027558
204	hsa-miR-4486	MIMAT0019020
205	hsa-miR-6805-3p	MIMAT0027511
206	hsa-miR-6826-5p	MIMAT0027552
207	hsa-miR-4508	MIMAT0019045
208	hsa-miR-1343-5p	MIMAT0027038
209	hsa-miR-7114-5p	MIMAT0028125
210	hsa-miR-3622a-5p	MIMAT0018003
211	hsa-miR-6765-5p	MIMAT0027430
212	hsa-miR-7845-5p	MIMAT0030420
213	hsa-miR-3960	MIMAT0019337
214	hsa-miR-6749-5p	MIMAT0027398
215	hsa-miR-1260b	MIMAT0015041
216	hsa-miR-6799-5p	MIMAT0027498
217	hsa-miR-4723-5p	MIMAT0019838
218	hsa-miR-6784-5p	MIMAT0027468
219	hsa-miR-5100	MIMAT0022259
220	hsa-miR-6769b-5p	MIMAT0027620
221	hsa-miR-1207-5p	MIMAT0005871
222	hsa-miR-642a-3p	MIMAT0020924
223	hsa-miR-4505	MIMAT0019041
224	hsa-miR-4270	MIMAT0016900
225	hsa-miR-6721-5p	MIMAT0025852
226	hsa-miR-7111-5p	MIMAT0028119
227	hsa-miR-6791-5p	MIMAT0027482
228	hsa-miR-7109-5p	MIMAT0028115
229	hsa-miR-4258	MIMAT0016879
230	hsa-miR-6515-3p	MIMAT0025487
231	hsa-miR-6851-5p	MIMAT0027602
232	hsa-miR-6125	MIMAT0024598
233	hsa-miR-4749-5p	MIMAT0019885
234	hsa-miR-4726-5p	MIMAT0019845
235	hsa-miR-4513	MIMAT0019050
236	hsa-miR-760	MIMAT0004957
237	hsa-miR-602	MIMAT0003270
238	hsa-miR-423-5p	MIMAT0004748
239	hsa-miR-92a-2-5p	MIMAT0004508
240	hsa-miR-16-5p	MIMAT0000069
241	hsa-miR-451a	MIMAT0001631
242	hsa-miR-135a-3p	MIMAT0004595
243	hsa-miR-486-5p	MIMAT0002177
244	hsa-miR-4257	MIMAT0016878
245	hsa-miR-92b-5p	MIMAT0004792
246	hsa-miR-1915-3p	MIMAT0007892
247	hsa-miR-718	MIMAT0012735
248	hsa-miR-940	MIMAT0004983
249	hsa-miR-296-5p	MIMAT0000690
250	hsa-miR-23b-3p	MIMAT0000418
251	hsa-miR-92a-3p	MIMAT0000092
252	hsa-miR-658	MIMAT0003336
253	hsa-miR-6842-5p	MIMAT0027586

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
5	254 hsa-miR-6124	MIMAT0024597
	255 hsa-miR-6765-3p	MIMAT0027431
	256 hsa-miR-7106-5p	MIMAT0028109
	257 hsa-miR-4534	MIMAT0019073
	258 hsa-miR-92b-3p	MIMAT0003218
	259 hsa-miR-3135b	MIMAT0018985
10	260 hsa-miR-4687-3p	MIMAT0019775
	261 hsa-miR-762	MIMAT0010313
	262 hsa-miR-3619-3p	MIMAT0019219
	263 hsa-miR-4467	MIMAT0018994
	264 hsa-miR-557	MIMAT0003221
	265 hsa-miR-1237-5p	MIMAT0022946
	266 hsa-miR-1908-5p	MIMAT0007881
15	267 hsa-miR-4286	MIMAT0016916
	268 hsa-miR-6885-5p	MIMAT0027670
	269 hsa-miR-6763-5p	MIMAT0027426
	270 hsa-mir-4783	MI0017428
	271 hsa-mir-4730	MI0017367
	272 hsa-mir-1307	MI0006444
20	273 hsa-mir-4634	MI0017261
	274 hsa-mir-663a	MI0003672
	275 hsa-mir-4532	MI0016899
	276 hsa-mir-7704	MI0025240
	277 hsa-mir-3178	MI0014212
	278 hsa-mir-6729	MI0022574
25	279 hsa-mir-6090	MI0020367
	280 hsa-mir-4732	MI0017369
	281 hsa-mir-3184	MI0014226
	282 hsa-mir-6727	MI0022572
	283 hsa-mir-6088	MI0020365
	284 hsa-mir-4674	MI0017305
30	285 hsa-mir-8073	MI0025909
	286 hsa-mir-4787	MI0017434
	287 hsa-mir-1469	MI0007074
	288 hsa-mir-125a	MI0000469
	289 hsa-mir-1233-1	MI0006323
	290 hsa-mir-1233-2	MI0015973
35	291 hsa-mir-885	MI0005560
	292 hsa-mir-6802	MI0022647
	293 hsa-mir-328	MI0000804
	294 hsa-mir-6787	MI0022632
	295 hsa-mir-8069	MI0025905
	296 hsa-mir-6875	MI0022722
40	297 hsa-mir-1246	MI0006381
	298 hsa-mir-4734	MI0017371
	299 hsa-mir-6757	MI0022602
	300 hsa-mir-6756	MI0022601
	301 hsa-mir-3665	MI0016066
	302 hsa-mir-6836	MI0022682
	303 hsa-mir-6821	MI0022666
45	304 hsa-mir-6805	MI0022650
	305 hsa-mir-4728	MI0017365
	306 hsa-mir-6726	MI0022571
	307 hsa-mir-197	MI0000239
	308 hsa-mir-149	MI0000478
	309 hsa-mir-6850	MI0022696
50	310 hsa-mir-4476	MI0016828
	311 hsa-mir-6858	MI0022704
	312 hsa-mir-564	MI0003570
	313 hsa-mir-4763	MI0017404
	314 hsa-mir-575	MI0003582
	315 hsa-mir-6771	MI0022616
55	316 hsa-mir-1231	MI0006321
	317 hsa-mir-1908	MI0008329
	318 hsa-mir-150	MI0000479
	319 hsa-mir-3937	MI0016593
	320 hsa-mir-887	MI0005562
	321 hsa-mir-3940	MI0016597
	322 hsa-mir-4741	MI0017379
60	323 hsa-mir-6808	MI0022653
	324 hsa-mir-6869	MI0022716
	325 hsa-mir-5090	MI0017979
	326 hsa-mir-615	MI0003628
	327 hsa-mir-8072	MI0025908
	328 hsa-mir-128-1	MI0000447
65	329 hsa-mir-1238	MI0006328
	330 hsa-mir-365a	MI0000767

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
331	hsa-mir-204	MI0000284
332	hsa-mir-4492	MI0016854
333	hsa-mir-6785	MI0022630
334	hsa-mir-6511a-1	MI0022223
335	hsa-mir-6511a-2	MI0023564
336	hsa-mir-6511a-3	MI0023565
337	hsa-mir-6511a-4	MI0023566
338	hsa-mir-4525	MI0016892
339	hsa-mir-1915	MI0008336
340	hsa-mir-3180-4	MI0016408
341	hsa-mir-3180-5	MI0016409
342	hsa-mir-6879	MI0022726
343	hsa-mir-1199	MI0020340
344	hsa-mir-6746	MI0022591
345	hsa-mir-711	MI0012488
346	hsa-mir-663b	MI0006336
347	hsa-mir-4707	MI0017340
348	hsa-mir-6893	MI0022740
349	hsa-mir-4675	MI0017306
350	hsa-mir-4638	MI0017265
351	hsa-mir-4651	MI0017279
352	hsa-mir-6087	MI0020364
353	hsa-mir-4665	MI0017295
354	hsa-mir-4758	MI0017399
355	hsa-mir-6887	MI0022734
356	hsa-mir-3620	MI0016011
357	hsa-mir-1909	MI0008330
358	hsa-mir-7641-1	MI0024975
359	hsa-mir-7641-2	MI0024976
360	hsa-mir-6724	MI0022559
361	hsa-mir-1343	MI0017320
362	hsa-mir-6780b	MI0022681
363	hsa-mir-4484	MI0016845
364	hsa-mir-4690	MI0017323
365	hsa-mir-4429	MI0016768
366	hsa-mir-1227	MI0006316
367	hsa-mir-4725	MI0017362
368	hsa-mir-6861	MI0022708
369	hsa-mir-6812	MI0022657
370	hsa-mir-3197	MI0014245
371	hsa-mir-8059	MI0025895
372	hsa-mir-3185	MI0014227
373	hsa-mir-4706	MI0017339
374	hsa-mir-4497	MI0016859
375	hsa-mir-3131	MI0014151
376	hsa-mir-6806	MI0022651
377	hsa-mir-187	MI0000274
378	hsa-mir-3180-1	MI0014214
379	hsa-mir-3180-2	MI0014215
380	hsa-mir-3180-3	MI0014217
381	hsa-mir-6848	MI0022694
382	hsa-mir-6820	MI0022665
383	hsa-mir-6800	MI0022645
384	hsa-mir-6717	MI0022551
385	hsa-mir-6795	MI0022640
386	hsa-mir-4632	MI0017259
387	hsa-mir-665	MI0005563
388	hsa-mir-6778	MI0022623
389	hsa-mir-3663	MI0016064
390	hsa-mir-4689	MI0017322
391	hsa-mir-211	MI0000287
392	hsa-mir-6511b-1	MI0022552
393	hsa-mir-6511b-2	MI0023431
394	hsa-mir-4750	MI0017389
395	hsa-mir-6126	MI0021260
396	hsa-mir-614	MI0003627
397	hsa-mir-7110	MI0022961
398	hsa-mir-744	MI0005559
399	hsa-mir-6769a	MI0022614
400	hsa-mir-4792	MI0017439
401	hsa-mir-5787	MI0019797
402	hsa-mir-6798	MI0022643
403	hsa-mir-6781	MI0022626
404	hsa-mir-4419b	MI0016861
405	hsa-mir-4446	MI0016789
406	hsa-mir-4259	MI0015858
407	hsa-mir-5572	MI0019117

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
408	hsa-mir-6075	MI0020352
409	hsa-mir-296	MI0000747
410	hsa-mir-6891	MI0022738
411	hsa-mir-4745	MI0017384
412	hsa-mir-6775	MI0022620
413	hsa-mir-6870	MI0022717
414	hsa-mir-920	MI0005712
415	hsa-mir-4530	MI0016897
416	hsa-mir-6819	MI0022664
417	hsa-mir-6825	MI0022670
418	hsa-mir-7847	MI0025517
419	hsa-mir-6131	MI0021276
420	hsa-mir-4433	MI0016773
421	hsa-mir-1228	MI0006318
422	hsa-mir-6743	MI0022588
423	hsa-mir-1268a	MI0006405
424	hsa-mir-3917	MI0016423
425	hsa-mir-6786	MI0022631
426	hsa-mir-3154	MI0014182
427	hsa-mir-638	MI0003653
428	hsa-mir-6741	MI0022586
429	hsa-mir-6889	MI0022736
430	hsa-mir-6840	MI0022686
431	hsa-mir-6510	MI0022222
432	hsa-mir-3188	MI0014232
433	hsa-mir-551b	MI0003575
434	hsa-mir-5001	MI0017867
435	hsa-mir-1268b	MI0016748
436	hsa-mir-7107	MI0022958
437	hsa-mir-6824	MI0022669
438	hsa-mir-6732	MI0022577
439	hsa-mir-371a	MI0000779
440	hsa-mir-6794	MI0022639
441	hsa-mir-6779	MI0022624
442	hsa-mir-4271	MI0015879
443	hsa-mir-5195	MI0018174
444	hsa-mir-6762	MI0022607
445	hsa-mir-939	MI0005761
446	hsa-mir-1247	MI0006382
447	hsa-mir-6777	MI0022622
448	hsa-mir-6722	MI0022557
449	hsa-mir-3656	MI0016056
450	hsa-mir-4688	MI0017321
451	hsa-mir-3195	MI0014240
452	hsa-mir-6766	MI0022611
453	hsa-mir-4447	MI0016790
454	hsa-mir-4656	MI0017284
455	hsa-mir-7108	MI0022959
456	hsa-mir-3191	MI0014236
457	hsa-mir-1273g	MI0018003
458	hsa-mir-4463	MI0016811
459	hsa-mir-2861	MI0013006
460	hsa-mir-3196	MI0014241
461	hsa-mir-6877	MI0022724
462	hsa-mir-3679	MI0016080
463	hsa-mir-4442	MI0016785
464	hsa-mir-6789	MI0022634
465	hsa-mir-6782	MI0022627
466	hsa-mir-486	MI0002470
467	hsa-mir-486-2	MI0023622
468	hsa-mir-6085	MI0020362
469	hsa-mir-4746	MI0017385
470	hsa-mir-619	MI0003633
471	hsa-mir-937	MI0005759
472	hsa-mir-6803	MI0022648
473	hsa-mir-4298	MI0015830
474	hsa-mir-4454	MI0016800
475	hsa-mir-4459	MI0016805
476	hsa-mir-7150	MI0023610
477	hsa-mir-6880	MI0022727
478	hsa-mir-4449	MI0016792
479	hsa-mir-8063	MI0025899
480	hsa-mir-4695	MI0017328
481	hsa-mir-6132	MI0021277
482	hsa-mir-6829	MI0022674
483	hsa-mir-4486	MI0016847
484	hsa-mir-6826	MI0022671

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TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
485	hsa-mir-4508	MI0016872
486	hsa-mir-7114	MI0022965
487	hsa-mir-3622a	MI0016013
488	hsa-mir-6765	MI0022610
489	hsa-mir-7845	MI0025515
490	hsa-mir-3960	MI0016964
491	hsa-mir-6749	MI0022594
492	hsa-mir-1260b	MI0014197
493	hsa-mir-6799	MI0022644
494	hsa-mir-4723	MI0017359
495	hsa-mir-6784	MI0022629
496	hsa-mir-5100	MI0019116
497	hsa-mir-6769b	MI0022706
498	hsa-mir-1207	MI0006340
499	hsa-mir-642a	MI0003657
500	hsa-mir-4505	MI0016868
501	hsa-mir-4270	MI0015878
502	hsa-mir-6721	MI0022556
503	hsa-mir-7111	MI0022962
504	hsa-mir-6791	MI0022636
505	hsa-mir-7109	MI0022960
506	hsa-mir-4258	MI0015857
507	hsa-mir-6515	MI0022227
508	hsa-mir-6851	MI0022697
509	hsa-mir-6125	MI0021259
510	hsa-mir-4749	MI0017388
511	hsa-mir-4726	MI0017363
512	hsa-mir-4513	MI0016879
513	hsa-mir-760	MI0005567
514	hsa-mir-602	MI0003615
515	hsa-mir-423	MI0001445
516	hsa-mir-92a-2	MI0000094
517	hsa-mir-16-1	MI0000070
518	hsa-mir-16-2	MI0000115
519	hsa-mir-451a	MI0001729
520	hsa-mir-135a	MI0000452
521	hsa-mir-4257	MI0015856
522	hsa-mir-92b	MI0003560
523	hsa-mir-718	MI0012489
524	hsa-mir-940	MI0005762
525	hsa-mir-23b	MI0000439
526	hsa-mir-92a-1	MI0000093
527	hsa-mir-92a-2	MI0000094
528	hsa-mir-658	MI0003682
529	hsa-mir-6842	MI0022688
530	hsa-mir-6124	MI0021258
531	hsa-mir-6765	MI0022610
532	hsa-mir-7106	MI0022957
533	hsa-mir-4534	MI0016901
534	hsa-mir-3135b	MI0016809
535	hsa-mir-4687	MI0017319
536	hsa-mir-762	MI0003892
537	hsa-mir-3619	MI0016009
538	hsa-mir-4467	MI0016818
539	hsa-mir-557	MI0003563
540	hsa-mir-1237	MI0006327
541	hsa-mir-1908	MI0008329
542	hsa-mir-4286	MI0015894
543	hsa-mir-6885	MI0022732
544	hsa-mir-6763	MI0022608
545	isomiR example 1 of SEQ ID NO: 1	—
546	isomiR example 2 of SEQ ID NO: 1	—
547	isomiR example 1 of SEQ ID NO: 2	—
548	isomiR example 2 of SEQ ID NO: 2	—
549	isomiR example 1 of SEQ ID NO: 3	—
550	isomiR example 2 of SEQ ID NO: 3	—
551	isomiR example 1 of SEQ ID NO: 4	—
552	isomiR example 2 of SEQ ID NO: 4	—
553	isomiR example 1 of SEQ ID NO: 5	—
554	isomiR example 2 of SEQ ID NO: 5	—
555	isomiR example 1 of SEQ ID NO: 6	—
556	isomiR example 2 of SEQ ID NO: 6	—
557	isomiR example 1 of SEQ ID NO: 8	—
558	isomiR example 2 of SEQ ID NO: 8	—
559	isomiR example 1 of SEQ ID NO: 11	—
560	isomiR example 2 of SEQ ID NO: 11	—
561	isomiR example 1 of SEQ ID NO: 14	—

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TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
562	isomiR example 2 of SEQ ID NO: 14	—
563	isomiR example 1 of SEQ ID NO: 15	—
564	isomiR example 2 of SEQ ID NO: 15	—
565	isomiR example 1 of SEQ ID NO: 19	—
566	isomiR example 2 of SEQ ID NO: 19	—
567	isomiR example 1 of SEQ ID NO: 20	—
568	isomiR example 2 of SEQ ID NO: 20	—
569	isomiR example 1 of SEQ ID NO: 21	—
570	isomiR example 2 of SEQ ID NO: 21	—
571	isomiR example 1 of SEQ ID NO: 23	—
572	isomiR example 2 of SEQ ID NO: 23	—
573	isomiR example 1 of SEQ ID NO: 27	—
574	isomiR example 2 of SEQ ID NO: 27	—
575	isomiR example 1 of SEQ ID NO: 28	—
576	isomiR example 2 of SEQ ID NO: 28	—
577	isomiR example 1 of SEQ ID NO: 31	—
578	isomiR example 2 of SEQ ID NO: 31	—
579	isomiR example 1 of SEQ ID NO: 35	—
580	isomiR example 2 of SEQ ID NO: 35	—
581	isomiR example 1 of SEQ ID NO: 37	—
582	isomiR example 2 of SEQ ID NO: 37	—
583	isomiR example 1 of SEQ ID NO: 38	—
584	isomiR example 2 of SEQ ID NO: 38	—
585	isomiR example 1 of SEQ ID NO: 40	—
586	isomiR example 2 of SEQ ID NO: 40	—
587	isomiR example 1 of SEQ ID NO: 42	—
588	isomiR example 2 of SEQ ID NO: 42	—
589	isomiR example 1 of SEQ ID NO: 43	—
590	isomiR example 2 of SEQ ID NO: 43	—
591	isomiR example 1 of SEQ ID NO: 47	—
592	isomiR example 2 of SEQ ID NO: 47	—
593	isomiR example 1 of SEQ ID NO: 48	—
594	isomiR example 2 of SEQ ID NO: 48	—
595	isomiR example 1 of SEQ ID NO: 50	—
596	isomiR example 2 of SEQ ID NO: 50	—
597	isomiR example 1 of SEQ ID NO: 51	—
598	isomiR example 2 of SEQ ID NO: 51	—
599	isomiR example 1 of SEQ ID NO: 52	—
600	isomiR example 2 of SEQ ID NO: 52	—
601	isomiR example 1 of SEQ ID NO: 55	—
602	isomiR example 2 of SEQ ID NO: 55	—
603	isomiR example 1 of SEQ ID NO: 56	—
604	isomiR example 2 of SEQ ID NO: 56	—
605	isomiR example 1 of SEQ ID NO: 58	—
606	isomiR example 2 of SEQ ID NO: 58	—
607	isomiR example 1 of SEQ ID NO: 60	—
608	isomiR example 2 of SEQ ID NO: 60	—
609	isomiR example 1 of SEQ ID NO: 61	—
610	isomiR example 2 of SEQ ID NO: 61	—
611	isomiR example 1 of SEQ ID NO: 62	—
612	isomiR example 2 of SEQ ID NO: 62	—
613	isomiR example 1 of SEQ ID NO: 64	—
614	isomiR example 2 of SEQ ID NO: 64	—
615	isomiR example 1 of SEQ ID NO: 65	—
616	isomiR example 2 of SEQ ID NO: 65	—
617	isomiR example 1 of SEQ ID NO: 66	—
618	isomiR example 2 of SEQ ID NO: 66	—
619	isomiR example 1 of SEQ ID NO: 67	—
620	isomiR example 2 of SEQ ID NO: 67	—
621	isomiR example 1 of SEQ ID NO: 71	—
622	isomiR example 2 of SEQ ID NO: 71	—
623	isomiR example 1 of SEQ ID NO: 72	—
624	isomiR example 2 of SEQ ID NO: 72	—
625	isomiR example 1 of SEQ ID NO: 73	—
626	isomiR example 2 of SEQ ID NO: 73	—
627	isomiR example 1 of SEQ ID NO: 76	—
628	isomiR example 2 of SEQ ID NO: 76	—
629	isomiR example 1 of SEQ ID NO: 77	—
630	isomiR example 2 of SEQ ID NO: 77	—
631	isomiR example 1 of SEQ ID NO: 78	—
632	isomiR example 2 of SEQ ID NO: 78	—
633	isomiR example 1 of SEQ ID NO: 79	—
634	isomiR example 2 of SEQ ID NO: 79	—
635	isomiR example 1 of SEQ ID NO: 80	—
636	isomiR example 2 of SEQ ID NO: 80	—
637	isomiR example 1 of SEQ ID NO: 82	—
638	isomiR example 2 of SEQ ID NO: 82	—

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.		SEQ ID NO:	Gene name	miRBase registration No.
639	isomiR example 1 of SEQ ID NO: 83	—	5	716	isomiR example 2 of SEQ ID NO: 149	—
640	isomiR example 2 of SEQ ID NO: 83	—		717	isomiR example 1 of SEQ ID NO: 153	—
641	isomiR example 1 of SEQ ID NO: 85	—		718	isomiR example 2 of SEQ ID NO: 153	—
642	isomiR example 2 of SEQ ID NO: 85	—		719	isomiR example 1 of SEQ ID NO: 154	—
643	isomiR example 1 of SEQ ID NO: 86	—		720	isomiR example 2 of SEQ ID NO: 154	—
644	isomiR example 2 of SEQ ID NO: 86	—	10	721	isomiR example 1 of SEQ ID NO: 155	—
645	isomiR example 1 of SEQ ID NO: 88	—		722	isomiR example 2 of SEQ ID NO: 155	—
646	isomiR example 2 of SEQ ID NO: 88	—		723	isomiR example 1 of SEQ ID NO: 156	—
647	isomiR example 1 of SEQ ID NO: 89	—		724	isomiR example 2 of SEQ ID NO: 156	—
648	isomiR example 2 of SEQ ID NO: 89	—		725	isomiR example 1 of SEQ ID NO: 157	—
649	isomiR example 1 of SEQ ID NO: 90	—	15	726	isomiR example 2 of SEQ ID NO: 157	—
650	isomiR example 2 of SEQ ID NO: 90	—		727	isomiR example 1 of SEQ ID NO: 161	—
651	isomiR example 1 of SEQ ID NO: 92	—		728	isomiR example 2 of SEQ ID NO: 161	—
652	isomiR example 2 of SEQ ID NO: 92	—		729	isomiR example 1 of SEQ ID NO: 164	—
653	isomiR example 1 of SEQ ID NO: 95	—		730	isomiR example 2 of SEQ ID NO: 164	—
654	isomiR example 2 of SEQ ID NO: 95	—	20	731	isomiR example 1 of SEQ ID NO: 165	—
655	isomiR example 1 of SEQ ID NO: 97	—		732	isomiR example 2 of SEQ ID NO: 165	—
656	isomiR example 2 of SEQ ID NO: 97	—		733	isomiR example 1 of SEQ ID NO: 167	—
657	isomiR example 1 of SEQ ID NO: 98	—		734	isomiR example 2 of SEQ ID NO: 167	—
658	isomiR example 2 of SEQ ID NO: 98	—		735	isomiR example 1 of SEQ ID NO: 168	—
659	isomiR example 1 of SEQ ID NO: 99	—	25	736	isomiR example 2 of SEQ ID NO: 168	—
660	isomiR example 2 of SEQ ID NO: 99	—		737	isomiR example 1 of SEQ ID NO: 171	—
661	isomiR example 1 of SEQ ID NO: 100	—		738	isomiR example 2 of SEQ ID NO: 171	—
662	isomiR example 2 of SEQ ID NO: 100	—		739	isomiR example 1 of SEQ ID NO: 172	—
663	isomiR example 1 of SEQ ID NO: 102	—		740	isomiR example 2 of SEQ ID NO: 172	—
664	isomiR example 2 of SEQ ID NO: 102	—	30	741	isomiR example 1 of SEQ ID NO: 173	—
665	isomiR example 1 of SEQ ID NO: 103	—		742	isomiR example 2 of SEQ ID NO: 173	—
666	isomiR example 2 of SEQ ID NO: 103	—		743	isomiR example 1 of SEQ ID NO: 178	—
667	isomiR example 1 of SEQ ID NO: 107	—		744	isomiR example 2 of SEQ ID NO: 178	—
668	isomiR example 2 of SEQ ID NO: 107	—		745	isomiR example 1 of SEQ ID NO: 179	—
669	isomiR example 1 of SEQ ID NO: 109	—	35	746	isomiR example 2 of SEQ ID NO: 179	—
670	isomiR example 2 of SEQ ID NO: 109	—		747	isomiR example 1 of SEQ ID NO: 180	—
671	isomiR example 1 of SEQ ID NO: 110	—		748	isomiR example 2 of SEQ ID NO: 180	—
672	isomiR example 2 of SEQ ID NO: 110	—		749	isomiR example 1 of SEQ ID NO: 181	—
673	isomiR example 1 of SEQ ID NO: 113	—		750	isomiR example 2 of SEQ ID NO: 181	—
674	isomiR example 2 of SEQ ID NO: 113	—	40	751	isomiR example 1 of SEQ ID NO: 182	—
675	isomiR example 1 of SEQ ID NO: 114	—		752	isomiR example 2 of SEQ ID NO: 182	—
676	isomiR example 2 of SEQ ID NO: 114	—		753	isomiR example 1 of SEQ ID NO: 184	—
677	isomiR example 1 of SEQ ID NO: 115	—		754	isomiR example 2 of SEQ ID NO: 184	—
678	isomiR example 2 of SEQ ID NO: 115	—		755	isomiR example 1 of SEQ ID NO: 185	—
679	isomiR example 1 of SEQ ID NO: 116	—	45	756	isomiR example 2 of SEQ ID NO: 185	—
680	isomiR example 2 of SEQ ID NO: 116	—		757	isomiR example 1 of SEQ ID NO: 188	—
681	isomiR example 1 of SEQ ID NO: 117	—		758	isomiR example 2 of SEQ ID NO: 188	—
682	isomiR example 2 of SEQ ID NO: 117	—		759	isomiR example 1 of SEQ ID NO: 191	—
683	isomiR example 1 of SEQ ID NO: 118	—		760	isomiR example 2 of SEQ ID NO: 191	—
684	isomiR example 2 of SEQ ID NO: 118	—	50	761	isomiR example 1 of SEQ ID NO: 192	—
685	isomiR example 1 of SEQ ID NO: 120	—		762	isomiR example 2 of SEQ ID NO: 192	—
686	isomiR example 2 of SEQ ID NO: 120	—		763	isomiR example 1 of SEQ ID NO: 194	—
687	isomiR example 1 of SEQ ID NO: 122	—		764	isomiR example 2 of SEQ ID NO: 194	—
688	isomiR example 2 of SEQ ID NO: 122	—		765	isomiR example 1 of SEQ ID NO: 195	—
689	isomiR example 1 of SEQ ID NO: 123	—	55	766	isomiR example 2 of SEQ ID NO: 195	—
690	isomiR example 2 of SEQ ID NO: 123	—		767	isomiR example 1 of SEQ ID NO: 196	—
691	isomiR example 1 of SEQ ID NO: 126	—		768	isomiR example 2 of SEQ ID NO: 196	—
692	isomiR example 2 of SEQ ID NO: 126	—		769	isomiR example 1 of SEQ ID NO: 199	—
693	isomiR example 1 of SEQ ID NO: 127	—		770	isomiR example 2 of SEQ ID NO: 199	—
694	isomiR example 2 of SEQ ID NO: 127	—	60	771	isomiR example 1 of SEQ ID NO: 201	—
695	isomiR example 1 of SEQ ID NO: 129	—		772	isomiR example 2 of SEQ ID NO: 201	—
696	isomiR example 2 of SEQ ID NO: 129	—		773	isomiR example 1 of SEQ ID NO: 202	—
697	isomiR example 1 of SEQ ID NO: 131	—		774	isomiR example 2 of SEQ ID NO: 202	—
698	isomiR example 2 of SEQ ID NO: 131	—		775	isomiR example 1 of SEQ ID NO: 204	—
699	isomiR example 1 of SEQ ID NO: 133	—	65	776	isomiR example 2 of SEQ ID NO: 204	—
700	isomiR example 2 of SEQ ID NO: 133	—		777	isomiR example 1 of SEQ ID NO: 207	—
701	isomiR example 1 of SEQ ID NO: 137	—		778	isomiR example 2 of SEQ ID NO: 207	—
702	isomiR example 2 of SEQ ID NO: 137	—		779	isomiR example 1 of SEQ ID NO: 210	—
703	isomiR example 1 of SEQ ID NO: 141	—		780	isomiR example 2 of SEQ ID NO: 210	—
704	isomiR example 2 of SEQ ID NO: 141	—	70	781	isomiR example 1 of SEQ ID NO: 213	—
705	isomiR example 1 of SEQ ID NO: 142	—		782	isomiR example 2 of SEQ ID NO: 213	—
706	isomiR example 2 of SEQ ID NO: 142	—		783	isomiR example 1 of SEQ ID NO: 215	—
707	isomiR example 1 of SEQ ID NO: 143	—		784	isomiR example 2 of SEQ ID NO: 215	—
708	isomiR example 2 of SEQ ID NO: 143	—		785	isomiR example 1 of SEQ ID NO: 217	—
709	isomiR example 1 of SEQ ID NO: 145	—	75	786	isomiR example 2 of SEQ ID NO: 217	—
710	isomiR example 2 of SEQ ID NO: 145	—		787	isomiR example 1 of SEQ ID NO: 219	—
711	isomiR example 1 of SEQ ID NO: 146	—		788	isomiR example 2 of SEQ ID NO: 219	—
712	isomiR example 2 of SEQ ID NO: 146	—		789	isomiR example 1 of SEQ ID NO: 222	—
713	isomiR example 1 of SEQ ID NO: 148	—		790	isomiR example 2 of SEQ ID NO: 222	—
714	isomiR example 2 of SEQ ID NO: 148	—	75	791	isomiR example 1 of SEQ ID NO: 223	—
715	isomiR example 1 of SEQ ID NO: 149	—		792	isomiR example 2 of SEQ ID NO: 223	—

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
793	isomiR example 1 of SEQ ID NO: 225	—
794	isomiR example 2 of SEQ ID NO: 225	—
795	isomiR example 1 of SEQ ID NO: 230	—
796	isomiR example 2 of SEQ ID NO: 230	—
797	isomiR example 1 of SEQ ID NO: 232	—
798	isomiR example 2 of SEQ ID NO: 232	—
799	isomiR example 1 of SEQ ID NO: 233	—
800	isomiR example 2 of SEQ ID NO: 233	—
801	isomiR example 1 of SEQ ID NO: 234	—
802	isomiR example 2 of SEQ ID NO: 234	—
803	isomiR example 1 of SEQ ID NO: 235	—
804	isomiR example 2 of SEQ ID NO: 235	—
805	isomiR example 1 of SEQ ID NO: 236	—
806	isomiR example 2 of SEQ ID NO: 236	—
807	isomiR example 1 of SEQ ID NO: 238	—
808	isomiR example 2 of SEQ ID NO: 238	—
809	isomiR example 1 of SEQ ID NO: 239	—
810	isomiR example 2 of SEQ ID NO: 239	—
811	isomiR example 1 of SEQ ID NO: 240	—
812	isomiR example 2 of SEQ ID NO: 240	—
813	isomiR example 1 of SEQ ID NO: 241	—
814	isomiR example 2 of SEQ ID NO: 241	—
815	isomiR example 1 of SEQ ID NO: 242	—
816	isomiR example 2 of SEQ ID NO: 242	—
817	isomiR example 1 of SEQ ID NO: 243	—
818	isomiR example 2 of SEQ ID NO: 243	—
819	isomiR example 1 of SEQ ID NO: 245	—
820	isomiR example 2 of SEQ ID NO: 245	—
821	isomiR example 1 of SEQ ID NO: 246	—
822	isomiR example 2 of SEQ ID NO: 246	—
823	isomiR example 1 of SEQ ID NO: 247	—
824	isomiR example 2 of SEQ ID NO: 247	—
825	isomiR example 1 of SEQ ID NO: 248	—
826	isomiR example 2 of SEQ ID NO: 248	—
827	isomiR example 1 of SEQ ID NO: 249	—
828	isomiR example 2 of SEQ ID NO: 249	—
829	isomiR example 1 of SEQ ID NO: 250	—
830	isomiR example 2 of SEQ ID NO: 250	—
831	isomiR example 1 of SEQ ID NO: 251	—
832	isomiR example 2 of SEQ ID NO: 251	—
833	isomiR example 1 of SEQ ID NO: 252	—
834	isomiR example 2 of SEQ ID NO: 252	—
835	isomiR example 1 of SEQ ID NO: 254	—
836	isomiR example 2 of SEQ ID NO: 254	—
837	isomiR example 1 of SEQ ID NO: 258	—
838	isomiR example 2 of SEQ ID NO: 258	—
839	isomiR example 1 of SEQ ID NO: 259	—
840	isomiR example 2 of SEQ ID NO: 259	—
841	isomiR example 1 of SEQ ID NO: 260	—
842	isomiR example 2 of SEQ ID NO: 260	—
843	isomiR example 1 of SEQ ID NO: 263	—
844	isomiR example 2 of SEQ ID NO: 263	—
845	isomiR example 1 of SEQ ID NO: 265	—
846	isomiR example 2 of SEQ ID NO: 265	—
847	isomiR example 1 of SEQ ID NO: 266	—
848	isomiR example 2 of SEQ ID NO: 266	—
849	isomiR example 1 of SEQ ID NO: 267	—
850	isomiR example 2 of SEQ ID NO: 267	—
851	hsa-miR-6089	MIMAT0023714
852	hsa-miR-6816-5p	MIMAT0027532
853	hsa-miR-4466	MIMAT0018993
854	hsa-miR-4488	MIMAT0019022
855	hsa-miR-6752-5p	MIMAT0027404
856	hsa-miR-4739	MIMAT0019868
857	hsa-mir-6089-1	MI0020366
858	hsa-mir-6089-2	MI0023563
859	hsa-mir-6816	MI0022661
860	hsa-mir-4466	MI0016817
861	hsa-mir-4488	MI0016849
862	hsa-mir-6752	MI0022597
863	hsa-mir-4739	MI0017377
864	isomiR example 1 of SEQ ID NO: 851	—
865	isomiR example 2 of SEQ ID NO: 851	—
866	isomiR example 1 of SEQ ID NO: 853	—
867	isomiR example 2 of SEQ ID NO: 853	—
868	isomiR example 1 of SEQ ID NO: 854	—
869	isomiR example 2 of SEQ ID NO: 854	—

TABLE 1-continued

SEQ ID NO:	Gene name	miRBase registration No.
870	isomiR example 1 of SEQ ID NO: 856	—
871	isomiR example 2 of SEQ ID NO: 856	—

The present specification encompasses the contents described in the specifications and/or drawings of Japanese Patent Application No. 2014-122672 and No. 2015-069321 from which the present application claims priorities.

Advantageous Effect of Invention

According to the present invention, breast cancer can be detected easily and in high accuracy. For example, the presence or absence of breast cancer in patients can be easily detected by using, as indicators, the measurement values of several miRNAs in bloods, sera, and/or plasmas of the patients, which can be collected with limitedly invasiveness.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 This figure shows the relationship between the nucleotide sequences of hsa-miR-6805-5p represented by SEQ ID NO: 34 and hsa-miR-6805-3p represented by SEQ ID NO: 205, which are produced from a precursor hsa-mir-6805 represented by SEQ ID NO: 304.

FIG. 2 Left diagram: the expression level measurement values of hsa-miR-4783-3p (SEQ ID NO: 1) in healthy subjects (100 persons) and breast cancer patients (62 persons) selected as a training cohort were each plotted on the ordinate. The horizontal line in the diagram depicts a threshold (6.63) that was optimized by Fisher's discriminant analysis and discriminated between the two groups. Right diagram: the expression level measurement values of hsa-miR-4783-3p (SEQ ID NO: 1) in healthy subjects (50 persons) and breast cancer patients (31 persons) selected as a validation cohort were each plotted on the ordinate. The horizontal line in the diagram depicts the threshold (6.63) that was set in the training cohort and discriminated between the two groups.

FIG. 3 Left diagram: the expression level measurement values of hsa-miR-4783-3p (SEQ ID NO: 1) in healthy subjects (100 persons, circles) and breast cancer patients (62 persons, triangles) selected as a training cohort were each plotted on the abscissa against their expression level measurement values of hsa-miR-4730 (SEQ ID NO: 2) on the ordinate. The line in the diagram depicts a discriminant function ($0=1.41x+y+0.77$) that was optimized by Fisher's discriminant analysis and discriminated between the two groups. Right diagram: the expression level measurement values of hsa-miR-4783-3p (SEQ ID NO: 1) in healthy subjects (50 persons, circles) and breast cancer patients (31 persons, triangles) selected as a validation cohort were each plotted on the abscissa against their expression level measurement values of hsa-miR-4730 (SEQ ID NO: 2) on the ordinate. The line in the diagram depicts the threshold ($0=1.41x+y+0.77$) that was set for the training cohort and discriminated between the two groups.

FIG. 4 Upper diagram: a discriminant ($1.87 \times \text{hsa-miR-4730} + 0.42 \times \text{hsa-miR-602} - 18.58$) was prepared by use of Fisher's discriminant analysis from the expression level measurement values of hsa-miR-602 (SEQ ID NO: 237) and hsa-miR-4730 (SEQ ID NO: 2) in 62 breast cancer patients, 102 healthy subjects, and 33 prostate cancer patients

selected as a training cohort, and discriminant scores obtained from the discriminant were plotted on the ordinate against the sample groups on the abscissa. The dotted line in the diagram depicts a discriminant boundary that offered a discriminant score of 0 and discriminated between the groups. Lower diagram: discriminant scores obtained from the discriminant prepared for the training cohort as to the expression level measurement values of hsa-miR-602 (SEQ ID NO: 237) and hsa-miR-4730 (SEQ ID NO: 2) in 31 breast cancer patients, 48 healthy subjects, and 19 prostate cancer patients selected as a validation cohort were plotted on the ordinate against the sample groups on the abscissa. The dotted line in the diagram depicts the discriminant boundary that offered a discriminant score of 0 and discriminated between the two groups.

DESCRIPTION OF EMBODIMENTS

Hereinafter, the present invention will be further described in detail.

1. Target Nucleic Acid for Breast Cancer

Primary target nucleic acids as breast cancer markers for detecting the presence and/or absence of breast cancer or breast cancer cells using the nucleic acid probes or the primers for the detection of breast cancer defined above according to the present invention, at least one miRNA selected from the group consisting of the following miRNAs: hsa-miR-4783-3p, hsa-miR-4730, hsa-miR-1307-3p, hsa-miR-4634, hsa-miR-663a, hsa-miR-4532, hsa-miR-7704, hsa-miR-3178, hsa-miR-6729-5p, hsa-miR-6090, hsa-miR-4732-5p, hsa-miR-3184-5p, hsa-miR-6727-5p, hsa-miR-6088, hsa-miR-4674, hsa-miR-8073, hsa-miR-4787-5p, hsa-miR-1469, hsa-miR-125a-3p, hsa-miR-1233-5p, hsa-miR-885-3p, hsa-miR-6802-5p, hsa-miR-328-5p, hsa-miR-6787-5p, hsa-miR-8069, hsa-miR-6875-5p, hsa-miR-1246, hsa-miR-4734, hsa-miR-6757-5p, hsa-miR-6756-5p, hsa-miR-3665, hsa-miR-6836-3p, hsa-miR-6821-5p, hsa-miR-6805-5p, hsa-miR-4728-5p, hsa-miR-6726-5p, hsa-miR-197-5p, hsa-miR-149-3p, hsa-miR-6850-5p, hsa-miR-4476, hsa-miR-6858-5p, hsa-miR-564, hsa-miR-4763-3p, hsa-miR-575, hsa-miR-6771-5p, hsa-miR-1231, hsa-miR-1908-3p, hsa-miR-150-3p, hsa-miR-3937, hsa-miR-887-3p, hsa-miR-3940-5p, hsa-miR-4741, hsa-miR-6808-5p, hsa-miR-6869-5p, hsa-miR-5090, hsa-miR-615-5p, hsa-miR-8072, hsa-miR-128-1-5p, hsa-miR-1238-5p, hsa-miR-365a-5p, hsa-miR-204-3p, hsa-miR-4492, hsa-miR-6785-5p, hsa-miR-6511a-5p, hsa-miR-4525, hsa-miR-1915-5p, hsa-miR-3180, hsa-miR-6879-5p, hsa-miR-1199-5p, hsa-miR-6746-5p, hsa-miR-711, hsa-miR-663b, hsa-miR-4707-3p, hsa-miR-6893-5p, hsa-miR-4675, hsa-miR-4638-5p, hsa-miR-4651, hsa-miR-6087, hsa-miR-4665-5p, hsa-miR-4758-5p, hsa-miR-6887-5p, hsa-miR-3620-5p, hsa-miR-1909-3p, hsa-miR-7641, hsa-miR-6724-5p, hsa-miR-1343-3p, hsa-miR-6780b-5p, hsa-miR-4484, hsa-miR-4690-5p, hsa-miR-4429, hsa-miR-1227-5p, hsa-miR-4725-3p, hsa-miR-6861-5p, hsa-miR-6812-5p, hsa-miR-3197, hsa-miR-8059, hsa-miR-3185, hsa-miR-4706, hsa-miR-4497, hsa-miR-3131, hsa-miR-6806-5p, hsa-miR-187-5p, hsa-miR-3180-3p, hsa-miR-6848-5p, hsa-miR-6820-5p, hsa-miR-6800-5p, hsa-miR-6717-5p, hsa-miR-6795-5p, hsa-miR-4632-5p, hsa-miR-665, hsa-miR-6778-5p, hsa-miR-3663-3p, hsa-miR-4689, hsa-miR-211-3p, hsa-miR-6511b-5p, hsa-miR-4750-5p, hsa-miR-6126, hsa-miR-614, hsa-miR-7110-5p, hsa-miR-744-5p, hsa-miR-6769a-5p, hsa-miR-4792, hsa-miR-5787, hsa-miR-6798-5p, hsa-miR-6781-5p, hsa-miR-4419b, hsa-miR-4446-3p, hsa-miR-4259, hsa-miR-5572, hsa-miR-6075, hsa-miR-296-3p, hsa-miR-6891-5p, hsa-miR-4745-

5p, hsa-miR-6775-5p, hsa-miR-6870-5p, hsa-miR-920, hsa-miR-4530, hsa-miR-6819-5p, hsa-miR-6825-5p, hsa-miR-7847-3p, hsa-miR-6131, hsa-miR-4433-3p, hsa-miR-1228-5p, hsa-miR-6743-5p, hsa-miR-1268a, hsa-miR-3917, hsa-miR-6786-5p, hsa-miR-3154, hsa-miR-638, hsa-miR-6741-5p, hsa-miR-6889-5p, hsa-miR-6840-3p, hsa-miR-6510-5p, hsa-miR-3188, hsa-miR-551b-5p, hsa-miR-5001-5p, hsa-miR-1268b, hsa-miR-7107-5p, hsa-miR-6824-5p, hsa-miR-6732-5p, hsa-miR-371a-5p, hsa-miR-6794-5p, hsa-miR-6779-5p, hsa-miR-4271, hsa-miR-5195-3p, hsa-miR-6762-5p, hsa-miR-939-5p, hsa-miR-1247-3p, hsa-miR-6777-5p, hsa-miR-6722-3p, hsa-miR-3656, hsa-miR-4688, hsa-miR-3195, hsa-miR-6766-5p, hsa-miR-4447, hsa-miR-4656, hsa-miR-7108-5p, hsa-miR-3191-3p, hsa-miR-1273g-3p, hsa-miR-4463, hsa-miR-2861, hsa-miR-3196, hsa-miR-6877-5p, hsa-miR-3679-5p, hsa-miR-4442, hsa-miR-6789-5p, hsa-miR-6782-5p, hsa-miR-486-3p, hsa-miR-6085, hsa-miR-4746-3p, hsa-miR-619-5p, hsa-miR-937-5p, hsa-miR-6803-5p, hsa-miR-4298, hsa-miR-4454, hsa-miR-4459, hsa-miR-7150, hsa-miR-6880-5p, hsa-miR-4449, hsa-miR-8063, hsa-miR-4695-5p, hsa-miR-6132, hsa-miR-6829-5p, hsa-miR-4486, hsa-miR-6805-3p, hsa-miR-6826-5p, hsa-miR-4508, hsa-miR-1343-5p, hsa-miR-7114-5p, hsa-miR-3622a-5p, hsa-miR-6765-5p, hsa-miR-7845-5p, hsa-miR-3960, hsa-miR-6749-5p, hsa-miR-1260b, hsa-miR-6799-5p, hsa-miR-4723-5p, hsa-miR-6784-5p, hsa-miR-5100, hsa-miR-6769b-5p, hsa-miR-1207-5p, hsa-miR-642a-3p, hsa-miR-4505, hsa-miR-4270, hsa-miR-6721-5p, hsa-miR-7111-5p, hsa-miR-6791-5p, hsa-miR-7109-5p, hsa-miR-4258, hsa-miR-6515-3p, hsa-miR-6851-5p, hsa-miR-6125, hsa-miR-4749-5p, hsa-miR-4726-5p, hsa-miR-4513, hsa-miR-6089, hsa-miR-6816-5p, hsa-miR-4466, hsa-miR-4488, hsa-miR-6752-5p and hsa-miR-4739 can be used. Furthermore, at least one miRNA selected from the group consisting of the following other breast cancer markers that can be combined with these miRNAs, i.e., hsa-miR-760, hsa-miR-602, hsa-miR-423-5p, hsa-miR-92a-2-5p, hsa-miR-16-5p, hsa-miR-451a, hsa-miR-135a-3p, hsa-miR-486-5p, hsa-miR-4257, hsa-miR-92b-5p, hsa-miR-1915-3p, hsa-miR-718, hsa-miR-940, hsa-miR-296-5p, hsa-miR-23b-3p and hsa-miR-92a-3p can also be preferably used as a target nucleic acid. Moreover, at least one miRNA selected from the group consisting of the following other breast cancer markers that can be combined with these miRNAs, i.e., hsa-miR-658, hsa-miR-6842-5p, hsa-miR-6124, hsa-miR-6765-3p, hsa-miR-7106-5p, hsa-miR-4534, hsa-miR-92b-3p, hsa-miR-3135b, hsa-miR-4687-3p, hsa-miR-762, hsa-miR-3619-3p, hsa-miR-4467, hsa-miR-557, hsa-miR-1237-5p, hsa-miR-1908-5p, hsa-miR-4286, hsa-miR-6885-5p and hsa-miR-6763-5p can also be preferably used as target nucleic acids.

These miRNAs include, for example, a human gene comprising a nucleotide sequence represented by any of SEQ ID NOS: 1 to 269, and 851 to 856 (i.e., hsa-miR-4783-3p, hsa-miR-4730, hsa-miR-1307-3p, hsa-miR-4634, hsa-miR-663a, hsa-miR-4532, hsa-miR-7704, hsa-miR-3178, hsa-miR-6729-5p, hsa-miR-6090, hsa-miR-4732-5p, hsa-miR-3184-5p, hsa-miR-6727-5p, hsa-miR-6088, hsa-miR-4674, hsa-miR-8073, hsa-miR-4787-5p, hsa-miR-1469, hsa-miR-125a-3p, hsa-miR-1233-5p, hsa-miR-885-3p, hsa-miR-6802-5p, hsa-miR-328-5p, hsa-miR-6787-5p, hsa-miR-8069, hsa-miR-6875-5p, hsa-miR-1246, hsa-miR-4734, hsa-miR-6757-5p, hsa-miR-6756-5p, hsa-miR-3665, hsa-miR-6836-3p, hsa-miR-6821-5p, hsa-miR-6805-5p, hsa-miR-4728-5p, hsa-miR-6726-5p, hsa-miR-197-5p, hsa-miR-149-3p, hsa-miR-6850-5p, hsa-miR-4476, hsa-miR-6858-5p, hsa-miR-564, hsa-miR-4763-3p, hsa-miR-575,

hsa-miR-6771-5p, hsa-miR-1231, hsa-miR-1908-3p, hsa-miR-150-3p, hsa-miR-3937, hsa-miR-887-3p, hsa-miR-3940-5p, hsa-miR-4741, hsa-miR-6808-5p, hsa-miR-6869-5p, hsa-miR-5090, hsa-miR-615-5p, hsa-miR-8072, hsa-miR-128-1-5p, hsa-miR-1238-5p, hsa-miR-365a-5p, hsa-miR-204-3p, hsa-miR-4492, hsa-miR-6785-5p, hsa-miR-6511a-5p, hsa-miR-4525, hsa-miR-1915-5p, hsa-miR-3180, hsa-miR-6879-5p, hsa-miR-1199-5p, hsa-miR-6746-5p, hsa-miR-711, hsa-miR-663b, hsa-miR-4707-3p, hsa-miR-6893-5p, hsa-miR-4675, hsa-miR-4638-5p, hsa-miR-4651, hsa-miR-6087, hsa-miR-4665-5p, hsa-miR-4758-5p, hsa-miR-6887-5p, hsa-miR-3620-5p, hsa-miR-1909-3p, hsa-miR-7641, hsa-miR-6724-5p, hsa-miR-1343-3p, hsa-miR-6780b-5p, hsa-miR-4484, hsa-miR-4690-5p, hsa-miR-4429, hsa-miR-1227-5p, hsa-miR-4725-3p, hsa-miR-6861-5p, hsa-miR-6812-5p, hsa-miR-3197, hsa-miR-8059, hsa-miR-3185, hsa-miR-4706, hsa-miR-4497, hsa-miR-3131, hsa-miR-6806-5p, hsa-miR-187-5p, hsa-miR-3180-3p, hsa-miR-6848-5p, hsa-miR-6820-5p, hsa-miR-6800-5p, hsa-miR-6717-5p, hsa-miR-6795-5p, hsa-miR-4632-5p, hsa-miR-665, hsa-miR-6778-5p, hsa-miR-3663-3p, hsa-miR-4689, hsa-miR-211-3p, hsa-miR-6511b-5p, hsa-miR-4750-5p, hsa-miR-6126, hsa-miR-614, hsa-miR-7110-5p, hsa-miR-744-5p, hsa-miR-6769a-5p, hsa-miR-4792, hsa-miR-5787, hsa-miR-6798-5p, hsa-miR-6781-5p, hsa-miR-4419b, hsa-miR-4446-3p, hsa-miR-4259, hsa-miR-5572, hsa-miR-6075, hsa-miR-296-3p, hsa-miR-6891-5p, hsa-miR-4745-5p, hsa-miR-6775-5p, hsa-miR-6870-5p, hsa-miR-920, hsa-miR-4530, hsa-miR-6819-5p, hsa-miR-6825-5p, hsa-miR-7847-3p, hsa-miR-6131, hsa-miR-4433-3p, hsa-miR-1228-5p, hsa-miR-6743-5p, hsa-miR-1268a, hsa-miR-3917, hsa-miR-6786-5p, hsa-miR-3154, hsa-miR-638, hsa-miR-6741-5p, hsa-miR-6889-5p, hsa-miR-6840-3p, hsa-miR-6510-5p, hsa-miR-3188, hsa-miR-551b-5p, hsa-miR-5001-5p, hsa-miR-1268b, hsa-miR-7107-5p, hsa-miR-6824-5p, hsa-miR-6732-5p, hsa-miR-371a-5p, hsa-miR-6794-5p, hsa-miR-6779-5p, hsa-miR-4271, hsa-miR-5195-3p, hsa-miR-6762-5p, hsa-miR-939-5p, hsa-miR-1247-3p, hsa-miR-6777-5p, hsa-miR-6722-3p, hsa-miR-3656, hsa-miR-4688, hsa-miR-3195, hsa-miR-6766-5p, hsa-miR-4447, hsa-miR-4656, hsa-miR-7108-5p, hsa-miR-3191-3p, hsa-miR-1273g-3p, hsa-miR-4463, hsa-miR-2861, hsa-miR-3196, hsa-miR-6877-5p, hsa-miR-3679-5p, hsa-miR-4442, hsa-miR-6789-5p, hsa-miR-6782-5p, hsa-miR-486-3p, hsa-miR-6085, hsa-miR-4746-3p, hsa-miR-619-5p, hsa-miR-937-5p, hsa-miR-6803-5p, hsa-miR-4298, hsa-miR-4454, hsa-miR-4459, hsa-miR-7150, hsa-miR-6880-5p, hsa-miR-4449, hsa-miR-8063, hsa-miR-4695-5p, hsa-miR-6132, hsa-miR-6829-5p, hsa-miR-4486, hsa-miR-6805-3p, hsa-miR-6826-5p, hsa-miR-4508, hsa-miR-1343-5p, hsa-miR-7114-5p, hsa-miR-3622a-5p, hsa-miR-6765-5p, hsa-miR-7845-5p, hsa-miR-3960, hsa-miR-6749-5p, hsa-miR-1260b, hsa-miR-6799-5p, hsa-miR-4723-5p, hsa-miR-6784-5p, hsa-miR-5100, hsa-miR-6769b-5p, hsa-miR-1207-5p, hsa-miR-642a-3p, hsa-miR-4505, hsa-miR-4270, hsa-miR-6721-5p, hsa-miR-7111-5p, hsa-miR-6791-5p, hsa-miR-7109-5p, hsa-miR-4258, hsa-miR-6515-3p, hsa-miR-6851-5p, hsa-miR-6125, hsa-miR-4749-5p, hsa-miR-4726-5p, hsa-miR-4513, hsa-miR-6089, hsa-miR-6816-5p, hsa-miR-4466, hsa-miR-4488, hsa-miR-6752-5p, hsa-miR-4739, hsa-miR-760, hsa-miR-602, hsa-miR-423-5p, hsa-miR-92a-2-5p, hsa-miR-16-5p, hsa-miR-451a, hsa-miR-135a-3p, hsa-miR-486-5p, hsa-miR-4257, hsa-miR-92b-5p, hsa-miR-1915-3p, hsa-miR-718, hsa-miR-940, hsa-miR-296-5p, hsa-miR-23b-3p, hsa-miR-92a-3p, hsa-miR-658, hsa-miR-6842-5p, hsa-miR-6124, hsa-miR-6765-3p, hsa-miR-7106-5p, hsa-miR-4534, hsa-miR-92b-3p, hsa-miR-3135b, hsa-miR-4687-3p, hsa-

miR-762, hsa-miR-3619-3p, hsa-miR-4467, hsa-miR-557, hsa-miR-1237-5p, hsa-miR-1908-5p, hsa-miR-4286, hsa-miR-6885-5p and hsa-miR-6763-5p, respectively), a congener thereof, a transcript thereof, or/and a variant or a derivative thereof. In this context, the gene, the congener, the transcript, the variant, and the derivative are as defined above.

The target nucleic acid is preferably a human gene comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 871 or a transcript thereof, more preferably the transcript, i.e., a miRNA or its precursor RNA (pri-miRNA or pre-miRNA).

The first target gene is the hsa-miR-4783-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The second target gene is the hsa-miR-4730 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The third target gene is the hsa-miR-1307-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The fourth target gene is the hsa-miR-4634 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The fifth target gene is the hsa-miR-663a gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The sixth target gene is the hsa-miR-4532 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The seventh target gene is the hsa-miR-7704 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The eighth target gene is the hsa-miR-3178 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The ninth target gene is the hsa-miR-6729-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The 10th target gene is the hsa-miR-6090 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The 11th target gene is the hsa-miR-4732-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The 25th target gene is the hsa-miR-8069 gene, a congener thereof, a transcript thereof, or a variant or a derivative

The 38th target gene is the hsa-miR-149-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports

The 78th target gene is the hsa-miR-6087 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that

The 91st target gene is the hsa-miR-1227-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The 105th target gene is the hsa-miR-6820-5p gene, a congener thereof, a transcript thereof, or a variant or a

The 158th target gene is the hsa-miR-7107-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports

The 171st target gene is the hsa-miR-3656 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

The 185th target gene is the hsa-miR-4442 gene, a congener thereof, a transcript thereof, or a variant or a derivative

The 238th target gene is the hsa-miR-423-5p gene, a congener thereof, a transcript thereof, or a variant or a

The 249th target gene is the hsa-miR-296-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. The previously known report shows that

change in the expression of the gene or the transcript thereof can serve as a marker for breast cancer.

2. Nucleic Acid Probe or Primer for Detection of Breast Cancer

In the present invention, nucleic acid(s) capable of specifically binding to any of the target nucleic acids as the breast cancer markers described above can be used as a nucleic acid, for example, a nucleic acid probe or a primer, for the detection or diagnosis of breast cancer.

In the present invention, the nucleic acid probes or the primers that can be used for detecting breast cancer or for diagnosing breast cancer enables qualitative and/or quantitative measurement of the presence, expression level, or existing amount (abundance) of any of the target nucleic acids as the breast cancer markers described above, for example, human-derived hsa-miR-4783-3p, hsa-miR-4730, hsa-miR-1307-3p, hsa-miR-4634, hsa-miR-663a, hsa-miR-4532, hsa-miR-7704, hsa-miR-3178, hsa-miR-6729-5p, hsa-miR-6090, hsa-miR-4732-5p, hsa-miR-3184-5p, hsa-miR-6727-5p, hsa-miR-6088, hsa-miR-4674, hsa-miR-8073, hsa-miR-4787-5p, hsa-miR-1469, hsa-miR-125a-3p, hsa-miR-1233-5p, hsa-miR-885-3p, hsa-miR-6802-5p, hsa-miR-328-5p, hsa-miR-6787-5p, hsa-miR-8069, hsa-miR-6875-5p, hsa-miR-1246, hsa-miR-4734, hsa-miR-6757-5p, hsa-miR-6756-5p, hsa-miR-3665, hsa-miR-6836-3p, hsa-miR-6821-5p, hsa-miR-6805-5p, hsa-miR-4728-5p, hsa-miR-6726-5p, hsa-miR-197-5p, hsa-miR-149-3p, hsa-miR-6850-5p, hsa-miR-4476, hsa-miR-6858-5p, hsa-miR-564, hsa-miR-4763-3p, hsa-miR-575, hsa-miR-6771-5p, hsa-miR-1231, hsa-miR-1908-3p, hsa-miR-150-3p, hsa-miR-3937, hsa-miR-887-3p, hsa-miR-3940-5p, hsa-miR-4741, hsa-miR-6808-5p, hsa-miR-6869-5p, hsa-miR-5090, hsa-miR-615-5p, hsa-miR-8072, hsa-miR-128-1-5p, hsa-miR-1238-5p, hsa-miR-365a-5p, hsa-miR-204-3p, hsa-miR-4492, hsa-miR-6785-5p, hsa-miR-6511a-5p, hsa-miR-4525, hsa-miR-1915-5p, hsa-miR-3180, hsa-miR-6879-5p, hsa-miR-1199-5p, hsa-miR-6746-5p, hsa-miR-711, hsa-miR-663b, hsa-miR-4707-3p, hsa-miR-6893-5p, hsa-miR-4675, hsa-miR-4638-5p, hsa-miR-4651, hsa-miR-6087, hsa-miR-4665-5p, hsa-miR-4758-5p, hsa-miR-6887-5p, hsa-miR-3620-5p, hsa-miR-1909-3p, hsa-miR-7641, hsa-miR-6724-5p, hsa-miR-1343-3p, hsa-miR-6780b-5p, hsa-miR-4484, hsa-miR-4690-5p, hsa-miR-4429, hsa-miR-1227-5p, hsa-miR-4725-3p, hsa-miR-6861-5p, hsa-miR-6812-5p, hsa-miR-3197, hsa-miR-8059, hsa-miR-3185, hsa-miR-4706, hsa-miR-4497, hsa-miR-3131, hsa-miR-6806-5p, hsa-miR-187-5p, hsa-miR-3180-3p, hsa-miR-6848-5p, hsa-miR-6820-5p, hsa-miR-6800-5p, hsa-miR-6717-5p, hsa-miR-6795-5p, hsa-miR-4632-5p, hsa-miR-665, hsa-miR-6778-5p, hsa-miR-3663-3p, hsa-miR-4689, hsa-miR-211-3p, hsa-miR-6511b-5p, hsa-miR-4750-5p, hsa-miR-6126, hsa-miR-614, hsa-miR-7110-5p, hsa-miR-744-5p, hsa-miR-6769a-5p, hsa-miR-4792, hsa-miR-5787, hsa-miR-6798-5p, hsa-miR-6781-5p, hsa-miR-4419b, hsa-miR-4446-3p, hsa-miR-4259, hsa-miR-5572, hsa-miR-6075, hsa-miR-296-3p, hsa-miR-6891-5p, hsa-miR-4745-5p, hsa-miR-6775-5p, hsa-miR-6870-5p, hsa-miR-920, hsa-miR-4530, hsa-miR-6819-5p, hsa-miR-6825-5p, hsa-miR-7847-3p, hsa-miR-6131, hsa-miR-4433-3p, hsa-miR-1228-5p, hsa-miR-6743-5p, hsa-miR-1268a, hsa-miR-3917, hsa-miR-6786-5p, hsa-miR-3154, hsa-miR-638, hsa-miR-6741-5p, hsa-miR-6889-5p, hsa-miR-6840-3p, hsa-miR-6510-5p, hsa-miR-3188, hsa-miR-551b-5p, hsa-miR-5001-5p, hsa-miR-1268b, hsa-miR-7107-5p, hsa-miR-6824-5p, hsa-miR-6732-5p, hsa-miR-371a-5p, hsa-miR-6794-5p, hsa-miR-6779-5p, hsa-miR-4271, hsa-miR-5195-3p, hsa-miR-6762-5p, hsa-miR-939-5p, hsa-miR-1247-3p, hsa-miR-6777-5p, hsa-miR-6722-3p, hsa-miR-

3656, hsa-miR-4688, hsa-miR-3195, hsa-miR-6766-5p, hsa-miR-4447, hsa-miR-4656, hsa-miR-7108-5p, hsa-miR-3191-3p, hsa-miR-1273g-3p, hsa-miR-4463, hsa-miR-2861, hsa-miR-3196, hsa-miR-6877-5p, hsa-miR-3679-5p, hsa-miR-4442, hsa-miR-6789-5p, hsa-miR-6782-5p, hsa-miR-486-3p, hsa-miR-6085, hsa-miR-4746-3p, hsa-miR-619-5p, hsa-miR-937-5p, hsa-miR-6803-5p, hsa-miR-4298, hsa-miR-4454, hsa-miR-4459, hsa-miR-7150, hsa-miR-6880-5p, hsa-miR-4449, hsa-miR-8063, hsa-miR-4695-5p, hsa-miR-6132, hsa-miR-6829-5p, hsa-miR-4486, hsa-miR-6805-3p, hsa-miR-6826-5p, hsa-miR-4508, hsa-miR-1343-5p, hsa-miR-7114-5p, hsa-miR-3622a-5p, hsa-miR-6765-5p, hsa-miR-7845-5p, hsa-miR-3960, hsa-miR-6749-5p, hsa-miR-1260b, hsa-miR-6799-5p, hsa-miR-4723-5p, hsa-miR-6784-5p, hsa-miR-5100, hsa-miR-6769b-5p, hsa-miR-1207-5p, hsa-miR-642a-3p, hsa-miR-4505, hsa-miR-4270, hsa-miR-6721-5p, hsa-miR-7111-5p, hsa-miR-6791-5p, hsa-miR-7109-5p, hsa-miR-4258, hsa-miR-6515-3p, hsa-miR-6851-5p, hsa-miR-6125, hsa-miR-4749-5p, hsa-miR-4726-5p, hsa-miR-4513, hsa-miR-6089, hsa-miR-6816-5p, hsa-miR-4466, hsa-miR-4488, hsa-miR-6752-5p and hsa-miR-4739 or a combination thereof, congeners thereof, transcripts thereof, or variants or derivatives thereof; and, optionally in combination therewith, hsa-miR-760, hsa-miR-602, hsa-miR-423-5p, hsa-miR-92a-2-5p, hsa-miR-16-5p, hsa-miR-451a, hsa-miR-135a-3p, hsa-miR-486-5p, hsa-miR-4257, hsa-miR-92b-5p, hsa-miR-1915-3p, hsa-miR-718, hsa-miR-940, hsa-miR-296-5p, hsa-miR-23b-3p and hsa-miR-92a-3p or a combination thereof, transcripts thereof, or variants or derivatives thereof; and, optionally in combination therewith, hsa-miR-658, hsa-miR-6842-5p, hsa-miR-6124, hsa-miR-6765-3p, hsa-miR-7106-5p, hsa-miR-4534, hsa-miR-92b-3p, hsa-miR-3135b, hsa-miR-4687-3p, hsa-miR-762, hsa-miR-3619-3p, hsa-miR-4467, hsa-miR-557, hsa-miR-1237-5p, hsa-miR-1908-5p, hsa-miR-4286, hsa-miR-6885-5p and hsa-miR-6763-5p or a combination thereof; congeners thereof; transcripts thereof; or variants or derivatives thereof.

The expression level of the target nucleic acids described above are increased or decreased (hereinafter, referred to as "increased/decreased") depending on the types of the target nucleic acids in subjects who have breast cancer, as compared with healthy subjects. Hence, the nucleic acid of the present invention can be effectively used for measuring expression levels of the target nucleic acids in body fluids derived from subjects (e.g., humans) suspected of having breast cancer and body fluids derived from healthy subjects and thereby detecting breast cancer through the comparison thereof.

The nucleic acid probe or the primer that can be used in the present invention is a nucleic acid probe capable of specifically binding to a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 1 to 235 and 851 to 856, or a primer for amplifying a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 1 to 235 and 851 to 856.

The nucleic acid probe or the primer that can be used in the present invention may further comprise a nucleic acid probe capable of specifically binding to a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 236 to 251, or a primer for amplifying a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 236 to 251.

The nucleic acid probe or the primer that can be used in the present invention may further comprise a nucleic acid probe capable of specifically binding to a polynucleotide

that consists of a nucleotide sequence represented by at least one of SEQ ID NOs: 252 to 269, or a primer for amplifying a polynucleotide that consists of a nucleotide sequence represented by at least one of SEQ ID NOs: 252 to 269.

Specifically, these nucleic acid probes or primers comprise a combination of one or more polynucleotides selected from: a group of polynucleotides that comprise nucleotide sequences represented by any of SEQ ID NOs: 1 to 871 or nucleotide sequences derived from the nucleotide sequences by the replacement of u with t, and a group of complementary polynucleotides thereof, a group of polynucleotides that respectively hybridize under stringent conditions (mentioned later) to DNAs consisting of nucleotide sequences complementary to these nucleotide sequences; and a group of complementary polynucleotides thereof, and a group of polynucleotides comprising 15 or more, preferably 17 or more consecutive nucleotides in the nucleotide sequences of these polynucleotides. These polynucleotides can be used as nucleic acid probes and primers for detecting the breast cancer markers as target nucleic acids.

More specifically, examples of the nucleic acid probes or the primers that can be used in the present invention include one or more polynucleotides selected from the group consisting of the following polynucleotides (a) to (e):

- (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (b) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t,
- (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

In addition to at least one polynucleotide selected from the group consisting of the polynucleotides (a) to (e), the nucleic acid probes or the primers that can be used in the present invention may further comprise a polynucleotide selected from the group consisting of the following polynucleotides (f) to (j):

- (f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251,
- (h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the

replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,

- (i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

In addition to at least one polynucleotide selected from the group consisting of (a) to (j) described above, the nucleic acid probes or the primers that can be used in the present invention may further comprise a polynucleotide selected from the group consisting of the following polynucleotides (k) to (o):

- (k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269,
- (m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).

For these polynucleotides, the “fragment thereof comprising 15 or more consecutive nucleotides” can comprise, but is not limited to, the number of nucleotides in the range from, for example, 15 consecutive nucleotides to less than the total number of nucleotides of the sequence, from 17 consecutive nucleotides to less than the total number of nucleotides of the sequence, or from 19 consecutive nucleotides to less than the total number of nucleotides of the sequence, in the nucleotide sequence of each polynucleotide.

These polynucleotides or fragments thereof used in the present invention may each be DNA or may each be RNA.

The polynucleotides that can be used in the present invention can each be prepared by use of a general technique such as a DNA recombination technique, a PCR method, or a method using an automatic DNA/RNA synthesizer.

The DNA recombination technique and the PCR method may employ techniques described in, for example, Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley & Sons, US (1993); and Sambrook et al., *Molecular Cloning—A Laboratory Manual*, Cold Spring Harbor Laboratory Press, US (1989).

The human-derived hsa-miR-4783-3p, hsa-miR-4730, hsa-miR-1307-3p, hsa-miR-4634, hsa-miR-663a, hsa-miR-4532, hsa-miR-7704, hsa-miR-3178, hsa-miR-6729-5p, hsa-miR-6090, hsa-miR-4732-5p, hsa-miR-3184-5p, hsa-miR-6727-5p, hsa-miR-6088, hsa-miR-4674, hsa-miR-8073, hsa-miR-4787-5p, hsa-miR-1469, hsa-miR-125a-3p, hsa-miR-1233-5p, hsa-miR-885-3p, hsa-miR-6802-5p, hsa-miR-328-5p, hsa-miR-6787-5p, hsa-miR-8069, hsa-miR-6875-5p,

hsa-miR-1246, hsa-miR-4734, hsa-miR-6757-5p, hsa-miR-6756-5p, hsa-miR-3665, hsa-miR-6836-3p, hsa-miR-6821-5p, hsa-miR-6805-5p, hsa-miR-4728-5p, hsa-miR-6726-5p, hsa-miR-197-5p, hsa-miR-149-3p, hsa-miR-6850-5p, hsa-miR-4476, hsa-miR-6858-5p, hsa-miR-564, hsa-miR-4763-3p, hsa-miR-575, hsa-miR-6771-5p, hsa-miR-1231, hsa-miR-1908-3p, hsa-miR-150-3p, hsa-miR-3937, hsa-miR-887-3p, hsa-miR-3940-5p, hsa-miR-4741, hsa-miR-6808-5p, hsa-miR-6869-5p, hsa-miR-5090, hsa-miR-615-5p, hsa-miR-8072, hsa-miR-128-1-5p, hsa-miR-1238-5p, hsa-miR-365a-5p, hsa-miR-204-3p, hsa-miR-4492, hsa-miR-6785-5p, hsa-miR-6511a-5p, hsa-miR-4525, hsa-miR-1915-5p, hsa-miR-3180, hsa-miR-6879-5p, hsa-miR-1199-5p, hsa-miR-6746-5p, hsa-miR-711, hsa-miR-663b, hsa-miR-4707-3p, hsa-miR-6893-5p, hsa-miR-4675, hsa-miR-4638-5p, hsa-miR-4651, hsa-miR-6087, hsa-miR-4665-5p, hsa-miR-4758-5p, hsa-miR-6887-5p, hsa-miR-3620-5p, hsa-miR-1909-3p, hsa-miR-7641, hsa-miR-6724-5p, hsa-miR-1343-3p, hsa-miR-6780b-5p, hsa-miR-4484, hsa-miR-4690-5p, hsa-miR-4429, hsa-miR-1227-5p, hsa-miR-4725-3p, hsa-miR-6861-5p, hsa-miR-6812-5p, hsa-miR-3197, hsa-miR-8059, hsa-miR-3185, hsa-miR-4706, hsa-miR-4497, hsa-miR-3131, hsa-miR-6806-5p, hsa-miR-187-5p, hsa-miR-3180-3p, hsa-miR-6848-5p, hsa-miR-6820-5p, hsa-miR-6800-5p, hsa-miR-6717-5p, hsa-miR-6795-5p, hsa-miR-4632-5p, hsa-miR-665, hsa-miR-6778-5p, hsa-miR-3663-3p, hsa-miR-4689, hsa-miR-211-3p, hsa-miR-6511b-5p, hsa-miR-4750-5p, hsa-miR-6126, hsa-miR-614, hsa-miR-7110-5p, hsa-miR-744-5p, hsa-miR-6769a-5p, hsa-miR-4792, hsa-miR-5787, hsa-miR-6798-5p, hsa-miR-6781-5p, hsa-miR-4419b, hsa-miR-4446-3p, hsa-miR-4259, hsa-miR-5572, hsa-miR-6075, hsa-miR-296-3p, hsa-miR-6891-5p, hsa-miR-4745-5p, hsa-miR-6775-5p, hsa-miR-6870-5p, hsa-miR-920, hsa-miR-4530, hsa-miR-6819-5p, hsa-miR-6825-5p, hsa-miR-7847-3p, hsa-miR-6131, hsa-miR-4433-3p, hsa-miR-1228-5p, hsa-miR-6743-5p, hsa-miR-1268a, hsa-miR-3917, hsa-miR-6786-5p, hsa-miR-3154, hsa-miR-638, hsa-miR-6741-5p, hsa-miR-6889-5p, hsa-miR-6840-3p, hsa-miR-6510-5p, hsa-miR-3188, hsa-miR-551b-5p, hsa-miR-5001-5p, hsa-miR-1268b, hsa-miR-7107-5p, hsa-miR-6824-5p, hsa-miR-6732-5p, hsa-miR-371a-5p, hsa-miR-6794-5p, hsa-miR-6779-5p, hsa-miR-4271, hsa-miR-5195-3p, hsa-miR-6762-5p, hsa-miR-939-5p, hsa-miR-1247-3p, hsa-miR-6777-5p, hsa-miR-6722-3p, hsa-miR-3656, hsa-miR-4688, hsa-miR-3195, hsa-miR-6766-5p, hsa-miR-4447, hsa-miR-4656, hsa-miR-7108-5p, hsa-miR-3191-3p, hsa-miR-1273g-3p, hsa-miR-4463, hsa-miR-2861, hsa-miR-3196, hsa-miR-6877-5p, hsa-miR-3679-5p, hsa-miR-4442, hsa-miR-6789-5p, hsa-miR-6782-5p, hsa-miR-486-3p, hsa-miR-6085, hsa-miR-4746-3p, hsa-miR-619-5p, hsa-miR-937-5p, hsa-miR-6803-5p, hsa-miR-4298, hsa-miR-4454, hsa-miR-4459, hsa-miR-7150, hsa-miR-6880-5p, hsa-miR-4449, hsa-miR-8063, hsa-miR-4695-5p, hsa-miR-6132, hsa-miR-6829-5p, hsa-miR-4486, hsa-miR-6805-3p, hsa-miR-6826-5p, hsa-miR-4508, hsa-miR-1343-5p, hsa-miR-7114-5p, hsa-miR-3622a-5p, hsa-miR-6765-5p, hsa-miR-7845-5p, hsa-miR-3960, hsa-miR-6749-5p, hsa-miR-1260b, hsa-miR-6799-5p, hsa-miR-4723-5p, hsa-miR-6784-5p, hsa-miR-5100, hsa-miR-6769b-5p, hsa-miR-1207-5p, hsa-miR-642a-3p, hsa-miR-4505, hsa-miR-4270, hsa-miR-6721-5p, hsa-miR-7111-5p, hsa-miR-6791-5p, hsa-miR-7109-5p, hsa-miR-4258, hsa-miR-6515-3p, hsa-miR-6851-5p, hsa-miR-6125, hsa-miR-4749-5p, hsa-miR-4726-5p, hsa-miR-4513, hsa-miR-6089, hsa-miR-6816-5p, hsa-miR-4466, hsa-miR-4488, hsa-miR-6752-5p, hsa-miR-4739, hsa-miR-760, hsa-miR-602, hsa-miR-423-5p, hsa-miR-92a-2-5p, hsa-miR-16-5p, hsa-miR-451a, hsa-miR-

135a-3p, hsa-miR-486-5p, hsa-miR-4257, hsa-miR-92b-5p, hsa-miR-1915-3p, hsa-miR-718, hsa-miR-940, hsa-miR-296-5p, hsa-miR-23b-3p, hsa-miR-92a-3p, hsa-miR-658, hsa-miR-6842-5p, hsa-miR-6124, hsa-miR-6765-3p, hsa-miR-7106-5p, hsa-miR-4534, hsa-miR-92b-3p, hsa-miR-3135b, hsa-miR-4687-3p, hsa-miR-762, hsa-miR-3619-3p, hsa-miR-4467, hsa-miR-557, hsa-miR-1237-5p, hsa-miR-1908-5p, hsa-miR-4286, hsa-miR-6885-5p and hsa-miR-6763-5p represented by SEQ ID NOs: 1 to 269, and 851 to 856 are known in the art, and their acquisition methods are also known as mentioned above. Therefore, each polynucleotide that can be used as a nucleic acid probe or a primer in the present invention can be prepared by cloning the gene.

Such nucleic acid probes or primers can be chemically synthesized using an automatic DNA synthesizer. In general, the phosphoramidite method is used in this synthesis, and single-stranded DNA up to approximately 100 nucleotides can be automatically synthesized by this method. The automatic DNA synthesizer is commercially available from, for example, Polygen GmbH, ABI, or Applied Biosystems, Inc.

Alternatively, the polynucleotides of the present invention can also be prepared by a cDNA cloning methods. The cDNA cloning technique may employ, for example, microRNA Cloning Kit Wako.

In this context, the sequences of the nucleic acid probes and the primers for detecting the polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 269, and 851 to 856 do not exist as miRNAs or precursors thereof in the living body or in vivo. For example, the nucleotide sequences represented by SEQ ID NO: 34 and SEQ ID NO: 205 are produced from the precursor represented by SEQ ID NO: 304. This precursor has a hairpin-like structure as shown in FIG. 1, and the nucleotide sequences represented by SEQ ID NO: 34 and SEQ ID NO: 205 have mismatch sequences with each other. As such, a nucleotide sequence completely complementary to the nucleotide sequence represented by SEQ ID NO: 34 or SEQ ID NO: 205 is not naturally produced in vivo. Likewise, the nucleic acid probe and the primer for detecting the nucleotide sequence represented by any of SEQ ID NOs: 1 to 269, and 851 to 856 have artificial nucleotide sequences that do not exist in the living body or in vivo.

3. Kit or Device for Detection of Breast Cancer

The present invention also provides a kit or a device for the detection of breast cancer, comprising one or more polynucleotides (which may include a variant, a fragment, or a derivative thereof; hereinafter, also referred to as polynucleotide for detection) that can be used as nucleic acid probes or primers in the present invention for measuring target nucleic acids as breast cancer markers.

The target nucleic acid as a breast cancer marker according to the present invention is selected from the following group 1:

miR-4783-3p, miR-4730, miR-1307-3p, miR-4634, miR-663a, miR-4532, miR-7704, miR-3178, miR-6729-5p, miR-6090, miR-4732-5p, miR-3184-5p, miR-6727-5p, miR-6088, miR-4674, miR-8073, miR-4787-5p, miR-1469, miR-125a-3p, miR-1233-5p, miR-885-3p, miR-6802-5p, miR-328-5p, miR-6787-5p, miR-8069, miR-6875-5p, miR-1246, miR-4734, miR-6757-5p, miR-6756-5p, miR-3665, miR-6836-3p, miR-6821-5p, miR-6805-5p, miR-4728-5p, miR-6726-5p, miR-197-5p, miR-149-3p, miR-6850-5p, miR-4476, miR-6858-5p, miR-564, miR-4763-3p, miR-575, miR-6771-5p, miR-1231, miR-1908-3p, miR-150-3p, miR-3937, miR-887-3p, miR-3940-5p, miR-4741, miR-6808-5p, miR-6869-5p, miR-5090, miR-615-5p, miR-8072,

miR-128-1-5p, miR-1238-5p, miR-365a-5p, miR-204-3p, miR-4492, miR-6785-5p, miR-6511a-5p, miR-4525, miR-1915-5p, miR-3180, miR-6879-5p, miR-1199-5p, miR-6746-5p, miR-711, miR-663b, miR-4707-3p, miR-6893-5p, miR-4675, miR-4638-5p, miR-4651, miR-6087, miR-4665-5p, miR-4758-5p, miR-6887-5p, miR-3620-5p, miR-1909-3p, miR-7641, miR-6724-5p, miR-1343-3p, miR-6780b-5p, miR-4484, miR-4690-5p, miR-4429, miR-1227-5p, miR-4725-3p, miR-6861-5p, miR-6812-5p, miR-3197, miR-8059, miR-3185, miR-4706, miR-4497, miR-3131, miR-6806-5p, miR-187-5p, miR-3180-3p, miR-6848-5p, miR-6820-5p, miR-6800-5p, miR-6717-5p, miR-6795-5p, miR-4632-5p, miR-665, miR-6778-5p, miR-3663-3p, miR-4689, miR-211-3p, miR-6511b-5p, miR-4750-5p, miR-6126, miR-614, miR-7110-5p, miR-744-5p, miR-6769a-5p, miR-4792, miR-5787, miR-6798-5p, miR-6781-5p, miR-4419b, miR-4446-3p, miR-4259, miR-5572, miR-6075, miR-296-3p, miR-6891-5p, miR-4745-5p, miR-6775-5p, miR-6870-5p, miR-920, miR-4530, miR-6819-5p, miR-6825-5p, miR-7847-3p, miR-6131, miR-4433-3p, miR-1228-5p, miR-6743-5p, miR-1268a, miR-3917, miR-6786-5p, miR-3154, miR-638, miR-6741-5p, miR-6889-5p, miR-6840-3p, miR-6510-5p, miR-3188, miR-551b-5p, miR-5001-5p, miR-1268b, miR-7107-5p, miR-6824-5p, miR-6732-5p, miR-371a-5p, miR-6794-5p, miR-6779-5p, miR-4271, miR-5195-3p, miR-6762-5p, miR-939-5p, miR-1247-3p, miR-6777-5p, miR-6722-3p, miR-3656, miR-4688, miR-3195, miR-6766-5p, miR-4447, miR-4656, miR-7108-5p, miR-3191-3p, miR-1273g-3p, miR-4463, miR-2861, miR-3196, miR-6877-5p, miR-3679-5p, miR-4442, miR-6789-5p, miR-6782-5p, miR-486-3p, miR-6085, miR-4746-3p, miR-619-5p, miR-937-5p, miR-6803-5p, miR-4298, miR-4454, miR-4459, miR-7150, miR-6880-5p, miR-4449, miR-8063, miR-4695-5p, miR-6132, miR-6829-5p, miR-4486, miR-6805-3p, miR-6826-5p, miR-4508, miR-1343-5p, miR-7114-5p, miR-3622a-5p, miR-6765-5p, miR-7845-5p, miR-3960, miR-6749-5p, miR-1260b, miR-6799-5p, miR-4723-5p, miR-6784-5p, miR-5100, miR-6769b-5p, miR-1207-5p, miR-642a-3p, miR-4505, miR-4270, miR-6721-5p, miR-7111-5p, miR-6791-5p, miR-7109-5p, miR-4258, miR-6515-3p, miR-6851-5p, miR-6125, miR-4749-5p, miR-4726-5p, miR-4513, miR-6089, miR-6816-5p, miR-4466, miR-4488, miR-6752-5p and miR-4739.

An additional target nucleic acid(s) that may be optionally used in the measurement is selected from the following group 2:

miR-760, miR-602, miR-423-5p, miR-92a-2-5p, miR-16-5p, miR-451a, miR-135a-3p, miR-486-5p, miR-4257, miR-92b-5p, miR-1915-3p, miR-718, miR-940, miR-296-5p, miR-23b-3p and miR-92a-3p.

An additional target nucleic acid(s) that may be optionally used in the measurement is further selected from the following group 3:

miR-658, miR-6842-5p, miR-6124, miR-6765-3p, miR-7106-5p, miR-4534, miR-92b-3p, miR-3135b, miR-4687-3p, miR-762, miR-3619-3p, miR-4467, miR-557, miR-1237-5p, miR-1908-5p, miR-4286, miR-6885-5p and miR-6763-5p.

The kit or the device of the present invention comprises nucleic acid(s) capable of specifically binding to any of the target nucleic acids as the breast cancer markers described above, preferably one or more polynucleotides selected from the nucleic acid probes or the primers described in the

preceding Section 2, specifically, the polynucleotides described in the preceding Section 2, or variants thereof, etc.

Specifically, the kit or the device of the present invention can comprise at least one polynucleotides comprising (or consisting of) a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, polynucleotide(s) comprising (or consisting of) a complementary sequence thereof, polynucleotide(s) hybridizing under stringent conditions to any of these polynucleotides, or variant(s) or fragment(s) comprising 15 or more consecutive nucleotides of any of these polynucleotide sequences.

The kit or the device of the present invention can further comprise one or more polynucleotides comprising (or consisting of) a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, polynucleotide(s) comprising (or consisting of) a complementary sequence thereof, polynucleotide(s) hybridizing under stringent conditions to any of these polynucleotides, variant(s) or fragment(s) comprising 15 or more consecutive nucleotides of any of these polynucleotide sequences.

The kit or the device of the present invention can further comprise one or more polynucleotide(s) comprising (or consisting of) a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, polynucleotide(s) comprising (or consisting of) a complementary sequence thereof, polynucleotide(s) hybridizing under stringent conditions to any of these polynucleotides, variant(s) or fragment(s) comprising 15 or more consecutive nucleotides of any of these polynucleotide sequences.

The fragment(s) that can be comprised in the kit or the device of the present invention is/are, for example, one or more polynucleotides, preferably two or more polynucleotides selected from the group consisting of the following polynucleotides (1) to (3):

- (1) a polynucleotide comprising 15 or more consecutive nucleotides in a nucleotide sequence that is derived from a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856 by the replacement of u with t, or a complementary sequence thereof;
- (2) a polynucleotide comprising 15 or more consecutive nucleotides in a nucleotide sequence that is derived from a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 by the replacement of u with t, or a complementary sequence thereof; and
- (3) a polynucleotide comprising 15 or more consecutive nucleotides in a nucleotide sequence that is derived from a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 by the replacement of u with t, or a complementary sequence thereof.

In a preferred embodiment, the polynucleotide is a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a polynucleotide consisting of a complementary sequence thereof, a polynucleotide hybridizing under stringent conditions to any of these polynucleotides, or a variant thereof comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

In a preferred embodiment, the polynucleotide is a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of

u with t, a polynucleotide consisting of a complementary sequence thereof, a polynucleotide hybridizing under stringent conditions to any of these polynucleotides, or a variant thereof comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

In a preferred embodiment, the polynucleotide is a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a polynucleotide consisting of a complementary sequence thereof, a polynucleotide hybridizing under stringent conditions to any of these polynucleotides, or a variant thereof comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

In a preferred embodiment, the fragment can be a polynucleotide comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

In the present invention, the size of the polynucleotide fragment is the number of nucleotides in the range from, for example, 15 consecutive nucleotides to less than the total number of nucleotides of the sequence, from 17 consecutive nucleotides to less than the total number of nucleotides of the sequence, or from 19 consecutive nucleotides to less than the total number of nucleotides of the sequence, in the nucleotide sequence of each polynucleotide.

Specific examples of the aforementioned polynucleotide combination that constitutes the kit or the device of the present invention can include combinations of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 269 and 851 to 856 shown in Table 1 described above. However, these are given merely for illustrative purposes, and all of various other possible combinations are included in the present invention.

The aforementioned combination constituting the kit or the device for discriminating a breast cancer patient from a healthy subject according to the present invention is desirably, for example, a combination of two or more polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs shown in Table 1. Usually, a combination of two of these polynucleotides can produce adequate performance. Specifically, any two of the aforementioned polynucleotides that consist of the aforementioned nucleotide sequences represented by SEQ ID NOs: 1 to 269 and 851 to 856 may be combined. For such a combination, it is preferred to select at least one of the newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 235 and 851 to 856. More specifically, the combination is more preferably a combination comprising at least one of polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 20, 24, 26, 27, 30, 33, 182, 194, 206, and 208, among the combinations of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 269 and 851 to 856.

The combination of polynucleotides with cancer type specificity capable of discriminating a breast cancer patient not only from a healthy subject but also from other cancer patients is preferably, for example, a combination of multiple polynucleotides comprising at least one polynucleotide selected from the group consisting of polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 41, 43, 45, 46, 47, 49, 50, 51, 52, 53, 54, 55, 58, 59, 60, 62, 63, 64, 65, 67, 68, 69, 70, 71, 72, 73, 75, 77, 79, 80, 81, 82, 83, 86, 88, 89, 90, 92, 93, 94, 96, 98, 99, 100, 103, 104, 106, 107, 108, 110, 111, 113, 114, 115, 116, 118, 119,

121, 122, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 135, 136, 139, 140, 143, 145, 146, 147, 149, 150, 155, 157, 160, 161, 165, 167, 171, 173, 174, 175, 177, 178, 181, 182, 186, 190, 193, 194, 199, 204, 205, 206, 208, 211, 218, 225, 232, 236, 237, 238, 239, 242, 243, 244, 246, 247, 252, 260, 265, 266, 851, 852, 853, 854, 855, 856 (hereinafter, this group is referred to as "cancer type-specific polynucleotide group 1"), with any of the polynucleotides consisting of nucleotide sequences represented by the other SEQ ID NOs.

The combination of polynucleotides with cancer type specificity capable of discriminating a breast cancer patient not only from a healthy subject but also from other cancer patients is more preferably a combination of multiple polynucleotides, any of which are selected from the cancer type-specific polynucleotide group 1 described above.

The combination of polynucleotides with cancer type specificity capable of discriminating a breast cancer patient not only from a healthy subject but also from other cancer patients is more preferably a combination comprising at least one polynucleotide selected from, particularly, the group consisting of polynucleotides of SEQ ID NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 41, 43, 45, 46, 47, 49, 50, 51, 52, 54, 55, 58, 59, 60, 62, 63, 64, 65, 67, 68, 69, 71, 72, 73, 75, 77, 79, 80, 82, 83, 86, 88, 92, 93, 96, 99, 103, 104, 106, 110, 111, 114, 116, 118, 119, 122, 124, 125, 127, 130, 132, 133, 135, 139, 143, 145, 147, 149, 157, 160, 173, 177, 181, 182, 186, 211, 218, 232, 236, 237, 238, 239, 242, 243, 246, 247, 260, 266, 851, 852, 853, 854 (hereinafter, this group is referred to as "cancer type-specific polynucleotide group 2"), among the combinations of multiple polynucleotides selected from the cancer type-specific polynucleotide group 1 described above.

The number of the polynucleotides with cancer type specificity in the combination described above can be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more and is preferably 2 or more for the combination.

Non-limiting examples of the combination of two polynucleotides that consist of one polynucleotide that consists of a nucleotide sequence selected from the cancer type-specific polynucleotide group 2 or a complementary sequence thereof and one polynucleotide that consists of a nucleotide sequence selected from the cancer type-specific polynucleotide group 1 or a complementary sequence thereof are listed below.

(1-1) a combination of SEQ ID NOs: 2 and 1 (markers: hsa-miR-4730 and hsa-miR-4783-3p);

(1-2) a combination of SEQ ID NOs: 2 and 237 (markers: hsa-miR-4730 and hsa-miR-602);

(1-3) a combination of SEQ ID NOs: 2 and 4 (markers: hsa-miR-4730 and hsa-miR-4634);

(1-4) a combination of SEQ ID NOs: 2 and 3 (markers: hsa-miR-4730 and hsa-miR-1307-3p);

(1-5) a combination of SEQ ID NOs: 2 and 51 (markers: hsa-miR-4730 and hsa-miR-3940-5p);

(2-1) a combination of SEQ ID NOs: 1 and 237 (markers: hsa-miR-4783-3p and hsa-miR-602);

(2-2) a combination of SEQ ID NOs: 1 and 4 (markers: hsa-miR-4783-3p and hsa-miR-4634);

(2-3) a combination of SEQ ID NOs: 1 and 3 (markers: hsa-miR-4783-3p and hsa-miR-1307-3p);

(2-4) a combination of SEQ ID NOs: 1 and 51 (markers: hsa-miR-4783-3p and hsa-miR-3940-5p);

(2-5) a combination of SEQ ID NOs: 1 and 6 (markers: hsa-miR-4783-3p and hsa-miR-4532);

(3-1) a combination of SEQ ID NOs: 4 and 237 (markers: hsa-miR-4634 and hsa-miR-602);

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- (3-2) a combination of SEQ ID NOs: 3 and 237 (markers: hsa-miR-1307-3p and hsa-miR-602);
- (3-3) a combination of SEQ ID NOs: 51 and 237 (markers: hsa-miR-3940-5p and hsa-miR-602);
- (3-4) a combination of SEQ ID NOs: 237 and 6 (markers: hsa-miR-602 and hsa-miR-4532); 5
- (3-5) a combination of SEQ ID NOs: 237 and 12 (markers: hsa-miR-602 and hsa-miR-3184-5p);
- (4-1) a combination of SEQ ID NOs: 3 and 4 (markers: hsa-miR-1307-3p and hsa-miR-4634); 10
- (4-2) a combination of SEQ ID NOs: 4 and 51 (markers: hsa-miR-4634 and hsa-miR-3940-5p);
- (4-3) a combination of SEQ ID NOs: 4 and 6 (markers: hsa-miR-4634 and hsa-miR-4532); 15
- (4-4) a combination of SEQ ID NOs: 4 and 12 (markers: hsa-miR-4634 and hsa-miR-3184-5p);
- (4-5) a combination of SEQ ID NOs: 4 and 15 (markers: hsa-miR-4634 and hsa-miR-4674);
- (5-1) a combination of SEQ ID NOs: 3 and 51 (markers: hsa-miR-1307-3p and hsa-miR-3940-5p); 20
- (5-2) a combination of SEQ ID NOs: 3 and 6 (markers: hsa-miR-1307-3p and hsa-miR-4532);
- (5-3) a combination of SEQ ID NOs: 3 and 12 (markers: hsa-miR-1307-3p and hsa-miR-3184-5p); 25
- (5-4) a combination of SEQ ID NOs: 3 and 15 (markers: hsa-miR-1307-3p and hsa-miR-4674);
- (5-5) a combination of SEQ ID NOs: 3 and 8 (markers: hsa-miR-1307-3p and hsa-miR-3178);
- (6-1) a combination of SEQ ID NOs: 51 and 6 (markers: hsa-miR-3940-5p and hsa-miR-4532); 30
- (6-2) a combination of SEQ ID NOs: 51 and 12 (markers: hsa-miR-3940-5p and hsa-miR-3184-5p);
- (6-3) a combination of SEQ ID NOs: 51 and 15 (markers: hsa-miR-3940-5p and hsa-miR-4674); 35
- (6-4) a combination of SEQ ID NOs: 51 and 8 (markers: hsa-miR-3940-5p and hsa-miR-3178);
- (6-5) a combination of SEQ ID NOs: 51 and 34 (markers: hsa-miR-3940-5p and hsa-miR-6805-5p);
- (7-1) a combination of SEQ ID NOs: 2 and 6 (markers: hsa-miR-4730 and hsa-miR-4532); 40
- (7-2) a combination of SEQ ID NOs: 12 and 6 (markers: hsa-miR-3184-5p and hsa-miR-4532);
- (7-3) a combination of SEQ ID NOs: 15 and 6 (markers: hsa-miR-4674 and hsa-miR-4532); 45
- (7-4) a combination of SEQ ID NOs: 8 and 6 (markers: hsa-miR-3178 and hsa-miR-4532);
- (7-5) a combination of SEQ ID NOs: 6 and 34 (markers: hsa-miR-4532 and hsa-miR-6805-5p);
- (8-1) a combination of SEQ ID NOs: 2 and 12 (markers: hsa-miR-4730 and hsa-miR-3184-5p); 50
- (8-2) a combination of SEQ ID NOs: 1 and 12 (markers: hsa-miR-4783-3p and hsa-miR-3184-5p);
- (8-3) a combination of SEQ ID NOs: 12 and 15 (markers: hsa-miR-3184-5p and hsa-miR-4674); 55
- (8-4) a combination of SEQ ID NOs: 8 and 12 (markers: hsa-miR-3178 and hsa-miR-3184-5p);
- (8-5) a combination of SEQ ID NOs: 12 and 34 (markers: hsa-miR-3184-5p and hsa-miR-6805-5p);
- (9-1) a combination of SEQ ID NOs: 2 and 15 (markers: hsa-miR-4730 and hsa-miR-4674); 60
- (9-2) a combination of SEQ ID NOs: 1 and 15 (markers: hsa-miR-4783-3p and hsa-miR-4674);
- (9-3) a combination of SEQ ID NOs: 237 and 15 (markers: hsa-miR-602 and hsa-miR-4674); 65
- (9-4) a combination of SEQ ID NOs: 8 and 15 (markers: hsa-miR-3178 and hsa-miR-4674);

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- (9-5) a combination of SEQ ID NOs: 15 and 34 (markers: hsa-miR-4674 and hsa-miR-6805-5p);
- (10-1) a combination of SEQ ID NOs: 2 and 8 (markers: hsa-miR-4730 and hsa-miR-3178);
- (10-2) a combination of SEQ ID NOs: 1 and 8 (markers: hsa-miR-4783-3p and hsa-miR-3178);
- (10-3) a combination of SEQ ID NOs: 237 and 8 (markers: hsa-miR-602 and hsa-miR-3178);
- (10-4) a combination of SEQ ID NOs: 4 and 8 (markers: hsa-miR-4634 and hsa-miR-3178);
- (10-5) a combination of SEQ ID NOs: 8 and 34 (markers: hsa-miR-3178 and hsa-miR-6805-5p);
- (11-1) a combination of SEQ ID NOs: 2 and 34 (markers: hsa-miR-4730 and hsa-miR-6805-5p);
- (11-2) a combination of SEQ ID NOs: 1 and 34 (markers: hsa-miR-4783-3p and hsa-miR-6805-5p);
- (11-3) a combination of SEQ ID NOs: 237 and 34 (markers: hsa-miR-602 and hsa-miR-6805-5p);
- (11-4) a combination of SEQ ID NOs: 4 and 34 (markers: hsa-miR-4634 and hsa-miR-6805-5p);
- (11-5) a combination of SEQ ID NOs: 3 and 34 (markers: hsa-miR-1307-3p and hsa-miR-6805-5p);
- (12-1) a combination of SEQ ID NOs: 2 and 9 (markers: hsa-miR-4730 and hsa-miR-6729-5p);
- (12-2) a combination of SEQ ID NOs: 1 and 9 (markers: hsa-miR-4783-3p and hsa-miR-6729-5p);
- (12-3) a combination of SEQ ID NOs: 9 and 237 (markers: hsa-miR-6729-5p and hsa-miR-602);
- (12-4) a combination of SEQ ID NOs: 4 and 9 (markers: hsa-miR-4634 and hsa-miR-6729-5p);
- (12-5) a combination of SEQ ID NOs: 3 and 9 (markers: hsa-miR-1307-3p and hsa-miR-6729-5p);
- (13-1) a combination of SEQ ID NOs: 2 and 143 (markers: hsa-miR-4730 and hsa-miR-1228-5p);
- (13-2) a combination of SEQ ID NOs: 1 and 143 (markers: hsa-miR-4783-3p and hsa-miR-1228-5p);
- (13-3) a combination of SEQ ID NOs: 237 and 143 (markers: hsa-miR-602 and hsa-miR-1228-5p);
- (13-4) a combination of SEQ ID NOs: 4 and 143 (markers: hsa-miR-4634 and hsa-miR-1228-5p);
- (13-5) a combination of SEQ ID NOs: 3 and 143 (markers: hsa-miR-1307-3p and hsa-miR-1228-5p);
- (14-1) a combination of SEQ ID NOs: 2 and 13 (markers: hsa-miR-4730 and hsa-miR-6727-5p);
- (14-2) a combination of SEQ ID NOs: 1 and 13 (markers: hsa-miR-4783-3p and hsa-miR-6727-5p);
- (14-3) a combination of SEQ ID NOs: 237 and 13 (markers: hsa-miR-602 and hsa-miR-6727-5p);
- (14-4) a combination of SEQ ID NOs: 4 and 13 (markers: hsa-miR-4634 and hsa-miR-6727-5p);
- (14-5) a combination of SEQ ID NOs: 3 and 13 (markers: hsa-miR-1307-3p and hsa-miR-6727-5p);
- (15-1) a combination of SEQ ID NOs: 2 and 125 (markers: hsa-miR-4730 and hsa-miR-6781-5p);
- (15-2) a combination of SEQ ID NOs: 1 and 125 (markers: hsa-miR-4783-3p and hsa-miR-6781-5p);
- (15-3) a combination of SEQ ID NOs: 237 and 125 (markers: hsa-miR-602 and hsa-miR-6781-5p);
- (15-4) a combination of SEQ ID NOs: 4 and 125 (markers: hsa-miR-4634 and hsa-miR-6781-5p);
- (15-5) a combination of SEQ ID NOs: 3 and 125 (markers: hsa-miR-1307-3p and hsa-miR-6781-5p);
- (16-1) a combination of SEQ ID NOs: 2 and 236 (markers: hsa-miR-4730 and hsa-miR-760);
- (16-2) a combination of SEQ ID NOs: 1 and 236 (markers: hsa-miR-4783-3p and hsa-miR-760);

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(16-3) a combination of SEQ ID NOs: 237 and 236 (markers: hsa-miR-602 and hsa-miR-760);
 (16-4) a combination of SEQ ID NOs: 4 and 236 (markers: hsa-miR-4634 and hsa-miR-760);
 (16-5) a combination of SEQ ID NOs: 3 and 236 (markers: hsa-miR-1307-3p and hsa-miR-760);
 (17-1) a combination of SEQ ID NOs: 2 and 46 (markers: hsa-miR-4730 and hsa-miR-1231);
 (17-2) a combination of SEQ ID NOs: 1 and 46 (markers: hsa-miR-4783-3p and hsa-miR-1231);
 (17-3) a combination of SEQ ID NOs: 237 and 46 (markers: hsa-miR-602 and hsa-miR-1231);
 (17-4) a combination of SEQ ID NOs: 4 and 46 (markers: hsa-miR-4634 and hsa-miR-1231);
 (17-5) a combination of SEQ ID NOs: 3 and 46 (markers: hsa-miR-1307-3p and hsa-miR-1231);
 (18-1) a combination of SEQ ID NOs: 2 and 32 (markers: hsa-miR-4730 and hsa-miR-6836-3p);
 (18-2) a combination of SEQ ID NOs: 1 and 32 (markers: hsa-miR-4783-3p and hsa-miR-6836-3p);
 (18-3) a combination of SEQ ID NOs: 237 and 32 (markers: hsa-miR-602 and hsa-miR-6836-3p);
 (18-4) a combination of SEQ ID NOs: 4 and 32 (markers: hsa-miR-4634 and hsa-miR-6836-3p);
 (18-5) a combination of SEQ ID NOs: 3 and 32 (markers: hsa-miR-1307-3p and hsa-miR-6836-3p);
 (19-1) a combination of SEQ ID NOs: 2 and 62 (markers: hsa-miR-4730 and hsa-miR-4492);
 (19-2) a combination of SEQ ID NOs: 1 and 62 (markers: hsa-miR-4783-3p and hsa-miR-4492);
 (19-3) a combination of SEQ ID NOs: 237 and 62 (markers: hsa-miR-602 and hsa-miR-4492);
 (19-4) a combination of SEQ ID NOs: 4 and 62 (markers: hsa-miR-4634 and hsa-miR-4492);
 (19-5) a combination of SEQ ID NOs: 3 and 62 (markers: hsa-miR-1307-3p and hsa-miR-4492);
 (20-1) a combination of SEQ ID NOs: 2 and 88 (markers: hsa-miR-4730 and hsa-miR-4484);
 (20-2) a combination of SEQ ID NOs: 1 and 88 (markers: hsa-miR-4783-3p and hsa-miR-4484);
 (20-3) a combination of SEQ ID NOs: 237 and 88 (markers: hsa-miR-602 and hsa-miR-4484);
 (20-4) a combination of SEQ ID NOs: 4 and 88 (markers: hsa-miR-4634 and hsa-miR-4484);
 (20-5) a combination of SEQ ID NOs: 3 and 88 (markers: hsa-miR-1307-3p and hsa-miR-4484);
 (21-1) a combination of SEQ ID NOs: 2 and 52 (markers: hsa-miR-4730 and hsa-miR-4741);
 (21-2) a combination of SEQ ID NOs: 1 and 52 (markers: hsa-miR-4783-3p and hsa-miR-4741);
 (21-3) a combination of SEQ ID NOs: 237 and 52 (markers: hsa-miR-602 and hsa-miR-4741);
 (21-4) a combination of SEQ ID NOs: 4 and 52 (markers: hsa-miR-4634 and hsa-miR-4741);
 (21-5) a combination of SEQ ID NOs: 3 and 52 (markers: hsa-miR-1307-3p and hsa-miR-4741);
 (22-1) a combination of SEQ ID NOs: 2 and 7 (markers: hsa-miR-4730 and hsa-miR-7704);
 (22-2) a combination of SEQ ID NOs: 1 and 7 (markers: hsa-miR-4783-3p and hsa-miR-7704);
 (22-3) a combination of SEQ ID NOs: 237 and 7 (markers: hsa-miR-602 and hsa-miR-7704);
 (22-4) a combination of SEQ ID NOs: 4 and 7 (markers: hsa-miR-4634 and hsa-miR-7704);
 (22-5) a combination of SEQ ID NOs: 3 and 7 (markers: hsa-miR-1307-3p and hsa-miR-7704);

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(23-1) a combination of SEQ ID NOs: 2 and 26 (markers: hsa-miR-4730 and hsa-miR-6875-5p);
 (23-2) a combination of SEQ ID NOs: 1 and 26 (markers: hsa-miR-4783-3p and hsa-miR-6875-5p);
 (23-3) a combination of SEQ ID NOs: 237 and 26 (markers: hsa-miR-602 and hsa-miR-6875-5p);
 (23-4) a combination of SEQ ID NOs: 4 and 26 (markers: hsa-miR-4634 and hsa-miR-6875-5p);
 (23-5) a combination of SEQ ID NOs: 3 and 26 (markers: hsa-miR-1307-3p and hsa-miR-6875-5p);
 (24-1) a combination of SEQ ID NOs: 2 and 25 (markers: hsa-miR-4730 and hsa-miR-8069);
 (24-2) a combination of SEQ ID NOs: 1 and 25 (markers: hsa-miR-4783-3p and hsa-miR-8069);
 (24-3) a combination of SEQ ID NOs: 237 and 25 (markers: hsa-miR-602 and hsa-miR-8069);
 (24-4) a combination of SEQ ID NOs: 4 and 25 (markers: hsa-miR-4634 and hsa-miR-8069);
 (24-5) a combination of SEQ ID NOs: 3 and 25 (markers: hsa-miR-1307-3p and hsa-miR-8069);
 (25-1) a combination of SEQ ID NOs: 2 and 54 (markers: hsa-miR-4730 and hsa-miR-6869-5p);
 (25-2) a combination of SEQ ID NOs: 1 and 54 (markers: hsa-miR-4783-3p and hsa-miR-6869-5p);
 (25-3) a combination of SEQ ID NOs: 237 and 54 (markers: hsa-miR-602 and hsa-miR-6869-5p);
 (25-4) a combination of SEQ ID NOs: 4 and 54 (markers: hsa-miR-4634 and hsa-miR-6869-5p);
 (25-5) a combination of SEQ ID NOs: 3 and 54 (markers: hsa-miR-1307-3p and hsa-miR-6869-5p);
 (26-1) a combination of SEQ ID NOs: 2 and 92 (markers: hsa-miR-4730 and hsa-miR-4725-3p);
 (26-2) a combination of SEQ ID NOs: 1 and 92 (markers: hsa-miR-4783-3p and hsa-miR-4725-3p);
 (26-3) a combination of SEQ ID NOs: 237 and 92 (markers: hsa-miR-602 and hsa-miR-4725-3p);
 (26-4) a combination of SEQ ID NOs: 4 and 92 (markers: hsa-miR-4634 and hsa-miR-4725-3p);
 (26-5) a combination of SEQ ID NOs: 3 and 92 (markers: hsa-miR-1307-3p and hsa-miR-4725-3p);
 (27-1) a combination of SEQ ID NOs: 2 and 14 (markers: hsa-miR-4730 and hsa-miR-6088);
 (27-2) a combination of SEQ ID NOs: 1 and 14 (markers: hsa-miR-4783-3p and hsa-miR-6088);
 (27-3) a combination of SEQ ID NOs: 237 and 14 (markers: hsa-miR-602 and hsa-miR-6088);
 (27-4) a combination of SEQ ID NOs: 4 and 14 (markers: hsa-miR-4634 and hsa-miR-6088);
 (27-5) a combination of SEQ ID NOs: 3 and 14 (markers: hsa-miR-1307-3p and hsa-miR-6088);
 (28-1) a combination of SEQ ID NOs: 2 and 242 (markers: hsa-miR-4730 and hsa-miR-135a-3p);
 (28-2) a combination of SEQ ID NOs: 1 and 242 (markers: hsa-miR-4783-3p and hsa-miR-135a-3p);
 (28-3) a combination of SEQ ID NOs: 237 and 242 (markers: hsa-miR-602 and hsa-miR-135a-3p);
 (28-4) a combination of SEQ ID NOs: 4 and 242 (markers: hsa-miR-4634 and hsa-miR-135a-3p);
 (28-5) a combination of SEQ ID NOs: 3 and 242 (markers: hsa-miR-1307-3p and hsa-miR-135a-3p);
 (29-1) a combination of SEQ ID NOs: 2 and 47 (markers: hsa-miR-4730 and hsa-miR-1908-3p);
 (29-2) a combination of SEQ ID NOs: 1 and 47 (markers: hsa-miR-4783-3p and hsa-miR-1908-3p);
 (29-3) a combination of SEQ ID NOs: 237 and 47 (markers: hsa-miR-602 and hsa-miR-1908-3p);

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(29-4) a combination of SEQ ID NOs: 4 and 47 (markers: hsa-miR-4634 and hsa-miR-1908-3p);
 (29-5) a combination of SEQ ID NOs: 3 and 47 (markers: hsa-miR-1307-3p and hsa-miR-1908-3p);
 (30-1) a combination of SEQ ID NOs: 2 and 45 (markers: hsa-miR-4730 and hsa-miR-6771-5p);
 (30-2) a combination of SEQ ID NOs: 1 and 45 (markers: hsa-miR-4783-3p and hsa-miR-6771-5p);
 (30-3) a combination of SEQ ID NOs: 237 and 45 (markers: hsa-miR-602 and hsa-miR-6771-5p);
 (30-4) a combination of SEQ ID NOs: 4 and 45 (markers: hsa-miR-4634 and hsa-miR-6771-5p);
 (30-5) a combination of SEQ ID NOs: 3 and 45 (markers: hsa-miR-1307-3p and hsa-miR-6771-5p);
 (31-1) a combination of SEQ ID NOs: 2 and 39 (markers: hsa-miR-4730 and hsa-miR-6850-5p);
 (31-2) a combination of SEQ ID NOs: 1 and 39 (markers: hsa-miR-4783-3p and hsa-miR-6850-5p);
 (31-3) a combination of SEQ ID NOs: 237 and 39 (markers: hsa-miR-602 and hsa-miR-6850-5p);
 (31-4) a combination of SEQ ID NOs: 4 and 39 (markers: hsa-miR-4634 and hsa-miR-6850-5p);
 (31-5) a combination of SEQ ID NOs: 3 and 39 (markers: hsa-miR-1307-3p and hsa-miR-6850-5p);
 (32-1) a combination of SEQ ID NOs: 2 and 21 (markers: hsa-miR-4730 and hsa-miR-885-3p);
 (32-2) a combination of SEQ ID NOs: 1 and 21 (markers: hsa-miR-4783-3p and hsa-miR-885-3p);
 (32-3) a combination of SEQ ID NOs: 237 and 21 (markers: hsa-miR-602 and hsa-miR-885-3p);
 (32-4) a combination of SEQ ID NOs: 4 and 21 (markers: hsa-miR-4634 and hsa-miR-885-3p);
 (32-5) a combination of SEQ ID NOs: 3 and 21 (markers: hsa-miR-1307-3p and hsa-miR-885-3p);
 (33-1) a combination of SEQ ID NOs: 2 and 17 (markers: hsa-miR-4730 and hsa-miR-4787-5p);
 (33-2) a combination of SEQ ID NOs: 1 and 17 (markers: hsa-miR-4783-3p and hsa-miR-4787-5p);
 (33-3) a combination of SEQ ID NOs: 237 and 17 (markers: hsa-miR-602 and hsa-miR-4787-5p);
 (33-4) a combination of SEQ ID NOs: 4 and 17 (markers: hsa-miR-4634 and hsa-miR-4787-5p);
 (33-5) a combination of SEQ ID NOs: 3 and 17 (markers: hsa-miR-1307-3p and hsa-miR-4787-5p);
 (34-1) a combination of SEQ ID NOs: 2 and 83 (markers: hsa-miR-4730 and hsa-miR-1909-3p);
 (34-2) a combination of SEQ ID NOs: 1 and 83 (markers: hsa-miR-4783-3p and hsa-miR-1909-3p);
 (34-3) a combination of SEQ ID NOs: 237 and 83 (markers: hsa-miR-602 and hsa-miR-1909-3p);
 (34-4) a combination of SEQ ID NOs: 4 and 83 (markers: hsa-miR-4634 and hsa-miR-1909-3p);
 (34-5) a combination of SEQ ID NOs: 3 and 83 (markers: hsa-miR-1307-3p and hsa-miR-1909-3p);
 (35-1) a combination of SEQ ID NOs: 2 and 149 (markers: hsa-miR-4730 and hsa-miR-638);
 (35-2) a combination of SEQ ID NOs: 1 and 149 (markers: hsa-miR-4783-3p and hsa-miR-638);
 (35-3) a combination of SEQ ID NOs: 237 and 149 (markers: hsa-miR-602 and hsa-miR-638);
 (35-4) a combination of SEQ ID NOs: 4 and 149 (markers: hsa-miR-4634 and hsa-miR-638);
 (35-5) a combination of SEQ ID NOs: 3 and 149 (markers: hsa-miR-1307-3p and hsa-miR-638);
 (36-1) a combination of SEQ ID NOs: 2 and 246 (markers: hsa-miR-4730 and hsa-miR-1915-3p);

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(36-2) a combination of SEQ ID NOs: 1 and 246 (markers: hsa-miR-4783-3p and hsa-miR-1915-3p);
 (36-3) a combination of SEQ ID NOs: 237 and 246 (markers: hsa-miR-602 and hsa-miR-1915-3p);
 (36-4) a combination of SEQ ID NOs: 4 and 246 (markers: hsa-miR-4634 and hsa-miR-1915-3p);
 (36-5) a combination of SEQ ID NOs: 3 and 246 (markers: hsa-miR-1307-3p and hsa-miR-1915-3p);
 (37-1) a combination of SEQ ID NOs: 2 and 22 (markers: hsa-miR-4730 and hsa-miR-6802-5p);
 (37-2) a combination of SEQ ID NOs: 1 and 22 (markers: hsa-miR-4783-3p and hsa-miR-6802-5p);
 (37-3) a combination of SEQ ID NOs: 237 and 22 (markers: hsa-miR-602 and hsa-miR-6802-5p);
 (37-4) a combination of SEQ ID NOs: 4 and 22 (markers: hsa-miR-4634 and hsa-miR-6802-5p);
 (37-5) a combination of SEQ ID NOs: 3 and 22 (markers: hsa-miR-1307-3p and hsa-miR-6802-5p);
 (38-1) a combination of SEQ ID NOs: 2 and 55 (markers: hsa-miR-4730 and hsa-miR-5090);
 (38-2) a combination of SEQ ID NOs: 1 and 55 (markers: hsa-miR-4783-3p and hsa-miR-5090);
 (38-3) a combination of SEQ ID NOs: 237 and 55 (markers: hsa-miR-602 and hsa-miR-5090);
 (38-4) a combination of SEQ ID NOs: 4 and 55 (markers: hsa-miR-4634 and hsa-miR-5090);
 (38-5) a combination of SEQ ID NOs: 3 and 55 (markers: hsa-miR-1307-3p and hsa-miR-5090);
 (39-1) a combination of SEQ ID NOs: 2 and 182 (markers: hsa-miR-4730 and hsa-miR-3196);
 (39-2) a combination of SEQ ID NOs: 1 and 182 (markers: hsa-miR-4783-3p and hsa-miR-3196);
 (39-3) a combination of SEQ ID NOs: 237 and 182 (markers: hsa-miR-602 and hsa-miR-3196);
 (39-4) a combination of SEQ ID NOs: 4 and 182 (markers: hsa-miR-4634 and hsa-miR-3196);
 (39-5) a combination of SEQ ID NOs: 3 and 182 (markers: hsa-miR-1307-3p and hsa-miR-3196);
 (40-1) a combination of SEQ ID NOs: 2 and 73 (markers: hsa-miR-4730 and hsa-miR-4707-3p);
 (40-2) a combination of SEQ ID NOs: 1 and 73 (markers: hsa-miR-4783-3p and hsa-miR-4707-3p);
 (40-3) a combination of SEQ ID NOs: 237 and 73 (markers: hsa-miR-602 and hsa-miR-4707-3p);
 (40-4) a combination of SEQ ID NOs: 4 and 73 (markers: hsa-miR-4634 and hsa-miR-4707-3p);
 (40-5) a combination of SEQ ID NOs: 3 and 73 (markers: hsa-miR-1307-3p and hsa-miR-4707-3p);
 (41-1) a combination of SEQ ID NOs: 2 and 77 (markers: hsa-miR-4730 and hsa-miR-4651);
 (41-2) a combination of SEQ ID NOs: 1 and 77 (markers: hsa-miR-4783-3p and hsa-miR-4651);
 (41-3) a combination of SEQ ID NOs: 237 and 77 (markers: hsa-miR-602 and hsa-miR-4651);
 (41-4) a combination of SEQ ID NOs: 4 and 77 (markers: hsa-miR-4634 and hsa-miR-4651);
 (41-5) a combination of SEQ ID NOs: 3 and 77 (markers: hsa-miR-1307-3p and hsa-miR-4651);
 (42-1) a combination of SEQ ID NOs: 2 and 24 (markers: hsa-miR-4730 and hsa-miR-6787-5p);
 (42-2) a combination of SEQ ID NOs: 1 and 24 (markers: hsa-miR-4783-3p and hsa-miR-6787-5p);
 (42-3) a combination of SEQ ID NOs: 237 and 24 (markers: hsa-miR-602 and hsa-miR-6787-5p);
 (42-4) a combination of SEQ ID NOs: 4 and 24 (markers: hsa-miR-4634 and hsa-miR-6787-5p);

(42-5) a combination of SEQ ID NOs: 3 and 24 (markers: hsa-miR-1307-3p and hsa-miR-6787-5p);
 (43-1) a combination of SEQ ID NOs: 2 and 103 (markers: hsa-miR-4730 and hsa-miR-3180-3p);
 (43-2) a combination of SEQ ID NOs: 1 and 103 (markers: hsa-miR-4783-3p and hsa-miR-3180-3p);
 (43-3) a combination of SEQ ID NOs: 237 and 103 (markers: hsa-miR-602 and hsa-miR-3180-3p);
 (43-4) a combination of SEQ ID NOs: 4 and 103 (markers: hsa-miR-4634 and hsa-miR-3180-3p);
 (43-5) a combination of SEQ ID NOs: 3 and 103 (markers: hsa-miR-1307-3p and hsa-miR-3180-3p);
 (44-1) a combination of SEQ ID NOs: 2 and 49 (markers: hsa-miR-4730 and hsa-miR-3937);
 (44-2) a combination of SEQ ID NOs: 1 and 49 (markers: hsa-miR-4783-3p and hsa-miR-3937);
 (44-3) a combination of SEQ ID NOs: 237 and 49 (markers: hsa-miR-602 and hsa-miR-3937);
 (44-4) a combination of SEQ ID NOs: 4 and 49 (markers: hsa-miR-4634 and hsa-miR-3937);
 (44-5) a combination of SEQ ID NOs: 3 and 49 (markers: hsa-miR-1307-3p and hsa-miR-3937);
 (45-1) a combination of SEQ ID NOs: 2 and 239 (markers: hsa-miR-4730 and hsa-miR-92a-2-5p);
 (45-2) a combination of SEQ ID NOs: 1 and 239 (markers: hsa-miR-4783-3p and hsa-miR-92a-2-5p);
 (45-3) a combination of SEQ ID NOs: 237 and 239 (markers: hsa-miR-602 and hsa-miR-92a-2-5p);
 (45-4) a combination of SEQ ID NOs: 4 and 239 (markers: hsa-miR-4634 and hsa-miR-92a-2-5p);
 (45-5) a combination of SEQ ID NOs: 3 and 239 (markers: hsa-miR-1307-3p and hsa-miR-92a-2-5p);
 (46-1) a combination of SEQ ID NOs: 2 and 23 (markers: hsa-miR-4730 and hsa-miR-328-5p);
 (46-2) a combination of SEQ ID NOs: 1 and 23 (markers: hsa-miR-4783-3p and hsa-miR-328-5p);
 (46-3) a combination of SEQ ID NOs: 237 and 23 (markers: hsa-miR-602 and hsa-miR-328-5p);
 (46-4) a combination of SEQ ID NOs: 4 and 23 (markers: hsa-miR-4634 and hsa-miR-328-5p);
 (46-5) a combination of SEQ ID NOs: 3 and 23 (markers: hsa-miR-1307-3p and hsa-miR-328-5p);
 (47-1) a combination of SEQ ID NOs: 2 and 58 (markers: hsa-miR-4730 and hsa-miR-128-1-5p);
 (47-2) a combination of SEQ ID NOs: 1 and 58 (markers: hsa-miR-4783-3p and hsa-miR-128-1-5p);
 (47-3) a combination of SEQ ID NOs: 237 and 58 (markers: hsa-miR-602 and hsa-miR-128-1-5p);
 (47-4) a combination of SEQ ID NOs: 4 and 58 (markers: hsa-miR-4634 and hsa-miR-128-1-5p);
 (47-5) a combination of SEQ ID NOs: 3 and 58 (markers: hsa-miR-1307-3p and hsa-miR-128-1-5p);
 (48-1) a combination of SEQ ID NOs: 2 and 211 (markers: hsa-miR-4730 and hsa-miR-6765-5p);
 (48-2) a combination of SEQ ID NOs: 1 and 211 (markers: hsa-miR-4783-3p and hsa-miR-6765-5p);
 (48-3) a combination of SEQ ID NOs: 237 and 211 (markers: hsa-miR-602 and hsa-miR-6765-5p);
 (48-4) a combination of SEQ ID NOs: 4 and 211 (markers: hsa-miR-4634 and hsa-miR-6765-5p);
 (48-5) a combination of SEQ ID NOs: 3 and 211 (markers: hsa-miR-1307-3p and hsa-miR-6765-5p);
 (49-1) a combination of SEQ ID NOs: 2 and 147 (markers: hsa-miR-4730 and hsa-miR-6786-5p);
 (49-2) a combination of SEQ ID NOs: 1 and 147 (markers: hsa-miR-4783-3p and hsa-miR-6786-5p);

(49-3) a combination of SEQ ID NOs: 237 and 147 (markers: hsa-miR-602 and hsa-miR-6786-5p);
 (49-4) a combination of SEQ ID NOs: 4 and 147 (markers: hsa-miR-4634 and hsa-miR-6786-5p);
 (49-5) a combination of SEQ ID NOs: 3 and 147 (markers: hsa-miR-1307-3p and hsa-miR-6786-5p);
 (50-1) a combination of SEQ ID NOs: 2 and 65 (markers: hsa-miR-4730 and hsa-miR-4525);
 (50-2) a combination of SEQ ID NOs: 1 and 65 (markers: hsa-miR-4783-3p and hsa-miR-4525);
 (50-3) a combination of SEQ ID NOs: 237 and 65 (markers: hsa-miR-602 and hsa-miR-4525);
 (50-4) a combination of SEQ ID NOs: 4 and 65 (markers: hsa-miR-4634 and hsa-miR-4525);
 (50-5) a combination of SEQ ID NOs: 3 and 65 (markers: hsa-miR-1307-3p and hsa-miR-4525);
 (51-1) a combination of SEQ ID NOs: 2 and 31 (markers: hsa-miR-4730 and hsa-miR-3665);
 (51-2) a combination of SEQ ID NOs: 1 and 31 (markers: hsa-miR-4783-3p and hsa-miR-3665);
 (51-3) a combination of SEQ ID NOs: 237 and 31 (markers: hsa-miR-602 and hsa-miR-3665);
 (51-4) a combination of SEQ ID NOs: 4 and 31 (markers: hsa-miR-4634 and hsa-miR-3665);
 (51-5) a combination of SEQ ID NOs: 3 and 31 (markers: hsa-miR-1307-3p and hsa-miR-3665);
 (52-1) a combination of SEQ ID NOs: 2 and 72 (markers: hsa-miR-4730 and hsa-miR-663b);
 (52-2) a combination of SEQ ID NOs: 1 and 72 (markers: hsa-miR-4783-3p and hsa-miR-663b);
 (52-3) a combination of SEQ ID NOs: 237 and 72 (markers: hsa-miR-602 and hsa-miR-663b);
 (52-4) a combination of SEQ ID NOs: 4 and 72 (markers: hsa-miR-4634 and hsa-miR-663b);
 (52-5) a combination of SEQ ID NOs: 3 and 72 (markers: hsa-miR-1307-3p and hsa-miR-663b);
 (53-1) a combination of SEQ ID NOs: 2 and 63 (markers: hsa-miR-4730 and hsa-miR-6785-5p);
 (53-2) a combination of SEQ ID NOs: 1 and 63 (markers: hsa-miR-4783-3p and hsa-miR-6785-5p);
 (53-3) a combination of SEQ ID NOs: 237 and 63 (markers: hsa-miR-602 and hsa-miR-6785-5p);
 (53-4) a combination of SEQ ID NOs: 4 and 63 (markers: hsa-miR-4634 and hsa-miR-6785-5p);
 (53-5) a combination of SEQ ID NOs: 3 and 63 (markers: hsa-miR-1307-3p and hsa-miR-6785-5p);
 (54-1) a combination of SEQ ID NOs: 2 and 80 (markers: hsa-miR-4730 and hsa-miR-4758-5p);
 (54-2) a combination of SEQ ID NOs: 1 and 80 (markers: hsa-miR-4783-3p and hsa-miR-4758-5p);
 (54-3) a combination of SEQ ID NOs: 237 and 80 (markers: hsa-miR-602 and hsa-miR-4758-5p);
 (54-4) a combination of SEQ ID NOs: 4 and 80 (markers: hsa-miR-4634 and hsa-miR-4758-5p);
 (54-5) a combination of SEQ ID NOs: 3 and 80 (markers: hsa-miR-1307-3p and hsa-miR-4758-5p);
 (55-1) a combination of SEQ ID NOs: 2 and 37 (markers: hsa-miR-4730 and hsa-miR-197-5p);
 (55-2) a combination of SEQ ID NOs: 1 and 37 (markers: hsa-miR-4783-3p and hsa-miR-197-5p);
 (55-3) a combination of SEQ ID NOs: 237 and 37 (markers: hsa-miR-602 and hsa-miR-197-5p);
 (55-4) a combination of SEQ ID NOs: 4 and 37 (markers: hsa-miR-4634 and hsa-miR-197-5p);
 (55-5) a combination of SEQ ID NOs: 3 and 37 (markers: hsa-miR-1307-3p and hsa-miR-197-5p);

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(56-1) a combination of SEQ ID NOs: 2 and 67 (markers: hsa-miR-4730 and hsa-miR-3180);
 (56-2) a combination of SEQ ID NOs: 1 and 67 (markers: hsa-miR-4783-3p and hsa-miR-3180);
 (56-3) a combination of SEQ ID NOs: 237 and 67 (markers: hsa-miR-602 and hsa-miR-3180);
 (56-4) a combination of SEQ ID NOs: 4 and 67 (markers: hsa-miR-4634 and hsa-miR-3180);
 (56-5) a combination of SEQ ID NOs: 3 and 67 (markers: hsa-miR-1307-3p and hsa-miR-3180);
 (57-1) a combination of SEQ ID NOs: 2 and 232 (markers: hsa-miR-4730 and hsa-miR-6125);
 (57-2) a combination of SEQ ID NOs: 1 and 232 (markers: hsa-miR-4783-3p and hsa-miR-6125);
 (57-3) a combination of SEQ ID NOs: 237 and 232 (markers: hsa-miR-602 and hsa-miR-6125);
 (57-4) a combination of SEQ ID NOs: 4 and 232 (markers: hsa-miR-4634 and hsa-miR-6125);
 (57-5) a combination of SEQ ID NOs: 3 and 232 (markers: hsa-miR-1307-3p and hsa-miR-6125);
 (58-1) a combination of SEQ ID NOs: 2 and 127 (markers: hsa-miR-4730 and hsa-miR-4446-3p);
 (58-2) a combination of SEQ ID NOs: 1 and 127 (markers: hsa-miR-4783-3p and hsa-miR-4446-3p);
 (58-3) a combination of SEQ ID NOs: 237 and 127 (markers: hsa-miR-602 and hsa-miR-4446-3p);
 (58-4) a combination of SEQ ID NOs: 4 and 127 (markers: hsa-miR-4634 and hsa-miR-4446-3p);
 (58-5) a combination of SEQ ID NOs: 3 and 127 (markers: hsa-miR-1307-3p and hsa-miR-4446-3p);
 (59-1) a combination of SEQ ID NOs: 2 and 145 (markers: hsa-miR-4730 and hsa-miR-1268a);
 (59-2) a combination of SEQ ID NOs: 1 and 145 (markers: hsa-miR-4783-3p and hsa-miR-1268a);
 (59-3) a combination of SEQ ID NOs: 237 and 145 (markers: hsa-miR-602 and hsa-miR-1268a);
 (59-4) a combination of SEQ ID NOs: 4 and 145 (markers: hsa-miR-4634 and hsa-miR-1268a);
 (59-5) a combination of SEQ ID NOs: 3 and 145 (markers: hsa-miR-1307-3p and hsa-miR-1268a);
 (60-1) a combination of SEQ ID NOs: 2 and 16 (markers: hsa-miR-4730 and hsa-miR-8073);
 (60-2) a combination of SEQ ID NOs: 1 and 16 (markers: hsa-miR-4783-3p and hsa-miR-8073);
 (60-3) a combination of SEQ ID NOs: 237 and 16 (markers: hsa-miR-602 and hsa-miR-8073);
 (60-4) a combination of SEQ ID NOs: 4 and 16 (markers: hsa-miR-4634 and hsa-miR-8073);
 (60-5) a combination of SEQ ID NOs: 3 and 16 (markers: hsa-miR-1307-3p and hsa-miR-8073);
 (61-1) a combination of SEQ ID NOs: 2 and 11 (markers: hsa-miR-4730 and hsa-miR-4732-5p);
 (61-2) a combination of SEQ ID NOs: 1 and 11 (markers: hsa-miR-4783-3p and hsa-miR-4732-5p);
 (61-3) a combination of SEQ ID NOs: 237 and 11 (markers: hsa-miR-602 and hsa-miR-4732-5p);
 (61-4) a combination of SEQ ID NOs: 4 and 11 (markers: hsa-miR-4634 and hsa-miR-4732-5p);
 (61-5) a combination of SEQ ID NOs: 3 and 11 (markers: hsa-miR-1307-3p and hsa-miR-4732-5p);
 (62-1) a combination of SEQ ID NOs: 2 and 186 (markers: hsa-miR-4730 and hsa-miR-6789-5p);
 (62-2) a combination of SEQ ID NOs: 1 and 186 (markers: hsa-miR-4783-3p and hsa-miR-6789-5p);
 (62-3) a combination of SEQ ID NOs: 237 and 186 (markers: hsa-miR-602 and hsa-miR-6789-5p);

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(62-4) a combination of SEQ ID NOs: 4 and 186 (markers: hsa-miR-4634 and hsa-miR-6789-5p);
 (62-5) a combination of SEQ ID NOs: 3 and 186 (markers: hsa-miR-1307-3p and hsa-miR-6789-5p);
 (63-1) a combination of SEQ ID NOs: 2 and 50 (markers: hsa-miR-4730 and hsa-miR-887-3p);
 (63-2) a combination of SEQ ID NOs: 1 and 50 (markers: hsa-miR-4783-3p and hsa-miR-887-3p);
 (63-3) a combination of SEQ ID NOs: 237 and 50 (markers: hsa-miR-602 and hsa-miR-887-3p);
 (63-4) a combination of SEQ ID NOs: 4 and 50 (markers: hsa-miR-4634 and hsa-miR-887-3p);
 (63-5) a combination of SEQ ID NOs: 3 and 50 (markers: hsa-miR-1307-3p and hsa-miR-887-3p);
 (64-1) a combination of SEQ ID NOs: 2 and 69 (markers: hsa-miR-4730 and hsa-miR-1199-5p);
 (64-2) a combination of SEQ ID NOs: 1 and 69 (markers: hsa-miR-4783-3p and hsa-miR-1199-5p);
 (64-3) a combination of SEQ ID NOs: 237 and 69 (markers: hsa-miR-602 and hsa-miR-1199-5p);
 (64-4) a combination of SEQ ID NOs: 4 and 69 (markers: hsa-miR-4634 and hsa-miR-1199-5p);
 (64-5) a combination of SEQ ID NOs: 3 and 69 (markers: hsa-miR-1307-3p and hsa-miR-1199-5p);
 (65-1) a combination of SEQ ID NOs: 2 and 33 (markers: hsa-miR-4730 and hsa-miR-6821-5p);
 (65-2) a combination of SEQ ID NOs: 1 and 33 (markers: hsa-miR-4783-3p and hsa-miR-6821-5p);
 (65-3) a combination of SEQ ID NOs: 237 and 33 (markers: hsa-miR-602 and hsa-miR-6821-5p);
 (65-4) a combination of SEQ ID NOs: 4 and 33 (markers: hsa-miR-4634 and hsa-miR-6821-5p);
 (65-5) a combination of SEQ ID NOs: 3 and 33 (markers: hsa-miR-1307-3p and hsa-miR-6821-5p);
 (66-1) a combination of SEQ ID NOs: 2 and 247 (markers: hsa-miR-4730 and hsa-miR-718);
 (66-2) a combination of SEQ ID NOs: 1 and 247 (markers: hsa-miR-4783-3p and hsa-miR-718);
 (66-3) a combination of SEQ ID NOs: 237 and 247 (markers: hsa-miR-602 and hsa-miR-718);
 (66-4) a combination of SEQ ID NOs: 4 and 247 (markers: hsa-miR-4634 and hsa-miR-718);
 (66-5) a combination of SEQ ID NOs: 3 and 247 (markers: hsa-miR-1307-3p and hsa-miR-718);
 (67-1) a combination of SEQ ID NOs: 2 and 36 (markers: hsa-miR-4730 and hsa-miR-6726-5p);
 (67-2) a combination of SEQ ID NOs: 1 and 36 (markers: hsa-miR-4783-3p and hsa-miR-6726-5p);
 (67-3) a combination of SEQ ID NOs: 237 and 36 (markers: hsa-miR-602 and hsa-miR-6726-5p);
 (67-4) a combination of SEQ ID NOs: 4 and 36 (markers: hsa-miR-4634 and hsa-miR-6726-5p);
 (67-5) a combination of SEQ ID NOs: 3 and 36 (markers: hsa-miR-1307-3p and hsa-miR-6726-5p);
 (68-1) a combination of SEQ ID NOs: 2 and 218 (markers: hsa-miR-4730 and hsa-miR-6784-5p);
 (68-2) a combination of SEQ ID NOs: 1 and 218 (markers: hsa-miR-4783-3p and hsa-miR-6784-5p);
 (68-3) a combination of SEQ ID NOs: 237 and 218 (markers: hsa-miR-602 and hsa-miR-6784-5p);
 (68-4) a combination of SEQ ID NOs: 4 and 218 (markers: hsa-miR-4634 and hsa-miR-6784-5p);
 (68-5) a combination of SEQ ID NOs: 3 and 218 (markers: hsa-miR-1307-3p and hsa-miR-6784-5p);
 (69-1) a combination of SEQ ID NOs: 2 and 43 (markers: hsa-miR-4730 and hsa-miR-4763-3p);

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(69-2) a combination of SEQ ID NOs: 1 and 43 (markers: hsa-miR-4783-3p and hsa-miR-4763-3p);
 (69-3) a combination of SEQ ID NOs: 237 and 43 (markers: hsa-miR-602 and hsa-miR-4763-3p);
 (69-4) a combination of SEQ ID NOs: 4 and 43 (markers: hsa-miR-4634 and hsa-miR-4763-3p);
 (69-5) a combination of SEQ ID NOs: 3 and 43 (markers: hsa-miR-1307-3p and hsa-miR-4763-3p);
 (70-1) a combination of SEQ ID NOs: 2 and 29 (markers: hsa-miR-4730 and hsa-miR-6757-5p);
 (70-2) a combination of SEQ ID NOs: 1 and 29 (markers: hsa-miR-4783-3p and hsa-miR-6757-5p);
 (70-3) a combination of SEQ ID NOs: 237 and 29 (markers: hsa-miR-602 and hsa-miR-6757-5p);
 (70-4) a combination of SEQ ID NOs: 4 and 29 (markers: hsa-miR-4634 and hsa-miR-6757-5p);
 (70-5) a combination of SEQ ID NOs: 3 and 29 (markers: hsa-miR-1307-3p and hsa-miR-6757-5p);
 (71-1) a combination of SEQ ID NOs: 2 and 110 (markers: hsa-miR-4730 and hsa-miR-665);
 (71-2) a combination of SEQ ID NOs: 1 and 110 (markers: hsa-miR-4783-3p and hsa-miR-665);
 (71-3) a combination of SEQ ID NOs: 237 and 110 (markers: hsa-miR-602 and hsa-miR-665);
 (71-4) a combination of SEQ ID NOs: 4 and 110 (markers: hsa-miR-4634 and hsa-miR-665);
 (71-5) a combination of SEQ ID NOs: 3 and 110 (markers: hsa-miR-1307-3p and hsa-miR-665);
 (72-1) a combination of SEQ ID NOs: 2 and 20 (markers: hsa-miR-4730 and hsa-miR-1233-5p);
 (72-2) a combination of SEQ ID NOs: 1 and 20 (markers: hsa-miR-4783-3p and hsa-miR-1233-5p);
 (72-3) a combination of SEQ ID NOs: 237 and 20 (markers: hsa-miR-602 and hsa-miR-1233-5p);
 (72-4) a combination of SEQ ID NOs: 4 and 20 (markers: hsa-miR-4634 and hsa-miR-1233-5p);
 (72-5) a combination of SEQ ID NOs: 3 and 20 (markers: hsa-miR-1307-3p and hsa-miR-1233-5p);
 (73-1) a combination of SEQ ID NOs: 2 and 157 (markers: hsa-miR-4730 and hsa-miR-1268b);
 (73-2) a combination of SEQ ID NOs: 1 and 157 (markers: hsa-miR-4783-3p and hsa-miR-1268b);
 (73-3) a combination of SEQ ID NOs: 237 and 157 (markers: hsa-miR-602 and hsa-miR-1268b);
 (73-4) a combination of SEQ ID NOs: 4 and 157 (markers: hsa-miR-4634 and hsa-miR-1268b);
 (73-5) a combination of SEQ ID NOs: 3 and 157 (markers: hsa-miR-1307-3p and hsa-miR-1268b);
 (74-1) a combination of SEQ ID NOs: 2 and 75 (markers: hsa-miR-4730 and hsa-miR-4675);
 (74-2) a combination of SEQ ID NOs: 1 and 75 (markers: hsa-miR-4783-3p and hsa-miR-4675);
 (74-3) a combination of SEQ ID NOs: 237 and 75 (markers: hsa-miR-602 and hsa-miR-4675);
 (74-4) a combination of SEQ ID NOs: 4 and 75 (markers: hsa-miR-4634 and hsa-miR-4675);
 (74-5) a combination of SEQ ID NOs: 3 and 75 (markers: hsa-miR-1307-3p and hsa-miR-4675);
 (75-1) a combination of SEQ ID NOs: 2 and 82 (markers: hsa-miR-4730 and hsa-miR-3620-5p);
 (75-2) a combination of SEQ ID NOs: 1 and 82 (markers: hsa-miR-4783-3p and hsa-miR-3620-5p);
 (75-3) a combination of SEQ ID NOs: 237 and 82 (markers: hsa-miR-602 and hsa-miR-3620-5p);
 (75-4) a combination of SEQ ID NOs: 4 and 82 (markers: hsa-miR-4634 and hsa-miR-3620-5p);

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(75-5) a combination of SEQ ID NOs: 3 and 82 (markers: hsa-miR-1307-3p and hsa-miR-3620-5p);
 (76-1) a combination of SEQ ID NOs: 2 and 106 (markers: hsa-miR-4730 and hsa-miR-6800-5p);
 (76-2) a combination of SEQ ID NOs: 1 and 106 (markers: hsa-miR-4783-3p and hsa-miR-6800-5p);
 (76-3) a combination of SEQ ID NOs: 237 and 106 (markers: hsa-miR-602 and hsa-miR-6800-5p);
 (76-4) a combination of SEQ ID NOs: 4 and 106 (markers: hsa-miR-4634 and hsa-miR-6800-5p);
 (76-5) a combination of SEQ ID NOs: 3 and 106 (markers: hsa-miR-1307-3p and hsa-miR-6800-5p);
 (77-1) a combination of SEQ ID NOs: 2 and 111 (markers: hsa-miR-4730 and hsa-miR-6778-5p);
 (77-2) a combination of SEQ ID NOs: 1 and 111 (markers: hsa-miR-4783-3p and hsa-miR-6778-5p);
 (77-3) a combination of SEQ ID NOs: 237 and 111 (markers: hsa-miR-602 and hsa-miR-6778-5p);
 (77-4) a combination of SEQ ID NOs: 4 and 111 (markers: hsa-miR-4634 and hsa-miR-6778-5p);
 (77-5) a combination of SEQ ID NOs: 3 and 111 (markers: hsa-miR-1307-3p and hsa-miR-6778-5p);
 (78-1) a combination of SEQ ID NOs: 2 and 96 (markers: hsa-miR-4730 and hsa-miR-8059);
 (78-2) a combination of SEQ ID NOs: 1 and 96 (markers: hsa-miR-4783-3p and hsa-miR-8059);
 (78-3) a combination of SEQ ID NOs: 237 and 96 (markers: hsa-miR-602 and hsa-miR-8059);
 (78-4) a combination of SEQ ID NOs: 4 and 96 (markers: hsa-miR-4634 and hsa-miR-8059);
 (78-5) a combination of SEQ ID NOs: 3 and 96 (markers: hsa-miR-1307-3p and hsa-miR-8059);
 (79-1) a combination of SEQ ID NOs: 2 and 266 (markers: hsa-miR-4730 and hsa-miR-1908-5p);
 (79-2) a combination of SEQ ID NOs: 1 and 266 (markers: hsa-miR-4783-3p and hsa-miR-1908-5p);
 (79-3) a combination of SEQ ID NOs: 237 and 266 (markers: hsa-miR-602 and hsa-miR-1908-5p);
 (79-4) a combination of SEQ ID NOs: 4 and 266 (markers: hsa-miR-4634 and hsa-miR-1908-5p);
 (79-5) a combination of SEQ ID NOs: 3 and 266 (markers: hsa-miR-1307-3p and hsa-miR-1908-5p);
 (80-1) a combination of SEQ ID NOs: 2 and 124 (markers: hsa-miR-4730 and hsa-miR-6798-5p);
 (80-2) a combination of SEQ ID NOs: 1 and 124 (markers: hsa-miR-4783-3p and hsa-miR-6798-5p);
 (80-3) a combination of SEQ ID NOs: 237 and 124 (markers: hsa-miR-602 and hsa-miR-6798-5p);
 (80-4) a combination of SEQ ID NOs: 4 and 124 (markers: hsa-miR-4634 and hsa-miR-6798-5p);
 (80-5) a combination of SEQ ID NOs: 3 and 124 (markers: hsa-miR-1307-3p and hsa-miR-6798-5p);
 (81-1) a combination of SEQ ID NOs: 2 and 68 (markers: hsa-miR-4730 and hsa-miR-6879-5p);
 (81-2) a combination of SEQ ID NOs: 1 and 68 (markers: hsa-miR-4783-3p and hsa-miR-6879-5p);
 (81-3) a combination of SEQ ID NOs: 237 and 68 (markers: hsa-miR-602 and hsa-miR-6879-5p);
 (81-4) a combination of SEQ ID NOs: 4 and 68 (markers: hsa-miR-4634 and hsa-miR-6879-5p);
 (81-5) a combination of SEQ ID NOs: 3 and 68 (markers: hsa-miR-1307-3p and hsa-miR-6879-5p);
 (82-1) a combination of SEQ ID NOs: 2 and 71 (markers: hsa-miR-4730 and hsa-miR-711);
 (82-2) a combination of SEQ ID NOs: 1 and 71 (markers: hsa-miR-4783-3p and hsa-miR-711);

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(82-3) a combination of SEQ ID NOs: 237 and 71 (markers: hsa-miR-602 and hsa-miR-711);
 (82-4) a combination of SEQ ID NOs: 4 and 71 (markers: hsa-miR-4634 and hsa-miR-711);
 (82-5) a combination of SEQ ID NOs: 3 and 71 (markers: hsa-miR-1307-3p and hsa-miR-711);
 (83-1) a combination of SEQ ID NOs: 2 and 35 (markers: hsa-miR-4730 and hsa-miR-4728-5p);
 (83-2) a combination of SEQ ID NOs: 1 and 35 (markers: hsa-miR-4783-3p and hsa-miR-4728-5p);
 (83-3) a combination of SEQ ID NOs: 237 and 35 (markers: hsa-miR-602 and hsa-miR-4728-5p);
 (83-4) a combination of SEQ ID NOs: 4 and 35 (markers: hsa-miR-4634 and hsa-miR-4728-5p);
 (83-5) a combination of SEQ ID NOs: 3 and 35 (markers: hsa-miR-1307-3p and hsa-miR-4728-5p);
 (84-1) a combination of SEQ ID NOs: 2 and 173 (markers: hsa-miR-4730 and hsa-miR-3195);
 (84-2) a combination of SEQ ID NOs: 1 and 173 (markers: hsa-miR-4783-3p and hsa-miR-3195);
 (84-3) a combination of SEQ ID NOs: 237 and 173 (markers: hsa-miR-602 and hsa-miR-3195);
 (84-4) a combination of SEQ ID NOs: 4 and 173 (markers: hsa-miR-4634 and hsa-miR-3195);
 (84-5) a combination of SEQ ID NOs: 3 and 173 (markers: hsa-miR-1307-3p and hsa-miR-3195);
 (85-1) a combination of SEQ ID NOs: 2 and 5 (markers: hsa-miR-4730 and hsa-miR-663a);
 (85-2) a combination of SEQ ID NOs: 1 and 5 (markers: hsa-miR-4783-3p and hsa-miR-663a);
 (85-3) a combination of SEQ ID NOs: 237 and 5 (markers: hsa-miR-602 and hsa-miR-663a);
 (85-4) a combination of SEQ ID NOs: 4 and 5 (markers: hsa-miR-4634 and hsa-miR-663a);
 (85-5) a combination of SEQ ID NOs: 3 and 5 (markers: hsa-miR-1307-3p and hsa-miR-663a);
 (86-1) a combination of SEQ ID NOs: 2 and 851 (markers: hsa-miR-4730 and hsa-miR-6089);
 (86-2) a combination of SEQ ID NOs: 1 and 851 (markers: hsa-miR-4783-3p and hsa-miR-6089);
 (86-3) a combination of SEQ ID NOs: 237 and 851 (markers: hsa-miR-602 and hsa-miR-6089);
 (86-4) a combination of SEQ ID NOs: 4 and 851 (markers: hsa-miR-4634 and hsa-miR-6089);
 (86-5) a combination of SEQ ID NOs: 3 and 851 (markers: hsa-miR-1307-3p and hsa-miR-6089);
 (87-1) a combination of SEQ ID NOs: 2 and 852 (markers: hsa-miR-4730 and hsa-miR-6816-5p);
 (87-2) a combination of SEQ ID NOs: 1 and 852 (markers: hsa-miR-4783-3p and hsa-miR-6816-5p);
 (87-3) a combination of SEQ ID NOs: 237 and 852 (markers: hsa-miR-602 and hsa-miR-6816-5p);
 (87-4) a combination of SEQ ID NOs: 4 and 852 (markers: hsa-miR-4634 and hsa-miR-6816-5p);
 (87-5) a combination of SEQ ID NOs: 3 and 852 (markers: hsa-miR-1307-3p and hsa-miR-6816-5p);
 (88-1) a combination of SEQ ID NOs: 2 and 30 (markers: hsa-miR-4730 and hsa-miR-6756-5p);
 (88-2) a combination of SEQ ID NOs: 1 and 30 (markers: hsa-miR-4783-3p and hsa-miR-6756-5p);
 (88-3) a combination of SEQ ID NOs: 237 and 30 (markers: hsa-miR-602 and hsa-miR-6756-5p);
 (88-4) a combination of SEQ ID NOs: 4 and 30 (markers: hsa-miR-4634 and hsa-miR-6756-5p);
 (88-5) a combination of SEQ ID NOs: 3 and 30 (markers: hsa-miR-1307-3p and hsa-miR-6756-5p);

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(89-1) a combination of SEQ ID NOs: 2 and 93 (markers: hsa-miR-4730 and hsa-miR-6861-5p);
 (89-2) a combination of SEQ ID NOs: 1 and 93 (markers: hsa-miR-4783-3p and hsa-miR-6861-5p);
 (89-3) a combination of SEQ ID NOs: 237 and 93 (markers: hsa-miR-602 and hsa-miR-6861-5p);
 (89-4) a combination of SEQ ID NOs: 4 and 93 (markers: hsa-miR-4634 and hsa-miR-6861-5p);
 (89-5) a combination of SEQ ID NOs: 3 and 93 (markers: hsa-miR-1307-3p and hsa-miR-6861-5p);
 (90-1) a combination of SEQ ID NOs: 2 and 27 (markers: hsa-miR-4730 and hsa-miR-1246);
 (90-2) a combination of SEQ ID NOs: 1 and 27 (markers: hsa-miR-4783-3p and hsa-miR-1246);
 (90-3) a combination of SEQ ID NOs: 237 and 27 (markers: hsa-miR-602 and hsa-miR-1246);
 (90-4) a combination of SEQ ID NOs: 4 and 27 (markers: hsa-miR-4634 and hsa-miR-1246);
 (90-5) a combination of SEQ ID NOs: 27 and 208 (markers: hsa-miR-1246 and hsa-miR-1343-5p);
 (91-1) a combination of SEQ ID NOs: 2 and 853 (markers: hsa-miR-4730 and hsa-miR-4466);
 (91-2) a combination of SEQ ID NOs: 1 and 853 (markers: hsa-miR-4783-3p and hsa-miR-4466);
 (91-3) a combination of SEQ ID NOs: 237 and 853 (markers: hsa-miR-602 and hsa-miR-4466);
 (91-4) a combination of SEQ ID NOs: 4 and 853 (markers: hsa-miR-4634 and hsa-miR-4466);
 (91-5) a combination of SEQ ID NOs: 3 and 853 (markers: hsa-miR-1307-3p and hsa-miR-4466);
 (92-1) a combination of SEQ ID NOs: 2 and 238 (markers: hsa-miR-4730 and hsa-miR-423-5p);
 (92-2) a combination of SEQ ID NOs: 1 and 238 (markers: hsa-miR-4783-3p and hsa-miR-423-5p);
 (92-3) a combination of SEQ ID NOs: 237 and 238 (markers: hsa-miR-602 and hsa-miR-423-5p);
 (92-4) a combination of SEQ ID NOs: 4 and 238 (markers: hsa-miR-4634 and hsa-miR-423-5p);
 (92-5) a combination of SEQ ID NOs: 3 and 238 (markers: hsa-miR-1307-3p and hsa-miR-423-5p);
 (93-1) a combination of SEQ ID NOs: 2 and 130 (markers: hsa-miR-4730 and hsa-miR-6075);
 (93-2) a combination of SEQ ID NOs: 1 and 130 (markers: hsa-miR-4783-3p and hsa-miR-6075);
 (93-3) a combination of SEQ ID NOs: 237 and 130 (markers: hsa-miR-602 and hsa-miR-6075);
 (93-4) a combination of SEQ ID NOs: 4 and 130 (markers: hsa-miR-4634 and hsa-miR-6075);
 (93-5) a combination of SEQ ID NOs: 3 and 130 (markers: hsa-miR-1307-3p and hsa-miR-6075);
 (94-1) a combination of SEQ ID NOs: 2 and 177 (markers: hsa-miR-4730 and hsa-miR-7108-5p);
 (94-2) a combination of SEQ ID NOs: 1 and 177 (markers: hsa-miR-4783-3p and hsa-miR-7108-5p);
 (94-3) a combination of SEQ ID NOs: 237 and 177 (markers: hsa-miR-602 and hsa-miR-7108-5p);
 (94-4) a combination of SEQ ID NOs: 4 and 177 (markers: hsa-miR-4634 and hsa-miR-7108-5p);
 (94-5) a combination of SEQ ID NOs: 3 and 177 (markers: hsa-miR-1307-3p and hsa-miR-7108-5p);
 (95-1) a combination of SEQ ID NOs: 2 and 64 (markers: hsa-miR-4730 and hsa-miR-6511a-5p);
 (95-2) a combination of SEQ ID NOs: 1 and 64 (markers: hsa-miR-4783-3p and hsa-miR-6511a-5p);
 (95-3) a combination of SEQ ID NOs: 237 and 64 (markers: hsa-miR-602 and hsa-miR-6511a-5p);

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(95-4) a combination of SEQ ID NOs: 4 and 64 (markers: hsa-miR-4634 and hsa-miR-6511a-5p);
 (95-5) a combination of SEQ ID NOs: 3 and 64 (markers: hsa-miR-1307-3p and hsa-miR-6511a-5p);
 (96-1) a combination of SEQ ID NOs: 2 and 114 (markers: hsa-miR-4730 and hsa-miR-211-3p);
 (96-2) a combination of SEQ ID NOs: 1 and 114 (markers: hsa-miR-4783-3p and hsa-miR-211-3p);
 (96-3) a combination of SEQ ID NOs: 237 and 114 (markers: hsa-miR-602 and hsa-miR-211-3p);
 (96-4) a combination of SEQ ID NOs: 4 and 114 (markers: hsa-miR-4634 and hsa-miR-211-3p);
 (96-5) a combination of SEQ ID NOs: 3 and 114 (markers: hsa-miR-1307-3p and hsa-miR-211-3p);
 (97-1) a combination of SEQ ID NOs: 2 and 119 (markers: hsa-miR-4730 and hsa-miR-7110-5p);
 (97-2) a combination of SEQ ID NOs: 1 and 119 (markers: hsa-miR-4783-3p and hsa-miR-7110-5p);
 (97-3) a combination of SEQ ID NOs: 237 and 119 (markers: hsa-miR-602 and hsa-miR-7110-5p);
 (97-4) a combination of SEQ ID NOs: 4 and 119 (markers: hsa-miR-4634 and hsa-miR-7110-5p);
 (97-5) a combination of SEQ ID NOs: 3 and 119 (markers: hsa-miR-1307-3p and hsa-miR-7110-5p);
 (98-1) a combination of SEQ ID NOs: 2 and 135 (markers: hsa-miR-4730 and hsa-miR-6870-5p);
 (98-2) a combination of SEQ ID NOs: 1 and 135 (markers: hsa-miR-4783-3p and hsa-miR-6870-5p);
 (98-3) a combination of SEQ ID NOs: 237 and 135 (markers: hsa-miR-602 and hsa-miR-6870-5p);
 (98-4) a combination of SEQ ID NOs: 4 and 135 (markers: hsa-miR-4634 and hsa-miR-6870-5p);
 (98-5) a combination of SEQ ID NOs: 3 and 135 (markers: hsa-miR-1307-3p and hsa-miR-6870-5p);
 (99-1) a combination of SEQ ID NOs: 2 and 243 (markers: hsa-miR-4730 and hsa-miR-486-5p);
 (99-2) a combination of SEQ ID NOs: 1 and 243 (markers: hsa-miR-4783-3p and hsa-miR-486-5p);
 (99-3) a combination of SEQ ID NOs: 237 and 243 (markers: hsa-miR-602 and hsa-miR-486-5p);
 (99-4) a combination of SEQ ID NOs: 4 and 243 (markers: hsa-miR-4634 and hsa-miR-486-5p);
 (99-5) a combination of SEQ ID NOs: 3 and 243 (markers: hsa-miR-1307-3p and hsa-miR-486-5p);
 (100-1) a combination of SEQ ID NOs: 2 and 122 (markers: hsa-miR-4730 and hsa-miR-4792);
 (100-2) a combination of SEQ ID NOs: 1 and 122 (markers: hsa-miR-4783-3p and hsa-miR-4792);
 (100-3) a combination of SEQ ID NOs: 237 and 122 (markers: hsa-miR-602 and hsa-miR-4792);
 (100-4) a combination of SEQ ID NOs: 4 and 122 (markers: hsa-miR-4634 and hsa-miR-4792);
 (100-5) a combination of SEQ ID NOs: 3 and 122 (markers: hsa-miR-1307-3p and hsa-miR-4792);
 (101-1) a combination of SEQ ID NOs: 2 and 260 (markers: hsa-miR-4730 and hsa-miR-4687-3p);
 (101-2) a combination of SEQ ID NOs: 1 and 260 (markers: hsa-miR-4783-3p and hsa-miR-4687-3p);
 (101-3) a combination of SEQ ID NOs: 237 and 260 (markers: hsa-miR-602 and hsa-miR-4687-3p);
 (101-4) a combination of SEQ ID NOs: 4 and 260 (markers: hsa-miR-4634 and hsa-miR-4687-3p);
 (101-5) a combination of SEQ ID NOs: 3 and 260 (markers: hsa-miR-1307-3p and hsa-miR-4687-3p);
 (102-1) a combination of SEQ ID NOs: 2 and 59 (markers: hsa-miR-4730 and hsa-miR-1238-5p);

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(102-2) a combination of SEQ ID NOs: 1 and 59 (markers: hsa-miR-4783-3p and hsa-miR-1238-5p);
 (102-3) a combination of SEQ ID NOs: 237 and 59 (markers: hsa-miR-602 and hsa-miR-1238-5p);
 (102-4) a combination of SEQ ID NOs: 4 and 59 (markers: hsa-miR-4634 and hsa-miR-1238-5p);
 (102-5) a combination of SEQ ID NOs: 3 and 59 (markers: hsa-miR-1307-3p and hsa-miR-1238-5p);
 (103-1) a combination of SEQ ID NOs: 2 and 854 (markers: hsa-miR-4730 and hsa-miR-4488);
 (103-2) a combination of SEQ ID NOs: 1 and 854 (markers: hsa-miR-4783-3p and hsa-miR-4488);
 (103-3) a combination of SEQ ID NOs: 237 and 854 (markers: hsa-miR-602 and hsa-miR-4488);
 (103-4) a combination of SEQ ID NOs: 4 and 854 (markers: hsa-miR-4634 and hsa-miR-4488);
 (103-5) a combination of SEQ ID NOs: 3 and 854 (markers: hsa-miR-1307-3p and hsa-miR-4488);
 (104-1) a combination of SEQ ID NOs: 2 and 132 (markers: hsa-miR-4730 and hsa-miR-6891-5p);
 (104-2) a combination of SEQ ID NOs: 1 and 132 (markers: hsa-miR-4783-3p and hsa-miR-6891-5p);
 (104-3) a combination of SEQ ID NOs: 237 and 132 (markers: hsa-miR-602 and hsa-miR-6891-5p);
 (104-4) a combination of SEQ ID NOs: 4 and 132 (markers: hsa-miR-4634 and hsa-miR-6891-5p);
 (104-5) a combination of SEQ ID NOs: 3 and 132 (markers: hsa-miR-1307-3p and hsa-miR-6891-5p);
 (105-1) a combination of SEQ ID NOs: 2 and 181 (markers: hsa-miR-4730 and hsa-miR-2861);
 (105-2) a combination of SEQ ID NOs: 1 and 181 (markers: hsa-miR-4783-3p and hsa-miR-2861);
 (105-3) a combination of SEQ ID NOs: 237 and 181 (markers: hsa-miR-602 and hsa-miR-2861);
 (105-4) a combination of SEQ ID NOs: 4 and 181 (markers: hsa-miR-4634 and hsa-miR-2861);
 (105-5) a combination of SEQ ID NOs: 3 and 181 (markers: hsa-miR-1307-3p and hsa-miR-2861);
 (106-1) a combination of SEQ ID NOs: 2 and 79 (markers: hsa-miR-4730 and hsa-miR-4665-5p);
 (106-2) a combination of SEQ ID NOs: 1 and 79 (markers: hsa-miR-4783-3p and hsa-miR-4665-5p);
 (106-3) a combination of SEQ ID NOs: 237 and 79 (markers: hsa-miR-602 and hsa-miR-4665-5p);
 (106-4) a combination of SEQ ID NOs: 4 and 79 (markers: hsa-miR-4634 and hsa-miR-4665-5p);
 (106-5) a combination of SEQ ID NOs: 3 and 79 (markers: hsa-miR-1307-3p and hsa-miR-4665-5p);
 (107-1) a combination of SEQ ID NOs: 2 and 133 (markers: hsa-miR-4730 and hsa-miR-4745-5p);
 (107-2) a combination of SEQ ID NOs: 1 and 133 (markers: hsa-miR-4783-3p and hsa-miR-4745-5p);
 (107-3) a combination of SEQ ID NOs: 237 and 133 (markers: hsa-miR-602 and hsa-miR-4745-5p);
 (107-4) a combination of SEQ ID NOs: 4 and 133 (markers: hsa-miR-4634 and hsa-miR-4745-5p);
 (107-5) a combination of SEQ ID NOs: 3 and 133 (markers: hsa-miR-1307-3p and hsa-miR-4745-5p);
 (108-1) a combination of SEQ ID NOs: 2 and 41 (markers: hsa-miR-4730 and hsa-miR-6858-5p);
 (108-2) a combination of SEQ ID NOs: 1 and 41 (markers: hsa-miR-4783-3p and hsa-miR-6858-5p);
 (108-3) a combination of SEQ ID NOs: 237 and 41 (markers: hsa-miR-602 and hsa-miR-6858-5p);
 (108-4) a combination of SEQ ID NOs: 4 and 41 (markers: hsa-miR-4634 and hsa-miR-6858-5p);

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(108-5) a combination of SEQ ID NOs: 3 and 41 (markers: hsa-miR-1307-3p and hsa-miR-6858-5p);
 (109-1) a combination of SEQ ID NOs: 2 and 139 (markers: hsa-miR-4730 and hsa-miR-6825-5p);
 (109-2) a combination of SEQ ID NOs: 1 and 139 (markers: hsa-miR-4783-3p and hsa-miR-6825-5p);
 (109-3) a combination of SEQ ID NOs: 237 and 139 (markers: hsa-miR-602 and hsa-miR-6825-5p);
 (109-4) a combination of SEQ ID NOs: 4 and 139 (markers: hsa-miR-4634 and hsa-miR-6825-5p);
 (109-5) a combination of SEQ ID NOs: 3 and 139 (markers: hsa-miR-1307-3p and hsa-miR-6825-5p);
 (110-1) a combination of SEQ ID NOs: 2 and 118 (markers: hsa-miR-4730 and hsa-miR-614);
 (110-2) a combination of SEQ ID NOs: 1 and 118 (markers: hsa-miR-4783-3p and hsa-miR-614);
 (110-3) a combination of SEQ ID NOs: 237 and 118 (markers: hsa-miR-602 and hsa-miR-614);
 (110-4) a combination of SEQ ID NOs: 4 and 118 (markers: hsa-miR-4634 and hsa-miR-614);
 (110-5) a combination of SEQ ID NOs: 3 and 118 (markers: hsa-miR-1307-3p and hsa-miR-614);
 (111-1) a combination of SEQ ID NOs: 2 and 86 (markers: hsa-miR-4730 and hsa-miR-1343-3p);
 (111-2) a combination of SEQ ID NOs: 1 and 86 (markers: hsa-miR-4783-3p and hsa-miR-1343-3p);
 (111-3) a combination of SEQ ID NOs: 237 and 86 (markers: hsa-miR-602 and hsa-miR-1343-3p);
 (111-4) a combination of SEQ ID NOs: 4 and 86 (markers: hsa-miR-4634 and hsa-miR-1343-3p);
 (111-5) a combination of SEQ ID NOs: 3 and 86 (markers: hsa-miR-1307-3p and hsa-miR-1343-3p);
 (112-1) a combination of SEQ ID NOs: 2 and 60 (markers: hsa-miR-4730 and hsa-miR-365a-5p);
 (112-2) a combination of SEQ ID NOs: 1 and 60 (markers: hsa-miR-4783-3p and hsa-miR-365a-5p);
 (112-3) a combination of SEQ ID NOs: 237 and 60 (markers: hsa-miR-602 and hsa-miR-365a-5p);
 (112-4) a combination of SEQ ID NOs: 4 and 60 (markers: hsa-miR-4634 and hsa-miR-365a-5p);
 (112-5) a combination of SEQ ID NOs: 3 and 60 (markers: hsa-miR-1307-3p and hsa-miR-365a-5p);
 (113-1) a combination of SEQ ID NOs: 2 and 116 (markers: hsa-miR-4730 and hsa-miR-4750-5p);
 (113-2) a combination of SEQ ID NOs: 1 and 116 (markers: hsa-miR-4783-3p and hsa-miR-4750-5p);
 (113-3) a combination of SEQ ID NOs: 237 and 116 (markers: hsa-miR-602 and hsa-miR-4750-5p);
 (113-4) a combination of SEQ ID NOs: 4 and 116 (markers: hsa-miR-4634 and hsa-miR-4750-5p);
 (113-5) a combination of SEQ ID NOs: 3 and 116 (markers: hsa-miR-1307-3p and hsa-miR-4750-5p);
 (114-1) a combination of SEQ ID NOs: 2 and 160 (markers: hsa-miR-4730 and hsa-miR-6732-5p);
 (114-2) a combination of SEQ ID NOs: 1 and 160 (markers: hsa-miR-4783-3p and hsa-miR-6732-5p);
 (114-3) a combination of SEQ ID NOs: 237 and 160 (markers: hsa-miR-602 and hsa-miR-6732-5p);
 (114-4) a combination of SEQ ID NOs: 4 and 160 (markers: hsa-miR-4634 and hsa-miR-6732-5p);
 (114-5) a combination of SEQ ID NOs: 3 and 160 (markers: hsa-miR-1307-3p and hsa-miR-6732-5p);
 (115-1) a combination of SEQ ID NOs: 2 and 38 (markers: hsa-miR-4730 and hsa-miR-149-3p);
 (115-2) a combination of SEQ ID NOs: 1 and 38 (markers: hsa-miR-4783-3p and hsa-miR-149-3p);

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(115-3) a combination of SEQ ID NOs: 237 and 38 (markers: hsa-miR-602 and hsa-miR-149-3p);
 (115-4) a combination of SEQ ID NOs: 4 and 38 (markers: hsa-miR-4634 and hsa-miR-149-3p);
 (115-5) a combination of SEQ ID NOs: 3 and 38 (markers: hsa-miR-1307-3p and hsa-miR-149-3p);
 (116-1) a combination of SEQ ID NOs: 2 and 99 (markers: hsa-miR-4730 and hsa-miR-4497);
 (116-2) a combination of SEQ ID NOs: 1 and 99 (markers: hsa-miR-4783-3p and hsa-miR-4497);
 (116-3) a combination of SEQ ID NOs: 237 and 99 (markers: hsa-miR-602 and hsa-miR-4497);
 (116-4) a combination of SEQ ID NOs: 4 and 99 (markers: hsa-miR-4634 and hsa-miR-4497);
 (116-5) a combination of SEQ ID NOs: 3 and 99 (markers: hsa-miR-1307-3p and hsa-miR-4497);
 (117-1) a combination of SEQ ID NOs: 2 and 104 (markers: hsa-miR-4730 and hsa-miR-6848-5p);
 (117-2) a combination of SEQ ID NOs: 1 and 104 (markers: hsa-miR-4783-3p and hsa-miR-6848-5p);
 (117-3) a combination of SEQ ID NOs: 237 and 104 (markers: hsa-miR-602 and hsa-miR-6848-5p);
 (117-4) a combination of SEQ ID NOs: 4 and 104 (markers: hsa-miR-4634 and hsa-miR-6848-5p); and
 (117-5) a combination of SEQ ID NOs: 3 and 104 (markers: hsa-miR-1307-3p and hsa-miR-6848-5p).

The kit or the device of the present invention may also comprise a polynucleotide(s) that is/are already known or that will be found in the future, to enable detection of breast cancer, in addition to the polynucleotide(s) (which may include variant(s), fragment(s), or derivative(s)) according to the present invention

The kit of the present invention may also comprise an antibody for measuring marker(s) for breast cancer examination known in the art, such as CEA, CA-15-3, and CA27-29, in addition to the polynucleotide(s) as described above.

The polynucleotides described above contained in the kit of the present invention may be packaged in different containers either individually or in any combination.

The kit of the present invention may comprise a kit for extracting nucleic acids (e.g., total RNA) from body fluids, cells, or tissues, a fluorescent material for labeling, an enzyme and a medium for nucleic acid amplification, an instruction manual, etc.

The device of the present invention is a device for cancer marker measurement in which nucleic acids such as the polynucleotides according to the present invention described above, variants thereof, derivatives thereof, or fragments thereof are bound or attached to, for example, a solid phase. Examples of the material for the solid phase include plastics, paper, glass, and silicon. The material for the solid phase is preferably a plastic from the viewpoint of easy processability. The solid phase has any shape and is, for example, square, round, reed-shaped, or film-shaped. The device of the present invention includes, for example, a device for measurement by a hybridization technique. Specific examples thereof include blotting devices and nucleic acid arrays (e.g., microarrays, DNA chips, and RNA chips).

The nucleic acid array technique is a technique which involves binding or attaching the nucleic acids one by one by use of a method [e.g., a method of spotting the nucleic acids using a high-density dispenser called spotter or arrayer onto the surface of the solid phase surface-treated, if necessary, by coating with L-lysine or the introduction of a functional group such as an amino group or a carboxyl group; a method of spraying the nucleic acids onto the solid phase using an

inkjet which injects very small liquid droplets by a piezo-electric element or the like from a nozzle; or a method of sequentially synthesizing nucleotides on the solid phase] to prepare an array such as a chip and measuring a target nucleic acid through the use of hybridization using this array.

The kit or the device of the present invention comprises nucleic acids capable of specifically binding to the polynucleotides of at least one, preferably at least two, more preferably at least three, most preferably at least five to all of the breast cancer marker miRNAs, respectively, of the group 1 described above. The kit or the device of the present invention may optionally further comprise nucleic acids capable of specifically binding to the polynucleotides of at least one, preferably at least two, more preferably at least three, most preferably at least five to all of the breast cancer marker miRNAs, respectively, of the group 2 described above. The kit or the device of the present invention may optionally further comprise nucleic acids capable of specifically binding to the polynucleotides of at least one, preferably at least two, more preferably at least three, most preferably at least five to all of the breast cancer marker miRNAs, respectively, of the group 3 described above. The kit or the device of the present invention can be used for detecting breast cancer as described in Section 4 below.

4. Method for Detecting Breast Cancer

The present invention further provides a method for detecting breast cancer, comprising using the kit or the device of the present invention (comprising the aforementioned nucleic acid(s) that can be used in the present invention) as described in Section “3. Kit or device for detection of breast cancer” to measure an expression level of one or more breast cancer-derived genes represented by: expression level(s) of breast cancer-derived genes selected from the following group of miRNAs, i.e., miR-4783-3p, miR-4730, miR-1307-3p, miR-4634, miR-663a, miR-4532, miR-7704, miR-3178, miR-6729-5p, miR-6090, miR-4732-5p, miR-3184-5p, miR-6727-5p, miR-6088, miR-4674, miR-8073, miR-4787-5p, miR-1469, miR-125a-3p, miR-1233-5p, miR-885-3p, miR-6802-5p, miR-328-5p, miR-6787-5p, miR-8069, miR-6875-5p, miR-1246, miR-4734, miR-6757-5p, miR-6756-5p, miR-3665, miR-6836-3p, miR-6821-5p, miR-6805-5p, miR-4728-5p, miR-6726-5p, miR-197-5p, miR-149-3p, miR-6850-5p, miR-4476, miR-6858-5p, miR-564, miR-4763-3p, miR-575, miR-6771-5p, miR-1231, miR-1908-3p, miR-150-3p, miR-3937, miR-887-3p, miR-3940-5p, miR-4741, miR-6808-5p, miR-6869-5p, miR-5090, miR-615-5p, miR-8072, miR-128-1-5p, miR-1238-5p, miR-365a-5p, miR-204-3p, miR-4492, miR-6785-5p, miR-6511a-5p, miR-4525, miR-1915-5p, miR-3180, miR-6879-5p, miR-1199-5p, miR-6746-5p, miR-711, miR-663b, miR-4707-3p, miR-6893-5p, miR-4675, miR-4638-5p, miR-4651, miR-6087, miR-4665-5p, miR-4758-5p, miR-6887-5p, miR-3620-5p, miR-1909-3p, miR-7641, miR-6724-5p, miR-1343-3p, miR-6780b-5p, miR-4484, miR-4690-5p, miR-4429, miR-1227-5p, miR-4725-3p, miR-6861-5p, miR-6812-5p, miR-3197, miR-8059, miR-3185, miR-4706, miR-4497, miR-3131, miR-6806-5p, miR-187-5p, miR-3180-3p, miR-6848-5p, miR-6820-5p, miR-6800-5p, miR-6717-5p, miR-6795-5p, miR-4632-5p, miR-665, miR-6778-5p, miR-3663-3p, miR-4689, miR-211-3p, miR-6511b-5p, miR-4750-5p, miR-6126, miR-614, miR-7110-5p, miR-744-5p, miR-6769a-5p, miR-4792, miR-5787, miR-6798-5p, miR-6781-5p, miR-4419b, miR-4446-3p, miR-4259, miR-5572, miR-6075, miR-296-3p, miR-6891-5p, miR-4745-5p, miR-6775-5p, miR-6870-5p, miR-920, miR-4530, miR-6819-5p, miR-6825-5p, miR-7847-3p,

miR-6131, miR-4433-3p, miR-1228-5p, miR-6743-5p, miR-1268a, miR-3917, miR-6786-5p, miR-3154, miR-638, miR-6741-5p, miR-6889-5p, miR-6840-3p, miR-6510-5p, miR-3188, miR-551b-5p, miR-5001-5p, miR-1268b, miR-7107-5p, miR-6824-5p, miR-6732-5p, miR-371a-5p, miR-6794-5p, miR-6779-5p, miR-4271, miR-5195-3p, miR-6762-5p, miR-939-5p, miR-1247-3p, miR-6777-5p, miR-6722-3p, miR-3656, miR-4688, miR-3195, miR-6766-5p, miR-4447, miR-4656, miR-7108-5p, miR-3191-3p, miR-1273g-3p, miR-4463, miR-2861, miR-3196, miR-6877-5p, miR-3679-5p, miR-4442, miR-6789-5p, miR-6782-5p, miR-486-3p, miR-6085, miR-4746-3p, miR-619-5p, miR-937-5p, miR-6803-5p, miR-4298, miR-4454, miR-4459, miR-7150, miR-6880-5p, miR-4449, miR-8063, miR-4695-5p, miR-6132, miR-6829-5p, miR-4486, miR-6805-3p, miR-6826-5p, miR-4508, miR-1343-5p, miR-7114-5p, miR-3622a-5p, miR-6765-5p, miR-7845-5p, miR-3960, miR-6749-5p, miR-1260b, miR-6799-5p, miR-4723-5p, miR-6784-5p, miR-5100, miR-6769b-5p, miR-1207-5p, miR-642a-3p, miR-4505, miR-4270, miR-6721-5p, miR-7111-5p, miR-6791-5p, miR-7109-5p, miR-4258, miR-6515-3p, miR-6851-5p, miR-6125, miR-4749-5p, miR-4726-5p, miR-4513, miR-6089, miR-6816-5p, miR-4466, miR-4488, miR-6752-5p and miR-4739; and optionally expression level(s) of breast cancer-derived gene(s) selected from the following group of miRNA: i.e., miR-760, miR-602, miR-423-5p, miR-92a-2-5p, miR-16-5p, miR-451a, miR-135a-3p, miR-486-5p, miR-4257, miR-92b-5p, miR-1915-3p, miR-718, miR-940, miR-296-5p, miR-23b-3p and miR-92a-3p, and optionally expression level(s) of breast cancer-derived gene(s) selected from the following group of miRNA: i.e., miR-658, miR-6842-5p, miR-6124, miR-6765-3p, miR-7106-5p, miR-4534, miR-92b-3p, miR-3135b, miR-4687-3p, miR-762, miR-3619-3p, miR-4467, miR-557, miR-1237-5p, miR-1908-5p, miR-4286, miR-6885-5p, and miR-6763-5p in a sample in vitro, further comparing, for example, the expression level(s) of the aforementioned gene(s) in the sample (e.g., blood, serum, or plasma) collected from a subject suspected of having breast cancer with a control expression level in the sample collected from a healthy subject (including a non-breast cancer patient(s)), and evaluating the subject as having breast cancer when the expression level(s) of the target nucleic acid(s) is different between the samples.

This method of the present invention enables limitedly-invasive early diagnosis of the breast cancer with high sensitivity and specificity and thereby brings about early treatment and improved prognosis. In addition, exacerbation of the disease or the effectiveness of surgical, radiotherapeutic, and chemotherapeutic treatments can be monitored.

The method for extracting the breast cancer-derived gene from the sample such as blood, serum, or plasma according to the present invention is particularly preferably a method in which the breast cancer-derived gene is prepared by the addition of a reagent for RNA extraction in 3D-Gener™ RNA extraction reagent from liquid sample kit (Toray Industries, Inc.). A general acidic phenol method (acid guanidinium-phenol-chloroform (AGPC)) may be used, or Trizol™ (Life Technologies Corp.) may be used. The breast cancer-derived gene(s) may be prepared by the addition of a reagent for RNA extraction containing acidic phenol, such as Trizol (Life Technologies Corp.) or Isogen (Nippon Gene Co., Ltd., Japan). Alternatively, a kit such as miRNeasy® Mini Kit (Qiagen N.V.) can be used, though the method is not limited thereto.

The present invention also provides use of the kit or the device of the present invention for detecting in vitro an

expression product(s) of a breast cancer-derived miRNA gene in a sample derived from a subject.

In the method of the present invention, the used kit or the device comprises a single polynucleotide or any possible combination of polynucleotides that can be used in the present invention as described above.

In the detection or (genetic) diagnosis of breast cancer according to the present invention, each polynucleotide contained in the kit or the device of the present invention can be used as a probe or a primer. In the case of using the polynucleotide as a primer, TaqMan™ MicroRNA Assays from Life Technologies Corp., miScript PCR System from Qiagen N.V., or the like can be used, though the method is not limited thereto.

The polynucleotide contained in the kit or the device of the present invention can be used as a primer or a probe according to a routine method in a method known in the art for specifically detecting the particular gene, for example, a hybridization technique such as Northern blot, Southern blot, in situ hybridization, Northern hybridization, or Southern hybridization, or a quantitative amplification technique such as quantitative RT-PCR. A body fluid such as blood, serum, plasma, or urine from a subject is collected as a sample to be assayed according to the type of the detection method used. Alternatively, total RNA prepared from such a body fluid by the method described above may be used, and various polynucleotides including cDNA prepared on the basis of the RNA may be used.

The kit or the device of the present invention is useful for the diagnosis of breast or the detection of the presence or absence of breast cancer. Specifically, the detection of breast cancer using the kit or the device can be performed by detecting in vitro an expression level of a gene using the nucleic acid probe or the primer contained in the kit or the device in a sample such as blood, serum, plasma, or urine from a subject suspected of having breast cancer. The subject suspected of having breast cancer can be evaluated as having breast cancer when the expression level(s) of a target miRNA marker(s) measured using a polynucleotide(s) (including variant(s), fragment(s), or derivative(s) thereof) consisting of a nucleotide sequence(s) represented by at least one of SEQ ID NOs: 1 to 235, and 851 to 856 or a complementary sequence(s) thereof, and optionally a nucleotide sequence(s) represented by one or more of SEQ ID NOs: 236 to 251 or a complementary sequence(s) thereof, and optionally a nucleotide sequence(s) represented by one or more of SEQ ID NOs: 252 to 269 or a complementary sequence thereof in the sample such as blood, serum, plasma, or urine of the subject has a statistically significantly higher or lower than the expression level(s) thereof in the sample such as blood, serum, or plasma, or urine of a healthy subject.

The method of the present invention can be combined with a diagnostic imaging method such as mammography, ultrasonography (echo examination), CT, MRI, abdominal ultrasonography, bone scintigraphy, or PET, or pathological examination which involves analyzing a lesion tissue under a microscope. The method of the present invention is capable of specifically detecting breast cancer and can substantially discriminate breast cancer from the other cancers.

The method for detecting the absence of an expression product(s) of a breast cancer-derived gene(s) or the presence of the expression product(s) of a breast cancer-derived gene in a sample using the kit or the device of the present invention comprises; collecting a body fluid such as blood, serum, plasma, or urine of a subject; measuring the expres-

sion level(s) of the target gene(s) contained therein using one or more polynucleotides (including variant(s), fragment(s), or derivative(s)) selected from the polynucleotide group of the present invention; and evaluating the presence or absence of breast cancer or to detect breast cancer. The method for detecting breast cancer according to the present invention can also evaluate or diagnose, for example, the presence or absence of amelioration of the disease or the degree of amelioration thereof in a breast cancer patient in the case that a therapeutic drug is administered to the patient for amelioration of the disease.

The method of the present invention can comprise, for example, the following steps (a), (b), and (c):

- (a) a step of contacting in vitro a sample from a subject with a polynucleotide(s) contained in the kit or the device of the present invention in vitro;
- (b) a step of measuring an expression level(s) of the target nucleic acid(s) in the sample using the polynucleotide(s) as a nucleic acid probe(s) or a primer(s); and
- (c) a step of evaluating the presence or absence of breast cancer (cells) in the subject on the basis of the measurement results in the step (b).

Specifically, the present invention provides a method for detecting breast cancer, comprising measuring an expression level(s) of a target nucleic acid(s) in a sample of a subject using nucleic acid(s) capable of specifically binding to at least one (preferably at least two) polynucleotides selected from the group consisting of miR-4783-3p, miR-4730, miR-1307-3p, miR-4634, miR-663a, miR-4532, miR-7704, miR-3178, miR-6729-5p, miR-6090, miR-4732-5p, miR-3184-5p, miR-6727-5p, miR-6088, miR-4674, miR-8073, miR-4787-5p, miR-1469, miR-125a-3p, miR-1233-5p, miR-885-3p, miR-6802-5p, miR-328-5p, miR-6787-5p, miR-8069, miR-6875-5p, miR-1246, miR-4734, miR-6757-5p, miR-6756-5p, miR-3665, miR-6836-3p, miR-6821-5p, miR-6805-5p, miR-4728-5p, miR-6726-5p, miR-197-5p, miR-149-3p, miR-6850-5p, miR-4476, miR-6858-5p, miR-564, miR-4763-3p, miR-575, miR-6771-5p, miR-1231, miR-1908-3p, miR-150-3p, miR-3937, miR-887-3p, miR-3940-5p, miR-4741, miR-6808-5p, miR-6869-5p, miR-5090, miR-615-5p, miR-8072, miR-128-1-5p, miR-1238-5p, miR-365a-5p, miR-204-3p, miR-4492, miR-6785-5p, miR-6511a-5p, miR-4525, miR-1915-5p, miR-3180, miR-6879-5p, miR-1199-5p, miR-6746-5p, miR-711, miR-663b, miR-4707-3p, miR-6893-5p, miR-4675, miR-4638-5p, miR-4651, miR-6087, miR-4665-5p, miR-4758-5p, miR-6887-5p, miR-3620-5p, miR-1909-3p, miR-7641, miR-6724-5p, miR-1343-3p, miR-6780b-5p, miR-4484, miR-4690-5p, miR-4429, miR-1227-5p, miR-4725-3p, miR-6861-5p, miR-6812-5p, miR-3197, miR-8059, miR-3185, miR-4706, miR-4497, miR-3131, miR-6806-5p, miR-187-5p, miR-3180-3p, miR-6848-5p, miR-6820-5p, miR-6800-5p, miR-6717-5p, miR-6795-5p, miR-4632-5p, miR-665, miR-6778-5p, miR-3663-3p, miR-4689, miR-211-3p, miR-6511b-5p, miR-4750-5p, miR-6126, miR-614, miR-7110-5p, miR-744-5p, miR-6769a-5p, miR-4792, miR-5787, miR-6798-5p, miR-6781-5p, miR-4419b, miR-4446-3p, miR-4259, miR-5572, miR-6075, miR-296-3p, miR-6891-5p, miR-4745-5p, miR-6775-5p, miR-6870-5p, miR-920, miR-4530, miR-6819-5p, miR-6825-5p, miR-7847-3p, miR-6131, miR-4433-3p, miR-1228-5p, miR-6743-5p, miR-1268a, miR-3917, miR-6786-5p, miR-3154, miR-638, miR-6741-5p, miR-6889-5p, miR-6840-3p, miR-6510-5p, miR-3188, miR-551b-5p, miR-5001-5p, miR-1268b, miR-7107-5p, miR-6824-5p, miR-6732-5p, miR-371a-5p, miR-6794-5p, miR-6779-5p, miR-4271, miR-5195-3p, miR-6762-5p, miR-939-5p, miR-1247-3p, miR-6777-5p, miR-6722-3p,

miR-3656, miR-4688, miR-3195, miR-6766-5p, miR-4447, miR-4656, miR-7108-5p, miR-3191-3p, miR-1273g-3p, miR-4463, miR-2861, miR-3196, miR-6877-5p, miR-3679-5p, miR-4442, miR-6789-5p, miR-6782-5p, miR-486-3p, miR-6085, miR-4746-3p, miR-619-5p, miR-937-5p, miR-6803-5p, miR-4298, miR-4454, miR-4459, miR-7150, miR-6880-5p, miR-4449, miR-8063, miR-4695-5p, miR-6132, miR-6829-5p, miR-4486, miR-6805-3p, miR-6826-5p, miR-4508, miR-1343-5p, miR-7114-5p, miR-3622a-5p, miR-6765-5p, miR-7845-5p, miR-3960, miR-6749-5p, miR-1260b, miR-6799-5p, miR-4723-5p, miR-6784-5p, miR-5100, miR-6769b-5p, miR-1207-5p, miR-642a-3p, miR-4505, miR-4270, miR-6721-5p, miR-7111-5p, miR-6791-5p, miR-7109-5p, miR-4258, miR-6515-3p, miR-6851-5p, miR-6125, miR-4749-5p, miR-4726-5p, miR-4513, miR-6089, miR-6816-5p, miR-4466, miR-4488, miR-6752-5p and miR-4739 and evaluating in vitro whether or not the subject has breast cancer in the subject using the above-measured expression levels and control expression levels of healthy subjects measured in the same way as above.

The term “evaluation” used herein is evaluation support based on results of in vitro examination, not physician’s judgment.

As described above, in a preferred embodiment of the method of the present invention, specifically, miR-4783-3p is hsa-miR-4783-3p, miR-4730 is hsa-miR-4730, miR-1307-3p is hsa-miR-1307-3p, miR-4634 is hsa-miR-4634, miR-663a is hsa-miR-663a, miR-4532 is hsa-miR-4532, miR-7704 is hsa-miR-7704, miR-3178 is hsa-miR-3178, miR-6729-5p is hsa-miR-6729-5p, miR-6090 is hsa-miR-6090, miR-4732-5p is hsa-miR-4732-5p, miR-3184-5p is hsa-miR-3184-5p, miR-6727-5p is hsa-miR-6727-5p, miR-6088 is hsa-miR-6088, miR-4674 is hsa-miR-4674, miR-8073 is hsa-miR-8073, miR-4787-5p is hsa-miR-4787-5p, miR-1469 is hsa-miR-1469, miR-125a-3p is hsa-miR-125a-3p, miR-1233-5p is hsa-miR-1233-5p, miR-885-3p is hsa-miR-885-3p, miR-6802-5p is hsa-miR-6802-5p, miR-328-5p is hsa-miR-328-5p, miR-6787-5p is hsa-miR-6787-5p, miR-8069 is hsa-miR-8069, miR-6875-5p is hsa-miR-6875-5p, miR-1246 is hsa-miR-1246, miR-4734 is hsa-miR-4734, miR-6757-5p is hsa-miR-6757-5p, miR-6756-5p is hsa-miR-6756-5p, miR-3665 is hsa-miR-3665, miR-6836-3p is hsa-miR-6836-3p, miR-6821-5p is hsa-miR-6821-5p, miR-6805-5p is hsa-miR-6805-5p, miR-4728-5p is hsa-miR-4728-5p, miR-6726-5p is hsa-miR-6726-5p, miR-197-5p is hsa-miR-197-5p, miR-149-3p is hsa-miR-149-3p, miR-6850-5p is hsa-miR-6850-5p, miR-4476 is hsa-miR-4476, miR-6858-5p is hsa-miR-6858-5p, miR-564 is hsa-miR-564, miR-4763-3p is hsa-miR-4763-3p, miR-575 is hsa-miR-575, miR-6771-5p is hsa-miR-6771-5p, miR-1231 is hsa-miR-1231, miR-1908-3p is hsa-miR-1908-3p, miR-150-3p is hsa-miR-150-3p, miR-3937 is hsa-miR-3937, miR-887-3p is hsa-miR-887-3p, miR-3940-5p is hsa-miR-3940-5p, miR-4741 is hsa-miR-4741, miR-6808-5p is hsa-miR-6808-5p, miR-6869-5p is hsa-miR-6869-5p, miR-5090 is hsa-miR-5090, miR-615-5p is hsa-miR-615-5p, miR-8072 is hsa-miR-8072, miR-128-1-5p is hsa-miR-128-1-5p, miR-1238-5p is hsa-miR-1238-5p, miR-365a-5p is hsa-miR-365a-5p, miR-204-3p is hsa-miR-204-3p, miR-4492 is hsa-miR-4492, miR-6785-5p is hsa-miR-6785-5p, miR-6511a-5p is hsa-miR-6511a-5p, miR-4525 is hsa-miR-4525, miR-1915-5p is hsa-miR-1915-5p, miR-3180 is hsa-miR-3180, miR-6879-5p is hsa-miR-6879-5p, miR-1199-5p is hsa-miR-1199-5p, miR-6746-5p is hsa-miR-6746-5p, miR-711 is hsa-miR-711, miR-663b is hsa-miR-663b, miR-4707-3p is hsa-miR-4707-3p, miR-6893-5p is hsa-miR-6893-5p,

miR-4675 is hsa-miR-4675, miR-4638-5p is hsa-miR-4638-5p, miR-4651 is hsa-miR-4651, miR-6087 is hsa-miR-6087, miR-4665-5p is hsa-miR-4665-5p, miR-4758-5p is hsa-miR-4758-5p, miR-6887-5p is hsa-miR-6887-5p, miR-3620-5p is hsa-miR-3620-5p, miR-1909-3p is hsa-miR-1909-3p, miR-7641 is hsa-miR-7641, miR-6724-5p is hsa-miR-6724-5p, miR-1343-3p is hsa-miR-1343-3p, miR-6780b-5p is hsa-miR-6780b-5p, miR-4484 is hsa-miR-4484, miR-4690-5p is hsa-miR-4690-5p, miR-4429 is hsa-miR-4429, miR-1227-5p is hsa-miR-1227-5p, miR-4725-3p is hsa-miR-4725-3p, miR-6861-5p is hsa-miR-6861-5p, miR-6812-5p is hsa-miR-6812-5p, miR-3197 is hsa-miR-3197, miR-8059 is hsa-miR-8059, miR-3185 is hsa-miR-3185, miR-4706 is hsa-miR-4706, miR-4497 is hsa-miR-4497, miR-3131 is hsa-miR-3131, miR-6806-5p is hsa-miR-6806-5p, miR-187-5p is hsa-miR-187-5p, miR-3180-3p is hsa-miR-3180-3p, miR-6848-5p is hsa-miR-6848-5p, miR-6820-5p is hsa-miR-6820-5p, miR-6800-5p is hsa-miR-6800-5p, miR-6717-5p is hsa-miR-6717-5p, miR-6795-5p is hsa-miR-6795-5p, miR-4632-5p is hsa-miR-4632-5p, miR-665 is hsa-miR-665, miR-6778-5p is hsa-miR-6778-5p, miR-3663-3p is hsa-miR-3663-3p, miR-4689 is hsa-miR-4689, miR-211-3p is hsa-miR-211-3p, miR-6511b-5p is hsa-miR-6511b-5p, miR-4750-5p is hsa-miR-4750-5p, miR-6126 is hsa-miR-6126, miR-614 is hsa-miR-614, miR-7110-5p is hsa-miR-7110-5p, miR-744-5p is hsa-miR-744-5p, miR-6769a-5p is hsa-miR-6769a-5p, miR-4792 is hsa-miR-4792, miR-5787 is hsa-miR-5787, miR-6798-5p is hsa-miR-6798-5p, miR-6781-5p is hsa-miR-6781-5p, miR-4419b is hsa-miR-4419b, miR-4446-3p is hsa-miR-4446-3p, miR-4259 is hsa-miR-4259, miR-5572 is hsa-miR-5572, miR-6075 is hsa-miR-6075, miR-296-3p is hsa-miR-296-3p, miR-6891-5p is hsa-miR-6891-5p, miR-4745-5p is hsa-miR-4745-5p, miR-6775-5p is hsa-miR-6775-5p, miR-6870-5p is hsa-miR-6870-5p, miR-920 is hsa-miR-920, miR-4530 is hsa-miR-4530, miR-6819-5p is hsa-miR-6819-5p, miR-6825-5p is hsa-miR-6825-5p, miR-7847-3p is hsa-miR-7847-3p, miR-6131 is hsa-miR-6131, miR-4433-3p is hsa-miR-4433-3p, miR-1228-5p is hsa-miR-1228-5p, miR-6743-5p is hsa-miR-6743-5p, miR-1268a is hsa-miR-1268a, miR-3917 is hsa-miR-3917, miR-6786-5p is hsa-miR-6786-5p, miR-3154 is hsa-miR-3154, miR-638 is hsa-miR-638, miR-6741-5p is hsa-miR-6741-5p, miR-6889-5p is hsa-miR-6889-5p, miR-6840-3p is hsa-miR-6840-3p, miR-6510-5p is hsa-miR-6510-5p, miR-3188 is hsa-miR-3188, miR-551b-5p is hsa-miR-551b-5p, miR-5001-5p is hsa-miR-5001-5p, miR-1268b is hsa-miR-1268b, miR-7107-5p is hsa-miR-7107-5p, miR-6824-5p is hsa-miR-6824-5p, miR-6732-5p is hsa-miR-6732-5p, miR-371a-5p is hsa-miR-371a-5p, miR-6794-5p is hsa-miR-6794-5p, miR-6779-5p is hsa-miR-6779-5p, miR-4271 is hsa-miR-4271, miR-5195-3p is hsa-miR-5195-3p, miR-6762-5p is hsa-miR-6762-5p, miR-939-5p is hsa-miR-939-5p, miR-1247-3p is hsa-miR-1247-3p, miR-6777-5p is hsa-miR-6777-5p, miR-6722-3p is hsa-miR-6722-3p, miR-3656 is hsa-miR-3656, miR-4688 is hsa-miR-4688, miR-3195 is hsa-miR-3195, miR-6766-5p is hsa-miR-6766-5p, miR-4447 is hsa-miR-4447, miR-4656 is hsa-miR-4656, miR-7108-5p is hsa-miR-7108-5p, miR-3191-3p is hsa-miR-3191-3p, miR-1273g-3p is hsa-miR-1273g-3p, miR-4463 is hsa-miR-4463, miR-2861 is hsa-miR-2861, miR-3196 is hsa-miR-3196, miR-6877-5p is hsa-miR-6877-5p, miR-3679-5p is hsa-miR-3679-5p, miR-4442 is hsa-miR-4442, miR-6789-5p is hsa-miR-6789-5p, miR-6782-5p is hsa-miR-6782-5p, miR-486-3p is hsa-miR-486-3p, miR-6085 is hsa-miR-6085, miR-4746-3p is hsa-miR-4746-3p, miR-619-5p is hsa-miR-619-5p, miR-937-5p is hsa-miR-937-5p, miR-6803-5p is

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hsa-miR-6803-5p, miR-4298 is hsa-miR-4298, miR-4454 is hsa-miR-4454, miR-4459 is hsa-miR-4459, miR-7150 is hsa-miR-7150, miR-6880-5p is hsa-miR-6880-5p, miR-4449 is hsa-miR-4449, miR-8063 is hsa-miR-8063, miR-4695-5p is hsa-miR-4695-5p, miR-6132 is hsa-miR-6132, miR-6829-5p is hsa-miR-6829-5p, miR-4486 is hsa-miR-4486, miR-6805-3p is hsa-miR-6805-3p, miR-6826-5p is hsa-miR-6826-5p, miR-4508 is hsa-miR-4508, miR-1343-5p is hsa-miR-1343-5p, miR-7114-5p is hsa-miR-7114-5p, miR-3622a-5p is hsa-miR-3622a-5p, miR-6765-5p is hsa-miR-6765-5p, miR-7845-5p is hsa-miR-7845-5p, miR-3960 is hsa-miR-3960, miR-6749-5p is hsa-miR-6749-5p, miR-1260b is hsa-miR-1260b, miR-6799-5p is hsa-miR-6799-5p, miR-4723-5p is hsa-miR-4723-5p, miR-6784-5p is hsa-miR-6784-5p, miR-5100 is hsa-miR-5100, miR-6769b-5p is hsa-miR-6769b-5p, miR-1207-5p is hsa-miR-1207-5p, miR-642a-3p is hsa-miR-642a-3p, miR-4505 is hsa-miR-4505, miR-4270 is hsa-miR-4270, miR-6721-5p is hsa-miR-6721-5p, miR-7111-5p is hsa-miR-7111-5p, miR-6791-5p is hsa-miR-6791-5p, miR-7109-5p is hsa-miR-7109-5p, miR-4258 is hsa-miR-4258, miR-6515-3p is hsa-miR-6515-3p, miR-6851-5p is hsa-miR-6851-5p, miR-6125 is hsa-miR-6125, miR-4749-5p is hsa-miR-4749-5p, miR-4726-5p is hsa-miR-4726-5p, miR-4513 is hsa-miR-4513, miR-6089 is hsa-miR-6089, miR-6816-5p is hsa-miR-6816-5p, miR-4466 is hsa-miR-4466, miR-4488 is hsa-miR-4488, miR-6752-5p is hsa-miR-6752-5p, and miR-4739 is hsa-miR-4739.

In a preferred embodiment of the method of the present invention, specifically, the nucleic acid (specifically, probe (s) or primer(s)) is selected from the group consisting of the following polynucleotides (a) to (e):

- (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (b) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856,
- (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

The method of the present invention can further employ nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of miR-760, miR-602, miR-423-5p, miR-92a-2-5p, miR-16-5p, miR-451a, miR-135a-3p, miR-486-5p, miR-4257, miR-92b-5p, miR-1915-3p, miR-718, miR-940, miR-296-5p, miR-23b-3p and miR-92a-3p.

As for such a nucleic acid, specifically, miR-760 is hsa-miR-760, miR-602 is hsa-miR-602, miR-423-5p is hsa-miR-423-5p, miR-92a-2-5p is hsa-miR-92a-2-5p, miR-16-5p is hsa-miR-16-5p, miR-451a is hsa-miR-451a, miR-135a-3p is hsa-miR-135a-3p, miR-486-5p is hsa-miR-486-

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5p, miR-4257 is hsa-miR-4257, miR-92b-5p is hsa-miR-92b-5p, miR-1915-3p is hsa-miR-1915-3p, miR-718 is hsa-miR-718, miR-940 is hsa-miR-940, miR-296-5p is hsa-miR-296-5p, miR-23b-3p is hsa-miR-23b-3p, and miR-92a-3p is hsa-miR-92a-3p.

In a preferred embodiment, such a nucleic acid is specifically selected from the group consisting of the following polynucleotides (f) to (j):

- (f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251,
- (h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and
- (j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

The nucleic acid(s) further used in the method of the present invention can comprise nucleic acid(s) capable of specifically binding to at least one polynucleotide selected from the group consisting of miR-658, miR-6842-5p, miR-6124, miR-6765-3p, miR-7106-5p, miR-4534, miR-92b-3p, miR-3135b, miR-4687-3p, miR-762, miR-3619-3p, miR-4467, miR-557, miR-1237-5p, miR-1908-5p, miR-4286, miR-6885-5p and miR-6763-5p.

Specifically, miR-658 is hsa-miR-658, miR-6842-5p is hsa-miR-6842-5p, miR-6124 is hsa-miR-6124, miR-6765-3p is hsa-miR-6765-3p, miR-7106-5p is hsa-miR-7106-5p, miR-4534 is hsa-miR-4534, miR-92b-3p is hsa-miR-92b-3p, miR-3135b is hsa-miR-3135b, miR-4687-3p is hsa-miR-4687-3p, miR-762 is hsa-miR-762, miR-3619-3p is hsa-miR-3619-3p, miR-4467 is hsa-miR-4467, miR-557 is hsa-miR-557, miR-1237-5p is hsa-miR-1237-5p, miR-1908-5p is hsa-miR-1908-5p, miR-4286 is hsa-miR-4286, miR-6885-5p is hsa-miR-6885-5p, and miR-6763-5p is hsa-miR-6763-5p.

Further, in a preferred embodiment, such nucleic acid(s) is/are specifically a polynucleotide selected from the group consisting of the following polynucleotides (k) to (o):

- (k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,
- (l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269,
- (m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides,

(n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and

(o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).

Examples of the sample used in the method of the present invention can include samples prepared from living tissues (preferably breast tissues) or body fluids such as blood, serum, plasma, and urine from the subjects. Specifically, for example, an RNA-containing sample prepared from the tissue, a polynucleotide-containing sample further prepared therefrom, a body fluid such as blood, serum, plasma, or urine, a portion or the whole of a living tissue collected from the subject by biopsy or the like, or a living tissue excised by surgery can be used, and the sample for measurement can be prepared therefrom.

The subject used herein refers to a mammal, for example, a human, a monkey, a mouse or a rat, and is preferably a human.

The steps of the method of the present invention can be changed according to the type of the sample to be assayed.

In the case of using RNA as an analyte, the detection of breast cancer (cells) can comprise, for example, the following steps (a), (b), and (c):

(a) a step of binding RNA prepared from a sample from a subject or complementary polynucleotides (cDNAs) transcribed from the RNA to a polynucleotide(s) in the kit or the device of the present invention;

(b) a step of measuring the sample-derived RNA or the cDNAs synthesized from the RNA, which is/are bound to the polynucleotide(s) by hybridization using the polynucleotide(s) as nucleic acid probe(s) or by quantitative RT-PCR using the polynucleotide(s) as primer (s); and

(c) a step of evaluating the presence or absence of breast cancer (or breast cancer-derived gene expression) on the basis of the measurement results of the step (b).

For example, various hybridization methods can be used for detecting, examining, evaluating, or diagnosing breast cancer (or breast cancer-derived gene expression) in vitro according to the present invention. For example, Northern blot, Southern blot, RT-PCR, DNA chip analysis, in situ hybridization, Northern hybridization, or Southern hybridization can be used as such a hybridization method.

In the case of using the Northern blot, the presence or absence of expression of each gene or the expression level thereof in the RNA can be detected or measured by use of the nucleic acid probe(s) that can be used in the present invention. Specific examples thereof can include a method which comprises labeling the nucleic acid probe (or a complementary strand) with a radioisotope (^{32}P , ^{33}P , ^{35}S , etc.), a fluorescent material, or the like, that hybridizes the labeled product with the tissue-derived RNA from a subject, which is transferred to a nylon membrane or the like according to a routine method, and then detecting and measuring a signal derived from the label (radioisotope or fluorescent material) on the formed DNA/RNA duplex using a radiation detector (examples thereof can include BAS-1800 II (Fujifilm Corp.)) or a fluorescence detector (examples thereof can include STORM 865 (GE Healthcare Japan Corp.)).

In the case of using the quantitative RT-PCR, the presence or absence of expression of each gene or the expression level thereof in the RNA can be detected or measured by use of the primer that can be used in the present invention. Specific

examples thereof can include a method which comprises; preparing cDNAs from the tissue-derived RNA of a subject according to a routine procedure; hybridizing a pair of primers (consisting of a plus strand and a reverse strand that bind to the cDNA) of the present invention with the cDNA such that the region of each target gene can be amplified with the cDNA as a template; and performing PCR according to a routine method to detect the obtained double-stranded DNA. The method for detecting the double-stranded DNA can include; a method of performing the PCR using the primers labeled in advance with a radioisotope or a fluorescent material; a method of electrophoresing the PCR product on an agarose gel and staining the double-stranded DNA with ethidium bromide or the like for detection; and a method of transferring the produced double-stranded DNA to a nylon membrane or the like according to a routine method and hybridizing the double-stranded DNA to a labeled nucleic acid probe for detection.

In the case of using the nucleic acid array analysis, an RNA chip or a DNA chip in which the nucleic acid probes (single-stranded or double-stranded) are attached to a substrate (solid phase) is used. Regions having the attached nucleic acid probes are referred to as probe spots, and regions having no attached nucleic acid probe are referred to as blank spots. A gene group immobilized on a solid-phase substrate is generally called a nucleic acid chip, a nucleic acid array, a microarray, or the like. The DNA or RNA array includes a DNA or RNA macroarray and a DNA or RNA microarray. The term "chip" used herein includes all of these arrays. 3D-Gene™ Human miRNA Oligo chip (Toray Industries, Inc.) can be used as the DNA chip, though the DNA chip is not limited thereto.

Examples of the measurement using the DNA chip can include, but are not limited to, a method of detecting and measuring a signal derived from the label on the nucleic acid probe using an image detector (examples thereof can include Typhoon 9410 (GE Healthcare) and 3D-Gene™ scanner (Toray Industries, Inc.)).

The "stringent conditions" used herein are, as mentioned above, conditions under which a nucleic acid probe hybridizes to its target sequence to a larger extent (e.g., a measurement value equal to or larger than "(a mean of background measurement values)+(a standard deviation of the background measurement values) $\times 2$ ") than that for other sequences.

The stringent conditions are defined by hybridization and subsequent washing conditions. Examples of the hybridization conditions include, but not limited to 30° C. to 60° C. for 1 to 24 hours in a solution containing SSC, a surfactant, formamide, dextran sulfate, a blocking agent(s), etc. In this context, 1×SSC is an aqueous solution (pH 7.0) containing 150 mM sodium chloride and 15 mM sodium citrate. The surfactant includes, for example, SDS (sodium dodecyl sulfate), Triton, or Tween. The hybridization conditions more preferably comprise 3-10×SSC and 0.1-1% SDS. Examples of the conditions for the washing, following the hybridization, which is another condition to define the stringent conditions, can include conditions comprising continuous washing at 30° C. in a solution containing 0.5×SSC and 0.1% SDS, at 30° C. in a solution containing 0.2×SSC and 0.1% SDS, and at 30° C. in a 0.05×SSC solution. It is desirable that the complementary strand should maintain its hybridized state with a target plus strand even by the washing under such conditions. Specifically, examples of such a complementary strand can include a strand consisting of a nucleotide sequence in a completely complementary relationship with the nucleotide sequence of the target plus

(+) strand, and a strand consisting of a nucleotide sequence having at least 80%, preferably at least 85%, more preferably at least 90% or at least 95%, for example, at least 98% or at least 99% identity to the strand.

Other examples of the “stringent conditions” for the hybridization are described in, for example, Sambrook, J. & Russel, D., *Molecular Cloning, A LABORATORY MANUAL*, Cold Spring Harbor Laboratory Press, published on Jan. 15, 2001, Vol. 1, 7.42 to 7.45 and Vol. 2, 8.9 to 8.17, and can be used in the present invention.

Examples of the conditions for carrying out PCR using polynucleotide fragments contained in the kit of the present invention as primers include treatment for approximately 15 seconds to 1 minute at 5 to 10° C. plus a Tm value calculated from the sequences of the primers, using a PCR buffer having composition such as 10 mM Tris-HCL (pH 8.3), 50 mM KCL, and 1 to 2 mM MgCl₂. Examples of the method for calculating such a Tm value include Tm value=2×(the number of adenine residues+the number of thymine residues)+4×(the number of guanine residues+the number of cytosine residues).

In the case of using the quantitative RT-PCR, a commercially available kit for measurement specially designed for quantitatively measuring miRNA, such as TaqMan™ MicroRNA Assays (Life Technologies Corp.); LNA™-based MicroRNA PCR (Exiqon); or Ncode™ miRNA qRT-PCT kit (Invitrogen Corp.) may be used.

For the calculation of gene expression levels, statistical analysis described in, for example, Statistical analysis of gene expression microarray data (Speed T., Chapman and Hall/CRC), and A beginner's guide Microarray gene expression data analysis (Causton H. C. et al., Blackwell publishing) can be used in the present invention, though the calculation method is not limited thereto. For example, twice, preferably 3 times, more preferably 6 times the standard deviation of the measurement values of the blank spots are added to the average measurement value of the blank spots on the DNA chip, and probe spots having a signal value equal to or larger than the resulting value can be regarded as detection spots. Alternatively, the average measurement value of the blank spots is regarded as a background and can be subtracted from the measurement values of the probe spots to determine gene expression levels. A missing value for a gene expression level can be excluded from the analyte, preferably replaced with the smallest value of the gene expression level in each DNA chip, or more preferably replaced with a value obtained by subtracting 0.1 from a logarithmic value of the smallest value of the gene expression level. In order to eliminate low-signal genes, only a gene that shows a gene expression level of 2⁶, preferably 2⁸, more preferably 2¹⁰, or larger, in 20% or more, preferably 50% or more, more preferably 80% or more of the number of measurement samples can be selected as the analyte. Examples of the normalization of the gene expression level include, but are not limited to, global normalization and quantile normalization (Bolstad, B. M. et al., 2003, *Bioinformatics*, Vol. 19, p. 185-193).

The present invention also provides a method comprising measuring target genes or gene expression levels in a sample from a subject using the polynucleotide, the kit, or the device (e.g., chip) for detection of the present invention, or a combination thereof, preparing a discriminant (discriminant function) with gene expression levels in a sample from a breast cancer patient and a sample from a healthy subject as supervising samples, and determining or evaluating the presence and/or absence of the breast cancer-derived genes in the sample.

Specifically, the present invention further provides the method comprising: a first step of measuring in vitro expression levels of target genes in multiple samples known to determine or evaluate the presence and/or absence of the breast cancer-derived genes in the samples, using the polynucleotides, the kit, or the device (e.g., chip) for detection of the present invention, or a combination thereof; a second step of preparing a discriminant with the measurement values of the expression levels of the target genes (target nucleic acid) obtained in the first step as supervising samples; a third step of measuring in vitro expression levels of the target genes in a sample derived from a subject in the same way as in the first step; and a fourth step of assigning the measurement values of the expression levels of the target genes obtained in the third step to the discriminant obtained in the second step, and determining or evaluating the presence and/or absence of the breast cancer-derived gene in the sample on the basis of the results obtained from the discriminant, wherein the target genes can be detected using the polynucleotides or using polynucleotides for detection contained in the kit or the device (e.g., chip). In this context, the discriminant can be prepared by use of Fisher's discriminant analysis, nonlinear discriminant analysis based on Mahalanobis' distance, neural network, Support Vector Machine (SVM), or the like, though the method is not limited thereto.

When a clustering boundary is a straight line or a hyperplane, the linear discriminant analysis is a method for determining the association of a cluster using Formula 1 as a discriminant. In this formula, x represents an explanatory variable, w represents a coefficient of the explanatory variable, and wo represents a constant term.

$$f(x) = w_0 + \sum_{i=1}^n w_i x_i \quad \text{Formula 1}$$

Values obtained from the discriminant are referred to as discriminant scores. The measurement values of a newly offered data set can be assigned as explanatory variables to the discriminant to determine clusters by the signs of the discriminant scores.

The Fisher's discriminant analysis, one type of linear discriminant analysis, is a dimension reduction method for selecting a dimension suitable for classification, and constructs a highly discriminating synthetic variable by focusing on the variance of synthetic variables and minimizing the variance of data having the same label (Venables, W. N. et al., *Modern Applied Statistics with S*. Fourth edition. Springer., 2002). In the Fisher's discriminant analysis, direction w of projection is determined so as to maximize Formula 2. In this formula, μ represents an average input, ng represents the number of data associated to class g, and μg represents an average input of the data associated to class g. The numerator and the denominator are the interclass variance and the intraclass variance, respectively, when each data is projected in the direction of the vector w. Discriminant coefficient wi is determined by maximizing this ratio (Takafumi Kanamori et al., “Pattern Recognition”, Kyoritsu Shuppan Co., Ltd. (2009); and Richard O. et al., *Pattern Classification Second Edition*., Wiley-Interscience, 2000).

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$$J(w) = \frac{\sum_{g=1}^G n_g (w^T \mu_g - w^T \mu) (w^T \mu_g - w^T \mu)^T}{\sum_{g=1}^G \sum_{i: y_i = g} (w^T x_i - w^T \mu_g) (w^T x_i - w^T \mu_g)}$$

$$\text{subject to } \mu = \sum_{i=1}^n \frac{x_i}{n}, \mu_g = \sum_{i: y_i = g} \frac{x_i}{n_g}$$

Formula 2

The Mahalanobis' distance is calculated according to Formula 3 in consideration of data correlation and can be used as nonlinear discriminant analysis for determining a cluster to which a data point is associated, based on a short Mahalanobis' distance from the data point to that cluster. In this formula, μ represents a central vector of each cluster, and S^{-1} represents an inverse matrix of the variance-covariance matrix of the cluster. The central vector is calculated from explanatory variable x , and an average vector, a median value vector, or the like can be used.

$$D(x, \mu) = \{(x - \mu)^T S^{-1} (x - \mu)\}^{1/2}$$

Formula 3

SVM is a discriminant analysis method devised by V. Vapnik (The Nature of Statistical Learning Theory, Springer, 1995). Particular data points of a data set having known classes are defined as explanatory variables, and classes are defined as objective variables. A boundary plane called hyperplane for correctly classifying the data set into the known classes is determined, and a discriminant for data classification is determined using the boundary plane. Then, the measurement values of a newly offered data set can be assigned as explanatory variables to the discriminant to determine classes. In this respect, the result of the discriminant analysis may be the associated class, may be a probability of being classified into correct classes, or may be the distance from the hyperplane. In SVM, a method of nonlinearly converting a feature vector to a high dimension and performing linear discriminant analysis in the space is known as a method for tackling nonlinear problems. An expression in which an inner product of two factors in a nonlinearly mapped space is expressed only by inputs in their original spaces is called kernel. Examples of the kernel can include a linear kernel, a RBF (Radial Basis Function) kernel, and a Gaussian kernel. While highly dimensional mapping is performed according to the kernel, the optimum discriminant, i.e., a discriminant, can be actually constructed by mere calculation according to the kernel, which avoids calculating features in the mapped space (e.g., Hideki Aso et al., Frontier of Statistical Science 6 "Statistics of pattern recognition and learning—New concepts and approaches", Iwanami Shoten, Publishers (2004); Nello Cristianini et al., Introduction to SVM, Kyoritsu Shuppan Co., Ltd. (2008)).

C-support vector classification (C-SVC), one type of SVM, comprises preparing a hyperplane by supervising a data set with the explanatory variables of two groups and classifying an unknown data set into either of the groups (C. Cortes et al., 1995, Machine Learning, Vol. 20, p. 273-297).

Exemplary calculation of the C-SVC discriminant that can be used in the method of the present invention will be given below. First, all subjects are divided into two groups, i.e., a breast cancer patient group and a healthy subject group. For example, breast tissue examination can be used for a reference under which each subject is confirmed either as a breast cancer patient or as a healthy subject.

Next, a data set consisting of comprehensive gene expression levels of serum-derived samples of the two divided

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groups (hereinafter, this data set is referred to as a training cohort) is prepared, and a C-SVC discriminant is determined by explanatory variables that are genes found to differ clearly in their gene expression levels between the two groups, and objective variables (e.g., -1 and $+1$) that are the grouping. An optimizing objective function is represented by Formula 4 wherein e represents all input vectors, y represents an objective variable, a represents a Lagrange's undetermined multiplier vector, Q represents a positive definite matrix, and C represents a parameter for adjusting constrained conditions.

$$\min_a \frac{1}{2} a^T Q a - e^T a$$

Formula 4

$$\text{subject to } y^T a = 0, 0 \leq a_i \leq C, i = 1, \dots, l,$$

Formula 5 is a finally obtained discriminant, and a group to which the data point is associated can be determined on the basis of the sign of a value obtained according to the discriminant. In this formula, x represents a support vector, y represents a label indicating the association of a group, a represents the corresponding coefficient, b represents a constant term, and K represents a kernel function.

$$f(x) = \text{sgn} \left(\sum_{i=1}^l y_i a_i K(x_i, x) + b \right)$$

Formula 5

For example, a RBF kernel defined by Formula 6 can be used as the kernel function. In this formula, x represents a support vector, and y represents a kernel parameter for adjusting the complexity of the hyperplane.

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2), \gamma > 0$$

Formula 6

In addition, an approach such as neural network, k-nearest neighbor algorithms, decision trees, or logistic regression analysis can be selected as a method for determining or evaluating the presence and/or absence of expression of a breast cancer-derived target gene in a sample derived from a subject, or for evaluating the expression level thereof by comparison with a control derived from a healthy subject.

The method of the present invention can comprise, for example, the following steps (a), (b), and (c):

- a step of measuring expression level(s) of target gene(s) in tissues containing breast cancer-derived genes derived from breast cancer patients and/or samples already known to be tissues containing no breast cancer-derived gene(s) derived from healthy subjects, using the polynucleotide(s), the kit, or the device (e.g., DNA chip) for detection according to the present invention;
- a step of preparing the discriminants of Formulas 1 to 3, 5, and 6 described above from the measurement values of the expression level measured in the step (a); and
- a step of measuring expression level(s) of the target gene(s) in a sample derived from a subject using the polynucleotide(s), the kit, or the device (e.g., DNA chip) for diagnosis (detection) according to the present invention, substituting the obtained measurement value into the discriminants prepared in the step (b), and determining or evaluating the presence and/or absence of the breast cancer-derived target gene in the sample, or evaluating the expression levels thereof by comparison.

son with a healthy subject-derived control, on the basis of the obtained results. In this context, in the discriminants of Formulas 1 to 3, 5, and 6, x represents an explanatory variable and includes a value obtained by measuring polynucleotide(s) selected from the polynucleotides described in the Section 2 above, or any fragment thereof. Specifically, the explanatory variable for discriminating a breast cancer patient from a healthy subject according to the present invention is gene expression level(s) selected from, for example, the following expression levels (1) to (3):

- (1) gene expression level(s) in the serum of a breast cancer patient or a healthy subject measured by any DNA comprising 15 or more consecutive nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235 and 851 to 856 or a complementary sequence thereof,
- (2) gene expression level(s) in the serum of a breast cancer patient or a healthy subject measured by any DNA comprising 15 or more consecutive nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 236 to 251 or a complementary sequence thereof,
- (3) gene expression level(s) in the serum of a breast cancer patient or a healthy subject measured by any DNA comprising 15 or more consecutive nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269 or a complementary sequence thereof.

As described above, for the method for determining or evaluating the presence and/or absence of breast cancer-derived gene(s) in a sample derived from a subject, the preparation of a discriminant requires a discriminant prepared from a training cohort. For enhancing the accuracy of the discriminant, it is necessary for the discriminant to use genes that show clear difference between two groups in the training cohort.

Each gene that is used for an explanatory variable in a discriminant is preferably determined as follows. First, comprehensive gene expression levels of a breast cancer patient group and comprehensive gene expression levels of a healthy subject group, both of which are in a training cohort, are used as a data set, the degree of difference in the expression level of each gene between the two groups is determined through the use of, for example, the P value of t test, which is parametric analysis, or the P value of Mann-Whitney's U test or Wilcoxon test, which is nonparametric analysis.

The gene can be regarded as being statistically significant when the critical rate (significance level) of the P value obtained by the test is smaller than, for example, 5%, 1%, or 0.01%.

In order to correct an increased probability of type I error attributed to the repetition of a test, a method known in the art, for example, Bonferroni or Holm method, can be used for the correction (e.g., Yasushi Nagata et al., "Basics of statistical multiple comparison methods", Scientist Press Co., Ltd. (2007)). As an example of the Bonferroni correction, for example, the P value obtained by a test is multiplied by the number of repetitions of the test, i.e., the number of genes used in the analysis, and the obtained value can be compared with a desired significance level to suppress a probability of causing type I error in the whole test.

Instead of the statistical test, the absolute value (fold change) of an expression ratio of a median value of each gene expression level between gene expression levels of a breast cancer patient group and gene expression levels of a healthy subject group may be calculated to select a gene that is used for an explanatory variable in a discriminant. Alter-

natively, ROC curves may be prepared using gene expression levels of a breast cancer patient group and a healthy subject group, and a gene that is used for an explanatory variable in a discriminant can be selected on the basis of an AUROC value.

Next, a discriminant that can be calculated by various methods described above is prepared using any number of genes that show large difference in their gene expression levels determined here. Examples of the method for constructing a discriminant that produces the largest discriminant accuracy include a method of constructing a discriminant in every combination of genes that satisfy the significance level of P value, and a method of repetitively evaluating the genes for use in the construction of a discriminant while increasing the number of genes one by one in a descending order of the difference in gene expression level (Furey T S. et al., 2000, Bioinformatics., Vol. 16, p. 906-14). A gene expression level of another independent breast cancer patient or healthy subject is assigned as an explanatory variable to this discriminant to calculate discriminant results of the group to which this independent breast cancer patient or healthy subject is associated. Specifically, the found gene set for diagnosis and the discriminant constructed using the gene set for diagnosis can be evaluated in an independent sample cohort to find a more universal gene set for diagnosis capable of detecting breast cancer and a more universal method for discriminating breast cancer.

Split-sample method is preferably used for evaluating the performance (generality) of the discriminant. Specifically, a data set is divided into a training cohort and a validation cohort, and gene selection by a statistical test and discriminant construction are performed using the training cohort. Accuracy, sensitivity, and specificity are calculated using results of discriminant analysis in the validation cohort according to the discriminant constructed and a true group to which the validation cohort is associated, to evaluate the discriminant performance of the discriminant. On the other hand, instead of dividing a data set, gene selection by a statistical test and discriminant construction may be performed using all of samples, and accuracy, sensitivity, and specificity can be calculated by the discriminant analysis using a newly prepared cohorts for evaluation of the performance of the discriminant.

The present invention provides polynucleotides for detection and disease diagnosis useful in the diagnosis and treatment of breast cancer, a method for detecting breast cancer using the polynucleotide(s), and a kit and a device for the detection of breast cancer, comprising the polynucleotide(s). Particularly, in order to select gene(s) for diagnosis and prepare a discriminant so as to exhibit accuracy beyond the breast cancer diagnosis method using an existing tumor marker CEA, a gene set for diagnosis and a discriminant for the method of the present invention can be constructed, which exhibit accuracy beyond CEA, for example, by comparing expressed genes in serum from a patient confirmed to be negative using CEA but finally found to have breast cancer by detailed examination such as computed tomography using a contrast medium, with genes expressed in serum from a patient having no breast cancer.

For example, the gene set for diagnosis is set to any combination selected from one or two or more of the polynucleotides based on a nucleotide sequence represented by any of SEQ ID NOs: 1 to 235, and 851 to 856, or a complementary sequence thereof as described above; and optionally one or two or more of the polynucleotides based on a nucleotide sequence represented by any of SEQ ID

NOs: 236 to 251, or a complementary sequence thereof; and optionally one or two or more of the polynucleotides based on a nucleotide sequence represented by any of SEQ ID NOs: 252 to 269, or a complementary sequence thereof. Further, a discriminant is constructed using expression levels of the gene set for diagnosis in samples from class I breast cancer patients as a result of tissue diagnosis and samples from class II healthy subjects as a result of tissue diagnosis. As a result, the presence or absence of breast cancer-derived genes in an unknown sample can be determined with 100% accuracy at the maximum by measuring expression levels of the gene set for diagnosis in an unknown sample.

EXAMPLES

Hereinafter, the present invention will be described further specifically with reference to Examples below. However, the scope of the present invention is not intended to be limited by these Examples.

Reference Example 1

<Collection of Samples from Breast Cancer Patients and Healthy Subjects>

Sera were collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from 100 healthy subjects (92 males and 8 females) and 62 breast cancer patients (20 cases with stage I, 24 cases with stage IIA, 7 cases with stage IIB, 2 cases with stage IIIA, 3 cases with stage IIIB, 1 case with stage IIIC, and 5 cases with stage IV) who were confirmed to have no primary cancer other than breast cancer after acquisition of informed consent, and used as a training cohort. Likewise, Sera were collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from 50 healthy subjects (44 males and 6 females) and 31 breast cancer patients (9 cases with stage I, 13 cases with stage IIA, 5 cases with stage IIB, 1 case with stage IIIA, 1 case with stage IIIB, 1 case with stage IIIC, and 1 case with stage IV) who were confirmed to have no primary cancer other than breast cancer after acquisition of informed consent, and used as a validation cohort.

<Extraction of Total RNA>

Total RNA was obtained from 300 μ L of the serum sample obtained from each of 243 persons in total of 150 healthy subjects and 93 breast cancer patients included in the training cohort and the validation cohort, using a reagent for RNA extraction in 3D-Gene™ RNA extraction reagent from liquid sample kit (Toray Industries, Inc.) according to the protocol provided by the manufacturer.

<Measurement of Gene Expression Level>

miRNAs in the total RNA obtained from the serum samples of each of 243 persons in total of 150 healthy subjects and 93 breast cancer patients included in the training cohort and the validation cohort were fluorescently labeled using 3D-Gene™ miRNA Labeling kit (Toray Industries, Inc.) according to the protocol (ver 2.20) provided by the manufacturer. The oligo DNA chip used was 3D-Gene™ Human miRNA Oligo chip (Toray Industries, Inc.) with attached probes having sequences complementary to 2,555 miRNAs among the miRNAs registered in miR-Base Release 20. Hybridization between the miRNAs in the total RNA and the probes on the DNA chip under stringent conditions and washing following the hybridization were performed according to the protocol provided by the manufacturer. The DNA chip was scanned using 3D-Gene™

scanner (Toray Industries, Inc.) to obtain images. Fluorescence intensity was digitized using 3D-Gene™ Extraction (Toray Industries, Inc.). The digitized fluorescence intensity was converted to a logarithmic value with a base of 2 and used as a gene expression level, from which a blank value was subtracted. A missing value was replaced with a value obtained by subtracting 0.1 from a logarithmic value of the smallest value of the gene expression level in each DNA chip. As a result, the comprehensive gene expression levels of the miRNAs in the sera were obtained for the 93 breast cancer patients and the 150 healthy subjects. Calculation and statistical analysis using the digitized gene expression levels of the miRNAs were carried out using R language 3.0.2 (R Development Core Team (2013). R: A language and environment for statistical computing. R Foundation for Statistical Computing, URL <http://www.R-project.org/>) and MASS package 7.3-30 (Venables, W. N. & Ripley, B. D. (2002) Modern Applied Statistics with S. Fourth Edition. Springer, New York. ISBN 0-387-95457-0).

Reference Example 2

<Collection of Samples from Patients with Cancer Other than Breast Cancer>

Sera were collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from each of 33 prostate cancer patients confirmed to have no cancer in other organs after acquisition of informed consent, and used as a training cohort together with the samples of 62 breast cancer patients and 102 healthy subjects of Reference Example 1. Likewise, sera were collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from each of 19 prostate cancer patients confirmed to have no cancer in other organs after acquisition of informed consent, and used as a validation cohort together with the samples of 31 breast cancer patients confirmed to have no cancer in organs other than the breast and 48 healthy subjects of Reference Example 1. Subsequent operations were conducted in the same way as in Reference Example 1.

Example 1

<Selection of Gene Marker Using the Training Cohort, and Method for Evaluating Breast Cancer Discriminant Performance of the Single Gene Marker Using the Validation Cohort>

In this Example, a gene marker for discriminating a breast cancer patient from a healthy subject was selected from the training cohort and studied in the validation cohort independent of the training cohort, for a method for evaluating the breast cancer discriminant performance of each selected gene marker alone.

Specifically, first, the miRNA expression levels of the training cohort and the validation cohort obtained in the preceding Reference Examples were combined and normalized by quantile normalization.

Next, genes for diagnosis were selected using the training cohort. Here, in order to acquire diagnostic markers with higher reliability, only genes that show the gene expression level of 2^6 or higher in 50% or more of the samples in either of the breast cancer patient group in the training cohort or the healthy subject group in the training cohort were selected. In order to further acquire statistically significant genes for discriminating a breast cancer patient group from a healthy subject group, the P value obtained by two-tailed t-test assuming equal variance as to each gene expression level was corrected by the Bonferroni method, and genes

that satisfied $p < 0.01$ were acquired as gene markers for use in explanatory variables of a discriminant. The obtained genes are described in Table 2.

In this way, hsa-miR-4783-3p, hsa-miR-4730, hsa-miR-1307-3p, hsa-miR-4634, hsa-miR-663a, hsa-miR-4532, hsa-miR-7704, hsa-miR-3178, hsa-miR-6729-5p, hsa-miR-6090, hsa-miR-4732-5p, hsa-miR-3184-5p, hsa-miR-6727-5p, hsa-miR-6088, hsa-miR-4674, hsa-miR-8073, hsa-miR-4787-5p, hsa-miR-1469, hsa-miR-125a-3p, hsa-miR-1233-5p, hsa-miR-885-3p, hsa-miR-6802-5p, hsa-miR-328-5p, hsa-miR-6787-5p, hsa-miR-8069, hsa-miR-6875-5p, hsa-miR-1246, hsa-miR-4734, hsa-miR-6757-5p, hsa-miR-6756-5p, hsa-miR-3665, hsa-miR-6836-3p, hsa-miR-6821-5p, hsa-miR-6805-5p, hsa-miR-4728-5p, hsa-miR-6726-5p, hsa-miR-197-5p, hsa-miR-149-3p, hsa-miR-6850-5p, hsa-miR-4476, hsa-miR-6858-5p, hsa-miR-564, hsa-miR-4763-3p, hsa-miR-575, hsa-miR-6771-5p, hsa-miR-1231, hsa-miR-1908-3p, hsa-miR-150-3p, hsa-miR-3937, hsa-miR-887-3p, hsa-miR-3940-5p, hsa-miR-4741, hsa-miR-6808-5p, hsa-miR-6869-5p, hsa-miR-5090, hsa-miR-615-5p, hsa-miR-8072, hsa-miR-128-1-5p, hsa-miR-1238-5p, hsa-miR-365a-5p, hsa-miR-204-3p, hsa-miR-4492, hsa-miR-6785-5p, hsa-miR-6511a-5p, hsa-miR-4525, hsa-miR-1915-5p, hsa-miR-3180, hsa-miR-6879-5p, hsa-miR-1199-5p, hsa-miR-6746-5p, hsa-miR-711, hsa-miR-663b, hsa-miR-4707-3p, hsa-miR-6893-5p, hsa-miR-4675, hsa-miR-4638-5p, hsa-miR-4651, hsa-miR-6087, hsa-miR-4665-5p, hsa-miR-4758-5p, hsa-miR-6887-5p, hsa-miR-3620-5p, hsa-miR-1909-3p, hsa-miR-7641, hsa-miR-6724-5p, hsa-miR-1343-3p, hsa-miR-6780b-5p, hsa-miR-4484, hsa-miR-4690-5p, hsa-miR-4429, hsa-miR-1227-5p, hsa-miR-4725-3p, hsa-miR-6861-5p, hsa-miR-6812-5p, hsa-miR-3197, hsa-miR-8059, hsa-miR-3185, hsa-miR-4706, hsa-miR-4497, hsa-miR-3131, hsa-miR-6806-5p, hsa-miR-187-5p, hsa-miR-3180-3p, hsa-miR-6848-5p, hsa-miR-6820-5p, hsa-miR-6800-5p, hsa-miR-6717-5p, hsa-miR-6795-5p, hsa-miR-4632-5p, hsa-miR-665, hsa-miR-6778-5p, hsa-miR-3663-3p, hsa-miR-4689, hsa-miR-211-3p, hsa-miR-6511b-5p, hsa-miR-4750-5p, hsa-miR-6126, hsa-miR-614, hsa-miR-7110-5p, hsa-miR-744-5p, hsa-miR-6769a-5p, hsa-miR-4792, hsa-miR-5787, hsa-miR-6798-5p, hsa-miR-6781-5p, hsa-miR-4419b, hsa-miR-4446-3p, hsa-miR-4259, hsa-miR-5572, hsa-miR-6075, hsa-miR-296-3p, hsa-miR-6891-5p, hsa-miR-4745-5p, hsa-miR-6775-5p, hsa-miR-6870-5p, hsa-miR-920, hsa-miR-4530, hsa-miR-6819-5p, hsa-miR-6825-5p, hsa-miR-7847-3p, hsa-miR-6131, hsa-miR-4433-3p, hsa-miR-1228-5p, hsa-miR-6743-5p, hsa-miR-1268a, hsa-miR-3917, hsa-miR-6786-5p, hsa-miR-3154, hsa-miR-638, hsa-miR-6741-5p, hsa-miR-6889-5p, hsa-miR-6840-3p, hsa-miR-6510-5p, hsa-miR-3188, hsa-miR-551b-5p, hsa-miR-5001-5p, hsa-miR-1268b, hsa-miR-7107-5p, hsa-miR-6824-5p, hsa-miR-6732-5p, hsa-miR-371a-5p, hsa-miR-6794-5p, hsa-miR-6779-5p, hsa-miR-4271, hsa-miR-5195-3p, hsa-miR-6762-5p, hsa-miR-939-5p, hsa-miR-1247-3p, hsa-miR-6777-5p, hsa-miR-6722-3p, hsa-miR-3656, hsa-miR-4688, hsa-miR-3195, hsa-miR-6766-5p, hsa-miR-4447, hsa-miR-4656, hsa-miR-7108-5p, hsa-miR-3191-3p, hsa-miR-1273g-3p, hsa-miR-4463, hsa-miR-2861, hsa-miR-3196, hsa-miR-6877-5p, hsa-miR-3679-5p, hsa-miR-4442, hsa-miR-6789-5p, hsa-miR-6782-5p, hsa-miR-486-3p, hsa-miR-6085, hsa-miR-4746-3p, hsa-miR-619-5p, hsa-miR-937-5p, hsa-miR-6803-5p, hsa-miR-4298, hsa-miR-4454, hsa-miR-4459, hsa-miR-7150, hsa-miR-6880-5p, hsa-miR-4449, hsa-miR-8063, hsa-miR-4695-5p, hsa-miR-6132, hsa-miR-6829-5p, hsa-miR-4486, hsa-miR-6805-3p, hsa-miR-6826-5p, hsa-miR-4508, hsa-miR-1343-5p, hsa-miR-7114-5p, hsa-miR-3622a-5p, hsa-miR-6765-

5p, hsa-miR-7845-5p, hsa-miR-3960, hsa-miR-6749-5p, hsa-miR-1260b, hsa-miR-6799-5p, hsa-miR-4723-5p, hsa-miR-6784-5p, hsa-miR-5100, hsa-miR-6769b-5p, hsa-miR-1207-5p, hsa-miR-642a-3p, hsa-miR-4505, hsa-miR-4270, hsa-miR-6721-5p, hsa-miR-7111-5p, hsa-miR-6791-5p, hsa-miR-7109-5p, hsa-miR-4258, hsa-miR-6515-3p, hsa-miR-6851-5p, hsa-miR-6125, hsa-miR-4749-5p, hsa-miR-4726-5p, hsa-miR-4513, hsa-miR-760, hsa-miR-602, hsa-miR-423-5p, hsa-miR-92a-2-5p, hsa-miR-16-5p, hsa-miR-451a, hsa-miR-135a-3p, hsa-miR-486-5p, hsa-miR-4257, hsa-miR-92b-5p, hsa-miR-1915-3p, hsa-miR-718, hsa-miR-940, hsa-miR-296-5p, hsa-miR-23b-3p and hsa-miR-92a-3p genes, and polynucleotides consisting of the nucleotide sequences of SEQ ID NOs: 1 to 251 related thereto were found.

Among them, genes newly found as markers for examining the presence or absence of breast cancer are polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 235.

A discriminant for determining the presence or absence of breast cancer was further prepared by Fisher's discriminant analysis with the expression levels of these genes as indicators. Specifically, any newly found polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 251 in the training cohort was applied for Formula 2 above to prepare a discriminant. Calculated accuracy, sensitivity, and specificity are shown in Table 3. In this respect, a discriminant coefficient and a constant term are shown in Table 4. Here, all of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251 were selected as markers capable of determining not only invasive ductal breast cancer (56 cases), which is a main type of breast cancer, but also invasive lobular cancer (3 cases) and unusual metastatic carcinoma with poor prognosis (1 case).

Accuracy, sensitivity, and specificity in the validation cohort were calculated using the discriminant thus prepared, and the discriminant performance of the selected polynucleotides was validated using the independent samples (Table 3). For example, the expression level measurement value of the gene that consists of the nucleotide sequence represented by SEQ ID NO: 1 was compared between the healthy subjects (100 persons) and the breast cancer patients (62 persons) in the training cohort. As a result, the expression level measurement values were found to be significantly lower in the breast cancer patient group than in the healthy subject group (see the left diagram of FIG. 2). These results were also reproducible for the healthy subjects (50 persons) and the breast cancer patients (31 persons) in the validation cohort (see the right diagram of FIG. 2). Likewise, the results obtained about the other polynucleotides shown in SEQ ID NOs: 2 to 251 showed that the expression level measurement values were significantly lower (−) or higher (+) in the breast cancer patient group than in the healthy subject group (Table 2). These results were able to be validated in the validation cohort. For example, as for this nucleotide sequence represented by SEQ ID NO: 1, the number of correctly or incorrectly identified samples in the detection of breast cancer was calculated using the threshold (6.63) that was set in the training cohort and discriminated between the two groups. As a result, 31 true positives, 50 true negatives, 0 false positives, and 0 false negatives were obtained. From these values, 100% accuracy, 100% sensitivity, and 100% specificity were obtained as the detection performance. In this way, the detection performance was calculated as to all of the polynucleotides shown in SEQ ID NOs: 1 to 251, and described in Table 3. Likewise, the

polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 2 to 251 shown in Table 2 exhibited sensitivity of 100%, 100%, 90.3%, 96.8%, 100%, 100%, 96.8%, 100%, 100%, 100%, 100%, 96.8%, 96.8%, 100%, 96.8%, 100%, 96.8%, 100%, 93.5%, 96.8%, 96.8%, 100%, 96.8%, 96.8%, 93.5%, 96.8%, 93.5%, 87.1%, 83.9%, 96.8%, 96.8%, 96.8%, 100%, 80.6%, 87.1%, 87.1%, 100%, 90.3%, 90.3%, 90.3%, 100%, 96.8%, 100%, 87.1%, 87.1%, 96.8%, 96.8%, 90.3%, 96.8%, 83.9%, 77.4%, 90%, 83.9%, 96.8%, 93.5%, 80.6%, 96.8%, 90.3%, 93.5%, 90.3%, 87.1%, 87.1%, 96.8%, 83.9%, 77.4%, 90.3%, 83.9%, 87.1%, 77.4%, 90.3%, 83.9%, 74.2%, 93.5%, 87.1%, 93.5%, 93.5%, 77.4%, 90.3%, 87.1%, 87.1%, 83.9%, 87.1%, 93.5%, 77.4%, 93.5%, 74.2%, 83.9%, 100%, 90.3%, 74.2%, 83.9%, 80.6%, 87.1%, 77.4%, 83.9%, 71%, 96.8%, 77.4%, 87.1%, 77.4%, 71%, 90.3%, 80.6%, 67.7%, 77.4%, 87.1%, 74.2%, 83.9%, 77.4%, 71%, 87.1%, 74.2%, 90.3%, 80.6%, 74.2%, 83.9%, 83.9%, 71%, 87.1%, 61.3%, 61.3%, 83.9%, 61.3%, 90.3%, 80.6%, 61.3%, 64.5%, 80.6%, 74.2%, 80.6%, 71%, 71%, 77.4%, 64.5%, 71%, 71%, 83.9%, 74.2%, 83.9%, 63.3%, 64.5%, 71%, 67.7%, 71%, 71%, 74.2%, 71%, 64.5%, 83.9%, 71%, 83.9%, 61.3%, 61.3%, 67.7%, 64.5%, 64.5%, 54.8%, 64.5%, 74.2%, 58.1%, 58.1%, 58.1%, 61.3%, 67.7%, 61.3%, 67.7%, 58.1%, 58.1%, 54.8%, 67.7%, 58.1%, 64.5%, 61.3%, 67.7%, 58.1%, 58.1%, 48.4%, 61.3%, 54.8%, 38.7%, 35.5%, 64.5%, 54.8%, 64.5%, 54.8%, 61.3%, 35.5%, 48.4%, 61.3%, 61.3%, 54.8%, 71%, 61.3%, 45.2%, 48.4%, 29%, 54.8%, 41.9%, 67.7%, 29%, 29%, 53.3%, 51.6%, 45.2%, 35.5%, 41.9%, 41.9%, 48.4%, 41.9%, 35.5%, 41.9%, 35.5%, 48.4%, 32.3%, 41.9%, 41.9%, 41.9%, 35.5%, 35.5%, 41.9%, 61.3%, 32.3%, 45.2%, 38.7%, 51.6%, 29%, 35.5%, 38.7%, 54.8%, 58.1%, 51.6%, 29%, 41.9%, 38.7%, 96.8%, 96.8%, 100%, 96.8%, 87.1%, 80.6%, 100%, 87.1%, 93.5%, 67.7%, 67.7%, 61.3%, 67.7%, 38.7% and 54.8%, respectively, in the validation cohort (Table 3). As seen from Comparative Example mentioned later, the existing marker CEA had sensitivity of 19.4% in the validation cohort (Table 5-2), demonstrating that all of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID Nos: 1 to 251 can discriminate, each alone, breast cancer in the validation cohort with sensitivity beyond CEA.

For example, the 83 polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 31, 32, 33, 34, 35, 36, 38, 40, 42, 44, 45, 47, 48, 49, 50, 55, 59, 64, 65, 66, 68, 70, 73, 75, 78, 79, 80, 81, 84, 88, 92, 93, 95, 97, 98, 99, 102, 109, 113, 119, 122, 124, 128, 130, 133, 145, 149, 169, 236, 237, 238, 239, 240, 241, 242, 243, 244 were able to correctly determine breast cancer in the 9 breast cancer samples of stage 1 contained in the validation cohort. Thus, these polynucleotides can detect even early breast cancer and contributes to the early diagnosis of breast cancer.

Example 2

<Method a for Evaluating Breast Cancer Discriminant Performance by Combination of Multiple Gene Markers Using the Samples in the Validation Cohort>

In this Example, a method for evaluating breast cancer discriminant performance by a combination of the gene markers selected in Example 1 was studied. Specifically, Fisher's discriminant analysis was conducted as to 31,255 polynucleotide combinations comprising at least one of the newly found polynucleotides consisting of the nucleotide

sequences represented by SEQ ID NOs: 1 to 235 among the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251 selected in Example 1, to construct a discriminant for determining the presence or absence of breast cancer. Next, accuracy, sensitivity, and specificity in the validation cohort were calculated using the discriminant thus prepared, and the discriminant performance of the selected polynucleotides was validated using the independent samples.

For example, the gene expression level measurement values of the nucleotide sequences represented by SEQ ID NO: 1 and SEQ ID NO: 2 were compared between the healthy subjects and the breast cancer patients. As a result, a scatter diagram that significantly separated the expression level measurement values of the breast cancer patient group from those of the healthy subject group was obtained (see the left diagram of FIG. 3). These results were also reproducible for the healthy subjects (50 persons) and the breast cancer patients (31 persons) in the validation cohort (see the right diagram of FIG. 3). Likewise, a scatter diagram that significantly separated the expression level measurement values of the breast cancer patient group from those of the healthy subject group was also obtained as to the other polynucleotide combinations comprising at least one of the newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 235 among the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251. These results were able to be validated in the validation cohort. For example, as for these nucleotide sequences represented by SEQ ID NO: 1 and SEQ ID NO: 2, the number of samples that were correctly or incorrectly identified breast cancer was calculated using the discriminant function ($0=1.41x+y+0.77$) that was set in the training cohort and discriminated between the two groups. As a result, 31 true positives, 50 true negatives, 0 false positives, and 0 false negatives were obtained. From these values, 100% accuracy, 100% sensitivity, and 100% specificity were obtained as the detection performance. In this way, the detection performance was calculated as to all of the polynucleotide combinations comprising at least one of the newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 235 among the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251. Among them, 250 combinations comprising the expression level measurement value of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 and the detection performance thereof were described in Table 6 as an example. For example, all of the combinations of the expression level measurement values of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 and 2, SEQ ID NOs: 1 and 3, SEQ ID NOs: 1 and 4, and SEQ ID NOs: 1 and 5 exhibited sensitivity of 100% in the validation cohort. Likewise, all of the combinations of two polynucleotides consisting of the nucleotide sequence represented by SEQ ID NO: 1 and a nucleotide sequence represented by any of SEQ ID NOs: 6 to 251 also exhibited sensitivity of 100%. In addition, the combinations of two polynucleotides consisting of the nucleotide sequences except for SEQ ID NO: 1 were described in Table 7 as an example. As the specific combinations of two polynucleotides, for example, the combinations represented by SEQ ID NOs: 3 and 20, SEQ ID NOs: 6 and 20, SEQ ID NOs: 7 and 20, SEQ ID NOs: 10 and 20, SEQ ID NOs: 20 and 22, SEQ ID NOs: 20 and 238, SEQ ID NOs: 20 and 239, SEQ ID NOs: 12 and 24, SEQ ID NOs: 20 and 24, SEQ ID NOs: 24 and 27, SEQ ID NOs: 24 and 33, SEQ ID NOs: 24 and

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236, SEQ ID NOs: 24 and 240, SEQ ID NOs: 3 and 26, SEQ ID NOs: 12 and 26, SEQ ID NOs: 13 and 26, SEQ ID NOs: 17 and 26, SEQ ID NOs: 19 and 26, SEQ ID NOs: 3 and 27, SEQ ID NOs: 5 and 27, SEQ ID NOs: 13 and 27, SEQ ID NOs: 20 and 27, SEQ ID NOs: 26 and 27, SEQ ID NOs: 27 and 120, SEQ ID NOs: 27 and 206, SEQ ID NOs: 27 and 237, SEQ ID NOs: 3 and 30, SEQ ID NOs: 17 and 30, SEQ ID NOs: 27 and 30, SEQ ID NOs: 27 and 33, SEQ ID NOs: 30 and 39, SEQ ID NOs: 30 and 117, SEQ ID NOs: 3 and 33, SEQ ID NOs: 7 and 33, SEQ ID NOs: 10 and 33, SEQ ID NOs: 11 and 33, SEQ ID NOs: 13 and 33, SEQ ID NOs: 25 and 33, SEQ ID NOs: 33 and 244, SEQ ID NOs: 3 and 182, SEQ ID NOs: 6 and 182, SEQ ID NOs: 7 and 182, SEQ ID NOs: 12 and 182, SEQ ID NOs: 27 and 182, SEQ ID NOs: 182 and 236, SEQ ID NOs: 2 and 194, SEQ ID NOs: 7 and 194, SEQ ID NOs: 27 and 194, SEQ ID NOs: 194 and 236, SEQ ID NOs: 2 and 206, SEQ ID NOs: 7 and 206, SEQ ID NOs: 206 and 236, SEQ ID NOs: 2 and 208, SEQ ID NOs: 7 and 208, SEQ ID NOs: 13 and 208, SEQ ID NOs: 20 and 208, and SEQ ID NOs: 27 and 208 exhibited accuracy of 96% or higher for discriminating the breast cancer patients from the healthy subjects in both of the training cohort and the validation cohort. Thus, the combinations of two of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251 also produced excellent detection sensitivity for breast cancer. Markers for the detection of breast cancer with better sensitivity are obtained by further combining 3, 4, 5, 6, 7, 8, 9, 10 or more of the measurement values of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251. For example, the newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 235 among the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251 selected in Example 1, were measured to obtain their expression levels between the healthy subject group and the breast cancer group in the validation cohort. All of the polynucleotides were ranked in the descending order of their P values based on the Student's t-test which indicates statistical significance of a difference between groups (i.e., one having the lowest P value was ranked in the first place), and breast cancer detection sensitivity was evaluated using combinations of one or more polynucleotides to which the polynucleotides were added one by one from the top to bottom according to the rank. In short, the order in which the polynucleotides were combined in this evaluation is in reverse in terms of SEQ ID NOs, such as SEQ ID NO: 235 to SEQ ID NOs: 234, 233, . . . shown in Table 2 in order. As a result, the sensitivity in the validation cohort was 38.7% for 1 polynucleotide (SEQ ID NO: 235), 48.4% for 2 polynucleotides (SEQ ID NOs: 234 and 235), 74.2% for 4 polynucleotides (SEQ ID NOs: 232 to 235), 87.1% for 6 polynucleotides (SEQ ID NOs: 230 to 235), 90.3% for 10 polynucleotides (SEQ ID NOs: 226 to 235), 93.5% for 13 polynucleotides (SEQ ID NOs: 223 to 235), 96.8% for 16 polynucleotides (SEQ ID NOs: 220 to 235), 100% for 20 polynucleotides (SEQ ID NOs: 216 to 235), 100% for 30 polynucleotides (SEQ ID NOs: 206 to 235), 100% for 50 polynucleotides (SEQ ID NOs: 186 to 235), 100% for 100 polynucleotides (SEQ ID NOs: 136 to 235), 100% for 200 polynucleotides (SEQ ID NOs: 36 to 235), and 100% for 235 polynucleotides (SEQ ID NOs: 1 to 235).

These results demonstrated that a combination of multiple polynucleotides can produce higher breast cancer discriminant performance than that of each polynucleotide alone or a combination of a fewer number of polynucleotides. In this

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context, the combinations of multiple polynucleotides are not limited to the combinations of the polynucleotides added in the order of statistically significant difference as described above, and any combination of multiple polynucleotides can be used in the detection of breast cancer.

From these results, it can be concluded that all of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 251 serve as excellent markers for breast cancer detection.

TABLE 2

SEQ ID NO:	Gene name	P value after Bonferoni correction	Expression level in breast cancer patient with respect to healthy subject
1	hsa-miR-4783-3p	1.88 · E-98	+
2	hsa-miR-4730	1.39 · E-95	-
3	hsa-miR-1307-3p	9.90 · E-81	+
4	hsa-miR-4634	1.73 · E-79	-
5	hsa-miR-663a	2.71 · E-76	+
6	hsa-miR-4532	2.24 · E-75	+
7	hsa-miR-7704	4.18 · E-74	-
8	hsa-miR-3178	2.19 · E-71	-
9	hsa-miR-6729-5p	1.88 · E-69	-
10	hsa-miR-6090	2.55 · E-68	+
11	hsa-miR-4732-5p	1.36 · E-67	+
12	hsa-miR-3184-5p	8.11 · E-67	-
13	hsa-miR-6727-5p	3.02 · E-64	-
14	hsa-miR-6088	3.54 · E-64	+
15	hsa-miR-4674	9.34 · E-64	-
16	hsa-miR-8073	1.47 · E-63	+
17	hsa-miR-4787-5p	5.85 · E-62	-
18	hsa-miR-1469	2.97 · E-61	+
19	hsa-miR-125a-3p	9.95 · E-61	-
20	hsa-miR-1233-5p	2.94 · E-60	+
21	hsa-miR-885-3p	6.54 · E-60	+
22	hsa-miR-6802-5p	4.56 · E-59	+
23	hsa-miR-328-5p	1.83 · E-58	+
24	hsa-miR-6787-5p	5.55 · E-58	+
25	hsa-miR-8069	2.26 · E-57	-
26	hsa-miR-6875-5p	2.72 · E-53	-
27	hsa-miR-1246	9.57 · E-53	+
28	hsa-miR-4734	2.06 · E-52	+
29	hsa-miR-6757-5p	4.71 · E-51	+
30	hsa-miR-6756-5p	5.35 · E-51	+
31	hsa-miR-3665	1.70 · E-50	-
32	hsa-miR-6836-3p	8.38 · E-50	-
33	hsa-miR-6821-5p	1.19 · E-49	+
34	hsa-miR-6805-5p	3.17 · E-49	-
35	hsa-miR-4728-5p	3.69 · E-49	+
36	hsa-miR-6726-5p	5.62 · E-49	-
37	hsa-miR-197-5p	1.43 · E-46	+
38	hsa-miR-149-3p	3.23 · E-46	+
39	hsa-miR-6850-5p	4.58 · E-46	-
40	hsa-miR-4476	5.54 · E-46	-
41	hsa-miR-6858-5p	9.61 · E-46	+
42	hsa-miR-564	1.62 · E-44	-
43	hsa-miR-4763-3p	2.01 · E-44	+
44	hsa-miR-575	3.61 · E-44	-
45	hsa-miR-6771-5p	2.62 · E-43	-
46	hsa-miR-1231	7.46 · E-43	-
47	hsa-miR-1908-3p	9.60 · E-43	-
48	hsa-miR-150-3p	2.19 · E-42	-
49	hsa-miR-3937	9.92 · E-42	-
50	hsa-miR-887-3p	2.38 · E-41	-
51	hsa-miR-3940-5p	3.44 · E-41	-
52	hsa-miR-4741	4.16 · E-41	-
53	hsa-miR-6808-5p	5.42 · E-41	+
54	hsa-miR-6869-5p	1.03 · E-40	-
55	hsa-miR-5090	5.03 · E-40	-
56	hsa-miR-615-5p	1.94 · E-39	-
57	hsa-miR-8072	2.71 · E-39	+
58	hsa-miR-128-l-5p	2.72 · E-39	-
59	hsa-miR-1238-5p	7.46 · E-39	+
60	hsa-miR-365a-5p	9.50 · E-39	+
61	hsa-miR-204-3p	1.32 · E-38	-
62	hsa-miR-4492	3.33 · E-37	-
63	hsa-miR-6785-5p	4.21 · E-37	-

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TABLE 2-continued

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject	
64	hsa-miR-6511a-5p	$7.68 \cdot E-37$	+	10
65	hsa-miR-4525	$1.16 \cdot E-36$	-	
66	hsa-miR-1915-5p	$1.34 \cdot E-36$	-	
67	hsa-miR-3180	$1.07 \cdot E-35$	-	
68	hsa-miR-6879-5p	$1.56 \cdot E-35$	+	
69	hsa-miR-1199-5p	$1.15 \cdot E-34$	-	
70	hsa-miR-6746-5p	$5.65 \cdot E-34$	+	
71	hsa-miR-711	$5.82 \cdot E-34$	-	
72	hsa-miR-663b	$1.42 \cdot E-33$	-	
73	hsa-miR-4707-3p	$2.19 \cdot E-33$	-	
74	hsa-miR-6893-5p	$6.31 \cdot E-33$	-	15
75	hsa-miR-4675	$6.39 \cdot E-33$	+	
76	hsa-miR-4638-5p	$6.40 \cdot E-33$	-	
77	hsa-miR-4651	$8.91 \cdot E-33$	-	
78	hsa-miR-6087	$1.91 \cdot E-32$	+	
79	hsa-miR-4665-5p	$3.57 \cdot E-32$	-	
80	hsa-miR-4758-5p	$4.55 \cdot E-32$	+	
81	hsa-miR-6887-5p	$4.45 \cdot E-31$	+	
82	hsa-miR-3620-5p	$4.64 \cdot E-31$	-	
83	hsa-miR-1909-3p	$5.74 \cdot E-31$	-	20
84	hsa-miR-7641	$8.30 \cdot E-31$	-	
85	hsa-miR-6724-5p	$1.02 \cdot E-30$	+	
86	hsa-miR-1343-3p	$1.19 \cdot E-30$	+	
87	hsa-miR-6780b-5p	$1.22 \cdot E-30$	+	
88	hsa-miR-4484	$2.77 \cdot E-30$	-	
89	hsa-miR-4690-5p	$3.50 \cdot E-30$	+	
90	hsa-miR-4429	$2.05 \cdot E-29$	+	
91	hsa-miR-1227-5p	$3.84 \cdot E-29$	+	
92	hsa-miR-4725-3p	$5.39 \cdot E-29$	-	30
93	hsa-miR-6861-5p	$5.43 \cdot E-29$	+	
94	hsa-miR-6812-5p	$7.48 \cdot E-29$	+	
95	hsa-miR-3197	$8.20 \cdot E-29$	+	
96	hsa-miR-8059	$9.29 \cdot E-29$	+	
97	hsa-miR-3185	$9.34 \cdot E-29$	+	
98	hsa-miR-4706	$1.69 \cdot E-28$	+	
99	hsa-miR-4497	$2.22 \cdot E-28$	-	
100	hsa-miR-3131	$3.64 \cdot E-28$	+	
101	hsa-miR-6806-5p	$9.04 \cdot E-28$	-	35
102	hsa-miR-187-5p	$4.89 \cdot E-27$	-	
103	hsa-miR-3180-3p	$7.10 \cdot E-27$	-	
104	hsa-miR-6848-5p	$7.26 \cdot E-27$	-	
105	hsa-miR-6820-5p	$7.77 \cdot E-27$	-	
106	hsa-miR-6800-5p	$1.80 \cdot E-26$	-	
107	hsa-miR-6717-5p	$1.97 \cdot E-26$	+	
108	hsa-miR-6795-5p	$5.16 \cdot E-26$	+	
109	hsa-miR-4632-5p	$8.43 \cdot E-26$	+	
110	hsa-miR-665	$2.56 \cdot E-25$	-	40
111	hsa-miR-6778-5p	$8.69 \cdot E-25$	-	
112	hsa-miR-3663-3p	$1.09 \cdot E-24$	+	
113	hsa-miR-4689	$3.36 \cdot E-24$	+	
114	hsa-miR-211-3p	$8.65 \cdot E-24$	+	
115	hsa-miR-6511b-5p	$9.67 \cdot E-24$	+	
116	hsa-miR-4750-5p	$1.07 \cdot E-23$	+	
117	hsa-miR-6126	$2.06 \cdot E-23$	+	
118	hsa-miR-614	$2.46 \cdot E-22$	+	
119	hsa-miR-7110-5p	$3.56 \cdot E-22$	-	50
120	hsa-miR-744-5p	$5.83 \cdot E-22$	+	
121	hsa-miR-6769a-5p	$1.44 \cdot E-21$	+	
122	hsa-miR-4792	$2.04 \cdot E-21$	-	
123	hsa-miR-5787	$3.93 \cdot E-21$	+	
124	hsa-miR-6798-5p	$5.13 \cdot E-21$	-	
125	hsa-miR-6781-5p	$2.43 \cdot E-20$	-	
126	hsa-miR-4419b	$2.59 \cdot E-20$	+	
127	hsa-miR-4446-3p	$7.52 \cdot E-20$	+	
128	hsa-miR-4259	$8.07 \cdot E-20$	-	55
129	hsa-miR-5572	$1.06 \cdot E-19$	-	
130	hsa-miR-6075	$1.78 \cdot E-19$	-	
131	hsa-miR-296-3p	$4.75 \cdot E-19$	+	
132	hsa-miR-6891-5p	$8.62 \cdot E-19$	-	
133	hsa-miR-4745-5p	$1.02 \cdot E-18$	-	
134	hsa-miR-6775-5p	$1.17 \cdot E-18$	+	
135	hsa-miR-6870-5p	$1.24 \cdot E-18$	-	
136	hsa-miR-920	$1.78 \cdot E-18$	+	
137	hsa-miR-4530	$3.26 \cdot E-18$	-	60
138	hsa-miR-6819-5p	$3.67 \cdot E-18$	+	

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TABLE 2-continued

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject	
139	hsa-miR-6825-5p	$5.28 \cdot E-18$	-	10
140	hsa-miR-7847-3p	$7.32 \cdot E-18$	+	
141	hsa-miR-6131	$1.09 \cdot E-17$	+	
142	hsa-miR-4433-3p	$1.35 \cdot E-17$	+	
143	hsa-miR-1228-5p	$9.82 \cdot E-17$	-	
144	hsa-miR-6743-5p	$1.40 \cdot E-16$	+	
145	hsa-miR-1268a	$2.20 \cdot E-16$	-	
146	hsa-miR-3917	$1.73 \cdot E-15$	+	
147	hsa-miR-6786-5p	$1.89 \cdot E-15$	-	
148	hsa-miR-3154	$2.12 \cdot E-15$	+	15
149	hsa-miR-638	$2.19 \cdot E-15$	-	
150	hsa-miR-6741-5p	$5.44 \cdot E-15$	+	
151	hsa-miR-6889-5p	$9.29 \cdot E-15$	-	
152	hsa-miR-6840-3p	$1.61 \cdot E-14$	-	
153	hsa-miR-6510-5p	$3.23 \cdot E-14$	-	
154	hsa-miR-3188	$4.44 \cdot E-14$	+	
155	hsa-miR-551b-5p	$1.71 \cdot E-13$	+	
156	hsa-miR-5001-5p	$2.07 \cdot E-13$	+	
157	hsa-miR-1268b	$2.24 \cdot E-13$	-	20
158	hsa-miR-7107-5p	$2.31 \cdot E-13$	-	
159	hsa-miR-6824-5p	$3.30 \cdot E-13$	+	
160	hsa-miR-6732-5p	$3.62 \cdot E-13$	-	
161	hsa-miR-371a-5p	$9.05 \cdot E-13$	+	
162	hsa-miR-6794-5p	$9.74 \cdot E-13$	+	
163	hsa-miR-6779-5p	$1.37 \cdot E-12$	-	
164	hsa-miR-4271	$1.69 \cdot E-12$	-	
165	hsa-miR-5195-3p	$1.79 \cdot E-12$	+	
166	hsa-miR-6762-5p	$3.61 \cdot E-12$	+	30
167	hsa-miR-939-5p	$4.78 \cdot E-12$	-	
168	hsa-miR-1247-3p	$7.37 \cdot E-12$	+	
169	hsa-miR-6777-5p	$9.79 \cdot E-12$	+	
170	hsa-miR-6722-3p	$1.21 \cdot E-11$	+	
171	hsa-miR-3656	$1.27 \cdot E-11$	+	
172	hsa-miR-4688	$1.86 \cdot E-11$	+	
173	hsa-miR-3195	$2.02 \cdot E-11$	-	
174	hsa-miR-6766-5p	$6.97 \cdot E-11$	+	
175	hsa-miR-4447	$1.08 \cdot E-10$	+	35
176	hsa-miR-4656	$1.12 \cdot E-10$	-	
177	hsa-miR-7108-5p	$1.51 \cdot E-10$	-	
178	hsa-miR-3191-3p	$2.67 \cdot E-10$	+	
179	hsa-miR-1273g-3p	$2.89 \cdot E-10$	-	
180	hsa-miR-4463	$4.62 \cdot E-10$	+	
181	hsa-miR-2861	$4.97 \cdot E-10$	+	
182	hsa-miR-3196	$5.22 \cdot E-10$	-	
183	hsa-miR-6877-5p	$6.47 \cdot E-10$	-	
184	hsa-miR-3679-5p	$1.33 \cdot E-09$	+	40
185	hsa-miR-4442	$1.56 \cdot E-09$	-	
186	hsa-miR-6789-5p	$1.93 \cdot E-09$	-	
187	hsa-miR-6782-5p	$1.97 \cdot E-09$	+	
188	hsa-miR-486-3p	$2.12 \cdot E-09$	-	
189	hsa-miR-6085	$4.04 \cdot E-09$	+	
190	hsa-miR-4746-3p	$8.57 \cdot E-09$	-	
191	hsa-miR-619-5p	$1.13 \cdot E-08$	-	
192	hsa-miR-937-5p	$1.65 \cdot E-08$	+	
193	hsa-miR-6803-5p	$2.32 \cdot E-08$	+	50
194	hsa-miR-4298	$2.33 \cdot E-08$	+	
195	hsa-miR-4454	$2.63 \cdot E-08$	+	
196	hsa-miR-4459	$1.83 \cdot E-07$	+	
197	hsa-miR-7150	$2.60 \cdot E-07$	+	
198	hsa-miR-6880-5p	$8.86 \cdot E-07$	-	
199	hsa-miR-4449	$9.44 \cdot E-07$	-	
200	hsa-miR-8063	$1.05 \cdot E-06$	+	
201	hsa-miR-4695-5p	$1.65 \cdot E-06$	+	
202	hsa-miR-6132	$1.93 \cdot E-06$	+	55
203	hsa-miR-6829-5p	$2.66 \cdot E-06$	+	
204	hsa-miR-4486	$2.83 \cdot E-06$	-	
205	hsa-miR-6805-3p	$3.24 \cdot E-06$	-	
206	hsa-miR-6826-5p	$4.59 \cdot E-06$	+	
207	hsa-miR-4508	$6.28 \cdot E-06$	+	
208	hsa-miR-1343-5p	$1.11 \cdot E-05$	-	
209	hsa-miR-7114-5p	$1.35 \cdot E-05$	+	
210	hsa-miR-3622a-5p	$1.53 \cdot E-05$	+	
211	hsa-miR-6765-5p	$1.77 \cdot E-05$	-	60
212	hsa-miR-7845-5p	$2.11 \cdot E-05$	-	
213	hsa-miR-3960	$2.70 \cdot E-05$	-	

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TABLE 2-continued

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject
214	hsa-miR-6749-5p	$4.51 \cdot E-05$	-
215	hsa-miR-1260b	$4.83 \cdot E-05$	+
216	hsa-miR-6799-5p	$5.38 \cdot E-05$	+
217	hsa-miR-4723-5p	$6.54 \cdot E-05$	+
218	hsa-miR-6784-5p	$7.88 \cdot E-05$	-
219	hsa-miR-5100	$8.28 \cdot E-05$	+
220	hsa-miR-6769b-5p	$9.25 \cdot E-05$	+
221	hsa-miR-1207-5p	$1.25 \cdot E-04$	+
222	hsa-miR-642a-3p	$1.38 \cdot E-04$	-
223	hsa-miR-4505	$1.49 \cdot E-04$	+
224	hsa-miR-4270	$1.79 \cdot E-04$	-
225	hsa-miR-6721-5p	$3.50 \cdot E-04$	-
226	hsa-miR-7111-5p	$5.29 \cdot E-04$	-
227	hsa-miR-6791-5p	$8.34 \cdot E-04$	+
228	hsa-miR-7109-5p	$1.07 \cdot E-03$	+
229	hsa-miR-4258	$1.55 \cdot E-03$	+
230	hsa-miR-6515-3p	$2.00 \cdot E-03$	+
231	hsa-miR-6851-5p	$2.15 \cdot E-03$	-
232	hsa-miR-6125	$2.94 \cdot E-03$	-
233	hsa-miR-4749-5p	$3.39 \cdot E-03$	+

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TABLE 2-continued

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject
234	hsa-miR-4726-5p	$6.77 \cdot E-03$	+
235	hsa-miR-4513	$9.77 \cdot E-03$	+
236	hsa-miR-760	$5.40 \cdot E-76$	-
237	hsa-miR-602	$3.27 \cdot E-58$	-
238	hsa-miR-423-5p	$4.16 \cdot E-57$	-
239	hsa-miR-92a-2-5p	$7.76 \cdot E-55$	-
240	hsa-miR-16-5p	$7.58 \cdot E-47$	-
241	hsa-miR-451a	$1.13 \cdot E-36$	-
242	hsa-miR-135a-3p	$1.83 \cdot E-35$	-
243	hsa-miR-486-5p	$8.56 \cdot E-34$	-
244	hsa-miR-4257	$4.39 \cdot E-31$	+
245	hsa-miR-92b-5p	$1.67 \cdot E-30$	+
246	hsa-miR-1915-3p	$3.95 \cdot E-18$	-
247	hsa-miR-718	$1.38 \cdot E-15$	-
248	hsa-miR-940	$4.63 \cdot E-15$	+
249	hsa-miR-296-5p	$2.67 \cdot E-10$	+
250	hsa-miR-23b-3p	$2.43 \cdot E-03$	+
251	hsa-miR-92a-3p	$3.86 \cdot E-03$	-

TABLE 3

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
1	99.4	98.4	100	100	100	100
2	99.4	98.4	100	100	100	100
3	99.4	98.4	100	100	100	100
4	99.4	98.4	100	96.3	90.3	100
5	98.8	100	98	98.8	96.8	100
6	99.4	98.4	100	100	100	100
7	100	100	100	100	100	100
8	99.4	98.4	100	98.8	96.8	100
9	96.9	91.9	100	98.8	96.8	100
10	98.8	96.8	100	97.5	100	96
11	99.4	98.4	100	97.5	100	96
12	100	100	100	100	100	100
13	99.4	98.4	100	100	100	100
14	98.1	96.8	99	98.8	96.8	100
15	96.3	90.3	100	98.8	96.8	100
16	98.1	95.2	100	98.8	100	98
17	98.8	96.8	100	100	100	100
18	98.8	96.8	100	96.3	96.8	96
19	99.4	100	99	100	100	100
20	98.8	98.4	99	97.5	93.5	100
21	98.1	98.4	98	96.3	96.8	96
22	98.1	95.2	100	98.8	96.8	100
23	96.9	93.5	99	100	100	100
24	97.5	93.5	100	98.8	96.8	100
25	97.5	93.5	100	96.3	96.8	96
26	96.3	90.3	100	96.3	93.5	98
27	96.9	93.5	99	96.3	96.8	96
28	95.7	88.7	100	97.5	93.5	100
29	96.9	95.2	98	95.1	87.1	100
30	95.1	87.1	100	91.4	83.9	96
31	96.9	91.9	100	98.8	96.8	100
32	95.1	88.7	99	98.8	96.8	100
33	94.4	96.8	93	98.8	96.8	100
34	96.9	91.9	100	100	100	100
35	96.9	95.2	98	88.9	80.6	94
36	95.1	88.7	99	95.1	87.1	100
37	96.3	91.9	99	95.1	87.1	100
38	96.3	93.5	98	100	100	100
39	95.7	91.9	98	95.1	90.3	98
40	96.9	95.2	98	95.1	90.3	98
41	94.4	88.7	98	93.8	90.3	96
42	97.5	98.4	97	98.8	100	98
43	98.1	98.4	98	97.5	96.8	98
44	97.5	93.5	100	100	100	100
45	94.4	88.7	98	95.1	87.1	100

TABLE 3-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
46	96.9	91.9	100	93.8	87.1	98
47	95.7	88.7	100	98.8	96.8	100
48	96.9	95.2	98	92.6	96.8	90
49	95.7	93.5	97	93.8	90.3	96
50	95.7	96.8	95	97.5	96.8	98
51	92.6	83.9	98	92.6	83.9	98
52	93.8	93.5	94	90.1	77.4	98
53	93.8	90.3	96	91.2	90	92
54	92.6	80.6	100	93.8	83.9	100
55	95.7	91.8	98	98.8	96.8	100
56	97.5	95.2	99	96.3	93.5	98
57	93.8	88.7	97	86.4	80.6	90
58	97.5	96.8	98	96.3	96.8	96
59	95.1	90.3	98	96.3	90.3	100
60	95.1	88.7	99	95.1	93.5	96
61	93.8	90.3	96	84	90.3	80
62	91.3	80.3	98	95.1	87.1	100
63	93.2	95.2	92	88.9	87.1	90
64	92.6	85.5	97	98.8	96.8	100
65	93.2	87.1	97	93.8	83.9	100
66	93.8	87.1	98	92.6	87.1	96
67	87.7	79	93	88.9	77.4	96
68	92.6	83.9	98	96.3	90.3	100
69	91.4	82.3	97	90.1	77.4	98
70	93.2	87.1	97	91.4	90.3	92
71	92.6	87.1	96	93.8	83.9	100
72	93.8	83.9	100	90.1	74.2	100
73	95.7	91.9	98	97.5	93.5	100
74	92.6	90.3	94	87.7	87.1	88
75	90.1	79	97	96.3	93.5	98
76	93.8	91.9	95	93.8	93.5	94
77	93.8	87.1	98	90.1	77.4	98
78	92.6	82.3	99	91.4	90.3	92
79	90.7	80.6	97	90.1	87.1	92
80	90.7	75.8	100	95.1	87.1	100
81	91.4	85.5	95	87.7	83.9	90
82	90.7	88.7	92	91.4	87.1	94
83	93.2	83.9	99	96.3	93.5	98
84	89.5	90.3	89	84	77.4	88
85	90.1	82.3	95	93.8	93.5	94
86	92.6	83.9	98	87.7	74.2	96
87	88.9	79	95	93.8	83.9	100
88	92	88.7	94	92.6	100	88
89	90.7	85.5	94	91.4	90.3	92
90	92.6	90.3	94	90.1	74.2	100
91	89.5	74.2	99	92.6	83.9	98
92	89.5	82.3	94	90.1	80.6	96
93	89.5	79	96	90.1	87.1	92
94	89.5	80.6	95	84	77.4	88
95	89.5	91.9	88	86.4	83.9	88
96	87.7	79	93	88.9	71	100
97	90.1	88.7	91	91.4	96.8	88
98	90.1	83.9	94	86.4	77.4	92
99	87.6	83.6	90	82.7	87.1	80
100	88.3	80.6	93	88.9	77.4	96
101	93.2	85.5	98	88.9	71	100
102	88.3	74.2	97	95.1	90.3	98
103	87.7	80.6	92	90.1	80.6	96
104	87.7	77.4	94	80.2	67.7	88
105	92	88.7	94	82.7	77.4	86
106	86.4	77.4	92	93.8	87.1	98
107	89.5	85.5	92	86.4	74.2	94
108	88.9	80.6	94	81.5	83.9	80
109	91.4	87.1	94	87.7	77.4	94
110	88.3	75.8	96	86.4	71	96
111	86.4	82.3	89	90.1	87.1	92
112	86.4	67.7	98	86.4	74.2	94
113	85.8	74.2	93	95.1	90.3	98
114	95.1	88.7	99	92.6	80.6	100
115	90.1	82.3	95	87.7	74.2	96
116	90.1	82.3	95	88.9	83.9	92
117	83.3	72.6	90	88.9	83.9	92
118	86.4	83.9	88	80.2	71	86
119	86.4	80.6	90	88.9	87.1	90
120	88.3	79	94	84	61.3	98

TABLE 3-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
121	84.6	72.6	92	75.3	61.3	84
122	84.6	74.2	91	85.2	83.9	86
123	82.7	79	85	77.8	61.3	88
124	84.6	80.6	87	85.2	90.3	82
125	92	82.3	98	91.4	80.6	98
126	82.7	69.4	91	84	61.3	98
127	88.9	72.6	99	86.4	64.5	100
128	88.9	83.9	92	91.4	80.6	98
129	87	82.3	90	84	74.2	90
130	84	66.1	95	86.4	80.6	90
131	79.6	69.4	86	80.2	71	86
132	84.6	71	93	84	71	92
133	81.5	71	88	85.2	77.4	90
134	81.5	61.3	94	81.5	64.5	92
135	82.1	66.1	92	82.7	71	90
136	83.3	74.2	89	82.7	71	90
137	83.3	79	86	77.8	83.9	74
138	80.2	67.7	88	79	74.2	82
139	84	74.2	90	86.4	83.9	88
140	81.5	71	88	85	63.3	98
141	82.1	67.7	91	82.7	64.5	94
142	79.6	69.4	86	80.2	71	86
143	83.3	75.8	88	76.5	67.7	82
144	87.7	74.2	96	85.2	71	94
145	83.3	79	86	80.2	71	86
146	81.5	69.4	89	81.5	74.2	86
147	84.6	69.4	94	84	71	92
148	85.8	72.6	94	77.8	64.5	86
149	85.2	66.1	97	93.8	83.9	100
150	79.6	62.9	90	80.2	71	86
151	80.9	67.7	89	86.4	83.9	88
152	82.7	67.7	92	79	61.3	90
153	83.3	67.7	93	79	61.3	90
154	79.6	64.5	89	84	67.7	94
155	79.6	66.1	88	82.7	64.5	94
156	78.4	59.7	90	81.5	64.5	92
157	77.2	66.1	84	71.6	54.8	82
158	77.8	61.3	88	79	64.5	88
159	84	67.7	94	85.2	74.2	92
160	80.2	69.4	87	71.6	58.1	80
161	77.2	56.5	90	76.5	58.1	88
162	79	61.3	90	70.4	58.1	78
163	79.6	64.5	89	81.5	58.1	96
164	82.1	62.9	94	84	61.3	98
165	77.8	59.7	89	82.7	67.7	92
166	82.1	56.5	98	82.7	61.3	96
167	77.8	69.4	83	79	67.7	86
168	72.8	56.5	83	70.4	58.1	78
169	80.2	64.5	90	72.8	58.1	82
170	80.2	53.2	97	79	54.8	94
171	83.3	74.2	89	81.5	67.7	90
172	79	69.4	85	74.1	58.1	84
173	79.6	54.8	95	84	64.5	96
174	77.2	56.5	90	80.2	61.3	92
175	77.2	54.8	91	76.5	67.7	82
176	83.3	72.6	90	79	58.1	92
177	77.2	59.7	88	77.8	58.1	90
178	77.8	59.7	89	77.8	48.4	96
179	78.4	56.5	92	80.2	61.3	92
180	72.2	45.2	89	80.2	54.8	96
181	74.7	50	90	70.4	38.7	90
182	75.9	59.7	86	61.7	35.5	78
183	77.2	61.3	87	76.5	64.5	84
184	75.9	58.1	87	77.8	54.8	92
185	74.7	59.7	84	75.3	64.5	82
186	77.8	54.8	92	77.8	54.8	92
187	77.2	64.5	85	70.4	61.3	76
188	78.4	54.8	93	74.1	35.5	98
189	72.8	61.3	80	72.8	48.4	88
190	77.8	54.8	92	75.3	61.3	84
191	83.3	62.9	96	81.5	61.3	94
192	74.7	50	90	80.2	54.8	96
193	69.8	43.5	86	74.1	71	76
194	74.7	53.2	88	81.5	61.3	94
195	75.3	54.8	88	69.1	45.2	84

TABLE 3-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
196	74.7	56.5	86	75.3	48.4	92
197	74.1	54.8	86	69.1	29	94
198	69.8	48.4	83	72.8	54.8	84
199	74.1	50	89	72.8	41.9	92
200	83.3	59.7	98	87.7	67.7	100
201	68.5	41.9	85	65.4	29	88
202	75.3	45.2	94	71.6	29	98
203	71.6	38.7	92	75	53.3	88
204	71.6	53.2	83	67.9	51.6	78
205	72.2	54.8	83	69.1	45.2	84
206	70.4	45.2	86	67.9	35.5	88
207	72.8	48.4	88	65.4	41.9	80
208	69.1	45.2	84	67.9	41.9	84
209	69.1	41.9	86	72.8	48.4	88
210	74.7	51.6	89	72.8	41.9	92
211	71	53.2	82	61.7	35.5	78
212	71	45.2	87	61.7	41.9	74
213	71	24.2	100	75.3	35.5	100
214	71	45.2	87	76.5	48.4	94
215	75.3	53.2	89	65.4	32.3	86
216	71.6	50	85	71.6	41.9	90
217	69.8	43.5	86	67.9	41.9	84
218	71	40.3	90	71.6	41.9	90
219	71	40.3	90	70.4	35.5	92
220	73.5	45.2	91	66.7	35.5	86
221	71.6	43.5	89	72.8	41.9	92
222	70.4	46.8	85	76.5	61.3	86
223	77.8	46.8	97	72.8	32.3	98
224	70.4	43.5	87	72.8	45.2	90
225	67.3	37.1	86	74.1	38.7	96
226	67.9	37.1	87	75.3	51.6	90
227	71	41.9	89	66.7	29	90
228	69.1	40.3	87	65.4	35.5	84
229	71.6	46.8	87	65.4	38.7	82
230	69.1	40.3	87	71.6	54.8	82
231	71	41.9	89	72.8	58.1	82
232	71.6	41.9	90	76.5	51.6	92
233	75.3	38.7	98	72.8	29	100
234	66	33.9	86	64.2	41.9	78
235	69.8	37.1	90	67.9	38.7	86
236	100	100	100	98.8	96.8	100
237	96.9	93.5	99	98.8	96.8	100
238	98.8	98.4	99	98.8	96.8	100
239	96.9	95.2	98	98.8	100	98
240	95.1	93.5	96	96.3	96.8	96
241	95	91.8	97	91.4	87.1	94
242	90.7	85.5	94	91.4	80.6	98
243	91.4	85.5	95	96.3	100	94
244	88.9	80.6	94	90.1	87.1	92
245	94.4	87.1	99	95.1	93.5	96
246	83.3	69.4	92	81.5	67.7	90
247	83.3	61.3	97	84	67.7	94
248	79.6	64.5	89	81.5	61.3	94
249	75.9	56.5	88	77.8	67.7	84
250	72.2	41.9	91	70.4	38.7	90
251	69.1	41.9	86	77.8	54.8	92

TABLE 4

SEQ ID NO:	Discriminant coefficient	Constant term
1	2.857	18.934
2	2.104	18.009
3	2.003	12.711
4	3.992	34.175
5	3.801	39.484
6	2.447	30.974
7	4.663	59.811
8	5.763	63.889
9	5.239	59.962
10	6.627	84.995

TABLE 4-continued

SEQ ID NO:	Discriminant coefficient	Constant term
11	1.955	14.548
12	2.330	14.910
13	4.150	48.634
14	3.013	32.040
15	2.055	18.255
16	2.774	19.077
17	5.084	64.746
18	4.563	46.280
19	1.821	9.652
20	2.266	26.872

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TABLE 4-continued

SEQ ID NO:	Discriminant coefficient	Constant term	
21	2.626	14.611	5
22	3.963	34.565	
23	4.250	47.336	
24	3.079	27.833	
25	6.783	80.858	
26	3.237	25.431	10
27	1.634	14.286	
28	2.807	33.656	
29	2.658	20.910	
30	4.607	39.043	
31	5.299	67.817	15
32	2.652	19.984	
33	4.337	38.009	
34	4.682	48.862	
35	5.352	38.578	
36	2.977	26.855	20
37	2.882	21.114	
38	5.933	55.485	
39	5.346	55.986	
40	1.915	11.901	
41	4.597	34.155	25
42	2.048	10.706	
43	4.172	34.713	
44	1.763	9.076	
45	5.446	45.658	
46	2.725	15.338	30
47	1.886	10.786	
48	2.535	15.378	
49	4.415	34.502	
50	2.583	15.891	
51	3.927	44.058	35
52	3.643	32.462	
53	6.086	41.540	
54	3.018	38.864	
55	3.179	22.917	
56	2.338	13.833	40
57	5.682	67.831	
58	2.838	18.281	
59	2.954	19.780	
60	3.642	21.511	
61	1.972	23.626	45
62	4.194	40.279	
63	3.430	28.886	
64	3.095	19.571	
65	2.739	17.283	
66	1.356	7.292	50
67	4.404	35.065	
68	3.358	27.546	
69	2.497	15.102	
70	3.754	25.584	
71	3.860	28.692	55
72	2.543	20.396	
73	2.485	13.944	
74	2.870	22.658	
75	2.136	17.183	
76	2.226	12.375	60
77	4.610	47.394	
78	3.174	36.978	
79	2.980	26.297	
80	3.396	29.485	
81	3.446	23.243	65
82	4.082	29.368	
83	3.358	28.190	
84	1.537	10.210	
85	4.715	45.652	
86	2.213	17.839	60
87	3.088	27.043	
88	2.399	24.629	
89	4.002	24.314	
90	2.863	17.406	
91	4.805	44.782	65
92	3.372	29.988	
93	4.132	30.521	
94	3.851	23.002	
95	3.959	36.256	
96	2.941	23.102	65
97	3.006	20.900	

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TABLE 4-continued

SEQ ID NO:	Discriminant coefficient	Constant term
98	3.728	29.308
99	2.595	30.882
100	2.755	20.339
101	3.144	19.162
102	1.952	17.956
103	4.489	37.034
104	4.734	32.179
105	3.519	24.011
106	4.202	32.975
107	2.654	17.634
108	4.562	28.615
109	4.900	37.966
110	2.427	15.293
111	2.332	16.477
112	2.960	35.679
113	2.931	27.885
114	2.409	15.430
115	2.659	15.558
116	2.722	15.656
117	3.039	32.084
118	2.066	14.991
119	2.114	14.125
120	2.812	19.320
121	4.273	28.046
122	1.987	11.296
123	4.888	61.938
124	3.116	29.551
125	3.831	36.923
126	3.114	20.097
127	1.704	11.244
128	2.708	16.078
129	2.902	17.111
130	3.522	27.388
131	2.509	16.024
132	3.797	26.247
133	2.381	26.930
134	5.446	44.750
135	2.830	18.893
136	2.243	13.704
137	2.742	24.223
138	5.572	41.312
139	2.328	13.203
140	4.269	27.904
141	1.790	19.364
142	3.970	28.366
143	3.831	42.123
144	2.470	23.361
145	3.643	37.749
146	3.590	21.954
147	6.096	72.341
148	4.450	26.391
149	2.450	28.929
150	3.765	26.311
151	3.107	20.676
152	3.191	26.734
153	2.247	12.949
154	3.520	21.019
155	1.939	10.899
156	3.714	28.667
157	3.379	30.812
158	3.869	28.692
159	4.714	30.302
160	3.800	29.719
161	3.111	23.178
162	4.700	38.234
163	5.628	38.292
164	2.984	23.210
165	2.953	20.742
166	2.411	18.314
167	2.915	19.822
168	4.774	29.161
169	3.435	22.933
170	4.014	33.267
171	4.788	52.969
172	4.261	30.393
173	2.698	20.419
174	3.739	22.190

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TABLE 4-continued

SEQ ID NO:	Discriminant coefficient	Constant term
175	3.142	19.164
176	3.424	23.274
177	4.230	35.956
178	3.270	18.860
179	2.287	16.040
180	3.760	39.967
181	5.991	71.910
182	6.044	68.999
183	4.325	29.700
184	3.000	19.877
185	3.823	34.710
186	3.882	35.815
187	3.870	23.356
188	2.281	17.158
189	5.818	59.167
190	2.555	15.132
191	1.200	8.000
192	4.027	33.373
193	6.532	69.738
194	3.836	23.124
195	2.198	25.446
196	3.531	27.965
197	3.619	27.108
198	2.671	18.647
199	3.253	19.608
200	1.449	12.340
201	4.621	33.285
202	3.174	24.106
203	3.280	19.895
204	3.339	22.048
205	2.930	20.016
206	2.705	17.147
207	5.111	63.738
208	4.111	39.848
209	4.418	30.103
210	3.749	22.673
211	4.949	48.983
212	3.159	19.444
213	3.351	49.006
214	4.185	40.143
215	2.257	19.312
216	4.261	34.145
217	2.984	26.230
218	3.064	35.877
219	2.311	23.637
220	4.200	26.644
221	3.404	21.751
222	3.076	22.488
223	3.273	26.709
224	5.275	40.615
225	3.461	24.300
226	5.789	41.289
227	4.602	40.407
228	5.697	41.534
229	2.323	20.843
230	3.839	25.014
231	3.506	21.544
232	4.082	45.878
233	3.909	30.033
234	4.028	26.498
235	4.173	25.119
236	3.887	30.454
237	2.014	10.053
238	2.607	16.589
239	1.925	14.853
240	1.301	6.443
241	1.152	8.583
242	1.774	11.079
243	2.154	13.220
244	3.309	23.530
245	3.226	25.503
246	3.387	34.563
247	2.652	16.404

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TABLE 4-continued

SEQ ID NO:	Discriminant coefficient	Constant term
248	3.534	22.326
249	3.779	28.687
250	1.490	9.528
251	2.072	13.756

TABLE 5-1

Training cohort		
Sample name	Cancer stage	CEA(ng/mL)
BB001	IV	1(-)
BB006	IIIB	14.4(+)
BB007	IIA	2.6(-)
BB009	IIIC	4.9(-)
BB010	IIA	1.2(-)
BB012	I	2.7(-)
BB013	I	14(-)
BB014	IIA	13.5(+)
BB015	I	3.5(-)
BB017	IIA	1.9(-)
BB018	IIA	2.1(-)
BB019	I	1.2(-)
BB023	IIA	2.8(-)
BB024	I	1.5(-)
BB026	I	3.8(-)
BB027	IIIB	3.4(-)
BB028	IV	7.4(+)
BB032	I	1.7(-)
BB033	IIIB	2.6(-)
BB035	IIA	2.7(-)
BB037	IIIB	1.2(-)
BB038	IV	0.8(-)
BB039	IIIA	4.7(-)
BB041	IIA	2.1(-)
BB042	IIIB	1.3(-)
BB043	IIIA	1.2(-)
BB046	I	1.7(-)
BB047	I	1(-)
BB048	I	1.9(-)
BB049	IIA	0.6(-)
BB051	IIA	1(-)
BB052	I	2.4(-)
BB054	IIA	1.3(-)
BB056	IIA	1.9(-)
BB057	IIIB	3460(+)
BB058	IIA	0.8(-)
BB060	I	0.5(-)
BB061	I	1.4(-)
BB064	IV	2.4(-)
BB065	IIIB	6.6(+)
BB068	IIA	0.6(-)
BB070	IIIB	0.7(-)
BB071	IIIB	1.8(-)
BB073	I	1.1(-)
BB076	I	2(-)
BB077	IIA	2.9(-)
BB078	IIA	2.3(-)
BB079	IIA	1.1(-)
BB082	I	1.5(-)
BB084	IV	435(+)
BB085	IIA	3.1(-)
BB086	IIA	1.5(-)
BB087	IIA	1.7(-)
BB089	I	2(-)
BB092	I	3.6(-)
BB093	IIIB	3.7(-)
BB094	I	6(+)
BB095	IIA	2.4(-)
BB096	I	4.6(-)

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TABLE 5-1-continued

Training cohort		
Sample name	Cancer stage	CEA(ng/mL)
BB097	IIA	1.8(-)
BB098	IIA	2.6(-)
BB099	IIA	4.2(-)
Sensitivity (%)		11.3

TABLE 5-2

Validation cohort		
Sample name	Cancer stage	CEA(ng/mL)
BB002	IIIC	1.3(-)
BB005	IIA	0.6(-)
BB008	IIA	0.7(-)
BB011	IIB	1.4(-)
BB016	I	2.9(-)
BB020	IIB	1.5(-)
BB021	IIB	1.5(-)
BB022	IIA	0.6(-)
BB025	IIA	4.1(-)
BB029	IIIA	2.7(-)
BB030	IIA	12.7(+)

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TABLE 5-2-continued

Validation cohort		
Sample name	Cancer stage	CEA(ng/mL)
BB031	IIA	1.4(-)
BB034	IIB	4.2(-)
BB036	I	3.5(-)
BB040	IIB	1.7(-)
BB050	IIIB	11.1(+)
BB053	I	0.8(-)
BB055	IIA	2.8(-)
BB059	IIA	10.2(+)
BB062	IIA	5.9(+)
BB063	I	0.7(-)
BB069	IIA	0.7(-)
BB072	IIA	1.1(-)
BB074	IIA	6.4(+)
BB075	I	0.5(-)
BB080	IV	6.6(+)
BB081	I	0.8(-)
BB083	I	0.7(-)
BB088	I	1(-)
BB090	I	5.4(+)
BB091	IIA	1.2(-)
Sensitivity (%)		19.4

For CEA, 5 ng/ml or lower was indicated by as “-”, while values exceeding these were indicated as “+”

TABLE 6

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
12	100	100	100	100	100	100
13	99.4	98.4	100	100	100	100
14	100	100	100	100	100	100
15	99.4	98.4	100	100	100	100
16	100	100	100	100	100	100
17	100	100	100	100	100	100
18	100	100	100	100	100	100
19	99.4	98.4	100	100	100	100
110	100	100	100	100	100	100
111	100	100	100	100	100	100
112	100	100	100	100	100	100
113	100	100	100	100	100	100
114	100	100	100	100	100	100
115	100	100	100	100	100	100
116	100	100	100	100	100	100
117	100	100	100	100	100	100
118	99.4	98.4	100	100	100	100
119	100	100	100	100	100	100
120	100	100	100	100	100	100
121	99.4	98.4	100	100	100	100
122	100	100	100	100	100	100
123	100	100	100	100	100	100
124	100	100	100	100	100	100
125	100	100	100	98.8	100	98
126	100	100	100	100	100	100
127	100	100	100	100	100	100
128	99.4	98.4	100	100	100	100
129	100	100	100	100	100	100
130	100	100	100	100	100	100
131	100	100	100	100	100	100
132	100	100	100	100	100	100
133	99.4	98.4	100	100	100	100
134	100	100	100	100	100	100
135	100	100	100	100	100	100
136	100	100	100	100	100	100
137	100	100	100	100	100	100
138	99.4	98.4	100	100	100	100
139	100	100	100	100	100	100
140	100	100	100	100	100	100
141	100	100	100	100	100	100
142	100	100	100	100	100	100
143	99.4	98.4	100	100	100	100

TABLE 6-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
144	100	100	100	100	100	100
145	100	100	100	100	100	100
146	99.4	98.4	100	100	100	100
147	100	100	100	100	100	100
148	99.4	98.4	100	100	100	100
149	99.4	98.4	100	100	100	100
150	99.4	98.4	100	100	100	100
151	100	100	100	100	100	100
152	100	100	100	100	100	100
153	99.4	98.4	100	100	100	100
154	99.4	98.4	100	100	100	100
155	100	100	100	100	100	100
156	99.4	98.4	100	100	100	100
157	100	100	100	100	100	100
158	100	100	100	100	100	100
159	100	100	100	100	100	100
160	99.4	98.4	100	100	100	100
161	99.4	98.4	100	100	100	100
162	100	100	100	100	100	100
163	99.4	98.4	100	100	100	100
164	100	100	100	100	100	100
165	99.4	98.4	100	100	100	100
166	99.4	98.4	100	100	100	100
167	100	100	100	100	100	100
168	100	100	100	100	100	100
169	100	100	100	100	100	100
170	100	100	100	100	100	100
171	99.4	98.4	100	100	100	100
172	100	100	100	100	100	100
173	99.4	98.4	100	100	100	100
174	99.4	98.4	100	100	100	100
175	99.4	98.4	100	100	100	100
176	100	100	100	100	100	100
177	100	100	100	100	100	100
178	100	100	100	100	100	100
179	99.4	98.4	100	100	100	100
180	100	100	100	100	100	100
181	100	100	100	100	100	100
182	99.4	98.4	100	100	100	100
183	99.4	98.4	100	100	100	100
184	99.4	98.4	100	100	100	100
185	99.4	98.4	100	100	100	100
186	100	100	100	100	100	100
187	99.4	98.4	100	100	100	100
188	99.4	98.4	100	100	100	100
189	99.4	98.4	100	100	100	100
190	100	100	100	100	100	100
191	99.4	98.4	100	100	100	100
192	99.4	98.4	100	100	100	100
193	99.4	98.4	100	100	100	100
194	100	100	100	100	100	100
195	99.4	98.4	100	100	100	100
196	100	100	100	100	100	100
197	99.4	98.4	100	100	100	100
198	99.4	98.4	100	100	100	100
199	99.4	98.4	100	100	100	100
1100	100	100	100	100	100	100
1101	99.4	98.4	100	100	100	100
1102	100	100	100	100	100	100
1103	99.4	98.4	100	100	100	100
1104	100	100	100	100	100	100
1105	99.4	98.4	100	100	100	100
1106	100	100	100	100	100	100
1107	100	100	100	100	100	100
1108	99.4	98.4	100	100	100	100
1109	100	100	100	100	100	100
1110	99.4	98.4	100	100	100	100
1111	99.4	98.4	100	100	100	100
1112	100	100	100	100	100	100
1113	100	100	100	100	100	100
1114	99.4	98.4	100	100	100	100
1115	99.4	98.4	100	100	100	100
1116	99.4	98.4	100	100	100	100
1117	99.4	98.4	100	100	100	100
1118	99.4	98.4	100	100	100	100

TABLE 6-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
1119	99.4	98.4	100	100	100	100
1120	99.4	98.4	100	100	100	100
1121	100	100	100	100	100	100
1122	99.4	98.4	100	100	100	100
1123	99.4	98.4	100	100	100	100
1124	99.4	98.4	100	100	100	100
1125	100	100	100	100	100	100
1126	99.4	98.4	100	100	100	100
1127	100	100	100	100	100	100
1128	99.4	98.4	100	100	100	100
1129	99.4	98.4	100	100	100	100
1130	99.4	98.4	100	100	100	100
1131	99.4	98.4	100	100	100	100
1132	99.4	98.4	100	100	100	100
1133	99.4	98.4	100	100	100	100
1134	99.4	98.4	100	100	100	100
1135	99.4	98.4	100	100	100	100
1136	99.4	98.4	100	100	100	100
1137	99.4	98.4	100	100	100	100
1138	99.4	98.4	100	100	100	100
1139	99.4	98.4	100	100	100	100
1140	99.4	98.4	100	100	100	100
1141	99.4	98.4	100	100	100	100
1142	99.4	98.4	100	100	100	100
1143	100	100	100	100	100	100
1144	99.4	98.4	100	100	100	100
1145	99.4	98.4	100	100	100	100
1146	99.4	98.4	100	100	100	100
1147	99.4	98.4	100	100	100	100
1148	100	100	100	100	100	100
1149	99.4	98.4	100	100	100	100
1150	99.4	98.4	100	100	100	100
1151	99.4	98.4	100	100	100	100
1152	100	100	100	100	100	100
1153	99.4	98.4	100	100	100	100
1154	99.4	98.4	100	100	100	100
1155	99.4	98.4	100	100	100	100
1156	99.4	98.4	100	100	100	100
1157	100	100	100	100	100	100
1158	99.4	98.4	100	100	100	100
1159	99.4	98.4	100	100	100	100
1160	99.4	98.4	100	100	100	100
1161	100	100	100	100	100	100
1162	100	100	100	100	100	100
1163	99.4	98.4	100	100	100	100
1164	99.4	98.4	100	100	100	100
1165	99.4	98.4	100	100	100	100
1166	99.4	98.4	100	100	100	100
1167	99.4	98.4	100	100	100	100
1168	99.4	98.4	100	100	100	100
1169	100	100	100	100	100	100
1170	99.4	98.4	100	100	100	100
1171	99.4	98.4	100	100	100	100
1172	99.4	98.4	100	100	100	100
1173	99.4	98.4	100	100	100	100
1174	99.4	98.4	100	100	100	100
1175	99.4	98.4	100	100	100	100
1176	99.4	98.4	100	100	100	100
1177	99.4	98.4	100	100	100	100
1178	99.4	98.4	100	100	100	100
1179	99.4	98.4	100	100	100	100
1180	99.4	98.4	100	100	100	100
1181	99.4	98.4	100	100	100	100
1182	99.4	98.4	100	100	100	100
1183	99.4	98.4	100	100	100	100
1184	99.4	98.4	100	100	100	100
1185	99.4	98.4	100	100	100	100
1186	100	100	100	100	100	100
1187	99.4	98.4	100	100	100	100
1188	100	100	100	100	100	100
1189	99.4	98.4	100	100	100	100
1190	99.4	98.4	100	100	100	100
1191	99.4	98.4	100	100	100	100
1192	99.4	98.4	100	100	100	100
1193	99.4	98.4	100	100	100	100

TABLE 6-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
1194	99.4	98.4	100	100	100	100
1195	100	100	100	100	100	100
1196	99.4	98.4	100	100	100	100
1197	99.4	98.4	100	100	100	100
1198	99.4	98.4	100	100	100	100
1199	99.4	98.4	100	100	100	100
1200	99.4	98.4	100	100	100	100
1201	99.4	98.4	100	100	100	100
1202	99.4	98.4	100	100	100	100
1203	99.4	98.4	100	100	100	100
1204	99.4	98.4	100	100	100	100
1205	99.4	98.4	100	100	100	100
1206	99.4	98.4	100	100	100	100
1207	99.4	98.4	100	100	100	100
1208	99.4	98.4	100	100	100	100
1209	99.4	98.4	100	100	100	100
1210	99.4	98.4	100	100	100	100
1211	99.4	98.4	100	100	100	100
1212	99.4	98.4	100	100	100	100
1213	99.4	98.4	100	100	100	100
1214	100	100	100	100	100	100
1215	100	100	100	100	100	100
1216	99.4	98.4	100	100	100	100
1217	99.4	98.4	100	100	100	100
1218	99.4	98.4	100	100	100	100
1219	100	100	100	100	100	100
1220	100	100	100	100	100	100
1221	99.4	98.4	100	100	100	100
1222	99.4	98.4	100	100	100	100
1223	99.4	98.4	100	100	100	100
1224	99.4	98.4	100	100	100	100
1225	100	100	100	100	100	100
1226	99.4	98.4	100	100	100	100
1227	99.4	98.4	100	100	100	100
1228	99.4	98.4	100	100	100	100
1229	99.4	98.4	100	100	100	100
1230	99.4	98.4	100	100	100	100
1231	99.4	98.4	100	100	100	100
1232	99.4	98.4	100	100	100	100
1233	99.4	98.4	100	100	100	100
1234	99.4	98.4	100	100	100	100
1235	99.4	98.4	100	100	100	100
1236	100	100	100	100	100	100
1237	100	100	100	100	100	100
1238	100	100	100	100	100	100
1239	100	100	100	100	100	100
1240	100	100	100	100	100	100
1241	100	100	100	100	100	100
1242	100	100	100	100	100	100
1243	100	100	100	100	100	100
1244	100	100	100	100	100	100
1245	99.4	98.4	100	100	100	100
1246	99.4	98.4	100	100	100	100
1247	99.4	98.4	100	100	100	100
1248	99.4	98.4	100	100	100	100
1249	99.4	98.4	100	100	100	100
1250	99.4	98.4	100	100	100	100
1251	99.4	98.4	100	100	100	100

TABLE 7

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
3_20	100	100	100	100	100	100
6_20	100	100	100	100	100	100
7_20	100	100	100	100	100	100
10_20	100	100	100	100	100	100
20_22	98.8	96.7	100	98.8	96.8	100
20_238	98.8	96.7	100	98.8	96.8	100

TABLE 7-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
20_239	100	100	100	100	100	100
12_24	100	100	100	98.8	96.8	100
20_24	100	100	100	98.8	96.8	100
24_27	98.8	96.8	100	98.8	96.8	100
24_33	100	100	100	100	100	100
24_236	100	100	100	98.8	96.8	100
24_240	100	100	100	100	100	100
3_26	100	100	100	100	100	100
12_26	100	100	100	98.8	96.8	100
13_26	100	100	100	98.8	96.8	100
17_26	100	100	100	98.8	100	98
19_26	100	100	100	98.8	100	98
3_27	100	100	100	100	100	100
5_27	100	100	100	100	100	100
13_27	99.4	98.4	100	100	100	100
20_27	98.1	96.7	99	100	100	100
26_27	99.4	98.4	100	96.3	93.5	98
27_120	98.8	98.4	99	100	100	100
27_206	98.1	95.2	100	97.5	96.8	98
27_237	99.4	98.4	100	100	100	100
3_30	100	100	100	100	100	100
17_30	99.4	98.4	100	100	100	100
27_30	100	100	100	98.8	96.8	100
27_33	98.8	98.4	99	100	100	100
30_39	99.4	98.4	100	100	100	100
30_117	99.4	98.4	100	98.8	96.8	100
3_33	99.4	98.4	100	100	100	100
7_33	100	100	100	100	100	100
10_33	100	100	100	100	100	100
11_33	100	100	100	98.8	100	98
13_33	100	100	100	100	100	100
25_33	100	100	100	98.8	100	98
33_244	100	100	100	100	100	100
3_182	99.4	98.4	100	100	100	100
6_182	99.4	98.4	100	100	100	100
7_182	100	100	100	100	100	100
12_182	100	100	100	100	100	100
27_182	98.1	96.8	99	97.5	96.8	98
182_236	100	100	100	98.8	96.8	100
2_194	99.4	98.4	100	100	100	100
7_194	100	100	100	100	100	100
27_194	98.1	95.2	100	97.5	96.8	98
194_236	100	100	100	98.8	96.8	100
2_206	99.4	98.4	100	100	100	100
7_206	100	100	100	100	100	100
206_236	100	100	100	98.8	96.8	100
2_208	99.4	98.4	100	100	100	100
7_208	100	100	100	100	100	100
13_208	99.4	98.4	100	100	100	100
20_208	98.1	96.7	99	97.5	93.5	100
27_208	98.1	95.2	100	98.8	96.8	100

Example 3

<Selection of Gene Markers Using all Samples and Method for Evaluating Breast Cancer Discriminant Performance of Acquired Gene Markers>

In this Example, the samples of the training cohort and the validation cohort used in Examples 1 and 2 were integrated, and selection of a gene marker and evaluation of its breast cancer discriminant performance were conducted using all of the samples.

Specifically, the miRNA expression levels in the sera of the 93 breast cancer patients and the 150 healthy subjects obtained in the preceding Reference Examples were normalized by quantile normalization. In order to acquire diagnostic markers with higher reliability, only genes having a gene expression level of 2^6 or higher in 50% or more of the samples in either of the breast cancer patient group or the healthy subject group, were selected in the gene marker

selection. In order to further acquire statistical significance for discriminating a breast cancer patient group from a healthy subject group, the P value obtained by two-tailed t-test assuming equal variance as to each gene expression level was corrected by the Bonferroni method, and genes that satisfied $p < 0.01$ were selected as gene markers for use in explanatory variables of a discriminant. The obtained genes are described in Table 7. In this way, hsa-miR-658, hsa-miR-6842-5p, hsa-miR-6124, hsa-miR-6765-3p, hsa-miR-7106-5p, hsa-miR-4534, hsa-miR-92b-3p, hsa-miR-3135b, hsa-miR-4687-3p, hsa-miR-762, hsa-miR-3619-3p, hsa-miR-4467, hsa-miR-557, hsa-miR-1237-5p, hsa-miR-1908-5p, hsa-miR-4286, hsa-miR-6885-5p and hsa-miR-6763-5p genes, and the nucleotide sequences of SEQ ID NOs: 252 to 269 related thereto were found in addition to the genes described in Table 2. As with the nucleotide sequences of SEQ ID NOs: 1 to 251, the results obtained about the polynucleotides represented by the nucleotide sequences of

SEQ ID NOs: 252 to 269 also showed that the measurement values were significantly lower (–) or higher (+) in the breast cancer patient group than in the healthy subject group (Table 8). These results were able to be validated in the validation cohort. Thus, the presence or absence of breast cancer in the newly obtained samples can be determined by the methods described in Examples 1 and 2 by using the gene expression level measurement values described in Table 8 either alone or in combination with the gene expression level measurement values described in Table 2

TABLE 8

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject
1	hsa-miR-4783-3p	1.81 · E-151	+
2	hsa-miR-4730	1.00 · E-148	–
3	hsa-miR-1307-3p	5.85 · E-126	+
4	hsa-miR-4634	9.92 · E-114	–
5	hsa-miR-663a	2.55 · E-112	–
6	hsa-miR-4532	8.96 · E-113	+
7	hsa-miR-7704	4.82 · E-114	+
8	hsa-miR-3178	5.69 · E-112	–
9	hsa-miR-6729-5p	3.52 · E-112	–
10	hsa-miR-6090	1.64 · E-100	–
11	hsa-miR-4732-5p	7.04 · E-96	–
12	hsa-miR-3184-5p	2.08 · E-103	+
13	hsa-miR-6727-5p	3.11 · E-99	+
14	hsa-miR-6088	4.52 · E-100	–
15	hsa-miR-4674	2.03 · E-94	+
16	hsa-miR-8073	5.49 · E-98	–
17	hsa-miR-4787-5p	1.53 · E-96	+
18	hsa-miR-1469	3.95 · E-87	+
19	hsa-miR-125a-3p	3.12 · E-91	–
20	hsa-miR-1233-5p	2.43 · E-93	+
21	hsa-miR-885-3p	4.92 · E-87	–
22	hsa-miR-6802-5p	5.06 · E-87	–
23	hsa-miR-328-5p	6.06 · E-96	+
24	hsa-miR-6787-5p	2.26 · E-85	+
25	hsa-miR-8069	5.26 · E-78	+
26	hsa-miR-6875-5p	2.38 · E-74	+
27	hsa-miR-1246	1.05 · E-75	+
28	hsa-miR-4734	6.82 · E-87	–
29	hsa-miR-6757-5p	9.57 · E-75	+
30	hsa-miR-6756-5p	4.28 · E-70	–
31	hsa-miR-3665	5.78 · E-78	–
32	hsa-miR-6836-3p	7.87 · E-76	–
33	hsa-miR-6821-5p	1.18 · E-80	–
34	hsa-miR-6805-5p	3.26 · E-79	–
35	hsa-miR-4728-5p	9.76 · E-65	+
36	hsa-miR-6726-5p	4.07 · E-72	+
37	hsa-miR-197-5p	5.63 · E-71	–
38	hsa-miR-149-3p	3.94 · E-70	+
39	hsa-miR-6850-5p	2.45 · E-70	–
40	hsa-miR-4476	8.32 · E-64	+
41	hsa-miR-6858-5p	4.09 · E-67	–
42	hsa-miR-564	5.81 · E-69	+
43	hsa-miR-4763-3p	2.83 · E-72	+
44	hsa-miR-575	1.96 · E-69	–
45	hsa-miR-6771-5p	1.90 · E-65	–
46	hsa-miR-1231	8.50 · E-61	–
47	hsa-miR-1908-3p	3.70 · E-64	+
48	hsa-miR-150-3p	2.41 · E-58	–
49	hsa-miR-3937	5.29 · E-66	–
50	hsa-miR-887-3p	1.78 · E-64	–
51	hsa-miR-3940-5p	2.92 · E-65	–
52	hsa-miR-4741	1.21 · E-57	+
53	hsa-miR-6808-5p	5.95 · E-62	–
54	hsa-miR-6869-5p	9.36 · E-66	–
55	hsa-miR-5090	2.20 · E-62	–
56	hsa-miR-615-5p	1.59 · E-59	–
57	hsa-miR-8072	1.93 · E-53	+
58	hsa-miR-128-l-5p	2.45 · E-57	+
59	hsa-miR-1238-5p	3.00 · E-60	–
60	hsa-miR-365a-5p	5.03 · E-59	+
61	hsa-miR-204-3p	1.93 · E-49	–

TABLE 8-continued

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject
62	hsa-miR-4492	6.71 · E-60	–
63	hsa-miR-6785-5p	3.81 · E-56	+
64	hsa-miR-6511a-5p	6.21 · E-62	+
65	hsa-miR-4525	2.38 · E-55	–
66	hsa-miR-1915-5p	8.81 · E-57	–
67	hsa-miR-3180	1.19 · E-50	–
68	hsa-miR-6879-5p	2.38 · E-58	–
69	hsa-miR-1199-5p	7.83 · E-51	–
70	hsa-miR-6746-5p	9.76 · E-50	–
71	hsa-miR-711	1.44 · E-49	+
72	hsa-miR-663b	4.49 · E-49	–
73	hsa-miR-4707-3p	2.18 · E-52	+
74	hsa-miR-6893-5p	2.49 · E-43	–
75	hsa-miR-4675	5.39 · E-56	+
76	hsa-miR-4638-5p	1.64 · E-51	+
77	hsa-miR-4651	6.25 · E-50	–
78	hsa-miR-6087	6.15 · E-53	–
79	hsa-miR-4665-5p	3.35 · E-48	–
80	hsa-miR-4758-5p	1.52 · E-54	–
81	hsa-miR-6887-5p	8.54 · E-46	–
82	hsa-miR-3620-5p	4.32 · E-42	–
83	hsa-miR-1909-3p	6.17 · E-53	+
84	hsa-miR-7641	2.63 · E-43	–
85	hsa-miR-6724-5p	3.58 · E-49	+
86	hsa-miR-1343-3p	1.04 · E-45	–
87	hsa-miR-6780b-5p	5.24 · E-50	–
88	hsa-miR-4484	5.63 · E-49	+
89	hsa-miR-4690-5p	2.68 · E-43	–
90	hsa-miR-4429	7.93 · E-44	–
91	hsa-miR-1227-5p	1.26 · E-46	–
92	hsa-miR-4725-3p	3.25 · E-44	+
93	hsa-miR-6861-5p	2.28 · E-46	+
94	hsa-miR-6812-5p	3.34 · E-37	+
95	hsa-miR-3197	8.32 · E-45	+
96	hsa-miR-8059	6.38 · E-41	+
97	hsa-miR-3185	7.20 · E-46	+
98	hsa-miR-4706	4.07 · E-39	+
99	hsa-miR-4497	7.65 · E-39	+
100	hsa-miR-3131	4.60 · E-42	–
101	hsa-miR-6806-5p	8.93 · E-34	+
102	hsa-miR-187-5p	5.69 · E-42	+
103	hsa-miR-3180-3p	1.37 · E-41	–
104	hsa-miR-6848-5p	9.99 · E-33	–
105	hsa-miR-6820-5p	2.48 · E-37	+
106	hsa-miR-6800-5p	2.58 · E-42	+
107	hsa-miR-6717-5p	2.75 · E-37	–
108	hsa-miR-6795-5p	2.83 · E-36	–
109	hsa-miR-4632-5p	1.50 · E-40	+
110	hsa-miR-665	3.04 · E-37	–
111	hsa-miR-6778-5p	3.57 · E-36	–
112	hsa-miR-3663-3p	2.48 · E-39	+
113	hsa-miR-4689	1.46 · E-43	+
114	hsa-miR-211-3p	1.36 · E-39	+
115	hsa-miR-6511b-5p	4.55 · E-39	+
116	hsa-miR-4750-5p	1.70 · E-32	+
117	hsa-miR-6126	6.52 · E-41	+
118	hsa-miR-614	1.36 · E-31	+
119	hsa-miR-7110-5p	2.57 · E-36	–
120	hsa-miR-744-5p	1.36 · E-32	–
121	hsa-miR-6769a-5p	2.68 · E-28	+
122	hsa-miR-4792	3.78 · E-29	–
123	hsa-miR-5787	5.62 · E-30	+
124	hsa-miR-6798-5p	8.69 · E-34	–
125	hsa-miR-6781-5p	1.34 · E-35	+
126	hsa-miR-4419b	3.49 · E-28	–
127	hsa-miR-4446-3p	2.83 · E-32	–
128	hsa-miR-4259	9.25 · E-30	–
129	hsa-miR-5572	7.92 · E-32	–
130	hsa-miR-6075	1.06 · E-34	–
131	hsa-miR-296-3p	7.28 · E-27	–
132	hsa-miR-6891-5p	1.19 · E-28	+
133	hsa-miR-4745-5p	1.47 · E-30	+
134	hsa-miR-6775-5p	1.24 · E-28	–
135	hsa-miR-6870-5p	1.50 · E-29	–
136	hsa-miR-920	2.07 · E-27	+

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TABLE 8-continued

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject
137	hsa-miR-4530	4.79 · E-26	+
138	hsa-miR-6819-5p	4.33 · E-27	-
139	hsa-miR-6825-5p	3.99 · E-29	+
140	hsa-miR-7847-3p	1.49 · E-26	+
141	hsa-miR-6131	3.83 · E-28	-
142	hsa-miR-4433-3p	8.75 · E-27	-
143	hsa-miR-1228-5p	4.12 · E-20	-
144	hsa-miR-6743-5p	4.05 · E-31	-
145	hsa-miR-1268a	1.93 · E-25	+
146	hsa-miR-3917	3.71 · E-24	-
147	hsa-miR-6786-5p	3.71 · E-27	+
148	hsa-miR-3154	1.49 · E-21	+
149	hsa-miR-638	2.61 · E-28	+
150	hsa-miR-6741-5p	2.54 · E-24	-
151	hsa-miR-6889-5p	3.14 · E-26	+
152	hsa-miR-6840-3p	4.16 · E-20	-
153	hsa-miR-6510-5p	2.15 · E-19	+
154	hsa-miR-3188	3.01 · E-21	+
156	hsa-miR-5001-5p	1.47 · E-22	-
157	hsa-miR-1268b	1.93 · E-20	+
158	hsa-miR-7107-5p	6.37 · E-21	-
159	hsa-miR-6824-5p	9.39 · E-21	-
160	hsa-miR-6732-5p	3.49 · E-17	-
161	hsa-miR-371a-5p	1.21 · E-19	-
162	hsa-miR-6794-5p	1.02 · E-15	+
163	hsa-miR-6779-5p	6.91 · E-17	+
164	hsa-miR-4271	2.14 · E-22	+
165	hsa-miR-5195-3p	1.17 · E-19	-
166	hsa-miR-6762-5p	5.65 · E-18	+
167	hsa-miR-939-5p	4.70 · E-21	+
168	hsa-miR-1247-3p	4.05 · E-14	+
169	hsa-miR-6777-5p	4.37 · E-16	-
170	hsa-miR-6722-3p	9.50 · E-18	-
171	hsa-miR-3656	1.77 · E-21	+
172	hsa-miR-4688	4.75 · E-17	-
173	hsa-miR-3195	3.33 · E-18	-
174	hsa-miR-6766-5p	1.29 · E-18	-
175	hsa-miR-4447	1.50 · E-14	+
176	hsa-miR-4656	7.31 · E-14	+
177	hsa-miR-7108-5p	1.06 · E-16	-
179	hsa-miR-1273g-3p	3.79 · E-17	+
180	hsa-miR-4463	4.63 · E-18	-
181	hsa-miR-2861	2.60 · E-15	+
182	hsa-miR-3196	6.55 · E-12	+
183	hsa-miR-6877-5p	2.44 · E-17	+
184	hsa-miR-3679-5p	6.45 · E-18	+
185	hsa-miR-4442	3.11 · E-15	-
186	hsa-miR-6789-5p	1.50 · E-14	-
187	hsa-miR-6782-5p	1.14 · E-13	+
188	hsa-miR-486-3p	8.80 · E-15	-
189	hsa-miR-6085	3.88 · E-15	+
190	hsa-miR-4746-3p	2.87 · E-13	-
191	hsa-miR-619-5p	2.89 · E-12	+
192	hsa-miR-937-5p	2.36 · E-13	+
193	hsa-miR-6803-5p	2.14 · E-12	+
194	hsa-miR-4298	1.25 · E-14	+
195	hsa-miR-4454	9.63 · E-11	+
196	hsa-miR-4459	3.66 · E-13	-
197	hsa-miR-7150	1.76 · E-10	+
198	hsa-miR-6880-5p	3.57 · E-11	-
199	hsa-miR-4449	4.62 · E-11	+
200	hsa-miR-8063	4.89 · E-13	+
201	hsa-miR-4695-5p	5.30 · E-09	-
202	hsa-miR-6132	2.15 · E-10	+
203	hsa-miR-6829-5p	3.96 · E-09	-
204	hsa-miR-4486	2.13 · E-07	+
205	hsa-miR-6805-3p	8.96 · E-10	+
206	hsa-miR-6826-5p	7.62 · E-08	-
207	hsa-miR-4508	6.79 · E-08	+
208	hsa-miR-1343-5p	8.96 · E-07	+
209	hsa-miR-7114-5p	1.55 · E-09	+
210	hsa-miR-3622a-5p	2.52 · E-08	-
211	hsa-miR-6765-5p	6.46 · E-07	+
212	hsa-miR-7845-5p	1.78 · E-04	-
213	hsa-miR-3960	2.10 · E-10	-

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TABLE 8-continued

SEQ ID NO:	Gene name	P value after Bonferroni correction	Expression level in breast cancer patient with respect to healthy subject
214	hsa-miR-6749-5p	1.32 · E-07	-
215	hsa-miR-1260b	1.44 · E-04	+
216	hsa-miR-6799-5p	1.14 · E-07	+
217	hsa-miR-4723-5p	6.30 · E-07	-
218	hsa-miR-6784-5p	4.08 · E-08	+
219	hsa-miR-5100	1.78 · E-06	-
220	hsa-miR-6769b-5p	6.44 · E-06	+
221	hsa-miR-1207-5p	8.21 · E-07	-
222	hsa-miR-642a-3p	2.06 · E-09	-
223	hsa-miR-4505	3.06 · E-06	+
224	hsa-miR-4270	9.73 · E-07	+
225	hsa-miR-6721-5p	1.84 · E-08	-
226	hsa-miR-7111-5p	3.24 · E-09	+
227	hsa-miR-6791-5p	2.88 · E-06	-
228	hsa-miR-7109-5p	7.96 · E-05	+
229	hsa-miR-4258	1.36 · E-05	+
230	hsa-miR-6515-3p	3.81 · E-03	+
231	hsa-miR-6851-5p	1.63 · E-06	-
232	hsa-miR-6125	1.36 · E-05	-
233	hsa-miR-4749-5p	7.65 · E-05	-
234	hsa-miR-4726-5p	5.88 · E-04	+
235	hsa-miR-4513	4.79 · E-06	-
236	hsa-miR-760	1.45 · E-111	+
237	hsa-miR-602	1.06 · E-88	-
238	hsa-miR-423-5p	9.24 · E-79	-
239	hsa-miR-92a-2-5p	2.54 · E-85	-
240	hsa-miR-16-5p	1.34 · E-69	+
241	hsa-miR-451a	5.22 · E-54	+
242	hsa-miR-135a-3p	8.02 · E-58	+
243	hsa-miR-486-5p	8.22 · E-52	+
244	hsa-miR-4257	4.02 · E-46	+
245	hsa-miR-92b-5p	1.86 · E-45	-
246	hsa-miR-1915-3p	4.13 · E-28	-
247	hsa-miR-718	1.29 · E-25	+
248	hsa-miR-940	5.23 · E-17	-
249	hsa-miR-296-5p	3.77 · E-16	-
250	hsa-miR-23b-3p	1.91 · E-04	+
251	hsa-miR-92a-3p	3.74 · E-07	+
252	hsa-miR-658	4.76 · E-43	+
253	hsa-miR-6842-5p	3.63 · E-12	-
254	hsa-miR-6124	8.99 · E-06	+
255	hsa-miR-6765-3p	2.30 · E-05	-
256	hsa-miR-7106-5p	4.48 · E-05	-
257	hsa-miR-4534	2.10 · E-04	-
258	hsa-miR-92b-3p	2.53 · E-04	-
259	hsa-miR-3135b	3.35 · E-04	+
260	hsa-miR-4687-3p	3.90 · E-04	+
261	hsa-miR-762	4.64 · E-04	-
262	hsa-miR-3619-3p	1.13 · E-03	+
263	hsa-miR-4467	1.30 · E-03	-
264	hsa-miR-557	1.49 · E-03	+
265	hsa-miR-1237-5p	1.57 · E-03	+
266	hsa-miR-1908-5p	4.45 · E-03	-
267	hsa-miR-4286	5.37 · E-03	-
268	hsa-miR-6885-5p	7.57 · E-03	+
269	hsa-miR-6763-5p	9.02 · E-03	+

Example 4

<Method for Evaluating Breast Cancer-Specific Discriminant Performance by Combination of Multiple Gene Markers Using Samples in the Validation Cohort>

In this Example, gene expression levels of miRNAs in sera were compared between breast cancer patients and a control group consisting of healthy subjects and prostate cancer patients in the same way as the method described in Example 1 using the gene markers selected in Example 1 with respect to the training cohort described in Reference Example 2, to select a gene marker for diagnosis. The polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOS: 851 to 856 thus newly selected

were further combined with the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 269 to study a method for evaluating breast cancer-specific discriminant performance.

Specifically, first, the miRNA expression levels in the training cohort and the validation cohort obtained in Reference Example 2 mentioned above were combined and normalized by quantile normalization. Next, Fisher's discriminant analysis was conducted as to combinations of 1 to 2 measurement values that comprise at least one of the measurement values of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 269, and 851 to 856 to construct a discriminant for determining the presence or absence of breast cancer. Next, accuracy, sensitivity, and specificity in the validation cohort were calculated using the discriminant thus prepared, with a positive sample group that is the breast cancer patient group and a negative sample group that is a combination of the healthy subject group and the prostate cancer patient group. The discriminant performance of the selected polynucleotides was validated using independent samples.

Most of polynucleotides consisting of the nucleotide sequences represented by these SEQ ID NOs (SEQ ID NOs: 1 to 269, and 851 to 856 corresponding to the miRNA markers of Table 1) or complementary sequences thereof mentioned above were able to provide relatively high accuracy, sensitivity, and specificity in the determination of the presence or absence of breast cancer, and furthermore, were able to specifically discriminate breast cancer from the other cancers.

At least one polynucleotide selected from the group consisting of polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 41, 43, 45, 46, 47, 49, 50, 51, 52, 53, 54, 55, 58, 59, 60, 62, 63, 64, 65, 67, 68, 69, 70, 71, 72, 73, 75, 77, 79, 80, 81, 82, 83, 86, 88, 89, 90, 92, 93, 94, 96, 98, 99, 100, 103, 104, 106, 107, 108, 110, 111, 113, 114, 115, 116, 118, 119, 121, 122, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 135, 136, 139, 140, 143, 145, 146, 147, 149, 150, 155, 157, 160, 161, 165, 167, 171, 173, 174, 175, 177, 178, 181, 182, 186, 190, 193, 194, 199, 204, 205, 206, 208, 211, 218, 225, 232, 236, 237, 238, 239, 242, 243, 244, 246, 247, 252, 260, 265, 266, 851, 852, 853, 854, 855 and 856, or complementary sequences thereof (the cancer type-specific polynucleotide group 1) was capable of specifically binding to the target marker. Among the combinations of multiple polynucleotides selected from the cancer type-specific polynucleotide group 1, particularly, combinations comprising at least one polynucleotide selected from the group consisting of polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 41, 43, 45, 46, 47, 49, 50, 51, 52, 54, 55, 58, 59, 60, 62, 63, 64, 65, 67, 68, 69, 71, 72, 73, 75, 77, 79, 80, 82, 83, 86, 88, 92, 93, 96, 99, 103, 104, 106, 110, 111, 114, 116, 118, 119, 122, 124, 125, 127, 130, 132, 133, 135, 139, 143, 145, 147, 149, 157, 160, 173, 177, 181, 182, 186, 211, 218, 232, 236, 237, 238, 239, 242, 243, 246, 247, 260, 266, 851, 852, 853 and 854, or complementary sequences thereof (the cancer type-specific polynucleotide group 2) were able to specifically discriminate breast cancer from the other cancer with high accuracy.

The number of the polynucleotides with cancer type specificity in the combination mentioned above can be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more for the combination. The

combinations of 2 or more of these polynucleotides were able to exhibit discriminant accuracy of 95% or higher.

Specifically, the discriminant accuracy of the measurement using combinations of one or two polynucleotide(s) that consists of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 269 and 851 to 856 or a complementary sequence thereof is shown in Table 9.

For example, the combinations of SEQ ID NOs: 2 and 1, SEQ ID NOs: 2 and 237, SEQ ID NOs: 2 and 4, SEQ ID NOs: 2 and 3, SEQ ID NOs: 2 and 51, SEQ ID NOs: 1 and 237, SEQ ID NOs: 1 and 4, SEQ ID NOs: 1 and 3, SEQ ID NOs: 1 and 51, SEQ ID NOs: 1 and 6, SEQ ID NOs: 4 and 237, SEQ ID NOs: 3 and 237, SEQ ID NOs: 51 and 237, SEQ ID NOs: 237 and 6, SEQ ID NOs: 237 and 12, SEQ ID NOs: 3 and 4, SEQ ID NOs: 4 and 51, SEQ ID NOs: 4 and 6, SEQ ID NOs: 4 and 12, SEQ ID NOs: 4 and 15, SEQ ID NOs: 3 and 51, SEQ ID NOs: 3 and 6, SEQ ID NOs: 3 and 12, SEQ ID NOs: 3 and 15, SEQ ID NOs: 3 and 8, SEQ ID NOs: 51 and 6, SEQ ID NOs: 51 and 12, SEQ ID NOs: 51 and 15, SEQ ID NOs: 51 and 8, SEQ ID NOs: 51 and 34, SEQ ID NOs: 2 and 6, SEQ ID NOs: 12 and 6, SEQ ID NOs: 15 and 6, SEQ ID NOs: 8 and 6, SEQ ID NOs: 6 and 34, SEQ ID NOs: 2 and 12, SEQ ID NOs: 1 and 12, SEQ ID NOs: 12 and 15, SEQ ID NOs: 8 and 12, SEQ ID NOs: 12 and 34, SEQ ID NOs: 2 and 15, SEQ ID NOs: 1 and 15, SEQ ID NOs: 237 and 15, SEQ ID NOs: 8 and 15, SEQ ID NOs: 15 and 34, SEQ ID NOs: 2 and 8, SEQ ID NOs: 1 and 8, SEQ ID NOs: 237 and 8, SEQ ID NOs: 4 and 8, SEQ ID NOs: 8 and 34, SEQ ID NOs: 2 and 34, SEQ ID NOs: 1 and 34, SEQ ID NOs: 237 and 34, SEQ ID NOs: 4 and 34, SEQ ID NOs: 3 and 34, SEQ ID NOs: 2 and 9, SEQ ID NOs: 1 and 9, SEQ ID NOs: 9 and 237, SEQ ID NOs: 4 and 9, SEQ ID NOs: 3 and 9, SEQ ID NOs: 2 and 143, SEQ ID NOs: 1 and 143, SEQ ID NOs: 237 and 143, SEQ ID NOs: 4 and 143, SEQ ID NOs: 3 and 143, SEQ ID NOs: 2 and 13, SEQ ID NOs: 1 and 13, SEQ ID NOs: 237 and 13, SEQ ID NOs: 4 and 13, SEQ ID NOs: 3 and 13, SEQ ID NOs: 2 and 125, SEQ ID NOs: 1 and 125, SEQ ID NOs: 237 and 125, SEQ ID NOs: 4 and 125, SEQ ID NOs: 3 and 125, SEQ ID NOs: 2 and 236, SEQ ID NOs: 1 and 236, SEQ ID NOs: 237 and 236, SEQ ID NOs: 4 and 236, SEQ ID NOs: 3 and 236, SEQ ID NOs: 2 and 46, SEQ ID NOs: 1 and 46, SEQ ID NOs: 237 and 46, SEQ ID NOs: 4 and 46, SEQ ID NOs: 3 and 46, SEQ ID NOs: 2 and 32, SEQ ID NOs: 1 and 32, SEQ ID NOs: 237 and 32, SEQ ID NOs: 4 and 32, SEQ ID NOs: 3 and 32, SEQ ID NOs: 2 and 62, SEQ ID NOs: 1 and 62, SEQ ID NOs: 237 and 62, SEQ ID NOs: 4 and 62, SEQ ID NOs: 3 and 62, SEQ ID NOs: 2 and 88, SEQ ID NOs: 1 and 88, SEQ ID NOs: 237 and 88, SEQ ID NOs: 4 and 88, SEQ ID NOs: 3 and 88, SEQ ID NOs: 2 and 52, SEQ ID NOs: 1 and 52, SEQ ID NOs: 237 and 52, SEQ ID NOs: 4 and 52, SEQ ID NOs: 3 and 52, SEQ ID NOs: 2 and 7, SEQ ID NOs: 1 and 7, SEQ ID NOs: 237 and 7, SEQ ID NOs: 4 and 7, SEQ ID NOs: 3 and 7, SEQ ID NOs: 2 and 26, SEQ ID NOs: 1 and 26, SEQ ID NOs: 237 and 26, SEQ ID NOs: 4 and 26, SEQ ID NOs: 3 and 26, SEQ ID NOs: 2 and 25, SEQ ID NOs: 1 and 25, SEQ ID NOs: 237 and 25, SEQ ID NOs: 4 and 25, SEQ ID NOs: 3 and 25, SEQ ID NOs: 2 and 54, SEQ ID NOs: 1 and 54, SEQ ID NOs: 237 and 54, SEQ ID NOs: 4 and 54, SEQ ID NOs: 3 and 54, SEQ ID NOs: 2 and 92, SEQ ID NOs: 1 and 92, SEQ ID NOs: 237 and 92, SEQ ID NOs: 4 and 92, SEQ ID NOs: 3 and 92, SEQ ID NOs: 2 and 14, SEQ ID NOs: 1 and 14, SEQ ID NOs: 237 and 14, SEQ ID NOs: 4 and 14, SEQ ID NOs: 3 and 14, SEQ ID NOs: 2 and 242, SEQ ID NOs: 1 and 242, SEQ ID NOs: 237 and 242, SEQ ID NOs: 4 and 242, SEQ ID NOs: 3 and 242, SEQ ID NOs: 2 and 47, SEQ ID NOs: 1 and 47, SEQ ID NOs: 237 and 47, SEQ ID NOs: 4 and 47, SEQ ID NOs:

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ID NOs: 237 and 130, SEQ ID NOs: 4 and 130, SEQ ID NOs: 3 and 130, SEQ ID NOs: 2 and 177, SEQ ID NOs: 1 and 177, SEQ ID NOs: 237 and 177, SEQ ID NOs: 4 and 177, SEQ ID NOs: 3 and 177, SEQ ID NOs: 2 and 64, SEQ ID NOs: 1 and 64, SEQ ID NOs: 237 and 64, SEQ ID NOs: 4 and 64, SEQ ID NOs: 3 and 64, SEQ ID NOs: 2 and 114, SEQ ID NOs: 1 and 114, SEQ ID NOs: 237 and 114, SEQ ID NOs: 4 and 114, SEQ ID NOs: 3 and 114, SEQ ID NOs: 2 and 119, SEQ ID NOs: 1 and 119, SEQ ID NOs: 237 and 119, SEQ ID NOs: 4 and 119, SEQ ID NOs: 3 and 119, SEQ ID NOs: 2 and 135, SEQ ID NOs: 1 and 135, SEQ ID NOs: 237 and 135, SEQ ID NOs: 4 and 135, SEQ ID NOs: 3 and 135, SEQ ID NOs: 2 and 243, SEQ ID NOs: 1 and 243, SEQ ID NOs: 237 and 243, SEQ ID NOs: 4 and 243, SEQ ID NOs: 3 and 243, SEQ ID NOs: 2 and 122, SEQ ID NOs: 1 and 122, SEQ ID NOs: 237 and 122, SEQ ID NOs: 4 and 122, SEQ ID NOs: 3 and 122, SEQ ID NOs: 2 and 260, SEQ ID NOs: 1 and 260, SEQ ID NOs: 237 and 260, SEQ ID NOs: 4 and 260, SEQ ID NOs: 3 and 260, SEQ ID NOs: 2 and 59, SEQ ID NOs: 1 and 59, SEQ ID NOs: 237 and 59, SEQ ID NOs: 4 and 59, SEQ ID NOs: 3 and 59, SEQ ID NOs: 2 and 854, SEQ ID NOs: 1 and 854, SEQ ID NOs: 237 and 854, SEQ ID NOs: 4 and 854, SEQ ID NOs: 3 and 854, SEQ ID NOs: 2 and 132, SEQ ID NOs: 1 and 132, SEQ ID NOs: 237 and 132, SEQ ID NOs: 4 and 132, SEQ ID NOs: 3 and 132, SEQ ID NOs: 2 and 181, SEQ ID NOs: 1 and 181, SEQ ID NOs: 237 and 181, SEQ ID NOs: 4 and 181, SEQ ID NOs: 3 and 181, SEQ ID NOs: 2 and 79, SEQ ID NOs: 1 and 79, SEQ ID NOs: 237 and 79, SEQ ID NOs: 4 and 79, SEQ ID NOs: 3 and 79, SEQ ID NOs: 2 and 133, SEQ ID NOs: 1 and 133, SEQ ID NOs: 237 and 133, SEQ ID NOs: 4 and 133, SEQ ID NOs: 3 and 133, SEQ ID NOs: 2 and 41,

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SEQ ID NOs: 1 and 41, SEQ ID NOs: 237 and 41, SEQ ID NOs: 4 and 41, SEQ ID NOs: 3 and 41, SEQ ID NOs: 2 and 139, SEQ ID NOs: 1 and 139, SEQ ID NOs: 237 and 139, SEQ ID NOs: 4 and 139, SEQ ID NOs: 3 and 139, SEQ ID NOs: 2 and 118, SEQ ID NOs: 1 and 118, SEQ ID NOs: 237 and 118, SEQ ID NOs: 4 and 118, SEQ ID NOs: 3 and 118, SEQ ID NOs: 2 and 86, SEQ ID NOs: 1 and 86, SEQ ID NOs: 237 and 86, SEQ ID NOs: 4 and 86, SEQ ID NOs: 3 and 86, SEQ ID NOs: 2 and 60, SEQ ID NOs: 1 and 60, SEQ ID NOs: 237 and 60, SEQ ID NOs: 4 and 60, SEQ ID NOs: 3 and 60, SEQ ID NOs: 2 and 116, SEQ ID NOs: 1 and 116, SEQ ID NOs: 237 and 116, SEQ ID NOs: 4 and 116, SEQ ID NOs: 3 and 116, SEQ ID NOs: 2 and 160, SEQ ID NOs: 1 and 160, SEQ ID NOs: 237 and 160, SEQ ID NOs: 4 and 160, SEQ ID NOs: 3 and 160, SEQ ID NOs: 2 and 38, SEQ ID NOs: 1 and 38, SEQ ID NOs: 237 and 38, SEQ ID NOs: 4 and 38, SEQ ID NOs: 3 and 38, SEQ ID NOs: 2 and 99, SEQ ID NOs: 1 and 99, SEQ ID NOs: 237 and 99, SEQ ID NOs: 4 and 99, SEQ ID NOs: 3 and 99, SEQ ID NOs: 2 and 104, SEQ ID NOs: 1 and 104, SEQ ID NOs: 237 and 104, SEQ ID NOs: 4 and 104, and SEQ ID NOs: 3 and 104 were able to produce breast cancer discriminant accuracy of 95% or higher.

The measurement values of the nucleotide sequences represented by SEQ ID NOs: 2 and 237 were further compared among 62 breast cancer patients, 102 healthy subjects, and 33 prostate cancer patients in the training cohort. As a result, a scatter diagram that significantly separated the discriminant scores of the breast cancer patient group from those of the other groups was obtained in the training cohort (see FIG. 4A). These results were also reproducible for the validation cohort (see FIG. 4B).

TABLE 9

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
2	99.5	99.5	99.5	99.5	99.5	99.5
2_1	100	100	100	100	100	100
2_237	99.5	98.4	100	100	100	100
2_4	99.5	98.4	100	100	100	100
2_3	100	100	100	100	100	100
2_51	99.5	98.4	100	100	100	100
1	99.5	99.5	99.5	99.5	99.5	99.5
1_237	100	100	100	100	100	100
1_4	100	100	100	100	100	100
1_3	99.5	98.4	100	100	100	100
1_51	100	100	100	100	100	100
1_6	99.5	98.4	100	100	100	100
237	98.5	98.5	98.5	98.5	98.5	98.5
4_237	99.5	98.4	100	99	96.8	100
3_237	100	100	100	100	100	100
51_237	99	96.8	100	100	100	100
237_6	100	100	100	100	100	100
237_12	100	100	100	99	96.8	100
4	99	99	99	99	99	99
3_4	100	100	100	100	100	100
4_51	99.5	98.4	100	100	100	100
4_6	100	100	100	100	100	100
4_12	100	100	100	99	96.8	100
4_15	99.5	98.4	100	99	96.8	100
3	99.5	99.5	99.5	99.5	99.5	99.5
3_51	100	100	100	100	100	100
3_6	99	96.8	100	100	100	100
3_12	100	100	100	100	100	100
3_15	100	100	100	100	100	100
3_8	100	100	100	100	100	100
51	99	99	99	99	99	99
51_6	99.5	100	99.3	100	100	100
51_12	100	100	100	100	100	100
51_15	99.5	98.4	100	100	100	100

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
51_8	100	100	100	100	100	100
51_34	99.5	98.4	100	100	100	100
6	98.5	98.5	98.5	98.5	98.5	98.5
2_6	100	100	100	100	100	100
12_6	100	100	100	100	100	100
15_6	100	100	100	100	100	100
8_6	100	100	100	100	100	100
6_34	100	100	100	100	100	100
12	100	100	100	100	100	100
2_12	100	100	100	100	100	100
1_12	100	100	100	100	100	100
12_15	100	100	100	99	96.8	100
8_12	100	100	100	100	100	100
12_34	100	100	100	100	100	100
15	98.5	98.5	98.5	98.5	98.5	98.5
2_15	99.5	98.4	100	100	100	100
1_15	100	100	100	100	100	100
237_15	99	96.8	100	99	96.8	100
8_15	99.5	98.4	100	100	100	100
15_34	99	96.8	100	100	100	100
8	99.5	99.5	99.5	99.5	99.5	99.5
2_8	99.5	98.4	100	100	100	100
1_8	100	100	100	100	100	100
237_8	100	100	100	100	100	100
4_8	100	100	100	100	100	100
8_34	99.5	98.4	100	99	100	98.5
34	99	99	99	99	99	99
2_34	99.5	98.4	100	100	100	100
1_34	100	100	100	100	100	100
237_34	100	100	100	99	96.8	100
4_34	100	100	100	100	100	100
3_34	100	100	100	100	100	100
9	99.5	99.5	99.5	99.5	99.5	99.5
2_9	100	100	100	100	100	100
1_9	99.5	98.4	100	100	100	100
9_237	100	100	100	100	100	100
4_9	100	100	100	100	100	100
3_9	100	100	100	100	100	100
143	98	98	98	98	98	98
2_143	100	100	100	100	100	100
1_143	100	100	100	100	100	100
237_143	100	100	100	99	96.8	100
4_143	100	100	100	100	100	100
3_143	100	100	100	100	100	100
13	99	99	99	99	99	99
2_13	99.5	98.4	100	100	100	100
1_13	99.5	98.4	100	100	100	100
237_13	100	100	100	100	100	100
4_13	99.5	98.4	100	100	100	100
3_13	100	100	100	100	100	100
125	98	98	98	98	98	98
2_125	99.5	98.4	100	100	100	100
1_125	100	100	100	100	100	100
237_125	99	96.8	100	99	96.8	100
4_125	99.5	98.4	100	99	96.8	100
3_125	99.5	98.4	100	100	100	100
236	100	100	100	100	100	100
2_236	100	100	100	100	100	100
1_236	100	100	100	100	100	100
237_236	99.5	98.4	100	99	96.8	100
4_236	100	100	100	99	96.8	100
3_236	100	100	100	100	100	100
46	95.4	95.4	95.4	95.4	95.4	95.4
2_46	99.5	98.4	100	100	100	100
1_46	99.5	98.4	100	100	100	100
237_46	99.5	98.4	100	98	93.5	100
4_46	100	100	100	99	96.8	100
3_46	100	100	100	100	100	100
32	97.5	97.5	97.5	97.5	97.5	97.5
2_32	99.5	98.4	100	100	100	100
1_32	100	100	100	100	100	100
237_32	99	96.8	100	99	96.8	100
4_32	99	96.8	100	98	93.5	100
3_32	100	100	100	100	100	100
62	95.4	95.4	95.4	95.4	95.4	95.4

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
2_62	99.5	98.4	100	100	100	100
162	99.5	98.4	100	100	100	100
237_62	99	96.7	100	99	96.8	100
4_62	99	96.7	100	98	93.5	100
3_62	100	100	100	99	96.8	100
88	94.9	94.9	94.9	94.9	94.9	94.9
2_88	100	100	100	100	100	100
1_88	99.5	98.4	100	100	100	100
237_88	100	100	100	99	96.8	100
4_88	100	100	100	100	100	100
3_88	100	100	100	100	100	100
52	95.4	95.4	95.4	95.4	95.4	95.4
2_52	100	100	100	100	100	100
1_52	100	100	100	100	100	100
237_52	100	100	100	99	96.8	100
4_52	99	96.8	100	99	96.8	100
3_52	100	100	100	100	100	100
7	98.5	98.5	98.5	98.5	98.5	98.5
2_7	99.5	98.4	100	100	100	100
1_7	99.5	98.4	100	100	100	100
237_7	99.5	98.4	100	100	100	100
4_7	99.5	98.4	100	99	96.8	100
3_7	100	100	100	100	100	100
26	94.4	94.4	94.4	94.4	94.4	94.4
2_26	99.5	98.4	100	100	100	100
1_26	100	100	100	100	100	100
237_26	99.5	98.4	100	99	96.8	100
4_26	100	100	100	99	96.8	100
3_26	100	100	100	100	100	100
25	99	99	99	99	99	99
2_25	99.5	98.4	100	100	100	100
1_25	100	100	100	100	100	100
237_25	99.5	98.4	100	100	100	100
4_25	100	100	100	100	100	100
3_25	100	100	100	100	100	100
54	95.9	95.9	95.9	95.9	95.9	95.9
2_54	100	100	100	100	100	100
1_54	99.5	98.4	100	100	100	100
237_54	99	96.8	100	99	96.8	100
4_54	99.5	98.4	100	99	96.8	100
3_54	99.5	98.4	100	100	100	100
92	94.4	94.4	94.4	94.4	94.4	94.4
2_92	100	100	100	100	100	100
1_92	99.5	98.4	100	100	100	100
237_92	99.5	98.4	100	98	93.5	100
4_92	99	96.8	100	99	96.8	100
3_92	99.5	98.4	100	100	100	100
14	96.4	96.4	96.4	96.4	96.4	96.4
2_14	100	100	100	100	100	100
1_14	100	100	100	100	100	100
237_14	100	100	100	100	100	100
4_14	100	100	100	100	100	100
3_14	100	100	100	100	100	100
242	95.4	95.4	95.4	95.4	95.4	95.4
2_242	99.5	98.4	100	100	100	100
1_242	100	100	100	100	100	100
237_242	98.5	95.2	100	99	96.8	100
4_242	99.5	98.4	100	99	96.8	100
3_242	99.5	98.4	100	100	100	100
47	97	97	97	97	97	97
2_47	99.5	98.4	100	100	100	100
1_47	100	100	100	100	100	100
237_47	98.5	95.2	100	98	93.5	100
4_47	99	96.8	100	98	93.5	100
3_47	100	100	100	100	100	100
45	95.9	95.9	95.9	95.9	95.9	95.9
2_45	100	100	100	100	100	100
1_45	100	100	100	100	100	100
237_45	99	96.8	100	98	93.5	100
4_45	100	100	100	99	96.8	100
3_45	100	100	100	100	100	100
39	98.5	98.5	98.5	98.5	98.5	98.5
2_39	99.5	98.4	100	100	100	100
1_39	99.5	98.4	100	100	100	100
237_39	99	96.8	100	100	100	100

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
4_39	99	96.8	100	100	100	100
3_39	100	100	100	100	100	100
21	95.9	95.9	95.9	95.9	95.9	95.9
2_21	100	100	100	100	100	100
1_21	99.5	98.4	100	100	100	100
237_21	99.5	100	99.3	100	100	100
4_21	99.5	100	99.3	100	100	100
3_21	99.5	98.4	100	100	100	100
17	98	98	98	98	98	98
2_17	99.5	98.4	100	100	100	100
1_17	99.5	98.4	100	100	100	100
237_17	99.5	98.4	100	100	100	100
4_17	99.5	98.4	100	100	100	100
3_17	99.5	98.4	100	100	100	100
83	96.4	96.4	96.4	96.4	96.4	96.4
2_83	99.5	98.4	100	100	100	100
1_83	99.5	98.4	100	100	100	100
237_83	99.5	98.4	100	99	96.8	100
4_83	100	100	100	100	100	100
3_83	99.5	98.4	100	100	100	100
149	94.9	94.9	94.9	94.9	94.9	94.9
2_149	99.5	98.4	100	100	100	100
1_149	99.5	98.4	100	100	100	100
237_149	99.5	98.4	100	99	96.8	100
4_149	99.5	98.4	100	99	96.8	100
3_149	99.5	98.4	100	100	100	100
246	94.4	94.4	94.4	94.4	94.4	94.4
2_246	99.5	98.4	100	100	100	100
1_246	99.5	98.4	100	100	100	100
237_246	98.5	95.2	100	98	93.5	100
4_246	99	96.8	100	98	93.5	100
3_246	100	100	100	100	100	100
22	94.9	94.9	94.9	94.9	94.9	94.9
2_22	100	100	100	100	100	100
1_22	99.5	98.4	100	100	100	100
237_22	99.5	98.4	100	100	100	100
4_22	99.5	98.4	100	99	96.8	100
3_22	100	100	100	100	100	100
55	98.5	98.5	98.5	98.5	98.5	98.5
2_55	99.5	98.4	100	100	100	100
1_55	100	100	100	100	100	100
237_55	99	96.8	100	100	100	100
4_55	99	96.8	100	100	100	100
3_55	100	100	100	100	100	100
1_82	97	97	97	97	97	97
2_182	99.5	98.4	100	100	100	100
1_182	99.5	98.4	100	100	100	100
237_182	100	100	100	100	100	100
4_182	100	100	100	100	100	100
3_182	100	100	100	100	100	100
73	98	98	98	98	98	98
2_73	99.5	98.4	100	100	100	100
1_73	100	100	100	100	100	100
237_73	99	96.8	100	99	96.8	100
4_73	99.5	98.4	100	99	96.8	100
3_73	100	100	100	100	100	100
77	93.9	93.9	93.9	93.9	93.9	93.9
2_77	99.5	98.4	100	100	100	100
1_77	99.5	98.4	100	100	100	100
237_77	99.5	98.4	100	99	96.8	100
4_77	99	96.8	100	98	93.5	100
3_77	100	100	100	100	100	100
24	94.4	94.4	94.4	94.4	94.4	94.4
2_24	100	100	100	100	100	100
1_24	99.5	98.4	100	100	100	100
237_24	100	100	100	100	100	100
4_24	100	100	100	100	100	100
3_24	99.5	98.4	100	99	96.8	100
103	92.4	92.4	92.4	92.4	92.4	92.4
2_103	100	100	100	100	100	100
1_103	100	100	100	100	100	100
237_103	99	96.8	100	98	93.5	100
4_103	99	96.8	100	99	96.8	100
3_103	100	100	100	100	100	100
49	94.4	94.4	94.4	94.4	94.4	94.4

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
2_49	100	100	100	100	100	100
1_49	100	100	100	100	100	100
237_49	99.5	98.4	100	98	93.5	100
4_49	100	100	100	99	96.8	100
3_49	100	100	100	100	100	100
239	98	98	98	98	98	98
2_239	100	100	100	100	100	100
1_239	100	100	100	100	100	100
237_239	100	100	100	100	100	100
4_239	100	100	100	100	100	100
3_239	100	100	100	100	100	100
23	89.3	89.3	89.3	89.3	89.3	89.3
2_23	100	100	100	100	100	100
1_23	100	100	100	100	100	100
237_23	99.5	98.4	100	100	100	100
4_23	100	100	100	100	100	100
3_23	100	100	100	100	100	100
58	94.4	94.4	94.4	94.4	94.4	94.4
2_58	99.5	98.4	100	100	100	100
1_58	100	100	100	100	100	100
237_58	100	100	100	99	96.8	100
4_58	100	100	100	99	96.8	100
3_58	100	100	100	100	100	100
211	93.4	93.4	93.4	93.4	93.4	93.4
2_211	99.5	98.4	100	100	100	100
1_211	99.5	98.4	100	100	100	100
237_211	100	100	100	96.9	90.3	100
4_211	99.5	98.4	100	99	96.8	100
3_211	99.5	98.4	100	100	100	100
147	93.4	93.4	93.4	93.4	93.4	93.4
2_147	100	100	100	100	100	100
1_147	99.5	98.4	100	100	100	100
237_147	100	100	100	100	100	100
4_147	99.5	98.4	100	100	100	100
3_147	99.5	98.4	100	100	100	100
65	90.9	90.9	90.9	90.9	90.9	90.9
2_65	99.5	98.4	100	100	100	100
1_65	99.5	98.4	100	100	100	100
237_65	99	96.8	100	98	93.5	100
4_65	99	96.8	100	98	93.5	100
3_65	99.5	98.4	100	100	100	100
31	91.9	91.9	91.9	91.9	91.9	91.9
2_31	99.5	98.4	100	100	100	100
1_31	99.5	98.4	100	100	100	100
237_31	99.5	98.4	100	99	96.8	100
4_31	99	96.8	100	99	96.8	100
3_31	99.5	98.4	100	100	100	100
72	90.4	90.4	90.4	90.4	90.4	90.4
2_72	99.5	98.4	100	100	100	100
1_72	100	100	100	100	100	100
237_72	98.5	95.2	100	100	100	100
4_72	99.5	98.4	100	100	100	100
3_72	100	100	100	100	100	100
63	87.8	87.8	87.8	87.8	87.8	87.8
2_63	100	100	100	100	100	100
1_63	99.5	98.4	100	100	100	100
237_63	99.5	98.4	100	98	93.5	100
4_63	99.5	98.4	100	99	96.8	100
3_63	99.5	98.4	100	100	100	100
80	88.3	88.3	88.3	88.3	88.3	88.3
2_80	100	100	100	100	100	100
1_80	99.5	98.4	100	100	100	100
237_80	100	100	100	100	100	100
4_80	100	100	100	99	96.8	100
3_80	99.5	98.4	100	100	100	100
37	97	97	97	97	97	97
2_37	99.5	98.4	100	100	100	100
1_37	99.5	98.4	100	100	100	100
237_37	99.5	98.4	100	100	100	100
4_37	99.5	98.4	100	99	96.8	100
3_37	100	100	100	100	100	100
67	92.4	92.4	92.4	92.4	92.4	92.4
2_67	99.5	98.4	100	100	100	100
1_67	100	100	100	100	100	100
237_67	98.5	95.2	100	99	96.8	100

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
4_67	99	96.8	100	99	96.8	100
3_67	100	100	100	100	100	100
232	90.9	90.9	90.9	90.9	90.9	90.9
2_232	99.5	98.4	100	100	100	100
1_232	99.5	98.4	100	100	100	100
237_232	99.5	98.4	100	98	93.5	100
4_232	99	96.8	100	99	96.8	100
3_232	99.5	98.4	100	100	100	100
127	83.8	83.8	83.8	83.8	83.8	83.8
2_127	99.5	98.4	100	100	100	100
1_127	100	100	100	100	100	100
237_127	98.5	95.2	100	98	93.5	100
4_127	99	96.8	100	98	93.5	100
3_127	100	100	100	99	96.8	100
145	91.9	91.9	91.9	91.9	91.9	91.9
2_145	100	100	100	100	100	100
1_145	99.5	98.4	100	100	100	100
237_145	100	100	100	99	96.8	100
4_145	99.5	98.4	100	100	100	100
3_145	100	100	100	100	100	100
16	98.5	98.5	98.5	98.5	98.5	98.5
2_16	100	100	100	100	100	100
1_16	99.5	98.4	100	100	100	100
237_16	100	100	100	100	100	100
4_16	100	100	100	99	96.8	100
3_16	99.5	98.4	100	99	96.8	100
11	99.5	99.5	99.5	99.5	99.5	99.5
2_11	100	100	100	100	100	100
1_11	99.5	98.4	100	100	100	100
237_11	100	100	100	100	100	100
4_11	100	100	100	100	100	100
3_11	100	100	100	100	100	100
186	87.3	87.3	87.3	87.3	87.3	87.3
2_186	99.5	98.4	100	100	100	100
1_186	99.5	98.4	100	100	100	100
237_186	98.5	95.2	100	96.9	90.3	100
4_186	99.5	98.4	100	99	96.8	100
3_186	100	100	100	100	100	100
50	98.5	98.5	98.5	98.5	98.5	98.5
2_50	100	100	100	100	100	100
1_50	99.5	98.4	100	100	100	100
237_50	100	100	100	100	100	100
4_50	100	100	100	100	100	100
3_50	100	100	100	100	100	100
69	90.9	90.9	90.9	90.9	90.9	90.9
2_69	99.5	98.4	100	100	100	100
1_69	99.5	98.4	100	100	100	100
237_69	98.5	95.2	100	98	93.5	100
4_69	99	96.8	100	98	93.5	100
3_69	100	100	100	100	100	100
33	81.2	81.2	81.2	81.2	81.2	81.2
2_33	100	100	100	100	100	100
1_33	99.5	98.4	100	100	100	100
237_33	100	100	100	100	100	100
4_33	100	100	100	100	100	100
3_33	99.5	98.4	100	100	100	100
247	89.3	89.3	89.3	89.3	89.3	89.3
2_247	100	100	100	100	100	100
1_247	100	100	100	100	100	100
237_247	98.5	95.2	100	98	93.5	100
4_247	99	96.8	100	99	96.8	100
3_247	100	100	100	100	100	100
36	93.9	93.9	93.9	93.9	93.9	93.9
2_36	99.5	98.4	100	100	100	100
1_36	99.5	98.4	100	100	100	100
237_36	100	100	100	99	96.8	100
4_36	99	96.8	100	98	93.5	100
3_36	100	100	100	100	100	100
218	89.3	89.3	89.3	89.3	89.3	89.3
2_218	99.5	98.4	100	100	100	100
1_218	99.5	98.4	100	100	100	100
237_218	99	96.8	100	100	100	100
4_218	99	96.8	100	99	96.8	100
3_218	100	100	100	100	100	100
43	82.7	82.7	82.7	82.7	82.7	82.7

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
2_43	99.5	98.4	100	100	100	100
1_43	99.5	98.4	100	100	100	100
237_43	100	100	100	99	96.8	100
4_43	99.5	98.4	100	99	96.8	100
3_43	99.5	98.4	100	100	100	100
29	96.4	96.4	96.4	96.4	96.4	96.4
2_29	99.5	98.4	100	100	100	100
1_29	99.5	98.4	100	100	100	100
237_29	98.5	95.2	100	100	100	100
4_29	99.5	98.4	100	100	100	100
3_29	100	100	100	100	100	100
110	83.8	83.8	83.8	83.8	83.8	83.8
2_110	99.5	98.4	100	100	100	100
1_110	99.5	98.4	100	100	100	100
237_110	99	96.8	100	96.9	90.3	100
4_110	99	96.8	100	99	96.8	100
3_110	99.5	98.4	100	99	96.8	100
20	95.9	95.9	95.9	95.9	95.9	95.9
2_20	100	100	100	100	100	100
1_20	100	100	100	100	100	100
237_20	99	96.7	100	99	96.8	100
4_20	99.5	98.4	100	98	93.5	100
3_20	100	100	100	100	100	100
157	89.3	89.3	89.3	89.3	89.3	89.3
2_157	100	100	100	100	100	100
1_157	99.5	98.4	100	100	100	100
237_157	100	100	100	98	93.5	100
4_157	99	96.8	100	99	96.8	100
3_157	100	100	100	100	100	100
75	90.9	90.9	90.9	90.9	90.9	90.9
2_75	100	100	100	100	100	100
1_75	99.5	98.4	100	100	100	100
237_75	99.5	98.4	100	100	100	100
4_75	100	100	100	100	100	100
3_75	99.5	98.4	100	100	100	100
82	89.3	89.3	89.3	89.3	89.3	89.3
2_82	100	100	100	100	100	100
1_82	99.5	98.4	100	100	100	100
237_82	99	96.8	100	98	93.5	100
4_82	100	100	100	99	96.8	100
3_82	100	100	100	100	100	100
106	89.3	89.3	89.3	89.3	89.3	89.3
2_106	99.5	98.4	100	100	100	100
1_106	100	100	100	100	100	100
237_106	99	96.8	100	99	96.8	100
4_106	99.5	98.4	100	100	100	100
3_106	100	100	100	100	100	100
111	81.7	81.7	81.7	81.7	81.7	81.7
2_111	99.5	98.4	100	100	100	100
1_111	99.5	98.4	100	100	100	100
237_111	98.5	95.2	100	98	93.5	100
4_111	99	96.8	100	99	96.8	100
3_111	99.5	98.4	100	100	100	100
96	86.3	86.3	86.3	86.3	86.3	86.3
2_96	99.5	98.4	100	100	100	100
1_96	99.5	98.4	100	100	100	100
237_96	99	96.8	100	99	96.8	100
4_96	99	96.8	100	99	96.8	100
3_96	99.5	98.4	100	100	100	100
266	87.8	87.8	87.8	87.8	87.8	87.8
2_266	99.5	98.4	100	100	100	100
1_266	99.5	98.4	100	100	100	100
237_266	99	96.8	100	98	93.5	100
4_266	99.5	98.4	100	99	96.8	100
3_266	99.5	98.4	100	99	96.8	100
124	91.9	91.9	91.9	91.9	91.9	91.9
2_124	100	100	100	100	100	100
1_124	99.5	98.4	100	100	100	100
237_124	100	100	100	100	100	100
4_124	100	100	100	100	100	100
3_124	99.5	98.4	100	100	100	100
68	81.2	81.2	81.2	81.2	81.2	81.2
2_68	99.5	98.4	100	100	100	100
1_68	99.5	98.4	100	100	100	100
237_68	99	96.8	100	99	96.8	100

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
4_68	99	96.8	100	99	96.8	100
3_68	99.5	98.4	100	100	100	100
71	86.8	86.8	86.8	86.8	86.8	86.8
2_71	100	100	100	100	100	100
1_71	99.5	98.4	100	100	100	100
237_71	99	96.8	100	96.9	90.3	100
4_71	99	96.8	100	99	96.8	100
3_71	100	100	100	99	96.8	100
35	90.4	90.4	90.4	90.4	90.4	90.4
2_35	99.5	98.4	100	100	100	100
1_35	99.5	98.4	100	100	100	100
237_35	99.5	98.4	100	100	100	100
4_35	99.5	98.4	100	100	100	100
3_35	99.5	98.4	100	100	100	100
173	82.2	82.2	82.2	82.2	82.2	82.2
2_173	99.5	98.4	100	100	100	100
1_173	99.5	98.4	100	100	100	100
237_173	98.5	95.2	100	98	93.5	100
4_173	99	96.8	100	98	93.5	100
3_173	99.5	98.4	100	100	100	100
5	94.4	94.4	94.4	94.4	94.4	94.4
2_5	100	100	100	100	100	100
1_5	99.5	98.4	100	100	100	100
237_5	100	100	100	100	100	100
4_5	100	100	100	100	100	100
3_5	99.5	98.4	100	100	100	100
851	81.7	81.7	81.7	81.7	81.7	81.7
2_851	99.5	98.4	100	100	100	100
1_851	99.5	98.4	100	100	100	100
237_851	99	96.8	100	98	93.5	100
4_851	99	96.8	100	99	96.8	100
3_851	99.5	98.4	100	100	100	100
852	85.8	85.8	85.8	85.8	85.8	85.8
2_852	100	100	100	100	100	100
1_852	99.5	98.4	100	100	100	100
237_852	99	96.8	100	98	93.5	100
4_852	99	96.8	100	99	96.8	100
3_852	100	100	100	100	100	100
30	93.4	93.4	93.4	93.4	93.4	93.4
2_30	100	100	100	100	100	100
1_30	99.5	98.4	100	100	100	100
237_30	99	96.8	100	100	100	100
4_30	99.5	98.4	100	99	96.8	100
3_30	99.5	98.4	100	100	100	100
93	88.8	88.8	88.8	88.8	88.8	88.8
2_93	99.5	98.4	100	100	100	100
1_93	99.5	98.4	100	100	100	100
237_93	99.5	98.4	100	100	100	100
4_93	100	100	100	100	100	100
3_93	99.5	98.4	100	100	100	100
27	98.5	98.5	98.5	98.5	98.5	98.5
2_27	100	100	100	100	100	100
1_27	99.5	98.4	100	100	100	100
237_27	100	100	100	100	100	100
4_27	99.5	98.4	100	99	96.8	100
27_208	99	98.4	99.3	98	96.8	98.5
853	79.2	79.2	79.2	79.2	79.2	79.2
2_853	99.5	98.4	100	100	100	100
1_853	99.5	98.4	100	100	100	100
237_853	98.5	95.2	100	98	93.5	100
4_853	99	96.8	100	99	96.8	100
3_853	99.5	98.4	100	100	100	100
238	97.5	97.5	97.5	97.5	97.5	97.5
2_238	100	100	100	100	100	100
1_238	100	100	100	100	100	100
237_238	99	96.8	100	99	96.8	100
4_238	99	96.8	100	99	96.8	100
3_238	100	100	100	100	100	100
130	80.7	80.7	80.7	80.7	80.7	80.7
2_130	100	100	100	100	100	100
1_130	99.5	98.4	100	100	100	100
237_130	99	96.8	100	99	96.8	100
4_130	99	96.8	100	99	96.8	100
3_130	99.5	98.4	100	100	100	100
177	81.7	81.7	81.7	81.7	81.7	81.7

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
2_177	100	100	100	100	100	100
1_177	99.5	98.4	100	100	100	100
237_177	99	96.8	100	99	96.8	100
4_177	99.5	98.4	100	99	96.8	100
3_177	99.5	98.4	100	99	96.8	100
64	87.3	87.3	87.3	87.3	87.3	87.3
2_64	99.5	98.4	100	100	100	100
1_64	99.5	98.4	100	100	100	100
237_64	99.5	98.4	100	100	100	100
4_64	99.5	98.4	100	100	100	100
3_64	99.5	98.4	100	100	100	100
114	95.4	95.4	95.4	95.4	95.4	95.4
2_114	100	100	100	99	100	98.5
1_114	99.5	98.4	100	99	100	98.5
237_114	99.5	98.4	100	99	100	98.5
4_114	100	100	100	99	100	98.5
3_114	99.5	98.4	100	99	100	98.5
119	92.4	92.4	92.4	92.4	92.4	92.4
2_119	100	100	100	100	100	100
1_119	99.5	98.4	100	100	100	100
237_119	100	100	100	99	96.8	100
4_119	100	100	100	100	100	100
3_119	99.5	98.4	100	100	100	100
135	87.8	87.8	87.8	87.8	87.8	87.8
2_135	99.5	98.4	100	100	100	100
1_135	99.5	98.4	100	100	100	100
237_135	99.5	98.4	100	96.9	90.3	100
4_135	99.5	98.4	100	99	96.8	100
3_135	99.5	98.4	100	99	96.8	100
243	93.9	93.9	93.9	93.9	93.9	93.9
2_243	99.5	98.4	100	100	100	100
1_243	99.5	98.4	100	100	100	100
237_243	99.5	98.4	100	99	96.8	100
4_243	99.5	98.4	100	100	100	100
3_243	100	100	100	100	100	100
122	87.8	87.8	87.8	87.8	87.8	87.8
2_122	99.5	98.4	100	100	100	100
1_122	99.5	98.4	100	100	100	100
237_122	98.5	95.2	100	96.9	90.3	100
4_122	99	96.8	100	99	96.8	100
3_122	100	100	100	100	100	100
260	90.9	90.9	90.9	90.9	90.9	90.9
2_260	99.5	98.4	100	100	100	100
1_260	99.5	98.4	100	100	100	100
237_260	99.5	98.4	100	96.9	90.3	100
4_260	99.5	98.4	100	99	96.8	100
3_260	99.5	98.4	100	99	96.8	100
59	91.9	91.9	91.9	91.9	91.9	91.9
2_59	99.5	98.4	100	100	100	100
1_59	99.5	98.4	100	100	100	100
237_59	99.5	98.4	100	100	100	100
4_59	99	96.8	100	99	96.8	100
3_59	99.5	98.4	100	100	100	100
854	77.7	77.7	77.7	77.7	77.7	77.7
2_854	99.5	98.4	100	100	100	100
1_854	99.5	98.4	100	100	100	100
237_854	99.5	98.4	100	99	96.8	100
4_854	99	96.8	100	99	96.8	100
3_854	99.5	98.4	100	100	100	100
132	91.4	91.4	91.4	91.4	91.4	91.4
2_132	100	100	100	100	100	100
1_132	99.5	98.4	100	100	100	100
237_132	99.5	98.4	100	98	93.5	100
4_132	99.5	98.4	100	99	96.8	100
3_132	99.5	98.4	100	99	96.8	100
181	79.2	79.2	79.2	79.2	79.2	79.2
2_181	99.5	98.4	100	100	100	100
1_181	99.5	98.4	100	100	100	100
237_181	99	96.8	100	99	96.8	100
4_181	99	96.8	100	98	93.5	100
3_181	99.5	98.4	100	100	100	100
79	92.4	92.4	92.4	92.4	92.4	92.4
2_79	100	100	100	100	100	100
1_79	99.5	98.4	100	100	100	100
237_79	99.5	98.4	100	98	93.5	100

TABLE 9-continued

SEQ ID NO:	Training cohort			Validation cohort		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
4_79	99.5	98.4	100	99	96.8	100
3_79	99.5	98.4	100	100	100	100
133	79.7	79.7	79.7	79.7	79.7	79.7
2_133	99.5	98.4	100	100	100	100
1_133	99.5	98.4	100	100	100	100
237_133	98.5	95.2	100	99	96.8	100
4_133	99	96.8	100	99	96.8	100
3_133	99.5	98.4	100	100	100	100
41	91.4	91.4	91.4	91.4	91.4	91.4
2_41	99.5	98.4	100	100	100	100
1_41	99.5	98.4	100	100	100	100
237_41	99	96.8	100	99	96.8	100
4_41	99.5	98.4	100	99	96.8	100
3_41	99.5	98.4	100	100	100	100
139	88.3	88.3	88.3	88.3	88.3	88.3
2_139	100	100	100	100	100	100
1_139	99.5	98.4	100	100	100	100
237_139	100	100	100	98	93.5	100
4_139	100	100	100	99	96.8	100
3_139	99.5	98.4	100	100	100	100
118	87.8	87.8	87.8	87.8	87.8	87.8
2_118	99.5	98.4	100	100	100	100
1_118	99.5	98.4	100	100	100	100
237_118	99	96.8	100	100	100	100
4_118	99.5	98.4	100	100	100	100
3_118	99.5	98.4	100	100	100	100
86	85.3	85.3	85.3	85.3	85.3	85.3
2_86	99.5	98.4	100	100	100	100
1_86	99.5	98.4	100	100	100	100
237_86	99	96.8	100	100	100	100
4_86	99	96.8	100	100	100	100
3_86	99.5	98.4	100	99	96.8	100
60	84.8	84.8	84.8	84.8	84.8	84.8
2_60	99.5	98.4	100	100	100	100
1_60	99.5	98.4	100	100	100	100
237_60	99	96.8	100	99	96.8	100
4_60	99.5	98.4	100	99	96.8	100
3_60	99.5	98.4	100	100	100	100
116	80.2	80.2	80.2	80.2	80.2	80.2
2_116	100	100	100	100	100	100
1_116	99.5	98.4	100	100	100	100
237_116	99.5	98.4	100	100	100	100
4_116	99.5	98.4	100	99	96.8	100
3_116	99.5	98.4	100	100	100	100
160	80.2	80.2	80.2	80.2	80.2	80.2
2_160	99.5	98.4	100	100	100	100
1_160	99.5	98.4	100	100	100	100
237_160	99.5	98.4	100	98	93.5	100
4_160	99	96.8	100	99	96.8	100
3_160	99.5	98.4	100	100	100	100
38	79.2	79.2	79.2	79.2	79.2	79.2
2_38	99.5	98.4	100	100	100	100
1_38	99.5	98.4	100	100	100	100
237_38	98.5	95.2	100	100	100	100
4_38	99	96.8	100	99	96.8	100
3_38	99.5	98.4	100	99	96.8	100
99	86.2	86.2	86.2	86.2	86.2	86.2
2_99	99.5	98.4	100	100	100	100
1_99	99.5	98.4	100	100	100	100
237_99	99	96.7	100	99	96.8	100
4_99	99	96.7	100	98	93.5	100
3_99	99.5	98.4	100	100	100	100
104	86.3	86.3	86.3	86.3	86.3	86.3
2_104	99.5	98.4	100	100	100	100
1_104	100	100	100	100	100	100
237_104	98.5	95.2	100	98	93.5	100
4_104	99	96.8	100	99	96.8	100
3_104	100	100	100	100	100	100

-continued

ccccggggag cccggcg	17
SEQ ID NO: 7	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 7	
cggggtcggc ggcgacgtg	19
SEQ ID NO: 8	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 8	
ggggcgcggc cggatcg	17
SEQ ID NO: 9	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 9	
tgggcgaggg cggctgagcg gc	22
SEQ ID NO: 10	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 10	
ggggagcgag gggcggggc	19
SEQ ID NO: 11	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 11	
tgtagagcag ggagcaggaa gct	23
SEQ ID NO: 12	moltype = RNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 12	
tgaggggcct cagaccgagc tttt	24
SEQ ID NO: 13	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 13	
ctcggggcag gcgctggga gcg	23
SEQ ID NO: 14	moltype = RNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 14	
agagatgaag cgggggggcg	20
SEQ ID NO: 15	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 15	
ctgggctcgg gacgcgcggc t	21
SEQ ID NO: 16	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 16
acctggcagc agggagcgtc gt 22

SEQ ID NO: 17      moltype = RNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 17
gcgggggtgg cggcgcatc cc 22

SEQ ID NO: 18      moltype = RNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 18
ctcgcgcggg ggcgcgggct cc 22

SEQ ID NO: 19      moltype = RNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 19
acaggtgagg ttcttgggag cc 22

SEQ ID NO: 20      moltype = RNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 20
agtgggaggc cagggcacgg ca 22

SEQ ID NO: 21      moltype = RNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 21
aggcagcggg gtgtagtgga ta 22

SEQ ID NO: 22      moltype = RNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 22
ctaggtgggg ggcttgaagc 20

SEQ ID NO: 23      moltype = RNA length = 23
FEATURE           Location/Qualifiers
source            1..23
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 23
gggggggcag gaggggctca ggg 23

SEQ ID NO: 24      moltype = RNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 24
tggcgggggt agagctggct gc 22

SEQ ID NO: 25      moltype = RNA length = 23
FEATURE           Location/Qualifiers
source            1..23
                  mol_type = unassigned RNA
                  organism = Homo sapiens

SEQUENCE: 25
ggatggttgg gggcggtcgg cgt 23

SEQ ID NO: 26      moltype = RNA length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = unassigned RNA

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SEQUENCE: 26	organism = Homo sapiens	
tgagggaccc aggacaggag a		21
SEQ ID NO: 27	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 27		
aatggatttt tggagcagg		19
SEQ ID NO: 28	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 28		
gctgcgggct gcggtcaggg cg		22
SEQ ID NO: 29	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 29		
tagggatggg aggccaggat ga		22
SEQ ID NO: 30	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 30		
aggggtgggac tggaggtggg gct		23
SEQ ID NO: 31	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 31		
agcaggtgcg gggcggcg		18
SEQ ID NO: 32	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 32		
atgcctcccc cggccccgca g		21
SEQ ID NO: 33	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 33		
gtgcgtgggtg gctcgaggcg ggg		23
SEQ ID NO: 34	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 34		
tagggggcgg cttgtggagt gt		22
SEQ ID NO: 35	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 35		
tgggagggga gaggcagcaa gca		23
SEQ ID NO: 36	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	

-continued

mol_type = unassigned RNA	
organism = Homo sapiens	
SEQUENCE: 36	
cgggagctgg ggtctgcagg t	21
SEQ ID NO: 37	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 37	
cgggtagaga gggcagtggg agg	23
SEQ ID NO: 38	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 38	
agggagggac gggggctgtg c	21
SEQ ID NO: 39	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 39	
gtgcggaacg ctggccgggg cg	22
SEQ ID NO: 40	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 40	
caggaaggat ttagggacag gc	22
SEQ ID NO: 41	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 41	
gtgaggaggg gctggcaggg ac	22
SEQ ID NO: 42	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 42	
aggcacggtg tcagcaggc	19
SEQ ID NO: 43	moltype = RNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 43	
aggcaggggc tgggtgctggg cggg	24
SEQ ID NO: 44	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 44	
gagccagttg gacaggagc	19
SEQ ID NO: 45	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 45	
ctcgggaggg catgggccag gc	22
SEQ ID NO: 46	moltype = RNA length = 20
FEATURE	Location/Qualifiers

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source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 46		
gtgtctgggc ggacagctgc		20
SEQ ID NO: 47	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 47		
ccggccgccg gctccgcccc g		21
SEQ ID NO: 48	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 48		
ctggtacagg cctgggggac ag		22
SEQ ID NO: 49	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 49		
acaggcggct gtagcaatgg ggg		23
SEQ ID NO: 50	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 50		
gtgaacgggc gccatcccga gg		22
SEQ ID NO: 51	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 51		
gtgggttggg gcgggctctg		20
SEQ ID NO: 52	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 52		
cgggctgtcc ggaggggtcg gct		23
SEQ ID NO: 53	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 53		
caggcaggga ggtgggacca tg		22
SEQ ID NO: 54	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 54		
gtgagtagtg gcgcgcggcg gc		22
SEQ ID NO: 55	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 55		
ccggggcaga ttggtgtagg gtg		23
SEQ ID NO: 56	moltype = RNA length = 22	

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FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 56		
gggggtcccc ggtgctcgga tc		22
SEQ ID NO: 57	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 57		
ggcggcgggg aggtaggcag		20
SEQ ID NO: 58	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 58		
cggggccgta gcactgtctg aga		23
SEQ ID NO: 59	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 59		
gtgagtggga gccccagtgt gtg		23
SEQ ID NO: 60	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 60		
agggactttt gggggcagat gtg		23
SEQ ID NO: 61	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 61		
gctgggaagg caaagggacg t		21
SEQ ID NO: 62	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 62		
ggggctgggc gcgcgcc		17
SEQ ID NO: 63	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 63		
tgggagggcg tggatgatgg tg		22
SEQ ID NO: 64	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 64		
caggcagaag tggggctgac agg		23
SEQ ID NO: 65	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 65		
ggggggatgt gcatgctggt t		21

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SEQ ID NO: 66	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 66		
accttgccctt gctgcccggg cc		22
SEQ ID NO: 67	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 67		
tggggcggag cttccggag		19
SEQ ID NO: 68	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 68		
cagggcaggg aaggtgggag ag		22
SEQ ID NO: 69	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 69		
cctgagcccg ggccgcgcag		20
SEQ ID NO: 70	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 70		
ccgggagaag gaggtggcct gg		22
SEQ ID NO: 71	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 71		
gggaccaggg gagagacgta ag		22
SEQ ID NO: 72	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 72		
ggtggcccgg ccgtgcctga gg		22
SEQ ID NO: 73	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 73		
agccgcgcc agccgaggtt ct		22
SEQ ID NO: 74	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 74		
caggcaggtg tagggtggag c		21
SEQ ID NO: 75	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 75		
ggggctgtga ttgaccagca gg		22

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SEQ ID NO: 76	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 76		
actcggctgc ggtggacaag t		21
SEQ ID NO: 77	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 77		
cggggtgggt gagtcgggc		20
SEQ ID NO: 78	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 78		
tgagcgggg gggcgagc		18
SEQ ID NO: 79	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 79		
ctgggggacg cgtgagcgcg agc		23
SEQ ID NO: 80	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 80		
gtgagtggga gccggtgggg ctg		23
SEQ ID NO: 81	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 81		
tggggggaca gatggagagg aca		23
SEQ ID NO: 82	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 82		
gtgggctggg ctgggctggg cc		22
SEQ ID NO: 83	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 83		
cgcaggggcc ggggtgctcac cg		22
SEQ ID NO: 84	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 84		
ttgatctcgg aagctaagc		19
SEQ ID NO: 85	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 85		

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ctgggccccgc ggcgggctg ggg	23
SEQ ID NO: 86	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 86	
ctcctggggc cgcactctc gc	22
SEQ ID NO: 87	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 87	
tggggaaggc ttggcagga aga	23
SEQ ID NO: 88	moltype = RNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 88	
aaaaggcggg agaagcccca	20
SEQ ID NO: 89	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 89	
gagcaggcga ggctgggctg aa	22
SEQ ID NO: 90	moltype = RNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 90	
aaaagctggg ctgagaggcg	20
SEQ ID NO: 91	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 91	
gtggggccag gcggtgg	17
SEQ ID NO: 92	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 92	
tggggaaggc gtcagtgtcg gg	22
SEQ ID NO: 93	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 93	
actgggtagg tggggctcca gg	22
SEQ ID NO: 94	moltype = RNA length = 25
FEATURE	Location/Qualifiers
source	1..25
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 94	
atggggtgag atggggagga gcagc	25
SEQ ID NO: 95	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 95
ggaggcgag gctcggaaag gcg 23

SEQ ID NO: 96 moltype = RNA length = 22
FEATURE Location/Qualifiers
source 1..22
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 96
ggggaactgt agatgaaaag gc 22

SEQ ID NO: 97 moltype = RNA length = 23
FEATURE Location/Qualifiers
source 1..23
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 97
agaagaaggc ggtcggctcg cgg 23

SEQ ID NO: 98 moltype = RNA length = 25
FEATURE Location/Qualifiers
source 1..25
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 98
agcggggagg aagtggggcg tgctt 25

SEQ ID NO: 99 moltype = RNA length = 17
FEATURE Location/Qualifiers
source 1..17
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 99
ctccgggacg gctgggc 17

SEQ ID NO: 100 moltype = RNA length = 23
FEATURE Location/Qualifiers
source 1..23
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 100
tcgaggactg gtggaagggc ctt 23

SEQ ID NO: 101 moltype = RNA length = 24
FEATURE Location/Qualifiers
source 1..24
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 101
tgtaggcacg aggcagggcc cagg 24

SEQ ID NO: 102 moltype = RNA length = 22
FEATURE Location/Qualifiers
source 1..22
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 102
ggctacaaca caggaccgg gc 22

SEQ ID NO: 103 moltype = RNA length = 22
FEATURE Location/Qualifiers
source 1..22
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 103
tggggcggag cttccggagg cc 22

SEQ ID NO: 104 moltype = RNA length = 23
FEATURE Location/Qualifiers
source 1..23
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 104
tgggggctgg gatgggcat ggt 23

SEQ ID NO: 105 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
 mol_type = unassigned RNA

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SEQUENCE: 105	organism = Homo sapiens	
tgcggcagag ctgggggtca		19
SEQ ID NO: 106	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 106		
gtaggtgaca gtcaggggcg g		21
SEQ ID NO: 107	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 107		
aggcgatgtg gggatgtaga ga		22
SEQ ID NO: 108	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 108		
tgggggggaca ggatgagagg ctgt		24
SEQ ID NO: 109	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 109		
gagggcgagcg tgggtgtggc gga		23
SEQ ID NO: 110	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 110		
accaggaggc tgaggccctt		20
SEQ ID NO: 111	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 111		
agtgggagga caggaggcag gt		22
SEQ ID NO: 112	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 112		
tgagcaccac acaggccggg cgc		23
SEQ ID NO: 113	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 113		
ttagggagac atggtggggg cc		22
SEQ ID NO: 114	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 114		
gcagggacag caaaggggtg c		21
SEQ ID NO: 115	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	

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	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 115		
ctgcaggcag aagtggggct gaca		24
SEQ ID NO: 116	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 116		
ctcgggcgga ggtggttag tg		22
SEQ ID NO: 117	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 117		
gtgaaggccc ggcggaga		18
SEQ ID NO: 118	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 118		
gaacgctgt tcttgccagg tgg		23
SEQ ID NO: 119	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 119		
tgggggtgtg gggagagaga g		21
SEQ ID NO: 120	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 120		
tcgggggcta gggctaacag ca		22
SEQ ID NO: 121	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 121		
aggtgggtat ggaggagccc t		21
SEQ ID NO: 122	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 122		
cggtgagcgc tcgctggc		18
SEQ ID NO: 123	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 123		
gggctggggc gcggggaggt		20
SEQ ID NO: 124	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 124		
ccaggggat gggcgagctt ggg		23
SEQ ID NO: 125	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	

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source                1..21
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 125
cgggccggag gtcaagggcg t                                21

SEQ ID NO: 126        moltype = RNA length = 18
FEATURE              Location/Qualifiers
source                1..18
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 126
gaggctgaag gaagatgg                                     18

SEQ ID NO: 127        moltype = RNA length = 22
FEATURE              Location/Qualifiers
source                1..22
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 127
cagggctggc agtgacatgg gt                                22

SEQ ID NO: 128        moltype = RNA length = 22
FEATURE              Location/Qualifiers
source                1..22
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 128
cagttgggtc taggggtcag ga                                22

SEQ ID NO: 129        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source                1..21
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 129
gttgggtgc aggggtctgc t                                  21

SEQ ID NO: 130        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source                1..21
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 130
acggcccagg cggcattggt g                                  21

SEQ ID NO: 131        moltype = RNA length = 22
FEATURE              Location/Qualifiers
source                1..22
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 131
gagggttggg tggaggctct cc                                22

SEQ ID NO: 132        moltype = RNA length = 19
FEATURE              Location/Qualifiers
source                1..19
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 132
taaggagggg gatgagggg                                     19

SEQ ID NO: 133        moltype = RNA length = 23
FEATURE              Location/Qualifiers
source                1..23
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 133
tgagtggggc tcccgggacg gcg                                23

SEQ ID NO: 134        moltype = RNA length = 25
FEATURE              Location/Qualifiers
source                1..25
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 134
tcggggcatg ggggaggag gctgg                               25

SEQ ID NO: 135        moltype = RNA length = 19

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FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 135		
tgggggagat gggggttga		19
SEQ ID NO: 136	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 136		
ggggagctgt ggaagcagta		20
SEQ ID NO: 137	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 137		
cccagcagga cgggagcg		18
SEQ ID NO: 138	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 138		
ttggggtgga gggccaagga gc		22
SEQ ID NO: 139	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 139		
tggggaggtg tggagtcagc at		22
SEQ ID NO: 140	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 140		
cgtggaggac gaggaggagg c		21
SEQ ID NO: 141	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 141		
ggctgggtcag atgggagtg		19
SEQ ID NO: 142	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 142		
acaggagtgg ggggtggaca t		21
SEQ ID NO: 143	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 143		
gtgggcgggg gcaggtgtgt g		21
SEQ ID NO: 144	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 144		
aaggggcagg gacgggtggc cc		22

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SEQ ID NO: 145	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 145		
cgggcgtggt ggtggggg		18
SEQ ID NO: 146	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 146		
gctcggactg agcagggtggg		20
SEQ ID NO: 147	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 147		
gcggtggggc cggaggggcg t		21
SEQ ID NO: 148	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 148		
cagaagggga gttgggagca ga		22
SEQ ID NO: 149	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 149		
agggatcgcg ggcgggtggc ggcct		25
SEQ ID NO: 150	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 150		
gtgggtgctg gtgggagccg tg		22
SEQ ID NO: 151	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 151		
tcggggagtc tggggtccgg aat		23
SEQ ID NO: 152	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 152		
gcccaggact ttgtgcgggg tg		22
SEQ ID NO: 153	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 153		
cagcagggga gagagaggag tc		22
SEQ ID NO: 154	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 154		
agaggcttg tgcggatacg ggg		23

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SEQ ID NO: 155	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 155		
gaaatcaagc gtgggtgaga cc		22
SEQ ID NO: 156	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 156		
agggctggac tcagcggcgg agct		24
SEQ ID NO: 157	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 157		
cgggcgtggt ggtgggggtg		20
SEQ ID NO: 158	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 158		
tcggcctggg gagggaggaag gg		22
SEQ ID NO: 159	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 159		
gtaggggagg ttggggccagg ga		22
SEQ ID NO: 160	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 160		
taggggggtg caggctggcc		20
SEQ ID NO: 161	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 161		
actcaaaactg tgggggcact		20
SEQ ID NO: 162	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 162		
caggggggact gggggtgagc		20
SEQ ID NO: 163	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 163		
ctgggagggg ctgggtttgg c		21
SEQ ID NO: 164	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 164		

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gggggaagaa aaggtgggg	19
SEQ ID NO: 165	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 165	
atccagttct ctgagggggc t	21
SEQ ID NO: 166	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 166	
cggggccatg gagcagcctg tgt	23
SEQ ID NO: 167	moltype = RNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 167	
tggggagctg aggctctggg ggtg	24
SEQ ID NO: 168	moltype = RNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 168	
ccccgggaac gtcgagactg gagc	24
SEQ ID NO: 169	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 169	
acgggggagtc aggcagtggt gga	23
SEQ ID NO: 170	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 170	
tgcaggggtc gggtagggcca gg	22
SEQ ID NO: 171	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 171	
ggcgggtgcg ggggtgg	17
SEQ ID NO: 172	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 172	
taggggcagc agaggacctg gg	22
SEQ ID NO: 173	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 173	
cgcgccgggc ccgggtt	17
SEQ ID NO: 174	moltype = RNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 174		
cgggtgggag cagatcttat tgag		24
SEQ ID NO: 175	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 175		
ggtgggggct gttgttt		17
SEQ ID NO: 176	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 176		
tgggctgagg gcaggaggcc tgt		23
SEQ ID NO: 177	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 177		
gtgtggccgg caggcgggtg g		21
SEQ ID NO: 178	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 178		
tggggacgta gctggccaga cag		23
SEQ ID NO: 179	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 179		
accactgcac tccagcctga g		21
SEQ ID NO: 180	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 180		
gagactgggg tggggcc		17
SEQ ID NO: 181	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 181		
ggggcctggc ggtgggcgg		19
SEQ ID NO: 182	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 182		
cggggcggca ggggcctc		18
SEQ ID NO: 183	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 183		
agggccgaag ggtggaagct gc		22
SEQ ID NO: 184	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	

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SEQUENCE: 184	organism = Homo sapiens	
tgaggatatg gcaggggaagg gga		23
SEQ ID NO: 185	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 185		
gccggacaag agggagag		17
SEQ ID NO: 186	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 186		
gtaggggcggt cccgggcgcg cggg		24
SEQ ID NO: 187	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 187		
taggggtggg ggaattcagg ggtgt		25
SEQ ID NO: 188	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 188		
cggggcagct cagtacagga t		21
SEQ ID NO: 189	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 189		
aaggggctgg gggagcaca		19
SEQ ID NO: 190	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 190		
agcggtgctc ctgcgggccc a		21
SEQ ID NO: 191	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 191		
gctgggatta caggcatgag cc		22
SEQ ID NO: 192	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 192		
gtgagtcagg gtggggctgg		20
SEQ ID NO: 193	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 193		
ctgggggtgg ggggctgggc gt		22
SEQ ID NO: 194	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	

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	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 194		
ctgggacagg aggaggaggc ag		22
SEQ ID NO: 195	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 195		
ggatccgagt cacggcacca		20
SEQ ID NO: 196	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 196		
ccaggaggcg gaggaggtgg ag		22
SEQ ID NO: 197	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 197		
ctggcagggg gagaggta		18
SEQ ID NO: 198	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 198		
tgggtggagga agagggcgagc tc		22
SEQ ID NO: 199	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 199		
cgtcccgggg ctgcgcgagg ca		22
SEQ ID NO: 200	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 200		
tcaaaatcag gagtccggggc tt		22
SEQ ID NO: 201	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 201		
caggaggcag tgggcgagca gg		22
SEQ ID NO: 202	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 202		
agcagggctg gggattgca		19
SEQ ID NO: 203	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 203		
tgggctgctg agaaggggca		20
SEQ ID NO: 204	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	

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source	1..17 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 204		
gctgggcgag gctggca		17
SEQ ID NO: 205	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 205		
ttgctctgct cccccgcccc cag		23
SEQ ID NO: 206	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 206		
tcaataggaa agaggtggga cct		23
SEQ ID NO: 207	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 207		
gcggggctgg gcgcgcg		17
SEQ ID NO: 208	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 208		
tggggagcgg cccccgggtg gg		22
SEQ ID NO: 209	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 209		
tctgtggagt ggggtgctg t		21
SEQ ID NO: 210	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 210		
caggcacggg agctcaggtg ag		22
SEQ ID NO: 211	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 211		
gtgaggcggg gccaggaggg tgtgt		25
SEQ ID NO: 212	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 212		
aaggacagg gagggtcgtg g		21
SEQ ID NO: 213	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 213		
ggcggcggcg gaggcggggg		20
SEQ ID NO: 214	moltype = RNA length = 22	

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FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 214		
tcgggcctgg ggttggggga gc		22
SEQ ID NO: 215	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 215		
atcccaccac tgccaccat		19
SEQ ID NO: 216	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 216		
ggggaggtgt gcagggtctg		20
SEQ ID NO: 217	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 217		
tgggggagcc atgagataag agca		24
SEQ ID NO: 218	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 218		
gccggggctt tgggtgaggg		20
SEQ ID NO: 219	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 219		
ttcagatccc agcgggtgcct ct		22
SEQ ID NO: 220	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 220		
tggtaggtgg ggaggagaag tgc		23
SEQ ID NO: 221	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 221		
tggcagggag gctgggaggg g		21
SEQ ID NO: 222	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 222		
agacacattt ggagaggga cc		22
SEQ ID NO: 223	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 223		
aggctgggct gggacgga		18

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SEQ ID NO: 224	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 224		
tcagggagtc aggggagggc		20
SEQ ID NO: 225	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 225		
tgggcagggg cttattgtag gag		23
SEQ ID NO: 226	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 226		
tgggggagga aggacaggcc at		22
SEQ ID NO: 227	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 227		
cccctggggc tgggcaggcg ga		22
SEQ ID NO: 228	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 228		
ctggggggag gagacctgc t		21
SEQ ID NO: 229	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 229		
ccccgccacc gccttg		17
SEQ ID NO: 230	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 230		
tctcttcac taccgccag		20
SEQ ID NO: 231	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 231		
aggaggtggt actaggggcc agc		23
SEQ ID NO: 232	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 232		
gcggaaggcg gagcgcgga		20
SEQ ID NO: 233	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 233		
tgcggggaca ggccagggca tc		22

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SEQ ID NO: 234	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 234		
agggccagag gagcctggag tgg		23
SEQ ID NO: 235	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 235		
agactgacgg ctggaggccc at		22
SEQ ID NO: 236	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 236		
cggctctggg tctgtgggga		20
SEQ ID NO: 237	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 237		
gacacgggcg acagctgggg ccc		23
SEQ ID NO: 238	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 238		
tgaggggcag agagcgagac ttt		23
SEQ ID NO: 239	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 239		
gggtggggat ttgttgcat ac		22
SEQ ID NO: 240	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 240		
tagcagcacg taaatattgg cg		22
SEQ ID NO: 241	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 241		
aaaccgttac cattactgag tt		22
SEQ ID NO: 242	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 242		
tatagggatt ggagccgtgg cg		22
SEQ ID NO: 243	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 243		

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tcctgtactg agctgccccg ag	22
SEQ ID NO: 244	moltype = RNA length = 18
FEATURE	Location/Qualifiers
source	1..18
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 244	
ccagaggtgg ggactgag	18
SEQ ID NO: 245	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 245	
agggacggga cgcggtgcag tg	22
SEQ ID NO: 246	moltype = RNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 246	
ccccagggcg acgcggcggg	20
SEQ ID NO: 247	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 247	
cttcgcccc gccgggcgtc g	21
SEQ ID NO: 248	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 248	
aaggcagggc ccccgctccc c	21
SEQ ID NO: 249	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 249	
agggcccccc ctcaatcctg t	21
SEQ ID NO: 250	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 250	
atcacattgc cagggattac c	21
SEQ ID NO: 251	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 251	
tattgcactt gtcccggcct gt	22
SEQ ID NO: 252	moltype = RNA length = 25
FEATURE	Location/Qualifiers
source	1..25
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 252	
ggcggaggga agtaggtccg ttggt	25
SEQ ID NO: 253	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 253		
tgggggtggt ctctagccaa gg		22
SEQ ID NO: 254	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 254		
gggaaaagga aggggggagga		20
SEQ ID NO: 255	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 255		
tcacctggct ggccccccca g		21
SEQ ID NO: 256	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 256		
tgggaggagg ggatcttggg		20
SEQ ID NO: 257	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 257		
ggatggagga ggggtct		17
SEQ ID NO: 258	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 258		
tattgcactc gtccccgct cc		22
SEQ ID NO: 259	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 259		
ggctggagcg agtgcagtgg tg		22
SEQ ID NO: 260	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 260		
tggctgttgg agggggcagg c		21
SEQ ID NO: 261	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 261		
ggggctgggg ccggggccga gc		22
SEQ ID NO: 262	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 262		
gggaccatcc tgctgtgt gg		22
SEQ ID NO: 263	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	

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                organism = Homo sapiens
SEQUENCE: 263
tggcggcggt agttatgggc tt                                22

SEQ ID NO: 264      moltype = RNA  length = 23
FEATURE            Location/Qualifiers
source             1..23
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 264
gtttgcacgg gtgggccttg tct                                23

SEQ ID NO: 265      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 265
cgggggcggg gccgaagcgc g                                  21

SEQ ID NO: 266      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 266
cggcggggac ggcgattggt c                                  21

SEQ ID NO: 267      moltype = RNA  length = 17
FEATURE            Location/Qualifiers
source             1..17
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 267
acccactcc tggtagc                                         17

SEQ ID NO: 268      moltype = RNA  length = 25
FEATURE            Location/Qualifiers
source             1..25
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 268
aggggggcac tgcgcaagca aagcc                               25

SEQ ID NO: 269      moltype = RNA  length = 19
FEATURE            Location/Qualifiers
source             1..19
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 269
ctggggagtg gctggggag                                       19

SEQ ID NO: 270      moltype = RNA  length = 82
FEATURE            Location/Qualifiers
source             1..82
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 270
gggaaagcgg agggcgcgcc cagctcccgg gctgattgcg ctaacagtgg ccccggtgtt 60
ggggcgcgtc tgccgctgcc cc                                     82

SEQ ID NO: 271      moltype = RNA  length = 76
FEATURE            Location/Qualifiers
source             1..76
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 271
cgcaggcctc tggcggagcc cattccatgc cagatgctga gcgatggctg gtgtgtgctg 60
ctccacaggc ctggtg                                           76

SEQ ID NO: 272      moltype = RNA  length = 149
FEATURE            Location/Qualifiers
source             1..149
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 272
catcaagacc cagctgagtc actgtcactg cctaccaatc tcgaccggac ctcgaccggc 60
tcgtctgtgt tgccaatcga ctggcgctgg cgtcggctgt ggtagatagg cggatcatgca 120
tacgaatttt cagctcttgt tctggtgac                             149

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SEQ ID NO: 273 moltype = RNA length = 54
 FEATURE Location/Qualifiers
 source 1..54
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 273
 ggacaaggcg ggcgcgaccg gcccggggct ctggggcgcg cgcgtttccc ctcc 54

SEQ ID NO: 274 moltype = RNA length = 93
 FEATURE Location/Qualifiers
 source 1..93
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 274
 ccttcggcg tcccaggcgg ggcgccgagg gaccgccctc gtgtctgtgg cggtgggac 60
 ccgcgggcgt gttttcctgg tggccggcc atg 93

SEQ ID NO: 275 moltype = RNA length = 51
 FEATURE Location/Qualifiers
 source 1..51
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 275
 acagaccccg gggagcccg cggtgaagct cctggtatcc tgggtgtctg a 51

SEQ ID NO: 276 moltype = RNA length = 59
 FEATURE Location/Qualifiers
 source 1..59
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 276
 cggggtcggc ggcgacgtgc tcagcttggc acccaagttc tgccgctccg acgcccggc 59

SEQ ID NO: 277 moltype = RNA length = 84
 FEATURE Location/Qualifiers
 source 1..84
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 277
 gaggtctggc ggggcgcgcc cggtacggtc gagagcgtcc tggtgatga cggtctcccg 60
 tgcccacgcc ccaaacgcag tctc 84

SEQ ID NO: 278 moltype = RNA length = 65
 FEATURE Location/Qualifiers
 source 1..65
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 278
 gaggggtggc gagggcggtc gagcggtcc atccccggc ctgctcatcc ccctcgccct 60
 ctacg 65

SEQ ID NO: 279 moltype = RNA length = 60
 FEATURE Location/Qualifiers
 source 1..60
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 279
 cgctgggtcc gcgcgccctg gcccgggcga tgtccgcttg ggggagcgag gggcggggcg 60

SEQ ID NO: 280 moltype = RNA length = 76
 FEATURE Location/Qualifiers
 source 1..76
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 280
 gagggagctg tagagcagg agcaggaagc tgtgtgtgtc cagccctgac ctgtcctgtt 60
 ctgccccag ccctc 76

SEQ ID NO: 281 moltype = RNA length = 75
 FEATURE Location/Qualifiers
 source 1..75
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 281
 aagcaagact gaggggcctc agaccgagct ttggaaaa agaaaagtct cgctctctgc 60
 ccctcagcct aactt 75

SEQ ID NO: 282 moltype = RNA length = 65

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FEATURE	Location/Qualifiers	
source	1..65	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 282		
gggtgctcgg ggcaggcggc tgggagcggc cctcacattg atggctcctg ccacctctc		60
cgcag		65
SEQ ID NO: 283	moltype = RNA length = 51	
FEATURE	Location/Qualifiers	
source	1..51	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 283		
agagatgaag cgggggggcg gggctcttgc ctattgccta cgctgatctc a		51
SEQ ID NO: 284	moltype = RNA length = 87	
FEATURE	Location/Qualifiers	
source	1..87	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 284		
cccaggcgcc cgctcccgac ccacgcccgc ccgcccgggc cctcctcccc ggagaggctg		60
ggctcgggac gcgcggctca gctcggg		87
SEQ ID NO: 285	moltype = RNA length = 72	
FEATURE	Location/Qualifiers	
source	1..72	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 285		
gatttcagt acctggcagc agggagcgtc gtcagtgttt gactgtttat ggtatgtcag		60
ggagctgggt cc		72
SEQ ID NO: 286	moltype = RNA length = 84	
FEATURE	Location/Qualifiers	
source	1..84	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 286		
cgggtccagac gtggcggggg tggcgcgccg atcccggacg gcctgtgagg gatgcgccg		60
ccactgcccc gcgcgcctg accg		84
SEQ ID NO: 287	moltype = RNA length = 47	
FEATURE	Location/Qualifiers	
source	1..47	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 287		
ctcgcgcgcg ggcgcgggct ccgggttggg gcgagccaac gccgggg		47
SEQ ID NO: 288	moltype = RNA length = 86	
FEATURE	Location/Qualifiers	
source	1..86	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 288		
tgccagtctc taggtccctg agacccttta acctgtgagg acatccaggg tcacaggtga		60
ggttcttggg agcctggcgt ctggcc		86
SEQ ID NO: 289	moltype = RNA length = 82	
FEATURE	Location/Qualifiers	
source	1..82	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 289		
gtgagtggga ggccagggca cggcaggggg agctgcaggg ctatgggagg ggccccagcg		60
tctgagccct gtctcccg ag		82
SEQ ID NO: 290	moltype = RNA length = 82	
FEATURE	Location/Qualifiers	
source	1..82	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 290		
gtgagtggga ggccagggca cggcaggggg agctgcaggg ctatgggagg ggccccagcg		60
tctgagccct gtctcccg ag		82
SEQ ID NO: 291	moltype = RNA length = 74	

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FEATURE	Location/Qualifiers	
source	1..74	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 291		
ccgcactctc tccattacac taccctgcct cttctccatg agaggcagcg ggggtgtagtg		60
gatatagcac ggggt		74
SEQ ID NO: 292	moltype = RNA length = 65	
FEATURE	Location/Qualifiers	
source	1..65	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 292		
gagggttagg tggggggcgt gaagccccga gatgcctcac gtcttcaccc ctctcaccta		60
agcag		65
SEQ ID NO: 293	moltype = RNA length = 75	
FEATURE	Location/Qualifiers	
source	1..75	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 293		
tggagtgggg gggcaggagg ggctcaggga gaaagtgcac acagccccctg gcctctctctg		60
cccttcctgc ccttg		75
SEQ ID NO: 294	moltype = RNA length = 61	
FEATURE	Location/Qualifiers	
source	1..61	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 294		
tgggtggcg ggggtagagc tggtgcagg cccggccccct ctacgtgct gcctctctca		60
g		61
SEQ ID NO: 295	moltype = RNA length = 86	
FEATURE	Location/Qualifiers	
source	1..86	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 295		
cgcttgagcg tgcagcagga catcttctctg acctggtaac aattaggtga gaaggatggt		60
tgggggcggt cggcgtaact caggga		86
SEQ ID NO: 296	moltype = RNA length = 72	
FEATURE	Location/Qualifiers	
source	1..72	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 296		
gagtcagagg gaccaggagc aggagaaggc ctatgggtgat ttgcattctt cctgccttgg		60
ctccatcttc ag		72
SEQ ID NO: 297	moltype = RNA length = 73	
FEATURE	Location/Qualifiers	
source	1..73	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 297		
tgtatccttg aatggatttt tggagcagga gtggacacct gacccaaagg aaatcaatcc		60
ataggctagc aat		73
SEQ ID NO: 298	moltype = RNA length = 70	
FEATURE	Location/Qualifiers	
source	1..70	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 298		
ctcggggccc accgcgccgg cccgcacctc ccggcccga gctgcgggct gcggtcaggg		60
cgatcccggg		70
SEQ ID NO: 299	moltype = RNA length = 69	
FEATURE	Location/Qualifiers	
source	1..69	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 299		
gggcttaggg atggaggcc aggatgaaga ttaatcccta atcccccaaca ctggccttgc		60
tatccccag		69

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SEQ ID NO: 300      moltype = RNA  length = 63
FEATURE            Location/Qualifiers
source              1..63
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 300
accctagggt ggggctggag gtggggctga ggctgagtct tctcccccct cctccctgcc 60
cag                                                63

SEQ ID NO: 301      moltype = RNA  length = 105
FEATURE            Location/Qualifiers
source              1..105
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 301
gcgggcgggc gcgggcgag cagcagcagg tgcggggcgg cgcccgcgct ggccgctcga 60
ctcccgagct gctcggtctg cttctccagc ttgcgcacca gctcc                105

SEQ ID NO: 302      moltype = RNA  length = 63
FEATURE            Location/Qualifiers
source              1..63
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 302
ggctccgcag ggccctggcg caggcatcca gacagcgggc gaatgcctcc cccggccccg 60
cag                                                63

SEQ ID NO: 303      moltype = RNA  length = 74
FEATURE            Location/Qualifiers
source              1..74
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 303
gtgcgtggtg gctcgaggcg ggggtggggg cctcgccctg cttgggacct cctgacctc 60
tccgctccgc acag                                                74

SEQ ID NO: 304      moltype = RNA  length = 62
FEATURE            Location/Qualifiers
source              1..62
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 304
tggcctaggg ggcggttctg ggagtgtatg ggctgagcct tgctctgctc ccccgcctcc 60
ag                                                62

SEQ ID NO: 305      moltype = RNA  length = 67
FEATURE            Location/Qualifiers
source              1..67
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 305
gtgggagggg agaggcagca agcacacagg gcttgggact agcatgctga cctccctcct 60
gccccag                                                67

SEQ ID NO: 306      moltype = RNA  length = 61
FEATURE            Location/Qualifiers
source              1..61
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 306
gggggcgggg gctggggtct gcaggttcgc actgatgcct gctcgccctg tctcccgcta 60
g                                                61

SEQ ID NO: 307      moltype = RNA  length = 75
FEATURE            Location/Qualifiers
source              1..75
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 307
ggctgtgccg ggtagagagg gcagtgaggag gtaagagctc ttcacccttc accaccttct 60
ccaccagca tggcc                                                75

SEQ ID NO: 308      moltype = RNA  length = 89
FEATURE            Location/Qualifiers
source              1..89
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 308

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gccggcgccc gagctctggc tccgtgtctt cactcccgtg cttgtccgag gagggaggga 60
gggacggggg ctgtgctggg gcagctgga 89

SEQ ID NO: 309      moltype = RNA length = 61
FEATURE            Location/Qualifiers
source              1..61
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 309
gtgcggaacg ctggccgggg cgggaggggg agggacgccc ggccggaacg ccgcactcac 60
g 61

SEQ ID NO: 310      moltype = RNA length = 70
FEATURE            Location/Qualifiers
source              1..70
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 310
aaaagcctgt ccctaagtcc ctcccagcct tccagagttg gtgccaggaa ggatttaggg 60
acaggctttg 70

SEQ ID NO: 311      moltype = RNA length = 67
FEATURE            Location/Qualifiers
source              1..67
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 311
gtgaggaggg gctggcaggg acccctccaa gttggggacg gcagccagcc cctgctcacc 60
cctcgcc 67

SEQ ID NO: 312      moltype = RNA length = 94
FEATURE            Location/Qualifiers
source              1..94
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 312
cgggcagcgg gtgccaggca cgggtgtcagc aggcaacatg gccgagaggc cggggcctcc 60
ggggcgggcc gtgtccgca ccgcgtaccc tgac 94

SEQ ID NO: 313      moltype = RNA length = 92
FEATURE            Location/Qualifiers
source              1..92
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 313
cctgtccctc ctgccctgcg cctgcccagc cctcctgctc tggtagctga ggaccgccag 60
gcaggggctg gtgctgggcg gggggcgggcg gg 92

SEQ ID NO: 314      moltype = RNA length = 94
FEATURE            Location/Qualifiers
source              1..94
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 314
aattcagccc tgccactggc ttatgtcatg accttgggct actcagctg tctgcacaat 60
gagccagttg gacaggagca gtgcactca actc 94

SEQ ID NO: 315      moltype = RNA length = 60
FEATURE            Location/Qualifiers
source              1..60
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 315
ggtgcctcgg gagggcatgg gccaggccac ataatgagcc aaaccctgt ctaccgcgag 60

SEQ ID NO: 316      moltype = RNA length = 92
FEATURE            Location/Qualifiers
source              1..92
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 316
gtcagtgtct gggcgacag ctgcaggaaa gggaagacca aggcttgctg tctgtccagt 60
ctgccaccct accctgtctg ttcttgccac ag 92

SEQ ID NO: 317      moltype = RNA length = 80
FEATURE            Location/Qualifiers
source              1..80
                    mol_type = unassigned RNA
                    organism = Homo sapiens

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SEQUENCE: 317
 cgggaatgcc gcggcgggga cggcgattgg tccgtatgtg tgggtgccacc ggccgcgggc 60
 tccgccccgg cccccgcccc 80

SEQ ID NO: 318 moltype = RNA length = 84
 FEATURE Location/Qualifiers
 source 1..84
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 318
 ctcccatgga cctgtctcc caacccttgt accagtgtg ggctcagacc ctggtacagg 60
 cctgggggac agggacctgg ggac 84

SEQ ID NO: 319 moltype = RNA length = 106
 FEATURE Location/Qualifiers
 source 1..106
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 319
 agaagaatgc ccaaccagcc ctcaattgct acagttccct gttgtttcag ctgcacaaca 60
 acaggcggct gtagcaatgg ggggctggat gggcatctca atgtgc 106

SEQ ID NO: 320 moltype = RNA length = 79
 FEATURE Location/Qualifiers
 source 1..79
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 320
 gtgcagatcc ttgggagccc tgttagactc tggattttac acttgagtg aacgggagcc 60
 atcccgaggc ttgcacag 79

SEQ ID NO: 321 moltype = RNA length = 102
 FEATURE Location/Qualifiers
 source 1..102
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 321
 gcttatcgag gaaaagatcg aggtgggttg gggcgggctc tggggatttg gtctcacagc 60
 cgggatccca gcccaacttac ctgggttact ctccttcctt ct 102

SEQ ID NO: 322 moltype = RNA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 322
 cgggcggggc gggtccggcc gcctccgagc ccggccggca gccccggcc ttaaagcgcg 60
 ggctgtccgg aggggtcggc ttcccaccg 90

SEQ ID NO: 323 moltype = RNA length = 59
 FEATURE Location/Qualifiers
 source 1..59
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 323
 ggggccaggc agggaggttg gaccatgggg gccttgctgt gtgaccaccg ttctgcag 59

SEQ ID NO: 324 moltype = RNA length = 62
 FEATURE Location/Qualifiers
 source 1..62
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 324
 gtgagtagtg gcgcgcggcg gctcggagta cctctgccgc cgcgcgcatc ggctcagcat 60
 gc 62

SEQ ID NO: 325 moltype = RNA length = 85
 FEATURE Location/Qualifiers
 source 1..85
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 325
 tctgaggtac ccggggcaga ttggtgtagg gtgcaaagcc tgcccccccc ctaagccttc 60
 tgcccccaac tccagcctgt cagga 85

SEQ ID NO: 326 moltype = RNA length = 96
 FEATURE Location/Qualifiers
 source 1..96
 mol_type = unassigned RNA

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                organism = Homo sapiens
SEQUENCE: 326
ctcgaggagg gcgaggagg ggtccccgt gctcgatct cgagggtgct tattgttcgg 60
tccgagcctg ggtctccctc tccccccaa ccccc 96

SEQ ID NO: 327      moltype = RNA length = 80
FEATURE            Location/Qualifiers
source             1..80
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 327
gcgtaaatat ggcggcgagg aggtaggcag agcaggacgc cgctgctgcc gccgccaccg 60
ccgctccgc tccagtgcgc 80

SEQ ID NO: 328      moltype = RNA length = 82
FEATURE            Location/Qualifiers
source             1..82
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 328
tgagctgttg gattcggggc cgtagcactg tctgagaggt ttacatttct cacagtgaac 60
cggctctttt ttcagtgcgc tc 82

SEQ ID NO: 329      moltype = RNA length = 83
FEATURE            Location/Qualifiers
source             1..83
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 329
gtgagtggga gccccagtg gtggttgggg ccatggcggg tgggcagccc agcctctgag 60
ccttcctcgt ctgtctgccc cag 83

SEQ ID NO: 330      moltype = RNA length = 87
FEATURE            Location/Qualifiers
source             1..87
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 330
accgcaggga aaatgaggga cttttggggg cagatgtgtt tccattccac tatcataatg 60
cccataaaaa tccttattgc tctgca 87

SEQ ID NO: 331      moltype = RNA length = 110
FEATURE            Location/Qualifiers
source             1..110
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 331
ggctacagtc tttcttcag tgactcgtgg acttcccttt gtcacccat gcctgagaat 60
atatgaagga ggctgggaag gcaaaggac gttcaattgt catcactggc 110

SEQ ID NO: 332      moltype = RNA length = 80
FEATURE            Location/Qualifiers
source             1..80
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 332
ctgcagcgtg cttctccagg ccccgcgcg gcacagacac acggacaagt cccgccaggg 60
gctgggcgcg cgcagccgg 80

SEQ ID NO: 333      moltype = RNA length = 81
FEATURE            Location/Qualifiers
source             1..81
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 333
ctccctggga gggcggtgat gatggtggga gaggagcccc actgtggaag tctgaccccc 60
acatcgcccc accttcccc g 81

SEQ ID NO: 334      moltype = RNA length = 67
FEATURE            Location/Qualifiers
source             1..67
                   mol_type = unassigned RNA
                   organism = Homo sapiens
SEQUENCE: 334
cctgcaggca gaagtggggc tgacagggca gaggggttgcg cccctccacc atcccttctg 60
cctgcag 67

SEQ ID NO: 335      moltype = RNA length = 67
FEATURE            Location/Qualifiers

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source                1..67
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 335
cctgcaggca gaagtggggc tgacagggca gagggttgcg cccctcacc atcccttctg 60
cctgcag                                                    67

SEQ ID NO: 336        moltype = RNA length = 67
FEATURE              Location/Qualifiers
source                1..67
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 336
cctgcaggca gaagtggggc tgacagggca gagggttgcg cccctcacc atcccttctg 60
cctgcag                                                    67

SEQ ID NO: 337        moltype = RNA length = 67
FEATURE              Location/Qualifiers
source                1..67
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 337
cctgcaggca gaagtggggc tgacagggca gagggttgcg cccctcacc atcccttctg 60
cctgcag                                                    67

SEQ ID NO: 338        moltype = RNA length = 75
FEATURE              Location/Qualifiers
source                1..75
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 338
gtcagagggg ggatgtgcat gctggttggg gtgggctgcc tgtggaccaa tcagcgtgca 60
cttccccacc ctgaa                                                    75

SEQ ID NO: 339        moltype = RNA length = 80
FEATURE              Location/Qualifiers
source                1..80
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 339
tgagaggccg caccttgctt tgtgccccg gccgtgcacc cgtgggcccc agggcgacgc 60
ggcgggggcg gccctagcga                                                    80

SEQ ID NO: 340        moltype = RNA length = 153
FEATURE              Location/Qualifiers
source                1..153
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 340
gctccgcccc acgtgcgatg cgccccggga acgcgtgggg cggagcttcc ggaggccccg 60
ctctgctgcc gacctgtgg agcggagggt gaagcctccg gatgccagtc cctcatcgct 120
ggcctggtcg cgctgtggcg aagggggcgg agc                                                    153

SEQ ID NO: 341        moltype = RNA length = 153
FEATURE              Location/Qualifiers
source                1..153
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 341
gctccgcccc acgtgcgatg cgccccggga acgcgtgggg cggagcttcc ggaggccccg 60
ccctgctgcc gacctgtgg agcggagggt gaagcctccg gatgccagtc cctcatcgct 120
ggcccggtcg cgctgtggcg aagggggcgg agc                                                    153

SEQ ID NO: 342        moltype = RNA length = 66
FEATURE              Location/Qualifiers
source                1..66
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 342
cagagcaggg cagggaaagt gggagagggg cccagctgac cctcctgtca cccgctcctt 60
gccccag                                                    66

SEQ ID NO: 343        moltype = RNA length = 119
FEATURE              Location/Qualifiers
source                1..119
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 343
agcctgcgcc ggagccgggg cctgagcccc ggccgcgcag gccgtgaact cgtcgagctg 60

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SEQUENCE: 352
 ggtgaggcgg gggggcgagc cctgaggggc tctcgcttct ggcgccaag 49

SEQ ID NO: 353 moltype = RNA length = 79
 FEATURE Location/Qualifiers
 source 1..79
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 353
 ctcgaggtgc tgggggacgc gtagagcgga gccgcttct caccgctcgg ccgcggcgcg 60
 tagcccccgc cacatcgagg 79

SEQ ID NO: 354 moltype = RNA length = 71
 FEATURE Location/Qualifiers
 source 1..71
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 354
 ggtgagtgagg agccgggtggg gctggagtaa gggcacgccc ggggctgccc caccctgtga 60
 ccaccctccc c 71

SEQ ID NO: 355 moltype = RNA length = 65
 FEATURE Location/Qualifiers
 source 1..65
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 355
 gagaatgggg ggacagatgg agaggacaca ggctggcact gaggtcccct ccactttcct 60
 cctag 65

SEQ ID NO: 356 moltype = RNA length = 79
 FEATURE Location/Qualifiers
 source 1..79
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 356
 gtgaggtggg ggccagcagg gagtgggctg ggctgggctg ggccaaggta caaggcctca 60
 ccctgcatcc cgcaccag 79

SEQ ID NO: 357 moltype = RNA length = 80
 FEATURE Location/Qualifiers
 source 1..80
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 357
 catccaggac aatgggtgagt gccgggtgcct gccctggggc cgtccctgcg caggggcccg 60
 gtgctcaccg catctgcccc 80

SEQ ID NO: 358 moltype = RNA length = 61
 FEATURE Location/Qualifiers
 source 1..61
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 358
 tctcgtttga tctcggaagc taagcagggt tgggcctggt tagtacttgg atgggaaact 60
 t 61

SEQ ID NO: 359 moltype = RNA length = 53
 FEATURE Location/Qualifiers
 source 1..53
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 359
 gtttgatctc ggaagctaag cagggtcggg cctggtagt acttgatgg gag 53

SEQ ID NO: 360 moltype = RNA length = 92
 FEATURE Location/Qualifiers
 source 1..92
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 360
 cgctgcgctt ctggggccgc ggcgggctg gggctgcccg ggccggtcga ccagcgccg 60
 gtagctcccg aggcccgagc cgcgacccgc gg 92

SEQ ID NO: 361 moltype = RNA length = 84
 FEATURE Location/Qualifiers
 source 1..84
 mol_type = unassigned RNA
 organism = Homo sapiens

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SEQUENCE: 361
gtggcgctcg gtgctgggga gggcccccg ggtgggcctc tgetctggcc cctcctgggg 60
ccgcactct cgctctgggc ccgc 84

SEQ ID NO: 362      moltype = RNA length = 79
FEATURE            Location/Qualifiers
source              1..79
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 362
cagcctgggg aaggtttggc agggaagaca catgagcagt gcctccactt cagcctctc 60
ccttgtctcc ttccctag 79

SEQ ID NO: 363      moltype = RNA length = 83
FEATURE            Location/Qualifiers
source              1..83
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 363
gggtttcttc tgcctttttt tccaatgaaa ataacgaaac ctgttatctc ccattgaggg 60
ggaaaaaggc gggagaagcc cca 83

SEQ ID NO: 364      moltype = RNA length = 60
FEATURE            Location/Qualifiers
source              1..60
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 364
gagcaggcga ggctgggctg aaccctgggg tgaggagtgc agcccagctg aggcctctgc 60

SEQ ID NO: 365      moltype = RNA length = 73
FEATURE            Location/Qualifiers
source              1..73
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 365
aggagaaaaa gctgggctga gaggcgactg gtgtctaatt tgtttgtctc tccaactcag 60
actgcctggc cca 73

SEQ ID NO: 366      moltype = RNA length = 88
FEATURE            Location/Qualifiers
source              1..88
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 366
gtggggccag gcggtggctg gcactgctgg ggtgggcaca gcagccatgc agagcgggca 60
ttgaccccg tgccaccctt tccccag 88

SEQ ID NO: 367      moltype = RNA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 367
gtgtctctct ggagaccctg cagccttccc acccaccagg gagctttcca tgggctgtgg 60
ggaaggcgtc agtgctgggt gagggaaac 90

SEQ ID NO: 368      moltype = RNA length = 64
FEATURE            Location/Qualifiers
source              1..64
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 368
gaggcaactgg gtagtggggg ctccagggct cctgacacct ggacctctcc tccccaggcc 60
caca 64

SEQ ID NO: 369      moltype = RNA length = 64
FEATURE            Location/Qualifiers
source              1..64
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 369
tgaggatggg gtgagatggg gaggagcagc cagtctctgc tcaccgtctc tcccctgacc 60
ccag 64

SEQ ID NO: 370      moltype = RNA length = 73
FEATURE            Location/Qualifiers
source              1..73
                    mol_type = unassigned RNA

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                organism = Homo sapiens
SEQUENCE: 370
ggcgagggga ggcgcaggct cggaaaggcg cgcgaggctc caggctcctt cccgatccac 60
cgctctcttc gct 73

SEQ ID NO: 371      moltype = RNA  length = 81
FEATURE            Location/Qualifiers
source             1..81
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 371
tacagggtgca ggggaactgt agatgaaaag gcttggcact tgagggaaaag cctcagttca 60
ttctcatttt gctcacctgt t 81

SEQ ID NO: 372      moltype = RNA  length = 68
FEATURE            Location/Qualifiers
source             1..68
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 372
gaatggaaga agaaggcggt cgggtctgctg gagccaggcc gcagagccat ccgccttctg 60
tccatgtc 68

SEQ ID NO: 373      moltype = RNA  length = 82
FEATURE            Location/Qualifiers
source             1..82
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 373
gctacgggga gcggggagga agtgggctgct gcttctgctg tatctggaag gagcagccca 60
ctctgtctct gggctctgtg gt 82

SEQ ID NO: 374      moltype = RNA  length = 89
FEATURE            Location/Qualifiers
source             1..89
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 374
acctccggga cggtctggcg ccggcgggccg ggagatccgc gcttctctgaa tcccggccgg 60
cccgcccggc gcccgctccgc ccgcgggctc 89

SEQ ID NO: 375      moltype = RNA  length = 63
FEATURE            Location/Qualifiers
source             1..63
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 375
gagtcgagga ctggtggaag ggcctttccc ctcagaccaa ggccctggcc ccagcttctt 60
ctc 63

SEQ ID NO: 376      moltype = RNA  length = 64
FEATURE            Location/Qualifiers
source             1..64
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 376
tgctctgtag gcatgaggga gggcccaggt tccatgtgat gctgaagctc tgacattcct 60
gcag 64

SEQ ID NO: 377      moltype = RNA  length = 109
FEATURE            Location/Qualifiers
source             1..109
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 377
ggtcgggctc accatgacac agtgtgagac ctcgggctac aacacaggac ccgggctgctg 60
ctctgacccc tcgtgtcttg tgtgcagcc ggagggacgc aggtccgca 109

SEQ ID NO: 378      moltype = RNA  length = 94
FEATURE            Location/Qualifiers
source             1..94
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 378
cagtgcgacg ggcggagctt ccagacgctc cgccccacgt cgcatgcgcc ccgggaaagc 60
gtggggcgga gcttccggag gccccgccct gctg 94

SEQ ID NO: 379      moltype = RNA  length = 88
FEATURE            Location/Qualifiers

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source                1..88
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 379
gcgacgggcg gagcttccag acgctccgcc ccacgtcgca tgcgccccgg gaaagcgtgg 60
ggcggagctt ccgaggcccc cgccctgc                                     88

SEQ ID NO: 380        moltype = RNA length = 94
FEATURE              Location/Qualifiers
source                1..94
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 380
cagtgcgacg ggccggagctt ccagacgctc cgcgccacgt cgcattgcgc ccgggaaagc 60
gtggggcgga gcttccggag gccccgccct gctg                                     94

SEQ ID NO: 381        moltype = RNA length = 70
FEATURE              Location/Qualifiers
source                1..70
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 381
gtccctgggg gctgggatgg gccatgggtg gctctgatcc cctgtgggc tcttggcccc 60
caggaactcc                                     70

SEQ ID NO: 382        moltype = RNA length = 62
FEATURE              Location/Qualifiers
source                1..62
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 382
ccttctgcgg cagagctggg gtcaccagcc ctcatgtact tgtgacttct cccctgccac 60
ag                                             62

SEQ ID NO: 383        moltype = RNA length = 82
FEATURE              Location/Qualifiers
source                1..82
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 383
acctgtaggt gacagtcagg ggccggggtg ggtggggctg gggtgggcc cctcctcaca 60
cctcctctgg catcgcccc ag                                             82

SEQ ID NO: 384        moltype = RNA length = 73
FEATURE              Location/Qualifiers
source                1..73
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 384
ctggtgtttg aggcgatgtg gggatgtaga gacaacttcc cagtctcatt tctcctcct 60
gccaggccac cat                                             73

SEQ ID NO: 385        moltype = RNA length = 68
FEATURE              Location/Qualifiers
source                1..68
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 385
agggttgggg ggacaggatg agaggtgtc ttcattccct cttgaccacc cctcgtttct 60
tccccag                                       68

SEQ ID NO: 386        moltype = RNA length = 61
FEATURE              Location/Qualifiers
source                1..61
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 386
gagggcagcg tgggtgtggc ggaggcaggc gtgaccgttt gccgccctct cgtgctctta 60
g                                             61

SEQ ID NO: 387        moltype = RNA length = 72
FEATURE              Location/Qualifiers
source                1..72
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 387
tctctcgag gggtctctgc ctctaccag gactctttca tgaccaggag gctgaggccc 60
ctcacaggcg gc                                             72

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SEQ ID NO: 388	moltype = RNA	length = 73	
FEATURE	Location/Qualifiers		
source	1..73		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 388			
gttcaagtgg	gaggacagga	ggcaggtgtg	gttgaggaa
gcagcctgaa	cctgcctccc	60	
tgacattcca	cag	73	
SEQ ID NO: 389	moltype = RNA	length = 97	
FEATURE	Location/Qualifiers		
source	1..97		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 389			
cccgggacct	tggtccaggc	gctgggtctgc	gtgggtgctcg
gggtggataag	tctgatctga	60	
gcaccacaca	ggccggggcgc	cgggaccaag	ggggctc
		97	
SEQ ID NO: 390	moltype = RNA	length = 70	
FEATURE	Location/Qualifiers		
source	1..70		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 390			
ggttttctct	tgaggagaca	tggtgggggc	cggtcaggca
gccccatgcca	tgtgtcctca	60	
tggagaggcc		70	
SEQ ID NO: 391	moltype = RNA	length = 110	
FEATURE	Location/Qualifiers		
source	1..110		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 391			
tcacctggcc	atgtgacttg	tgggcttccc	tttgtcatcc
ttcgctagg	gctctgagca	60	
gggcagggac	agcaaaaggg	tgctcagttg	tcacttccca
cagcacggag		110	
SEQ ID NO: 392	moltype = RNA	length = 85	
FEATURE	Location/Qualifiers		
source	1..85		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 392			
gggacggggc	ctgcaggcag	aagtggggct	gacagggcag
agggttgctc	cccctcacca	60	
ccccttctgc	ctgcagcggt	gggct	85
SEQ ID NO: 393	moltype = RNA	length = 71	
FEATURE	Location/Qualifiers		
source	1..71		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 393			
ggggcctgca	ggcagaagtg	gggctgacag	ggcagagggt
tgcgccccct	caccaccctc	60	
tctgcctgca	g	71	
SEQ ID NO: 394	moltype = RNA	length = 56	
FEATURE	Location/Qualifiers		
source	1..56		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 394			
cgctcgggcg	gaggtgggtg	agtgccgact	ggcgctgac
ccacccccctc	cgcgag	56	
SEQ ID NO: 395	moltype = RNA	length = 89	
FEATURE	Location/Qualifiers		
source	1..89		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 395			
agcctgtggg	aaagagaaga	gcagggcagg	gtgaaggccc
ggcggagaca	ctctgcccac	60	
cccacacctc	gcctatgggc	cacacagct	89
SEQ ID NO: 396	moltype = RNA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = unassigned RNA		
	organism = Homo sapiens		
SEQUENCE: 396			
tctaagaaac	gcagtggtct	ctgaagcctg	caggggcagg
ccagccctgc	actgaacgcc	60	
tqttcttqcc	aqqtqccaga	aaqttqctqc	90

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SEQ ID NO: 397      moltype = RNA  length = 86
FEATURE            Location/Qualifiers
source              1..86
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 397
ggggctgggg gtgtggggag agagagtgca cagccagctc agggattaaa gctctttctc 60
tctctctctc tcccacttcc ctgcag                                     86

SEQ ID NO: 398      moltype = RNA  length = 98
FEATURE            Location/Qualifiers
source              1..98
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 398
tgggcaagg tgcggggcta gggctaacag cagtcttact gaagggttcc tggaaaccac 60
gcacatgctg ttgccactaa cctcaacctt actcggtc                               98

SEQ ID NO: 399      moltype = RNA  length = 73
FEATURE            Location/Qualifiers
source              1..73
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 399
aggccaggtg ggtatggagg agccctcata tggcagttgg cgaggggcca gtgagcccct 60
ctctgctctc cag                                           73

SEQ ID NO: 400      moltype = RNA  length = 74
FEATURE            Location/Qualifiers
source              1..74
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 400
gcagcccggt gagcgctcgc tggcctggca gtgcgtcgga agaacagggc gggtgggggc 60
gcgcacatct ctgc                                           74

SEQ ID NO: 401      moltype = RNA  length = 55
FEATURE            Location/Qualifiers
source              1..55
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 401
gggggctggg gcgcggggag gtgctaggtc ggctctggct cccgcgcgcg acccc 55

SEQ ID NO: 402      moltype = RNA  length = 67
FEATURE            Location/Qualifiers
source              1..67
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 402
ggcagccagg gggatgggag agcttggggc cattccttc cttaccctac ccccatccc 60
cctgtag                                           67

SEQ ID NO: 403      moltype = RNA  length = 64
FEATURE            Location/Qualifiers
source              1..64
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 403
aaccgccggc cggagggtcaa gggcgctcgt tctcccta atgtgcctctt ttccacggcc 60
tcag                                           64

SEQ ID NO: 404      moltype = RNA  length = 68
FEATURE            Location/Qualifiers
source              1..68
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 404
ctcaggctca gtgggtgcatg cttatagtcc cagccactct ggaggctgaa ggaagatggc 60
ttgagcct                                           68

SEQ ID NO: 405      moltype = RNA  length = 67
FEATURE            Location/Qualifiers
source              1..67
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 405
ctggtccatt tccctgcat tcccttggt tcaatttact ccagggctg gcagtgacat 60

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gggtcaa 67

SEQ ID NO: 406      moltype = RNA  length = 101
FEATURE            Location/Qualifiers
source             1..101
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 406
gatgggcccc ttgtgtcctg aattgggtgg gggctctgag tggggaaagt gggggcctag 60
gggaggtcac agttgggtct aggggtcagg agggcccagg a 101

SEQ ID NO: 407      moltype = RNA  length = 137
FEATURE            Location/Qualifiers
source             1..137
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 407
agccagacaa gagggcatg gggagtcact gtcaaccag agcaggcact gcccctgcga 60
ccagcctggg gcacgggttg ggggtcaggg gtctgtctgt gatgctttcc atctctttgc 120
ttgtcctga ttgtagc 137

SEQ ID NO: 408      moltype = RNA  length = 95
FEATURE            Location/Qualifiers
source             1..95
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 408
gacaccacat gctcctccag gctgcctgc cctccaggtc atgttccagt gtcccacaga 60
tgcagcacca cggcccaggc ggcattggtg tcacc 95

SEQ ID NO: 409      moltype = RNA  length = 80
FEATURE            Location/Qualifiers
source             1..80
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 409
aggacccttc cagagggccc cccctcaatc ctgttgtgcc taattcagag ggttgggtgg 60
aggctctcct gaagggctct 80

SEQ ID NO: 410      moltype = RNA  length = 93
FEATURE            Location/Qualifiers
source             1..93
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 410
gtaaggaggg ggatgagggg tcatatctct tctcagggaa agcaggagcc cttcagcagg 60
gtcagggccc ctcattcttc cctcctttcc cag 93

SEQ ID NO: 411      moltype = RNA  length = 62
FEATURE            Location/Qualifiers
source             1..62
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 411
gtgagtgggg ctccggggac ggcgcccgcc ctggccctgg cccggcgacg tctcacggtc 60
cc 62

SEQ ID NO: 412      moltype = RNA  length = 69
FEATURE            Location/Qualifiers
source             1..69
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 412
gaacctcggg gcattgggga gggaggctgg acaggagagg gctcaccag gccctgtcct 60
ctgccccag 69

SEQ ID NO: 413      moltype = RNA  length = 60
FEATURE            Location/Qualifiers
source             1..60
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 413
caaggtgggg gagatggggg ttgaacttca tttctcatgc tcatccccat ctcccttcag 60

SEQ ID NO: 414      moltype = RNA  length = 75
FEATURE            Location/Qualifiers
source             1..75
                   mol_type = unassigned RNA
                   organism = Homo sapiens

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SEQUENCE: 414
 gtagttgttc tacagaagac ctggatgtgt aggagctaag acacactcca ggggagctgt 60
 ggaagcagta acacg 75

SEQ ID NO: 415 moltype = RNA length = 56
 FEATURE Location/Qualifiers
 source 1..56
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 415
 cgaccgcacc cgccgaagc tgggtcaagg agccagcag gacgggagcg cggcgc 56

SEQ ID NO: 416 moltype = RNA length = 61
 FEATURE Location/Qualifiers
 source 1..61
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 416
 gaggggttggg gtggagggcc aaggagctgg gtgggggtgcc aagcctctgt cccacccca 60
 g 61

SEQ ID NO: 417 moltype = RNA length = 66
 FEATURE Location/Qualifiers
 source 1..66
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 417
 gggcatgggg aggtgtggag tcagcatggg gctaggaggc cccgcgtga cccgccttct 60
 ccgcag 66

SEQ ID NO: 418 moltype = RNA length = 103
 FEATURE Location/Qualifiers
 source 1..103
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 418
 gtgtcggctg tggcgtgact gtccctctgt gtcccccact agggccactg ctcaagtggag 60
 cgtggaggac gaggaggagg ccgtccacga gcaatgccag cat 103

SEQ ID NO: 419 moltype = RNA length = 109
 FEATURE Location/Qualifiers
 source 1..109
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 419
 tcccgcattc cctctgcttt ggtcagggtg tgccctcctt ccatgggtag agccagagat 60
 ggtgggttct ggctggtcag atgggagtgg acagagaccc ggggtcctc 109

SEQ ID NO: 420 moltype = RNA length = 81
 FEATURE Location/Qualifiers
 source 1..81
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 420
 catcctcctt acgtcccacc cccactcct gtttctggtg aaatattcaa acaggagtgg 60
 ggggtgggaca taaggaggat a 81

SEQ ID NO: 421 moltype = RNA length = 73
 FEATURE Location/Qualifiers
 source 1..73
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 421
 gtgggcgggg gcagggtgtg ggtgggtggt ggcctgcggt gacgagggcc ctcacacctg 60
 cctcgccccc cag 73

SEQ ID NO: 422 moltype = RNA length = 71
 FEATURE Location/Qualifiers
 source 1..71
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 422
 gggtaaaggg gcagggacgg gtggccccag gaagaagggc ctggtggagc cgctcttctc 60
 cctgcccaca g 71

SEQ ID NO: 423 moltype = RNA length = 52
 FEATURE Location/Qualifiers
 source 1..52
 mol_type = unassigned RNA

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                organism = Homo sapiens
SEQUENCE: 423
tagccgggcg tgggtggtggg ggcctgtggt cccagctact ttggaggtcg ag      52

SEQ ID NO: 424      moltype = RNA length = 93
FEATURE            Location/Qualifiers
source             1..93
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 424
ggcgcttttg tgcgcgcccg ggtctgttgg tgctcagagt gtggtcaggc ggctcggact  60
gagcaggtgg gtgcggggct cggaggaggc ggc                                93

SEQ ID NO: 425      moltype = RNA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 425
gcccgggtggg gcggggcgcc ctcaggaggg gccagctcc cctggatgtg ctgcggtggg  60
gcccggagggg cgtcacgtgc acccaagtga ccccccttct gattctgcct cag      113

SEQ ID NO: 426      moltype = RNA length = 84
FEATURE            Location/Qualifiers
source             1..84
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 426
ggccccctct tctcagcccc agctcccgtc caccctgcc acgtcaaagg aggcagaagg  60
ggagttggga gcagagaggg gacc                                84

SEQ ID NO: 427      moltype = RNA length = 100
FEATURE            Location/Qualifiers
source             1..100
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 427
gtgagcgggc gcggcaggga tcgcggggcg gtggcggcct agggcgcgga gggcggaccg  60
ggaatggcgc gccgtgcgcc gccggcgtaa ctgcggcgct                                100

SEQ ID NO: 428      moltype = RNA length = 63
FEATURE            Location/Qualifiers
source             1..63
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 428
aatgggtggg tgctggtggg agccgtgccc tggccactca ttcggctctc tccctcaccc  60
tag                                                63

SEQ ID NO: 429      moltype = RNA length = 59
FEATURE            Location/Qualifiers
source             1..59
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 429
ctgtgtcggg gagtctgggg tccggaattc tccagagcct ctgtgcccct acttcccag  59

SEQ ID NO: 430      moltype = RNA length = 71
FEATURE            Location/Qualifiers
source             1..71
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 430
tgaccacccc cgggcaaaga cctgcagatc cctgttaga gacggggcca ggactttgtg  60
cggggtgccc a                                                71

SEQ ID NO: 431      moltype = RNA length = 54
FEATURE            Location/Qualifiers
source             1..54
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 431
agcagcaggg gagagagagg agtcctctag acaccgactc tgtctcctgc agat      54

SEQ ID NO: 432      moltype = RNA length = 85
FEATURE            Location/Qualifiers
source             1..85
                   mol_type = unassigned RNA
                   organism = Homo sapiens

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SEQUENCE: 432
ggcgctctct gctctgctgt gccgccaggg cctcccctag cgcgcttct ggagaggctt 60
tgtgcggata cgggggtgga ggctt 85

SEQ ID NO: 433      moltype = RNA  length = 96
FEATURE            Location/Qualifiers
source             1..96
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 433
agatgtgctc tcctggccca tgaatcaag cgtgggtgag acctgggtgca gaacgggaag 60
gcgaccata cttggtttca gaggctgtga gaataa 96

SEQ ID NO: 434      moltype = RNA  length = 100
FEATURE            Location/Qualifiers
source             1..100
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 434
agctcagggc ggctgcgcag agggctggac tcagcggcgg agctgggtgc tggcctcagt 60
tctgcctctg tccaggtcct tgtgaccgcg cgcctctcct 100

SEQ ID NO: 435      moltype = RNA  length = 50
FEATURE            Location/Qualifiers
source             1..50
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 435
acccgggcgt ggtgggtggg gtgggtgcct gtaattccag ctagtggga 50

SEQ ID NO: 436      moltype = RNA  length = 80
FEATURE            Location/Qualifiers
source             1..80
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 436
tgccgtcggc ctggggagga ggaaggga gtcctaaagg atacagtgg tctgttcatt 60
ctctcttttt ggctacaag 80

SEQ ID NO: 437      moltype = RNA  length = 63
FEATURE            Location/Qualifiers
source             1..63
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 437
gaggtgtagg ggaggttggg ccagggatgc cttcactgtg tctctctggt cttgccaccc 60
cag 63

SEQ ID NO: 438      moltype = RNA  length = 60
FEATURE            Location/Qualifiers
source             1..60
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 438
aggcctaggg ggtggcaggc tggccatcag tgtgggctaa ccctgtcctc tccctccag 60

SEQ ID NO: 439      moltype = RNA  length = 67
FEATURE            Location/Qualifiers
source             1..67
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 439
gtggcactca aactgtgggg gcactttctg ctctctggtg aaagtgcgc catcttttga 60
gtgttac 67

SEQ ID NO: 440      moltype = RNA  length = 68
FEATURE            Location/Qualifiers
source             1..68
                   mol_type = unassigned RNA
                   organism = Homo sapiens

SEQUENCE: 440
gggcgcaggg ggactggggg tgagcaggcc cagaaccag ctcgtgctca ctctcagtc 60
ctccctag 68

SEQ ID NO: 441      moltype = RNA  length = 64
FEATURE            Location/Qualifiers
source             1..64
                   mol_type = unassigned RNA
                   organism = Homo sapiens

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SEQUENCE: 441
gagctctggg aggggctggg ttggcagga cagtttccaa gccctgtctc ctcccatctt 60
ccag 64

SEQ ID NO: 442 moltype = RNA length = 67
FEATURE Location/Qualifiers
source 1..67
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 442
aaatctctct ccatatcttt cctgcagccc ccaggtgggg gggaagaaaa ggtggggaat 60
tagattc 67

SEQ ID NO: 443 moltype = RNA length = 115
FEATURE Location/Qualifiers
source 1..115
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 443
gagcaaaaac cagagaacaa catgggagcg ttcctaacc ctaaggcaac tggatgggag 60
aactgaccca tccagttctc tgagggggct cttgtgtgtt ctacaagggt gttca 115

SEQ ID NO: 444 moltype = RNA length = 86
FEATURE Location/Qualifiers
source 1..86
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 444
agagccgggg ccatggagca gccctgttag acggggacct gccctgcatg ggcaccccct 60
cactggctgc ttccttgggt ctccag 86

SEQ ID NO: 445 moltype = RNA length = 82
FEATURE Location/Qualifiers
source 1..82
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 445
tgtgggcagg gccctgggga gctgaggctc tgggggtggc cggggctgac cctgggcctc 60
tgctcccag tgtctgaccg cg 82

SEQ ID NO: 446 moltype = RNA length = 136
FEATURE Location/Qualifiers
source 1..136
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 446
ccgcttgctt cgcccagcgc agccccggcc gctgggcgca cccgtccgt tcgtccccgg 60
acgttgctct ctaccccggg aacgtcgaga ctggagcgcc cgaactgagc caccttcgcy 120
gaccccagaga gcggcg 136

SEQ ID NO: 447 moltype = RNA length = 66
FEATURE Location/Qualifiers
source 1..66
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 447
tcaagacggg gagtcaggca gtggtggaga tggagagccc tgagcctcca ctctcctggc 60
ccccag 66

SEQ ID NO: 448 moltype = RNA length = 78
FEATURE Location/Qualifiers
source 1..78
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 448
ggcctcaggc aggcgcaccc gaccacatgc atggctggtg gcggcgtgca ggggtcgggt 60
gggcaggct gtggggcg 78

SEQ ID NO: 449 moltype = RNA length = 69
FEATURE Location/Qualifiers
source 1..69
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 449
ctttcgcca gcgggacggc atccgaggtg ggctaggctc gggcccgtgg cgggtgcggg 60
gtggggagg 69

SEQ ID NO: 450 moltype = RNA length = 83
FEATURE Location/Qualifiers

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SEQ ID NO: 459 moltype = RNA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 459
ggcgccctctg cagctccggc tccccctggc ctctcgggaa ctacaagtcc cagggggcct 60
ggcgggtggc ggccggcgga agaggcgggg 90

SEQ ID NO: 460 moltype = RNA length = 64
FEATURE Location/Qualifiers
source 1..64
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 460
gggtgggggc gggcgggcag gggcctcccc cagtgccagg cccattctg cttctctccc 60
agct 64

SEQ ID NO: 461 moltype = RNA length = 64
FEATURE Location/Qualifiers
source 1..64
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 461
agttcagggc cgaagggtgg aagctgctgg tgctcatctc agcctctgcc cttggcctcc 60
ccag 64

SEQ ID NO: 462 moltype = RNA length = 68
FEATURE Location/Qualifiers
source 1..68
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 462
cgtggtgagg atatggcagg gaaggggagt ttccctctat tcccttcccc ccagtaatct 60
tcacatg 68

SEQ ID NO: 463 moltype = RNA length = 67
FEATURE Location/Qualifiers
source 1..67
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 463
gcgccctccc tctctccccg gtgtgcaaat gtgtgtgtgc ggtgttatgc cggacaagag 60
ggaggtg 67

SEQ ID NO: 464 moltype = RNA length = 98
FEATURE Location/Qualifiers
source 1..98
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 464
cgaggtaggg gcgtcccgcc cgcgcggggc ggtcccaggc tgggccctc ggaggccggg 60
tgctcactgc ccggtcccg cgcctgtgtc tctccag 98

SEQ ID NO: 465 moltype = RNA length = 69
FEATURE Location/Qualifiers
source 1..69
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 465
tggggtaggg gtgggggaat tcaggggtgt cgaactcatg gctgccacct ttgtgtcccc 60
atcctgcag 69

SEQ ID NO: 466 moltype = RNA length = 68
FEATURE Location/Qualifiers
source 1..68
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 466
gcacctgtga ctgagctgcc ccgaggccct tcattgctgcc cagctcgggg cagctcagta 60
caggatac 68

SEQ ID NO: 467 moltype = RNA length = 64
FEATURE Location/Qualifiers
source 1..64
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 467
tcctgtactg agctgccccg agctgggcag catgaagggc ctccgggcag ctcagtacag 60

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gatg 64

SEQ ID NO: 468 moltype = RNA length = 110
 FEATURE Location/Qualifiers
 source 1..110
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 468
 gtctaccagg tgtgggcccc gctttacata gttcatgctg aggccgggat ttcatacaga 60
 aaactggttg caaaaggtgc tgaaggggct gggggagcac aagggaagaag 110

SEQ ID NO: 469 moltype = RNA length = 71
 FEATURE Location/Qualifiers
 source 1..71
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 469
 gtgtctgtgc cgggccagg agaactgca gaggcacggtg gtcagcgggt ctctgcggg 60
 ccgacactca c 71

SEQ ID NO: 470 moltype = RNA length = 99
 FEATURE Location/Qualifiers
 source 1..99
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 470
 cggccacctc agcctcccaa aatgctggga ttacaggcat gagccactgc ggtcgacccat 60
 gacctggaca tggttggtcc cagtactgtc agtttgag 99

SEQ ID NO: 471 moltype = RNA length = 86
 FEATURE Location/Qualifiers
 source 1..86
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 471
 agcactgccc ccggtgagtc aggggtgggc tggccccctg ctctgtgccc atccgcgctc 60
 tgactctctg cccacctgca ggagct 86

SEQ ID NO: 472 moltype = RNA length = 65
 FEATURE Location/Qualifiers
 source 1..65
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 472
 ctctctggtg ggtggggggc tgggcgtggt ggacagcgat gcacccctcg ccttctcacc 60
 ctgag 65

SEQ ID NO: 473 moltype = RNA length = 73
 FEATURE Location/Qualifiers
 source 1..73
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 473
 ggggaggtac ctgggacagg agggaggagc agccttgctt cagaaccaa actgtcaaaa 60
 gtgtaggttc cac 73

SEQ ID NO: 474 moltype = RNA length = 55
 FEATURE Location/Qualifiers
 source 1..55
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 474
 ccggatccga gtcacggcac caaatttcat gcgtgtccgt gtgaagagac cacca 55

SEQ ID NO: 475 moltype = RNA length = 66
 FEATURE Location/Qualifiers
 source 1..66
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 475
 acccaggagg cggaggaggt ggagggtgca gtgagccaag atcgtggcac tgactccagc 60
 ctgggg 66

SEQ ID NO: 476 moltype = RNA length = 94
 FEATURE Location/Qualifiers
 source 1..94
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 476

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cacgggtgtcc cctgggtggaa cctggcaggg ggagaggttaa ggtctttcag cctctccaaa 60
gcccattggtc aggtactcag gtgggggagc cctg 94

SEQ ID NO: 477      moltype = RNA length = 62
FEATURE            Location/Qualifiers
source              1..62
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 477
gaggggtggg gaggaagagg gcagctccca tgactgcctg accgccttct ctcctccccc 60
ag 62

SEQ ID NO: 478      moltype = RNA length = 66
FEATURE            Location/Qualifiers
source              1..66
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 478
acgagccctc ggcgggcccgg gggggggggcg gcggtgcccg tcccgggggt gcgcgaggca 60
caggcg 66

SEQ ID NO: 479      moltype = RNA length = 81
FEATURE            Location/Qualifiers
source              1..81
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 479
tagaggcagt ttcaacagat gtgtagactt ttgatatgag aaattggtt caaaatcagg 60
agtcggggct ttactgctt t 81

SEQ ID NO: 480      moltype = RNA length = 74
FEATURE            Location/Qualifiers
source              1..74
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 480
cctgcaggag gcagtggggcg agcaggcggg gcagcccaat gccatgggccc tgatctcacc 60
gtgcctcct tccc 74

SEQ ID NO: 481      moltype = RNA length = 109
FEATURE            Location/Qualifiers
source              1..109
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 481
tgctattgtc ttactgctac agcaggggctg gggattgcag tatccgctgt tgctgctgct 60
cccagtctcg cccctgctgc tacctagtcc agcctcaccg catcccaga 109

SEQ ID NO: 482      moltype = RNA length = 67
FEATURE            Location/Qualifiers
source              1..67
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 482
cagcgtgggc tgctgagaag gggcagggtc ctccagctca ttcctcctgc ctctccgtg 60
gctcag 67

SEQ ID NO: 483      moltype = RNA length = 63
FEATURE            Location/Qualifiers
source              1..63
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 483
gcatgctggg cgaggctggc atctagcaca ggcggtagat gcttgctctt gccattgcaa 60
tga 63

SEQ ID NO: 484      moltype = RNA length = 98
FEATURE            Location/Qualifiers
source              1..98
                    mol_type = unassigned RNA
                    organism = Homo sapiens

SEQUENCE: 484
cttgggtcaat aggaaagagg tgggacctcc tggcttttcc tctgcagcat ggctcggacc 60
tagtgcaatg tttaagctcc cctctcttcc ctgttcag 98

SEQ ID NO: 485      moltype = RNA length = 70
FEATURE            Location/Qualifiers
source              1..70
                    mol_type = unassigned RNA

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                                organism = Homo sapiens
SEQUENCE: 485
aggagccagc ggggctgggc gcgcggagca gcgctgggtg cagcgctgc gccggcagct 60
gcaagggccg                                     70

SEQ ID NO: 486          moltype = RNA  length = 61
FEATURE                Location/Qualifiers
source                1..61
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 486
tccgctctgt ggagtggggt gccctgtcccc tgccactggg tgacccaccc ctctccacca 60
g                                     61

SEQ ID NO: 487          moltype = RNA  length = 83
FEATURE                Location/Qualifiers
source                1..83
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 487
aatagagggt gcacaggcac gggagctcag gtgaggcagg gagctgagct cacctgacct 60
cccatgcctg tgcacctctt att                                     83

SEQ ID NO: 488          moltype = RNA  length = 87
FEATURE                Location/Qualifiers
source                1..87
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 488
gtgagggcgg gccaggaggg tgtgtggcgt ggggtgctcg gggccgctcag ggtgcctgcg 60
ggacgctcac ctggtggcc cgccag                                     87

SEQ ID NO: 489          moltype = RNA  length = 99
FEATURE                Location/Qualifiers
source                1..99
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 489
gcaagggaca gggagggtcg tggcgacct cgcgccagct cccgggacgg ctgggctcgg 60
gctggtcgcc gacctccgac cctccactag atgcctggc                                     99

SEQ ID NO: 490          moltype = RNA  length = 91
FEATURE                Location/Qualifiers
source                1..91
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 490
ggcgccccgg ctccccgcgc ccccgatcgg ggccgcccgt agtagtggcg gcggcgagg 60
cgggggcagc ggcggcgggc gcggaggcgc c                                     91

SEQ ID NO: 491          moltype = RNA  length = 69
FEATURE                Location/Qualifiers
source                1..69
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 491
ggccctcggg cctgggggttg ggggagctct gtctgtctc actcattgct cctcccctgc 60
ctggcccag                                     69

SEQ ID NO: 492          moltype = RNA  length = 89
FEATURE                Location/Qualifiers
source                1..89
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 492
tctccgttta tcccaccact gccaccatta ttgctactgt tcagcagggt ctgctgggtg 60
tgatggtgat agtctggttg gggcggttg                                     89

SEQ ID NO: 493          moltype = RNA  length = 69
FEATURE                Location/Qualifiers
source                1..69
                     mol_type = unassigned RNA
                     organism = Homo sapiens
SEQUENCE: 493
gaggagggga ggtgtgcagg gctgggggtca ctgactctgc tccccctgcc ctgcatggtg 60
tcccacag                                     69

SEQ ID NO: 494          moltype = RNA  length = 81
FEATURE                Location/Qualifiers

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source                1..81
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 494
agttgggtggg ggagccatga gataagagca cctcctagag aatgttgaac taaagggtgcc 60
ctctctggct cctccccaaa g                                         81

SEQ ID NO: 495        moltype = RNA length = 67
FEATURE              Location/Qualifiers
source                1..67
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 495
tacaggccgg ggctttgggt gagggacccc cggagtctgt cacgggtctca ccccaactct 60
gccccag                                         67

SEQ ID NO: 496        moltype = RNA length = 119
FEATURE              Location/Qualifiers
source                1..119
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 496
ccatgaggag ctggcagtgg gatggcctgg gggtaggagc gtggcttctg gagctagacc 60
acatgggttc agatcccagc ggtgcctcta actggccaca ggaccttggg cagtcagct 119

SEQ ID NO: 497        moltype = RNA length = 62
FEATURE              Location/Qualifiers
source                1..62
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 497
cttcctggtg ggtggggagg agaagtgccg tctcatgag ccctctctg tcccacccat 60
ag                                         62

SEQ ID NO: 498        moltype = RNA length = 87
FEATURE              Location/Qualifiers
source                1..87
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 498
gcagggttgg cagggaggct gggaggggct ggctgggtct ggtagtgggc atcagctggc 60
cctcatttct taagacagca cttctgt                                         87

SEQ ID NO: 499        moltype = RNA length = 97
FEATURE              Location/Qualifiers
source                1..97
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 499
atctgagttg ggagggtccc tctccaaatg tgtcttgggg tgggggatca agacacattt 60
ggagagggaa cctcccaact cggcctctgc catcatt                                         97

SEQ ID NO: 500        moltype = RNA length = 73
FEATURE              Location/Qualifiers
source                1..73
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 500
ggaggctggg ctgggacgga caccggcct ccactttctg tggcaggtag ctctccatg 60
tcggcccgcc ttg                                         73

SEQ ID NO: 501        moltype = RNA length = 70
FEATURE              Location/Qualifiers
source                1..70
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 501
acaaatagct tcagggagtc aggggagggc agaaatagat ggccttcccc tgetgggaag 60
aaagtgggtc                                         70

SEQ ID NO: 502        moltype = RNA length = 87
FEATURE              Location/Qualifiers
source                1..87
                      mol_type = unassigned RNA
                      organism = Homo sapiens

SEQUENCE: 502
ccctcatctc tgggcagggg cttattgtag gagtctctga agagagctgt ggactgacct 60
gctttaacct ttcccagggt tcccatt                                         87

```

SEQ ID NO: 503	moltype = RNA	length = 72
FEATURE	Location/Qualifiers	
source	1..72	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 503		
ctgggggagg aaggacaggc	catctgctat	tcgctccacca acctgacttg atcctctctt
ccctcctccc ag		
SEQ ID NO: 504	moltype = RNA	length = 67
FEATURE	Location/Qualifiers	
source	1..67	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 504		
ccagaccctt ggggctgggc	aggcggaag	aggtctgaac tgcctctgcc tccttgggtct
ccggcag		
SEQ ID NO: 505	moltype = RNA	length = 65
FEATURE	Location/Qualifiers	
source	1..65	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 505		
gtctcctggg gggaggagac	cctgctctcc	ctggcagcaa gcctctcctg cccttccaga
ttagc		
SEQ ID NO: 506	moltype = RNA	length = 91
FEATURE	Location/Qualifiers	
source	1..91	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 506		
acgcccccg ccccgccacc	gccttggagg	ctgacctctt accttcggtc ggtctctctt
cctgggcttg gtttgggggc	gggggagatgt	c
SEQ ID NO: 507	moltype = RNA	length = 57
FEATURE	Location/Qualifiers	
source	1..57	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 507		
cattggaggg tgtggaagac	atctggggcca	actctgatct cttcatctac cccccag
SEQ ID NO: 508	moltype = RNA	length = 67
FEATURE	Location/Qualifiers	
source	1..67	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 508		
cagggaggag gtggtactag	gggcccagcaa	cctgattacc cctctttggc cctttgtacc
cctccag		
SEQ ID NO: 509	moltype = RNA	length = 96
FEATURE	Location/Qualifiers	
source	1..96	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 509		
gctctggggc gtgccgccgc	cgctcgctgc	acctccccta ccgctagtgg aagaagatgg
cggaaaggcg agcggcggat	ctggacaccc	agcggt
SEQ ID NO: 510	moltype = RNA	length = 61
FEATURE	Location/Qualifiers	
source	1..61	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 510		
cctcggggga caggccaggg	catctaggct	gtgcacatg acgccccctc tgccccccaca
g		
SEQ ID NO: 511	moltype = RNA	length = 58
FEATURE	Location/Qualifiers	
source	1..58	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 511		
aggqccaaq qacccctgaq	tggctgggct	gaactgaacc aggttccttc tqccqca

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SEQ ID NO: 512 moltype = RNA length = 86
 FEATURE Location/Qualifiers
 source 1..86
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 512
 attctaggtg gggagactga cggtctggagg cccataagct gtctaaaact tcggcccca 60
 gatttctggt ctccccactt cagaac 86

SEQ ID NO: 513 moltype = RNA length = 80
 FEATURE Location/Qualifiers
 source 1..80
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 513
 ggcgcgctgc cccctcagt ccaccagagc ccggatacct cagaaattcg gctctgggtc 60
 tgtggggagc gaaatgcaac 80

SEQ ID NO: 514 moltype = RNA length = 98
 FEATURE Location/Qualifiers
 source 1..98
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 514
 ttctcaccgc cgctgacac gggcgacagc tgcggccgc tgtgttact cgggcccagc 60
 gcgtctctcg tcaggcaagg gagagcagag cccccctg 98

SEQ ID NO: 515 moltype = RNA length = 94
 FEATURE Location/Qualifiers
 source 1..94
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 515
 ataaaggaag ttaggctgag gggcagagag cgagactttt ctattttcca aaagctcggc 60
 ctgaggcccc tcagtcttgc ttctaaccg gcgc 94

SEQ ID NO: 516 moltype = RNA length = 75
 FEATURE Location/Qualifiers
 source 1..75
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 516
 tcattccctg gtggggattt gttgcattac ttgtgttcta tataaagat tgcacttgct 60
 cggcctgtg gaaga 75

SEQ ID NO: 517 moltype = RNA length = 89
 FEATURE Location/Qualifiers
 source 1..89
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 517
 gtcagcagtg ccttagcagc acgtaaatat tggcgtaag attctaaat tatctccagt 60
 attaactgtg ctgctgaagt aaggttgac 89

SEQ ID NO: 518 moltype = RNA length = 81
 FEATURE Location/Qualifiers
 source 1..81
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 518
 gttccactct agcagcacgt aaatatggc gtagtgaaat atatattaaa caccaatatt 60
 actgtgctgc tttagtgtga c 81

SEQ ID NO: 519 moltype = RNA length = 72
 FEATURE Location/Qualifiers
 source 1..72
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 519
 cttgggaatg gcaaggaaac cgttaccatt actgagttta gtaatggtaa tggttctctt 60
 gctataccca ga 72

SEQ ID NO: 520 moltype = RNA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 520
 aggcctcgct gttctctatg gctttttatt cctatgtgat tctactgctc actcatatag 60

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ggattggagc cgtggcgcac ggcgggggaca 90

SEQ ID NO: 521 moltype = RNA length = 86
 FEATURE Location/Qualifiers
 source 1..86
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 521
 ggcttagaaa cagtccctag gtaggatttg gggaggagct aagaagcccc tacagggccc 60
 agaggtgggg actgagcctt agttgg 86

SEQ ID NO: 522 moltype = RNA length = 96
 FEATURE Location/Qualifiers
 source 1..96
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 522
 cgggccccgg gcgggcggga gggacgggac gcggtgcagt gttgtttttt cccccgcaa 60
 tattgcactc gtcccgccct ccggcccccc cggccc 96

SEQ ID NO: 523 moltype = RNA length = 70
 FEATURE Location/Qualifiers
 source 1..70
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 523
 ggcccgcgcg cgcaagatgg cggcgggccc gggcaccgcc ccttcgcccg cgccgggct 60
 cgcacgaggc 70

SEQ ID NO: 524 moltype = RNA length = 94
 FEATURE Location/Qualifiers
 source 1..94
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 524
 gtgaggtgtg ggcccggccc caggagcggg gcctgggcag ccccggtgtg tgaggaagga 60
 aggcagggcc ccgctcccc gggcctgacc ccac 94

SEQ ID NO: 525 moltype = RNA length = 97
 FEATURE Location/Qualifiers
 source 1..97
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 525
 ctacaggtgct ctggctgctt gggttcctgg catgctgatt tgtgacttaa gattaaaatc 60
 acattgccag ggattaccac gcaaccacga ccttggc 97

SEQ ID NO: 526 moltype = RNA length = 78
 FEATURE Location/Qualifiers
 source 1..78
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 526
 ctttctacac aggttgggat cgggtgcaat gctgtgttct tgatggtat tgcacttgct 60
 ccggcctgtt gagtttgg 78

SEQ ID NO: 527 moltype = RNA length = 75
 FEATURE Location/Qualifiers
 source 1..75
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 527
 tcatccctgg gtggggattt gttgcattac ttgtgttcta tataaagtat tgcacttgct 60
 ccggcctgtg gaaga 75

SEQ ID NO: 528 moltype = RNA length = 100
 FEATURE Location/Qualifiers
 source 1..100
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 528
 gctcggttgc cgtggttgcg ggcctgccc gcccgccagc tcgctgacag cagcactcag 60
 ggcggaggga agtaggtccg ttggtcggtc gggaacgagg 100

SEQ ID NO: 529 moltype = RNA length = 65
 FEATURE Location/Qualifiers
 source 1..65
 mol_type = unassigned RNA
 organism = Homo sapiens

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SEQUENCE: 529
agccctgggg gtggtctcta gccaaaggctc tggggtctca cccttggtg gtctctgctc 60
cgacag 65

SEQ ID NO: 530 moltype = RNA length = 85
FEATURE Location/Qualifiers
source 1..85
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 530
ggggaggtag ggaaaaggaa gggggaggag aaggtagagac caatgtcctg ggtgccactc 60
ctgcccagtg cctcccttcc tcgtt 85

SEQ ID NO: 531 moltype = RNA length = 87
FEATURE Location/Qualifiers
source 1..87
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 531
gtgaggcggg gccaggagg tgtgtggcgt ggggtgctgcg gggccgtcag ggtgcctgcg 60
ggacgctcac ctggctggcc cgcccag 87

SEQ ID NO: 532 moltype = RNA length = 65
FEATURE Location/Qualifiers
source 1..65
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 532
gcttctggga ggaggggatc ttgggagtga tcccaacagc tgagctccct gaatccctgt 60
cccag 65

SEQ ID NO: 533 moltype = RNA length = 60
FEATURE Location/Qualifiers
source 1..60
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 533
tgtgaatgac ccccttccag agccaaaatc accagggatg gaggaggggt ctgggtact 60

SEQ ID NO: 534 moltype = RNA length = 68
FEATURE Location/Qualifiers
source 1..68
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 534
tgcccaggct ggagcgagt cagtgggtgca gtcagtccta gctcactgca gcctcgaact 60
cctgggct 68

SEQ ID NO: 535 moltype = RNA length = 80
FEATURE Location/Qualifiers
source 1..80
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 535
acctgaggag ccagccctcc tcccgcaccc aaacttggag cacttgacct ttggctgttg 60
gagggggcag gctcgcggggt 80

SEQ ID NO: 536 moltype = RNA length = 83
FEATURE Location/Qualifiers
source 1..83
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 536
ggcccggtc cgggtctcgg cccgtacagt ccggccggcc atgctggcgg ggctggggcc 60
ggggccgagc ccgcgcggg gcc 83

SEQ ID NO: 537 moltype = RNA length = 83
FEATURE Location/Qualifiers
source 1..83
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 537
acggcatctt tgcactcagc aggcaggctg gtgcagcccg tgggtggggga ccatacctgcc 60
tgctgtgggg taaggacggc tgt 83

SEQ ID NO: 538 moltype = RNA length = 63
FEATURE Location/Qualifiers
source 1..63
 mol_type = unassigned RNA

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                organism = Homo sapiens
SEQUENCE: 538
tggtggcggc ggtagttatg ggcttctctt tctcaccagc agccctctggg ccgccgcctc 60
cct                                                    63

SEQ ID NO: 539      moltype = RNA  length = 98
FEATURE            Location/Qualifiers
source              1..98
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 539
agaatgggca aatgaacagt aaatttgag gctggggcc ctccctgtctg ctggagaagt 60
gtttgcacgg gtgggccttg tctttgaaag gagtgga                98

SEQ ID NO: 540      moltype = RNA  length = 102
FEATURE            Location/Qualifiers
source              1..102
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 540
gtgggagggc ccaggcggcg gcagggggtgg ggggggcaga gcgctgtccc gggggcgggg 60
ccgaagcgcg gcgaccgtaa ctcttctgc tccgtcccc ag                102

SEQ ID NO: 541      moltype = RNA  length = 80
FEATURE            Location/Qualifiers
source              1..80
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 541
cggaatgcc gcggcgggga cggcgattgg tccgtatgtg tggcgccacc ggccgcggcg 60
tccgccccgg ccccgcccc                80

SEQ ID NO: 542      moltype = RNA  length = 93
FEATURE            Location/Qualifiers
source              1..93
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 542
tacttatggc acccactcc tggaccata gtcataagt aggagatgtt agagctgtga 60
gtaccatgac ttaagtgtgg tggcttaaac atg                93

SEQ ID NO: 543      moltype = RNA  length = 66
FEATURE            Location/Qualifiers
source              1..66
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 543
cctggagggg ggccactgcgc aagcaaagcc agggaccctg agaggctttg ctccctgctc 60
ccctag                                                    66

SEQ ID NO: 544      moltype = RNA  length = 65
FEATURE            Location/Qualifiers
source              1..65
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 544
ttctcctggg gagtggctgg ggagcagaca gacccaacct catgctcccc ggctctgccc 60
cccag                                                    65

SEQ ID NO: 545      moltype = RNA  length = 22
FEATURE            Location/Qualifiers
source              1..22
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 545
ccccggtggt ggggcgcgtc tg                22

SEQ ID NO: 546      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = unassigned RNA
                    organism = Homo sapiens
SEQUENCE: 546
cccggtgttg gggcgcgtct g                21

SEQ ID NO: 547      moltype = RNA  length = 22
FEATURE            Location/Qualifiers
source              1..22
                    mol_type = unassigned RNA

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SEQUENCE: 547	organism = Homo sapiens	
tggcggagcc cattccatgc ca		22
SEQ ID NO: 548	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 548		
ctggcggagc ccattccatg c		21
SEQ ID NO: 549	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 549		
actcggcgtg gcgtcggtcg tggta		25
SEQ ID NO: 550	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 550		
actcggcgtg gcgtc		15
SEQ ID NO: 551	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 551		
cggcgcgacc ggcccgggg		19
SEQ ID NO: 552	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 552		
cggcgcgacc ggcccgggg		19
SEQ ID NO: 553	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 553		
cgggtgggatc ccgcggccgt gttttc		26
SEQ ID NO: 554	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 554		
ggggcgcgcg gggac		15
SEQ ID NO: 555	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 555		
ccccggggag cccggcggtg		20
SEQ ID NO: 556	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 556		
accccgggga gcccg		15
SEQ ID NO: 557	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	

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	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 557 gatcggtcga gacggtcctg gctg		24
SEQ ID NO: 558 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 558 gctgggcggg gcgcg		15
SEQ ID NO: 559 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 559 tgtagagcag ggagcaggaa gct		23
SEQ ID NO: 560 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 560 caggagcag gaagc		15
SEQ ID NO: 561 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 561 tgaagcggg gggcg		15
SEQ ID NO: 562 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 562 tgaagcggg gggcg		15
SEQ ID NO: 563 FEATURE source	moltype = RNA length = 22 Location/Qualifiers 1..22 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 563 ctgggctcgg gacgcgcgc tc		22
SEQ ID NO: 564 FEATURE source	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 564 ctgggctcgg gacgcgcgc		19
SEQ ID NO: 565 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 565 cacaggtgag gttcttgga gcc		23
SEQ ID NO: 566 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 566 acagtgagg ttctt		15
SEQ ID NO: 567 FEATURE	moltype = RNA length = 19 Location/Qualifiers	

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source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 567		
agtgggaggc cagggcacg		19
SEQ ID NO: 568	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 568		
agggggagct gcagg		15
SEQ ID NO: 569	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 569		
aggcagcggg gtgtagtgga ta		22
SEQ ID NO: 570	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 570		
aggcagcggg gtgtagtgga t		21
SEQ ID NO: 571	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 571		
ggggggcagg aggggctcag gg		22
SEQ ID NO: 572	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 572		
gtgggggggc aggagg		16
SEQ ID NO: 573	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 573		
gaatggattt ttggagcagg a		21
SEQ ID NO: 574	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 574		
gaatggattt ttgga		15
SEQ ID NO: 575	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 575		
gctgcgggct gcggtcaggg cgat		24
SEQ ID NO: 576	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 576		
gctgcgggct gcggtcaggg		20
SEQ ID NO: 577	moltype = RNA length = 18	

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FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 577		
gcggcggcgg cggcagca		18
SEQ ID NO: 578	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 578		
gcggcggcgg gcggc		15
SEQ ID NO: 579	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 579		
tgggagggga gaggcagcaa gc		22
SEQ ID NO: 580	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 580		
tgggagggga gaggcagcaa gc		22
SEQ ID NO: 581	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 581		
cgggtagaga gggcagtggg aggtaa		26
SEQ ID NO: 582	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 582		
cgggtagaga gggca		15
SEQ ID NO: 583	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 583		
gagggagggga cgggggctgt gct		23
SEQ ID NO: 584	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 584		
gaggagggag ggagg		15
SEQ ID NO: 585	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 585		
caggaaggat ttagggacag gcttt		25
SEQ ID NO: 586	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 586		
caggaaggat ttagggaca		19

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SEQ ID NO: 587	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 587		
ctccgggagg cgccgtgt		18
SEQ ID NO: 588	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 588		
ctccgggagg cgccgtgt		18
SEQ ID NO: 589	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 589		
aggcaggggc tgggtgctggg cggg		24
SEQ ID NO: 590	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 590		
gggcgggggg cggcg		15
SEQ ID NO: 591	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 591		
cggccgccc gctccggc c		21
SEQ ID NO: 592	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 592		
cggccgccc gctccgc		17
SEQ ID NO: 593	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 593		
ctggtacagg cctgggggac aggg		24
SEQ ID NO: 594	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 594		
ctggtacagg cctggggg		18
SEQ ID NO: 595	moltype = RNA length = 27	
FEATURE	Location/Qualifiers	
source	1..27	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 595		
gtgaacgggc gccatccga ggctttg		27
SEQ ID NO: 596	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 596		
gtgaacgggc gccatc		16

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SEQ ID NO: 597	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 597		
gtgggttggg gcgggctct		19
SEQ ID NO: 598	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 598		
gtgggttggg gcgggctct		19
SEQ ID NO: 599	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 599		
gcgggctgtc cggaggggtc ggcttt		26
SEQ ID NO: 600	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 600		
gtgtccgga ggggtc		16
SEQ ID NO: 601	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 601		
ccggggcag attggtgtag ggtg		24
SEQ ID NO: 602	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 602		
cggggcagat tgggtga		17
SEQ ID NO: 603	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 603		
gggggtcccc ggtgctcgga tct		23
SEQ ID NO: 604	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 604		
tcgggagggg cgggag		16
SEQ ID NO: 605	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 605		
cggggccgta gcactgtctg aga		23
SEQ ID NO: 606	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 606		

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cggggccgta gcaactgtctg	20
SEQ ID NO: 607	moltype = RNA length = 27
FEATURE	Location/Qualifiers
source	1..27
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 607	
tgagggactt ttgggggcag atgtgtt	27
SEQ ID NO: 608	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 608	
ggacttttgg gggcaga	17
SEQ ID NO: 609	moltype = RNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 609	
gaggctggga aggcaaaggg acgt	24
SEQ ID NO: 610	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 610	
gaaggaggct gggaa	15
SEQ ID NO: 611	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 611	
aggggctggg cgcgcg	17
SEQ ID NO: 612	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 612	
caggggctgg gcgcg	15
SEQ ID NO: 613	moltype = RNA length = 25
FEATURE	Location/Qualifiers
source	1..25
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 613	
tgcaggcaga agtggggctg acagg	25
SEQ ID NO: 614	moltype = RNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 614	
ctgcaggcag aagtggggct	20
SEQ ID NO: 615	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 615	
ggggggatgt gcatgctggt tgg	23
SEQ ID NO: 616	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 616 atcagcgtgc acttc	15
SEQ ID NO: 617 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 617 caccttgctt tgctgcccgg gcc	23
SEQ ID NO: 618 FEATURE source	moltype = RNA length = 22 Location/Qualifiers 1..22 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 618 caccttgctt tgctgcccgg gc	22
SEQ ID NO: 619 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 619 tggggcggag cttccggagg ccc	23
SEQ ID NO: 620 FEATURE source	moltype = RNA length = 17 Location/Qualifiers 1..17 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 620 tggggcggag cttccgg	17
SEQ ID NO: 621 FEATURE source	moltype = RNA length = 16 Location/Qualifiers 1..16 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 621 ggaccaggag agagac	16
SEQ ID NO: 622 FEATURE source	moltype = RNA length = 16 Location/Qualifiers 1..16 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 622 ggaccaggag agagac	16
SEQ ID NO: 623 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 623 ggcccgccg tgctgaggt ttc	23
SEQ ID NO: 624 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 624 ggcgtggga tcccg	15
SEQ ID NO: 625 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 625 cagccgccc cagccgaggt tct	23
SEQ ID NO: 626 FEATURE source	moltype = RNA length = 17 Location/Qualifiers 1..17 mol_type = unassigned RNA

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SEQUENCE: 626	organism = Homo sapiens	
agcccgcccc agccgag		17
SEQ ID NO: 627	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 627		
actcgggtgc ggtggacaag tc		22
SEQ ID NO: 628	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 628		
actcgggtgc ggtggacaag		20
SEQ ID NO: 629	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 629		
ggtgggtgag gtcgggcccc aag		23
SEQ ID NO: 630	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 630		
cggggtgggt gagtcgggc		20
SEQ ID NO: 631	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 631		
gaggggtctt cgcttctggc gccaaag		26
SEQ ID NO: 632	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 632		
ggtgaggcgg ggggg		15
SEQ ID NO: 633	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 633		
ctgggggacg cgtgagcgcg agc		23
SEQ ID NO: 634	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 634		
ctgggggacg cgtgagcgcg a		21
SEQ ID NO: 635	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 635		
gtgagtggga gccggtgggg ctgg		24
SEQ ID NO: 636	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	

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SEQUENCE: 636	mol_type = unassigned RNA	
ggggctggag taagg	organism = Homo sapiens	15
SEQ ID NO: 637	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 637		
gtgggctggg ctgggctggg cca		23
SEQ ID NO: 638	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 638		
gggctgggct gggct		15
SEQ ID NO: 639	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 639		
tgcgcagggg ccgggtgctc acc		23
SEQ ID NO: 640	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 640		
cgcagggggcc ggggtgctca		19
SEQ ID NO: 641	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 641		
ttctggggccc gcggcgggcg tgggg		25
SEQ ID NO: 642	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 642		
cgcggcgggc gtggg		15
SEQ ID NO: 643	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 643		
ctcctggggc ccgcactctc gct		23
SEQ ID NO: 644	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 644		
ctcctggggc ccgcactc		18
SEQ ID NO: 645	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 645		
gaaaaaggcg ggagaagccc ca		22
SEQ ID NO: 646	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	

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source	1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 646 gaaaaaggcg ggaga		15
SEQ ID NO: 647 FEATURE source	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 647 gcaggcgagg ctgggctga		19
SEQ ID NO: 648 FEATURE source	moltype = RNA length = 16 Location/Qualifiers 1..16 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 648 aggcgaggct gggctg		16
SEQ ID NO: 649 FEATURE source	moltype = RNA length = 22 Location/Qualifiers 1..22 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 649 aaaagctggg ctgagaggcg ac		22
SEQ ID NO: 650 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 650 aaagctgggc tgaga		15
SEQ ID NO: 651 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 651 tggggaaggc gtcagtgctg ggt		23
SEQ ID NO: 652 FEATURE source	moltype = RNA length = 16 Location/Qualifiers 1..16 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 652 tggggaaggc gtcagt		16
SEQ ID NO: 653 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 653 ggaggcgag gctcgaaaag gcg		23
SEQ ID NO: 654 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 654 gcaggctcgg aaagg		15
SEQ ID NO: 655 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 655 agaagaaggc ggtcggtctg cgg		23
SEQ ID NO: 656	moltype = RNA length = 21	

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 656		
aagaaggcgg tcggtctgcg g		21
SEQ ID NO: 657	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 657		
agcggggagg aagtggggcg tcgtt		25
SEQ ID NO: 658	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 658		
agcggggagg aagtggggcg t		21
SEQ ID NO: 659	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 659		
cctccgggac ggctggg		17
SEQ ID NO: 660	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 660		
ctccgggacg gctgg		15
SEQ ID NO: 661	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 661		
tcgaggactg gtggaaggcg cttt		24
SEQ ID NO: 662	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 662		
tcgaggactg gtggaa		16
SEQ ID NO: 663	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 663		
ggctacaaca caggaccgg gcg		23
SEQ ID NO: 664	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 664		
ggctacaaca caggaccgg g		21
SEQ ID NO: 665	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 665		
tggggcggag cttccggagg ccc		23

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SEQ ID NO: 666	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 666		
gccccgggaa agcgt		15
SEQ ID NO: 667	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 667		
gagggcgatgt ggggatgtag a		21
SEQ ID NO: 668	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 668		
cccagtctca tttcctcatc		20
SEQ ID NO: 669	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 669		
gagggcagcg tgggtgtggc g		21
SEQ ID NO: 670	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 670		
gagggcagcg tgggtgtggc g		21
SEQ ID NO: 671	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 671		
accaggaggc tgaggccct ca		22
SEQ ID NO: 672	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 672		
accaggaggc tgagg		15
SEQ ID NO: 673	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 673		
ttgaggagac atggtggggg c		21
SEQ ID NO: 674	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 674		
ttgaggagac atggt		15
SEQ ID NO: 675	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 675		
ggcagggaca gcaaagggt gc		22

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SEQ ID NO: 676	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 676		
gcaggacag caaagggg		18
SEQ ID NO: 677	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 677		
ctgcaggcag aagtggggct gacag		25
SEQ ID NO: 678	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 678		
caggcagaag tggggctga		19
SEQ ID NO: 679	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 679		
cgctcgggcg gaggtggttg agtg		24
SEQ ID NO: 680	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 680		
tcgggcggag gtggttg		17
SEQ ID NO: 681	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 681		
gtgaaggccc ggcgga		16
SEQ ID NO: 682	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 682		
gtgaaggccc ggcgg		15
SEQ ID NO: 683	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 683		
tgcaggggca ggcagc		17
SEQ ID NO: 684	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 684		
tgcaggggca ggcagc		17
SEQ ID NO: 685	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 685		

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tcgctgggcta gggctaacag cagtc	25
SEQ ID NO: 686	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 686	
tcgctgggcta gggct	15
SEQ ID NO: 687	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 687	
ggtgagcgct cgctggc	17
SEQ ID NO: 688	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 688	
cggtgagcgc tcgct	15
SEQ ID NO: 689	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 689	
ggcgcgggga ggtgc	15
SEQ ID NO: 690	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 690	
ggcgcgggga ggtgc	15
SEQ ID NO: 691	moltype = RNA length = 18
FEATURE	Location/Qualifiers
source	1..18
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 691	
gaggctgaag gaagatgg	18
SEQ ID NO: 692	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 692	
gaggctgaag gaaga	15
SEQ ID NO: 693	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 693	
ccagggtgg cagtgcacatg ggt	23
SEQ ID NO: 694	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 694	
cagggtggc agtgacatg	19
SEQ ID NO: 695	moltype = RNA length = 16
FEATURE	Location/Qualifiers
source	1..16
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 695 tgctggtgat gctttc	16
SEQ ID NO: 696 FEATURE source	moltype = RNA length = 16 Location/Qualifiers 1..16 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 696 tgctggtgat gctttc	16
SEQ ID NO: 697 FEATURE source	moltype = RNA length = 22 Location/Qualifiers 1..22 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 697 gaggggtggg tggaggctct cc	22
SEQ ID NO: 698 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 698 gaggggtggg tggag	15
SEQ ID NO: 699 FEATURE source	moltype = RNA length = 20 Location/Qualifiers 1..20 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 699 tgagtggggc tcccgggacg	20
SEQ ID NO: 700 FEATURE source	moltype = RNA length = 20 Location/Qualifiers 1..20 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 700 tgagtggggc tcccgggacg	20
SEQ ID NO: 701 FEATURE source	moltype = RNA length = 21 Location/Qualifiers 1..21 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 701 cccagcagga cgggagcgcg g	21
SEQ ID NO: 702 FEATURE source	moltype = RNA length = 16 Location/Qualifiers 1..16 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 702 aagctgggtc aaggag	16
SEQ ID NO: 703 FEATURE source	moltype = RNA length = 20 Location/Qualifiers 1..20 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 703 ggctgggtcag atgggagtgg	20
SEQ ID NO: 704 FEATURE source	moltype = RNA length = 20 Location/Qualifiers 1..20 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 704 ggctgggtcag atgggagtgg	20
SEQ ID NO: 705 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA

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SEQUENCE: 705	organism = Homo sapiens	
acaggagtgg ggggtgggaca taa		23
SEQ ID NO: 706	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 706		
acaggagtgg ggggtgggaca		20
SEQ ID NO: 707	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 707		
gtgggcgggg gcaggtgtgt gg		22
SEQ ID NO: 708	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 708		
cgggggcagg tgtgt		15
SEQ ID NO: 709	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 709		
gccgggcgtg gtggtggggg c		21
SEQ ID NO: 710	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 710		
tagccgggcg tggtg		15
SEQ ID NO: 711	moltype = RNA length = 29	
FEATURE	Location/Qualifiers	
source	1..29	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 711		
ggtcaggcgg ctcggactga gcaggtggg		29
SEQ ID NO: 712	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 712		
agagtgtggt caggc		15
SEQ ID NO: 713	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 713		
cagaagggga gttgggagca ga		22
SEQ ID NO: 714	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 714		
gaaggggagt tgggag		16
SEQ ID NO: 715	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	

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SEQUENCE: 715	mol_type = unassigned RNA	
ggcgcggagg gcgac	organism = Homo sapiens	16
SEQ ID NO: 716	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 716		15
ggcgcggagg gcgga		
SEQ ID NO: 717	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 717		21
cagcagggga gagagaggag t		
SEQ ID NO: 718	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 718		20
cagcagggga gagagaggag		
SEQ ID NO: 719	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 719		26
ccttctggag aggccttggtg cggata		
SEQ ID NO: 720	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 720		15
ccttctggag aggct		
SEQ ID NO: 721	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 721		23
gaaatcaagc gtgggtgaga cct		
SEQ ID NO: 722	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 722		20
gaaatcaagc gtgggtgaga		
SEQ ID NO: 723	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 723		26
agggctggac tcagcggcgg agctgg		
SEQ ID NO: 724	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 724		17
gcggcggagc tggctgc		
SEQ ID NO: 725	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	

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source	1..24 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 725		
cgggcgtggt ggtgggggtg ggtg		24
SEQ ID NO: 726	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 726		
cgggcgtggt ggtgg		15
SEQ ID NO: 727	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 727		
actcaaactg tgggggcact tt		22
SEQ ID NO: 728	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 728		
actcaaactg tgggggcac		19
SEQ ID NO: 729	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 729		
tgggggggaa gaaaag		16
SEQ ID NO: 730	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 730		
tgggggggaa gaaaag		16
SEQ ID NO: 731	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 731		
atccagttct ctgagggggc t		21
SEQ ID NO: 732	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 732		
atccagttct ctgagggggc t		21
SEQ ID NO: 733	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 733		
tggggagctg aggctctggg ggtg		24
SEQ ID NO: 734	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 734		
ggccctgggg agctg		15
SEQ ID NO: 735	moltype = RNA length = 22	

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FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 735		
cggggaacgt cgagactgga gc		22
SEQ ID NO: 736	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 736		
cgggaacgtc gagac		15
SEQ ID NO: 737	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 737		
tggcgggtgc ggggggtggg		19
SEQ ID NO: 738	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 738		
tggcgggtgc ggggg		15
SEQ ID NO: 739	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 739		
taggggcagc agaggacctg ggc		23
SEQ ID NO: 740	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 740		
taggggcagc agaggacctg		20
SEQ ID NO: 741	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 741		
ggggcggggg cgggggc		17
SEQ ID NO: 742	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 742		
cgcgccgggc ccggg		15
SEQ ID NO: 743	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 743		
tggggacgta gctggccaga cag		23
SEQ ID NO: 744	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 744		
tggggacgta gctggccaga		20

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SEQ ID NO: 745	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 745		
cagcctgagt gacagagcaa g		21
SEQ ID NO: 746	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 746		
actgcactcc agcct		15
SEQ ID NO: 747	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 747		
gagactgggg tggggcct		18
SEQ ID NO: 748	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 748		
agactggggt ggggcc		16
SEQ ID NO: 749	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 749		
ggcgtgggc ggcggg		16
SEQ ID NO: 750	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 750		
ggcctctcgg gaact		15
SEQ ID NO: 751	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 751		
gcggggcggc aggggcc		17
SEQ ID NO: 752	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 752		
gggggcgggg cggca		15
SEQ ID NO: 753	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 753		
tgaggatatg gcagggaagg gga		23
SEQ ID NO: 754	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 754		
tgaggatatg gcagggaag		19

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SEQ ID NO: 755	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 755		
ctccccggtg tgcaaatgtg		20
SEQ ID NO: 756	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 756		
gtgtgcggtg ttatg		15
SEQ ID NO: 757	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 757		
cggggcagct cagtacagga tac		23
SEQ ID NO: 758	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 758		
agctcagtac aggat		15
SEQ ID NO: 759	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 759		
cccaaatgc tgggattaca ggca		24
SEQ ID NO: 760	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 760		
gcccacctca gcctc		15
SEQ ID NO: 761	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 761		
gtgagtcagg gtggggctgg c		21
SEQ ID NO: 762	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 762		
gtgagtcagg gtggggctgg c		21
SEQ ID NO: 763	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 763		
aggaggagga ggcag		15
SEQ ID NO: 764	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 764		

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aggaggagga ggcag	15
SEQ ID NO: 765	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 765	
cggatccgag tcacggcacc a	21
SEQ ID NO: 766	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 766	
ggatccgagt cacgg	15
SEQ ID NO: 767	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 767	
ccaggaggcg gaggaggtgg agg	23
SEQ ID NO: 768	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 768	
acccaggagg cggag	15
SEQ ID NO: 769	moltype = RNA length = 26
FEATURE	Location/Qualifiers
source	1..26
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 769	
gtcccggggc tgcgcgaggc acaggc	26
SEQ ID NO: 770	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 770	
ggcccggggg gcggg	15
SEQ ID NO: 771	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 771	
aggaggcagt gggcgagcag g	21
SEQ ID NO: 772	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 772	
aggaggcagt gggcgagcag g	21
SEQ ID NO: 773	moltype = RNA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 773	
acagcagggc tggggattgc agt	23
SEQ ID NO: 774	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 774 tgctgctccc agtcctgcc	19
SEQ ID NO: 775 FEATURE source	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 775 gctgggcgag gctggcatc	19
SEQ ID NO: 776 FEATURE source	moltype = RNA length = 17 Location/Qualifiers 1..17 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 776 gctgggcgag gctggca	17
SEQ ID NO: 777 FEATURE source	moltype = RNA length = 18 Location/Qualifiers 1..18 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 777 cagcggggct gggcgcg	18
SEQ ID NO: 778 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 778 cagcggggct gggcg	15
SEQ ID NO: 779 FEATURE source	moltype = RNA length = 22 Location/Qualifiers 1..22 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 779 caggcacggg agtcaggtag ag	22
SEQ ID NO: 780 FEATURE source	moltype = RNA length = 17 Location/Qualifiers 1..17 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 780 caggcacggg agtcag	17
SEQ ID NO: 781 FEATURE source	moltype = RNA length = 18 Location/Qualifiers 1..18 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 781 ggcagcggcg gcggcggc	18
SEQ ID NO: 782 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 782 gtccccgcg ccccc	15
SEQ ID NO: 783 FEATURE source	moltype = RNA length = 20 Location/Qualifiers 1..20 mol_type = unassigned RNA organism = Homo sapiens
SEQUENCE: 783 atcccaccac tgccaccatt	20
SEQ ID NO: 784 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA

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SEQUENCE: 784	organism = Homo sapiens	
atcccaccac tgcca		15
SEQ ID NO: 785	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 785		
gggggagcca tgagataaga gcacc		25
SEQ ID NO: 786	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 786		
tgggggagcc atgagataag		20
SEQ ID NO: 787	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 787		
gatcccagcg gtgcctc		17
SEQ ID NO: 788	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 788		
gatcccagcg gtgcc		15
SEQ ID NO: 789	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 789		
agacacattt ggagagggaa cctc		24
SEQ ID NO: 790	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 790		
agacacattt ggagag		16
SEQ ID NO: 791	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 791		
gctgggctgg gacggacacc cggcctccac		30
SEQ ID NO: 792	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 792		
gaggctgggc tgggacgga		19
SEQ ID NO: 793	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 793		
tgggcagggg cttattgtag gagtc		25
SEQ ID NO: 794	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	

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SEQUENCE: 794	mol_type = unassigned RNA organism = Homo sapiens	
tgggcagggg cttattgta		19
SEQ ID NO: 795	moltype = RNA length = 21 Location/Qualifiers	
FEATURE	1..21	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 795		
caactctgat ctcttcacatc a		21
SEQ ID NO: 796	moltype = RNA length = 20 Location/Qualifiers	
FEATURE	1..20	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 796		
tctcttcacatc taccocccag		20
SEQ ID NO: 797	moltype = RNA length = 23 Location/Qualifiers	
FEATURE	1..23	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 797		
ctagtgggaag aagatggcgg aag		23
SEQ ID NO: 798	moltype = RNA length = 15 Location/Qualifiers	
FEATURE	1..15	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 798		
tagtgggaaga agatg		15
SEQ ID NO: 799	moltype = RNA length = 24 Location/Qualifiers	
FEATURE	1..24	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 799		
ctgcgggggac aggccagggc atct		24
SEQ ID NO: 800	moltype = RNA length = 20 Location/Qualifiers	
FEATURE	1..20	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 800		
ctgcgggggac aggccagggc		20
SEQ ID NO: 801	moltype = RNA length = 27 Location/Qualifiers	
FEATURE	1..27	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 801		
agggccagag gagcctggag tggtcgg		27
SEQ ID NO: 802	moltype = RNA length = 23 Location/Qualifiers	
FEATURE	1..23	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 802		
agggccagag gagcctggag tgg		23
SEQ ID NO: 803	moltype = RNA length = 18 Location/Qualifiers	
FEATURE	1..18	
source	mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 803		
tctaggtggg gagactga		18
SEQ ID NO: 804	moltype = RNA length = 16 Location/Qualifiers	
FEATURE		

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source	1..16 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 804 gtggggagac tgacgg		16
SEQ ID NO: 805 FEATURE source	moltype = RNA length = 23 Location/Qualifiers 1..23 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 805 tcggctctgg gtctgtgggg agc		23
SEQ ID NO: 806 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 806 gccccgatac ctacg		15
SEQ ID NO: 807 FEATURE source	moltype = RNA length = 30 Location/Qualifiers 1..30 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 807 tgaggggcag agagcgagac ttttctattt		30
SEQ ID NO: 808 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 808 tgaggggcag agagc		15
SEQ ID NO: 809 FEATURE source	moltype = RNA length = 25 Location/Qualifiers 1..25 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 809 gggtggggat ttgttgcatt acttg		25
SEQ ID NO: 810 FEATURE source	moltype = RNA length = 20 Location/Qualifiers 1..20 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 810 gggtggggat ttgttgcatt		20
SEQ ID NO: 811 FEATURE source	moltype = RNA length = 27 Location/Qualifiers 1..27 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 811 tagcagcacg taaatattgg cgttaag		27
SEQ ID NO: 812 FEATURE source	moltype = RNA length = 15 Location/Qualifiers 1..15 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 812 tagcagcacg taaat		15
SEQ ID NO: 813 FEATURE source	moltype = RNA length = 27 Location/Qualifiers 1..27 mol_type = unassigned RNA organism = Homo sapiens	
SEQUENCE: 813 aaaccgttac cattactgag tttagta		27
SEQ ID NO: 814	moltype = RNA length = 15	

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FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 814		
gaaaccgtta ccatt		15
SEQ ID NO: 815	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 815		
atatagggat tggagccgtg gc		22
SEQ ID NO: 816	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 816		
atatagggat tggagccgtg		20
SEQ ID NO: 817	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 817		
tcctgtactg agctgccccg aggcc		25
SEQ ID NO: 818	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 818		
tcctgtactg agctg		15
SEQ ID NO: 819	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 819		
agggacggga cgcggtgcag tgttgt		26
SEQ ID NO: 820	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 820		
ggcgggcggg agggga		15
SEQ ID NO: 821	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 821		
ccccagggcg acgcggcggg		20
SEQ ID NO: 822	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 822		
cgcggcgggg gcggc		15
SEQ ID NO: 823	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 823		
ggcggcgggc ccggg		15

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SEQ ID NO: 824	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 824		
ggcggcgggc ccggg		15
SEQ ID NO: 825	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 825		
aaggcagggc ccccgctccc cgggc		25
SEQ ID NO: 826	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 826		
gtgtgttgag gaagg		15
SEQ ID NO: 827	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 827		
gagggccccc cctcaatcct gtt		23
SEQ ID NO: 828	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 828		
aggggccccc ctcaat		16
SEQ ID NO: 829	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 829		
aaaatcacat tgccagggat taccac		26
SEQ ID NO: 830	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 830		
aatcacattg ccagg		15
SEQ ID NO: 831	moltype = RNA length = 28	
FEATURE	Location/Qualifiers	
source	1..28	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 831		
gtatggatt gcacttgccc cggcctgt		28
SEQ ID NO: 832	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 832		
tattgcactt gtccc		15
SEQ ID NO: 833	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 833		
gcacgactca gggcggagg aa		22

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SEQ ID NO: 834	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 834		
agggcggagg gaagt		15
SEQ ID NO: 835	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 835		
aaaaggaagg gggaggag		18
SEQ ID NO: 836	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 836		
aaggaagggg gagggag		16
SEQ ID NO: 837	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 837		
aatattgcac tcgtcccggc ctcc		24
SEQ ID NO: 838	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 838		
tattgcactc gtccc		15
SEQ ID NO: 839	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 839		
cccaggctgg agcgagtga g		21
SEQ ID NO: 840	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 840		
agctcactgc agcct		15
SEQ ID NO: 841	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 841		
tggctgttgg agggggcagg		20
SEQ ID NO: 842	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 842		
ggagggggca ggctc		15
SEQ ID NO: 843	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 843		

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tggcggcggt agttatgggc ttctc	25
SEQ ID NO: 844	moltype = RNA length = 25
FEATURE	Location/Qualifiers
source	1..25
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 844	
tggcggcggt agttatgggc ttctc	25
SEQ ID NO: 845	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 845	
tggcagagcg ctgtc	15
SEQ ID NO: 846	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 846	
tggcagagcg ctgtc	15
SEQ ID NO: 847	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 847	
cgcggcgggg acggcgattg gt	22
SEQ ID NO: 848	moltype = RNA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 848	
cggcggggac ggcgatt	17
SEQ ID NO: 849	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 849	
acccactcc tggtagcata gt	22
SEQ ID NO: 850	moltype = RNA length = 15
FEATURE	Location/Qualifiers
source	1..15
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 850	
acccactcc tggta	15
SEQ ID NO: 851	moltype = RNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 851	
ggaggcggg gtggggcggg gcg	24
SEQ ID NO: 852	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = unassigned RNA
	organism = Homo sapiens
SEQUENCE: 852	
tggggcggg caggtccctg c	21
SEQ ID NO: 853	moltype = RNA length = 18
FEATURE	Location/Qualifiers
source	1..18
	mol_type = unassigned RNA
	organism = Homo sapiens

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SEQUENCE: 853
 gggtgcgggc cggcgggg 18

SEQ ID NO: 854 moltype = RNA length = 18
 FEATURE Location/Qualifiers
 source 1..18
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 854
 agggggcggg ctccggcg 18

SEQ ID NO: 855 moltype = RNA length = 22
 FEATURE Location/Qualifiers
 source 1..22
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 855
 ggggggtgtg gagccagggg gc 22

SEQ ID NO: 856 moltype = RNA length = 25
 FEATURE Location/Qualifiers
 source 1..25
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 856
 aaggaggag gagcggaggg gccct 25

SEQ ID NO: 857 moltype = RNA length = 64
 FEATURE Location/Qualifiers
 source 1..64
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 857
 ccccgggccc ggcgttcct ccccttcct gcgccagtgg aggcggggg ggggcggggc 60
 gggg 64

SEQ ID NO: 858 moltype = RNA length = 64
 FEATURE Location/Qualifiers
 source 1..64
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 858
 ccccgggccc ggcgttcct ccccttcct gcgccagtgg aggcggggg ggggcggggc 60
 gggg 64

SEQ ID NO: 859 moltype = RNA length = 66
 FEATURE Location/Qualifiers
 source 1..66
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 859
 ccgagtggg cggggcaggt ccctgcagg actgtgacac tgaaggacct gcaccttcgc 60
 ccacag 66

SEQ ID NO: 860 moltype = RNA length = 54
 FEATURE Location/Qualifiers
 source 1..54
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 860
 acgcggtgc gggccggcgg ggtagaagcc acccgggccc gcccgggccc gcga 54

SEQ ID NO: 861 moltype = RNA length = 62
 FEATURE Location/Qualifiers
 source 1..62
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 861
 ggtaggggc gggctccggc gctgggaccc cactagggtg gcgccttggc cccgccccgc 60
 cc 62

SEQ ID NO: 862 moltype = RNA length = 71
 FEATURE Location/Qualifiers
 source 1..71
 mol_type = unassigned RNA
 organism = Homo sapiens

SEQUENCE: 862
 atggagggg gtgtgagcc agggggccca ggtctacag ttctcccgc tccctgcccc 60
 catactccca g 71

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SEQ ID NO: 863	moltype = RNA length = 74	
FEATURE	Location/Qualifiers	
source	1..74	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 863		
gggaggaaga agggaggagg agcggagggg cccttgcttt cccagagcct ctcccttcct		60
ccctccccc tccc		74
SEQ ID NO: 864	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 864		
cgggcccggc gttccc		16
SEQ ID NO: 865	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 865		
ccgggcccg cgttc		15
SEQ ID NO: 866	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 866		
gggtgcgggc cggcggggt		19
SEQ ID NO: 867	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 867		
tcggggccgg cgggg		15
SEQ ID NO: 868	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 868		
agggggcggg ctccggcgc		19
SEQ ID NO: 869	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 869		
gtagggggcg ggctc		15
SEQ ID NO: 870	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 870		
aaggaggag gagcggagg gcc		23
SEQ ID NO: 871	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = unassigned RNA	
	organism = Homo sapiens	
SEQUENCE: 871		
gggaggagga gcgga		15

The invention claimed is:

1. A method for detecting breast cancer in a human subject, comprising:

measuring expression level of miR-6757-5p in a blood, serum or plasma sample from the subject,

comparing the measured expression level of hsa-miR-6757-5p to a control expression level for a healthy subject;

detecting an increased level of hsa-miR-6757-5p in the sample from the subject as compared to the control expression level from the sample from the healthy subject;

wherein the increased level of hsa-miR-6757-5p indicates that the subject has breast cancer; and

wherein the method further comprises treating the subject for the breast cancer or performing a diagnostic procedure on the subject with the breast cancer;

wherein the treatment comprises surgery, radiotherapy, chemotherapy or a combination thereof; and

wherein the diagnostic procedure comprises mammography, ultrasonography, CT, MRI, abdominal ultrasonography, bone scintigraphy, PET, pathological examination which involves analyzing a lesion tissue under a microscope, or a combination thereof.

2. The method according to claim 1, wherein the expression level of hsa-miR-6757-5p in the sample is measured by using a kit comprising a nucleic acid(s) that specifically binds to hsa-miR-6757-5p.

3. The method according to claim 2, wherein the kit further comprises at least one nucleic acid capable of specifically binding to at least one polynucleotide selected from the group consisting of other breast cancer markers: miR-1307-3p, miR-4634, miR-663a, miR-4532, miR-7704, miR-3178, miR-6729-5p, miR-6090, miR-4732-5p, miR-3184-5p, miR-6727-5p, miR-6088, miR-4674, miR-8073, miR-4787-5p, miR-1469, miR-125a-3p, miR-885-3p, miR-6802-5p, miR-328-5p, miR-6787-5p, miR-8069, miR-6875-5p, miR-1246, miR-4734, miR-6756-5p, miR-3665, miR-6836-3p, miR-6821-5p, miR-6805-5p, miR-4728-5p, miR-6726-5p, miR-197-5p, miR-149-3p, miR-6850-5p, miR-4476, miR-6858-5p, miR-564, miR-4763-3p, miR-575, miR-6771-5p, miR-1231, miR-1908-3p, miR-150-3p, miR-3937, miR-887-3p, miR-3940-5p, miR-4741, miR-6808-5p, miR-6869-5p, miR-5090, miR-615-5p, miR-8072, miR-128-1-5p, miR-1238-5p, miR-365a-5p, miR-204-3p, miR-4492, miR-6785-5p, miR-6511a-5p, miR-4525, miR-1915-5p, miR-3180, miR-6879-5p, miR-1199-5p, miR-6746-5p, miR-711, miR-663b, miR-4707-3p, miR-6893-5p, miR-4675, miR-4638-5p, miR-4651, miR-6087, miR-4665-5p, miR-4758-5p, miR-6887-5p, miR-3620-5p, miR-1909-3p, miR-7641, miR-6724-5p, miR-1343-3p, miR-6780b-5p, miR-4484, miR-4690-5p, miR-4429, miR-1227-5p, miR-4725-3p, miR-6861-5p, miR-6812-5p, miR-3197, miR-8059, miR-3185, miR-4706, miR-4497, miR-3131, miR-6806-5p, miR-187-5p, miR-3180-3p, miR-6848-5p, miR-6820-5p, miR-6800-5p, miR-6717-5p, miR-6795-5p, miR-4632-5p, miR-665, miR-6778-5p, miR-3663-3p, miR-4689, miR-211-3p, miR-6511b-5p, miR-4750-5p, miR-6126, miR-614, miR-7110-5p, miR-744-5p, miR-6769a-5p, miR-4792, miR-5787, miR-6798-5p, miR-6781-5p, miR-4419b, miR-4446-3p, miR-4259, miR-5572, miR-6075, miR-296-3p, miR-6891-5p, miR-4745-5p, miR-6775-5p, miR-6870-5p, miR-920, miR-4530, miR-6819-5p, miR-6825-5p, miR-7847-3p, miR-6131, miR-4433-3p, miR-1228-5p, miR-6743-5p, miR-1268a, miR-3917, miR-6786-5p, miR-3154, miR-638, miR-6741-5p, miR-6889-5p, miR-6840-3p, miR-6510-5p, miR-3188, miR-551b-5p, miR-5001-5p, miR-

1268b, miR-7107-5p, miR-6824-5p, miR-6732-5p, miR-371a-5p, miR-6794-5p, miR-6779-5p, miR-4271, miR-5195-3p, miR-6762-5p, miR-939-5p, miR-1247-3p, miR-6777-5p, miR-6722-3p, miR-3656, miR-4688, miR-3195, miR-6766-5p, miR-4447, miR-4656, miR-7108-5p, miR-3191-3p, miR-1273g-3p, miR-4463, miR-2861, miR-3196, miR-6877-5p, miR-3679-5p, miR-4442, miR-6789-5p, miR-6782-5p, miR-486-3p, miR-6085, miR-4746-3p, miR-619-5p, miR-937-5p, miR-6803-5p, miR-4298, miR-4454, miR-4459, miR-7150, miR-6880-5p, miR-4449, miR-8063, miR-4695-5p, miR-6132, miR-6829-5p, miR-4486, miR-6805-3p, miR-6826-5p, miR-4508, miR-1343-5p, miR-7114-5p, miR-3622a-5p, miR-6765-5p, miR-7845-5p, miR-3960, miR-6749-5p, miR-1260b, miR-6799-5p, miR-4723-5p, miR-6784-5p, miR-5100, miR-6769b-5p, miR-1207-5p, miR-642a-3p, miR-4505, miR-4270, miR-6721-5p, miR-7111-5p, miR-6791-5p, miR-7109-5p, miR-4258, miR-6515-3p, miR-6851-5p, miR-6125, miR-4749-5p, miR-4726-5p, miR-4513, miR-6089, miR-6816-5p, miR-4466, miR-4488, miR-6752-5p and miR-4739, and/or miR-760, miR-602, miR-423-5p, miR-92a-2-5p, miR-16-5p, miR-451a, miR-135a-3p, miR-486-5p, miR-4257, miR-92b-5p, miR-1915-3p, miR-718, miR-940, miR-296-5p, miR-23b-3p, miR-92a-3p, miR-658, miR-6842-5p, miR-6124, miR-6765-3p, miR-7106-5p, miR-4534, miR-92b-3p, miR-3135b, miR-4687-3p, miR-762, miR-3619-3p, miR-4467, miR-557, miR-1237-5p, miR-1908-5p, miR-4286, miR-6885-5p and miR-6763-5p.

4. The method according to claim 1, comprising performing the diagnostic procedure on the subject.

5. The method according to claim 1, wherein the expression level of hsa-miR-6757-5p in the sample is measured by using a device comprising a nucleic acid(s) that specifically binds to hsa-miR-6757-5p.

6. The method according to claim 5, wherein the device further comprises at least one nucleic acid capable of specifically binding to at least one polynucleotide selected from the group consisting of other breast cancer markers: miR-1307-3p, miR-4634, miR-663a, miR-4532, miR-7704, miR-3178, miR-6729-5p, miR-6090, miR-4732-5p, miR-3184-5p, miR-6727-5p, miR-6088, miR-4674, miR-8073, miR-4787-5p, miR-1469, miR-125a-3p, miR-885-3p, miR-6802-5p, miR-328-5p, miR-6787-5p, miR-8069, miR-6875-5p, miR-1246, miR-4734, miR-6756-5p, miR-3665, miR-6836-3p, miR-6821-5p, miR-6805-5p, miR-4728-5p, miR-6726-5p, miR-197-5p, miR-149-3p, miR-6850-5p, miR-4476, miR-6858-5p, miR-564, miR-4763-3p, miR-575, miR-6771-5p, miR-1231, miR-1908-3p, miR-150-3p, miR-3937, miR-887-3p, miR-3940-5p, miR-4741, miR-6808-5p, miR-6869-5p, miR-5090, miR-615-5p, miR-8072, miR-128-1-5p, miR-1238-5p, miR-365a-5p, miR-204-3p, miR-4492, miR-6785-5p, miR-6511a-5p, miR-4525, miR-1915-5p, miR-3180, miR-6879-5p, miR-1199-5p, miR-6746-5p, miR-711, miR-663b, miR-4707-3p, miR-6893-5p, miR-4675, miR-4638-5p, miR-4651, miR-6087, miR-4665-5p, miR-4758-5p, miR-6887-5p, miR-3620-5p, miR-1909-3p, miR-7641, miR-6724-5p, miR-1343-3p, miR-6780b-5p, miR-4484, miR-4690-5p, miR-4429, miR-1227-5p, miR-4725-3p, miR-6861-5p, miR-6812-5p, miR-3197, miR-8059, miR-3185, miR-4706, miR-4497, miR-3131, miR-6806-5p, miR-187-5p, miR-3180-3p, miR-6848-5p, miR-6820-5p, miR-6800-5p, miR-6717-5p, miR-6795-5p, miR-4632-5p, miR-665, miR-6778-5p, miR-3663-3p, miR-4689, miR-211-3p, miR-6511b-5p, miR-4750-5p, miR-6126, miR-614, miR-7110-5p, miR-744-5p, miR-6769a-5p, miR-4792, miR-5787, miR-6798-5p, miR-6781-5p, miR-4419b, miR-4446-3p, miR-4259, miR-5572, miR-6075, miR-296-3p,

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miR-6891-5p, miR-4745-5p, miR-6775-5p, miR-6870-5p, miR-920, miR-4530, miR-6819-5p, miR-6825-5p, miR-7847-3p, miR-6131, miR-4433-3p, miR-1228-5p, miR-6743-5p, miR-1268a, miR-3917, miR-6786-5p, miR-3154, miR-638, miR-6741-5p, miR-6889-5p, miR-6840-3p, miR-6510-5p, miR-3188, miR-551b-5p, miR-5001-5p, miR-1268b, miR-7107-5p, miR-6824-5p, miR-6732-5p, miR-371a-5p, miR-6794-5p, miR-6779-5p, miR-4271, miR-5195-3p, miR-6762-5p, miR-939-5p, miR-1247-3p, miR-6777-5p, miR-6722-3p, miR-3656, miR-4688, miR-3195, miR-6766-5p, miR-4447, miR-4656, miR-7108-5p, miR-3191-3p, miR-1273g-3p, miR-4463, miR-2861, miR-3196, miR-6877-5p, miR-3679-5p, miR-4442, miR-6789-5p, miR-6782-5p, miR-486-3p, miR-6085, miR-4746-3p, miR-619-5p, miR-937-5p, miR-6803-5p, miR-4298, miR-4454, miR-4459, miR-7150, miR-6880-5p, miR-4449, miR-8063, miR-4695-5p, miR-6132, miR-6829-5p, miR-4486, miR-6805-3p, miR-6826-5p, miR-4508, miR-1343-5p, miR-

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7114-5p, miR-3622a-5p, miR-6765-5p, miR-7845-5p, miR-3960, miR-6749-5p, miR-1260b, miR-6799-5p, miR-4723-5p, miR-6784-5p, miR-5100, miR-6769b-5p, miR-1207-5p, miR-642a-3p, miR-4505, miR-4270, miR-6721-5p, miR-7111-5p, miR-6791-5p, miR-7109-5p, miR-4258, miR-6515-3p, miR-6851-5p, miR-6125, miR-4749-5p, miR-4726-5p, miR-4513, miR-6089, miR-6816-5p, miR-4466, miR-4488, miR-6752-5p and miR-4739, and/or miR-760, miR-602, miR-423-5p, miR-92a-2-5p, miR-16-5p, miR-451a, miR-135a-3p, miR-486-5p, miR-4257, miR-92b-5p, miR-1915-3p, miR-718, miR-940, miR-296-5p, miR-23b-3p, miR-92a-3p, miR-658, miR-6842-5p, miR-6124, miR-6765-3p, miR-7106-5p, miR-4534, miR-92b-3p, miR-3135b, miR-4687-3p, miR-762, miR-3619-3p, miR-4467, miR-557, miR-1237-5p, miR-1908-5p, miR-4286, miR-6885-5p and miR-6763-5p.

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